LAN Switch Operations and Maintenance

From the Enterasys legacy to the current CLI of eight vendors



4	4	17	8
days	modules	chapters	vendors

Rodolfo Nützmann

2026 Edition

Prepared for: [Client] · In partnership with: [Partner]

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MODULE 01 · DAY 1

Access, operation and basic maintenance



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MODULE 01 · DAY 1

Course agenda

Day 1

TODAY

Access, operation and basic maintenance

- Platforms: 2007 legacy and current
- CLI, users and session
- Clock, identity and connectivity
- Save, erase and reboot
- Password recovery
- Files, backups and firmware

Day 2

NEXT

Layer 2 switching

- Interfaces and naming
- Speed, duplex and jumbo frames
- Storm control, isolation, mirroring
- LACP link aggregation
- 802.1Q VLANs
- Spanning Tree and BPDU protection

Day 3

NEXT

IP routing

- Addressing, masks and subnets
- Route determination and ARP
- Proxy-ARP
- L3 interfaces and management
- Static routes, RIP and OSPF
- Routing table, ECMP and ARP

Day 4

NEXT

Redundancy, security and management

- OSPF and VRRP
- Access control lists
- Syslog, SNMP and AAA
- Multicast: IGMP and PIM
- Troubleshooting and support
- Final lab and assessment

Introductions

Rodolfo Nützmann

Senior Technical Instructor and Advisor, Red Education

Enterprise networking and security since 1996

Full-time technical instructor since 2020

F5 DevCentral MVP 2022, 2023 and 2024

Authorized instructor

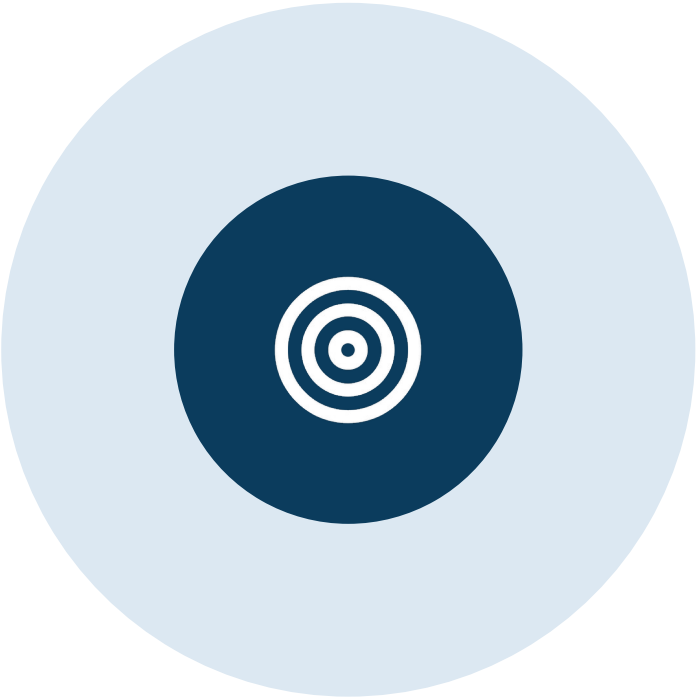
- F5
- Extreme Networks
- Fortinet
- Netskope

And you? Name, role, the equipment you operate today and what you expect from the course.

LEGACY 2007

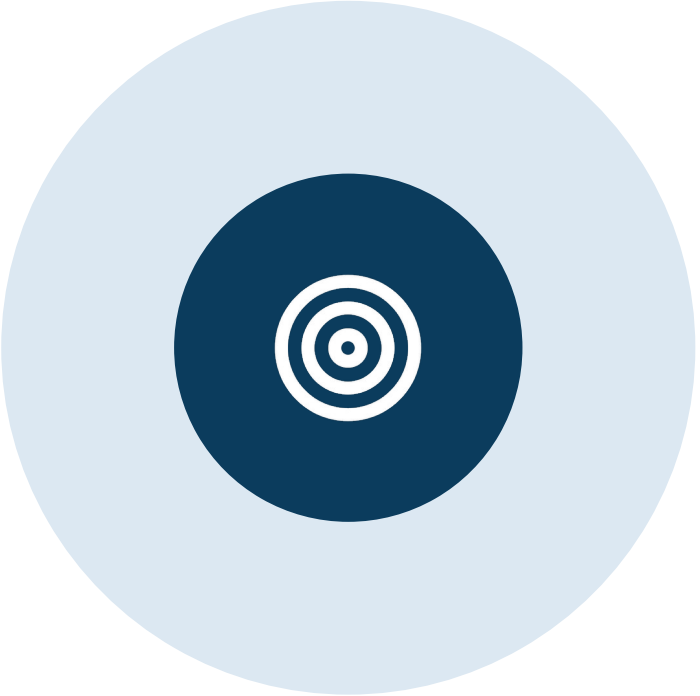
Course objectives

- By the end of the training, participants will be familiar with:
- **Features**, **configuration** and **maintenance** of enterprise LAN switches
  - Historical reference (2007): Enterasys **SecureStack** and **Matrix N** with **DFE-P**
  - Current equivalents: **Extreme Switch Engine**, **Extreme Fabric Engine**, **Cisco IOS XE**, **Aruba AOS-CX**
  - **Juniper Junos**, **MikroTik RouterOS**, **Arista EOS** and **Huawei VRP**



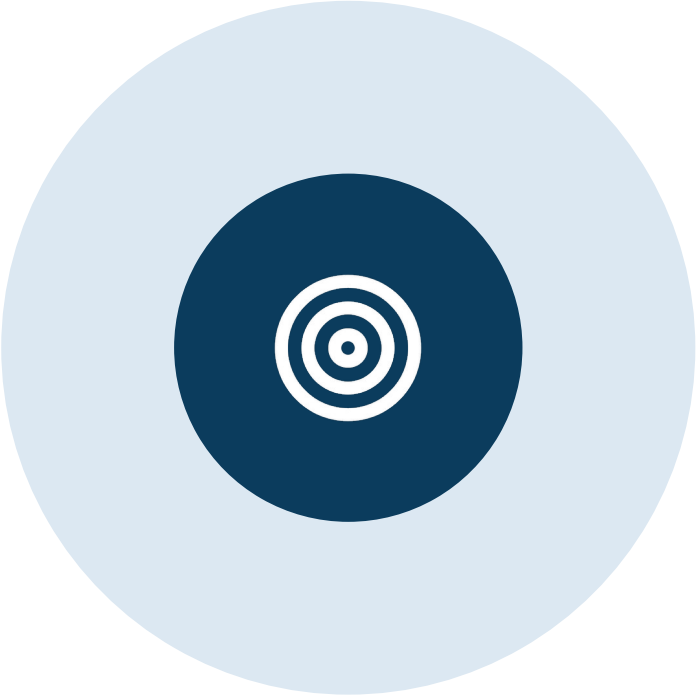
Course objectives

- Participants will be able to:
- Access, configure and manage the devices
- Upgrade the switch firmware
- Manage configuration files
- Reset the configuration to factory defaults
- Recover access to devices with an unknown password



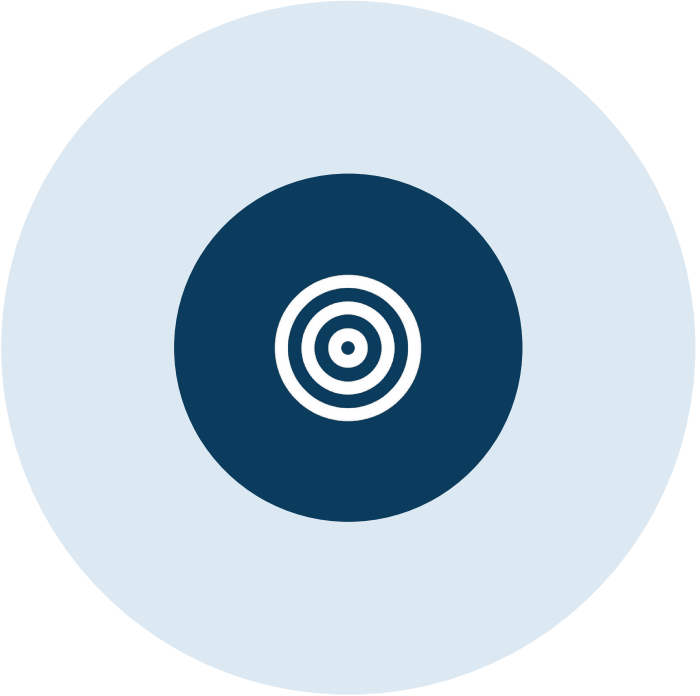
Course objectives

- Participants will be able to configure features for:
- **Switching:** Ports, VLANs, Spanning Tree, Link Aggregation
- **Routing:** IP interfaces and routes, OSPF, VRRP
- **Management:** SNMP, Traps, Syslog, RADIUS
- **Security:** ACLs and switch protection features
- **Multicast:** IGMP



Course objectives

- And finally, participants will be able to:
- Identify and fix network problems
  - E.g.: poor application performance, connectivity problems
- Identify and fix operational problems on the devices
  - E.g.: misconfigurations, operational failures
- Use a protocol analyzer
- Get support from the vendor (TAC) and from [Partner]



Course format

- The training will be more hands-on than theoretical
- Interaction and discussion of the topics are encouraged
- Working directly with the equipment is encouraged
- Please bring **questions and real situations** from your day-to-day operations, maintenance and network planning
- Labs: Throughout the course, following the theory
- Final lab: Covering all the topics addressed



1.1

Platforms and vendors



From 2007 to 2026: the eight course vendors and how to read the MODERN CLI slides

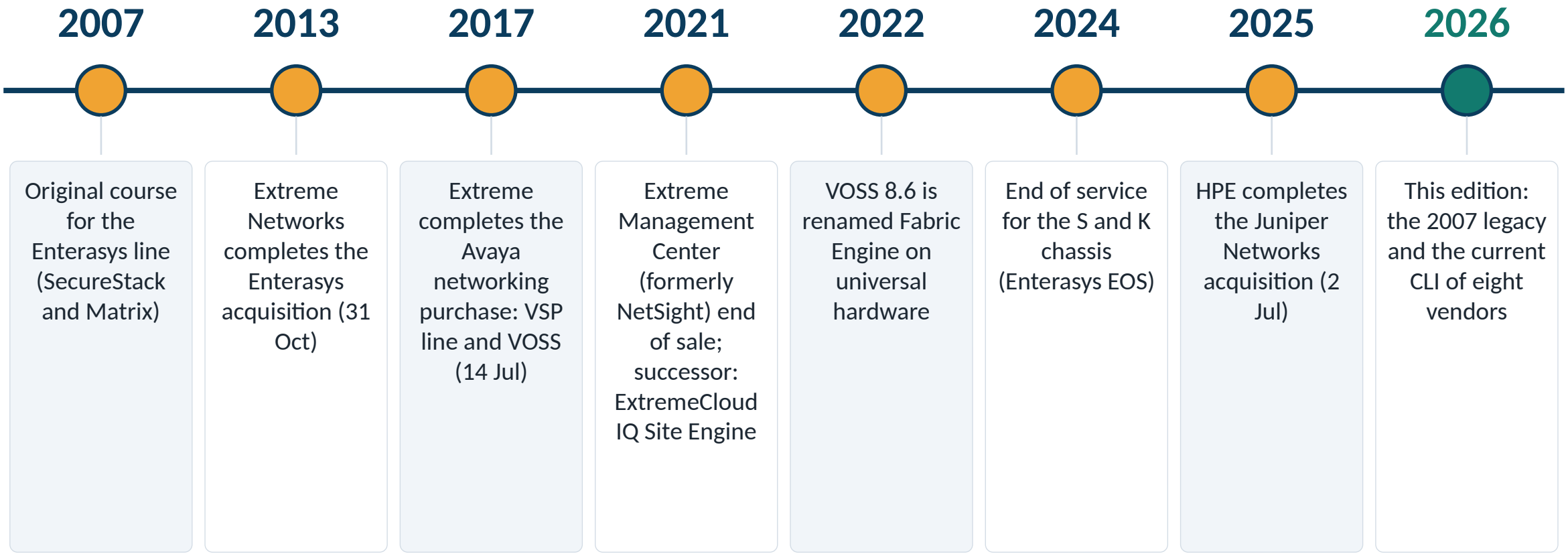
- From 2007 to 2026: what changed
- Reference platforms and the eight vendor profiles
- CLI families: what carries over to other vendors
- Cloud-managed switching
- How to read the MODERN CLI slides
- Initial knowledge assessment

Chapter 1.1 agenda

1. From 2007 to 2026: what changed
2. Reference platforms and the eight vendor profiles
3. CLI families: what carries over to other vendors
4. Cloud-managed switching
5. How to read the MODERN CLI slides
6. Initial knowledge assessment

NEW

From 2007 to 2026: what changed



NEW

Reference platforms of the MODERN CLI slides

Extreme Switch Engine

Extreme Switch Engine 33.7.1 on universal hardware 5320/5420/5520/5720; maintenance 32.7.4.15-Patch1-14

*Verb-object CLI (configure/show/enable/disable/create/delete); no config mode; a * in the prompt flags unsaved changes.*

Extreme Fabric Engine

Extreme 5520-48T running Fabric Engine 9.4.1.0

IOS-like modal CLI: enable, configure terminal, interface vlan 10. Prompt Switch:1(config)#; routing sits under the interface or 'router ospf'.

Cisco IOS XE

Cisco Catalyst 9300 running IOS XE 17.18.4 and 26.1.x

IOS-style CLI: user EXEC (>), privileged EXEC (#), global and sub-modes; ? gives help; no reverts and default restores the default.

Aruba AOS-CX

HPE Aruba Networking CX 6300M, AOS-CX 10.16

Hierarchical IOS-like CLI: operator (>), manager (#) and config contexts; 'do' runs manager commands from config; '?' for help.

Juniper Junos

EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx

Junos CLI: operational mode (show, ping, request) and configuration mode (configure; set, delete, edit on a statement hierarchy).

MikroTik RouterOS

MikroTik CRS354-48G-4S+2Q+RM running RouterOS 7.23.8; documentation: manual.mikrotik.com 7.24

Tree of menus: type the full path (/interface bridge port add ...) or move into a menu; ? and Tab help.

Arista EOS

Arista EOS 4.36.x: campus 720XP and 7000-series switches

IOS-like modal CLI: exec (>), enable (#), configure terminal; configuration sessions (commit/abort) and checkpoints.

Huawei VRP

Huawei S5700/S6700 campus switches running VRP. Newer train: V600R026C00.

Prompt <HUAWEI> (user view); system-view enters configuration; undo negates, display shows; quit and return leave views.

VENDOR PROFILE

Extreme Switch Engine

MODERN CLI

Extreme Switch Engine (formerly ExtremeXOS / EXOS on universal hardware)

Extreme Switch Engine 33.7.1 (recommended 33.7.1.6-Patch1-32) on universal hardware 5320/5420/5520/5720; maintenance 32.7.4.15-Patch1-14

Lineage

ExtremeXOS (EXOS), renamed Switch Engine in release 31.6 on Universal hardware; VOSS became Fabric Engine.

CLI style

Verb-object CLI (configure/show/enable/disable/create/delete); no config mode; a * in the prompt flags unsaved changes.

How the config is saved

save configuration writes named .cfg (XML) files; auto-save is disabled by default; .xsf is the text form.

Console

RJ-45 serial and Micro-B USB (5420; check each model's guide); 115200 baud, 8N1, XON/OFF flow control.

Out-of-band management

Mgmt port in VLAN Mgmt (VR-Mgmt); most 5320 have none (except 5320-24T-4X-XT and -24T-24S-4XE-XT); front ports work.

First login

Default accounts admin and user; admin has no password (press Enter). First login asks safe-defaults and Switch Engine or Fabric Engine.

Lab image

Official: 'Virtual Switch Engine (EXOS)' / EXOS-VM (GitHub extremenetworks/Virtual_EXOS), up to 33.6.1; not GTAC-supported, features partly missing.

Milestones

Release 31.6 renamed ExtremeXOS to Switch Engine. Management Center ended sale 30 Sep 2021; legacy EOS S/K/7100 ended service in 2024.

VENDOR PROFILE

Extreme Fabric Engine

MODERN CLI

Extreme Fabric Engine (formerly VOSS)

Extreme 5520-48T (universal hardware) running Fabric Engine 9.4.1.0 (v9.4 documentation)

Lineage

VOSS (VSP Operating System Software). Since 8.6 it is called Fabric Engine on universal hardware; non-universal hardware keeps the VOSS name.

CLI style

IOS-like modal CLI: enable, configure terminal, interface vlan 10. Prompt Switch:1(config)#; routing sits under the interface or 'router ospf'.

How the config is saved

Settings are retained only after 'save config' (Privileged EXEC), which writes the configuration file.

Console

RJ-45 serial (some models also have USB or micro USB): 115200 baud, 8 data bits, 1 stop bit, no parity, no flow control.

Out-of-band management

OOB port: L1/L2 under 'interface mgmtEthernet mgmt'; IP and route under 'mgmt oob' (segmented Management Instance), if the model has the port.

First login

The 5520 ships with Switch Engine; a first-login question lets you switch to Fabric Engine. Default CLI login: rwa/rwa.

Lab image

Virtual VOSS for GNS3 and VMware on Extreme's GitHub, up to 9.4.0; not supported by GTAC (lab use).

Milestones

2017-07-14: Extreme completed buying Avaya's networking business (VSP). From 8.6 (notes of March 2022) VOSS is called Fabric Engine on universal hardware.

VENDOR PROFILE

Cisco IOS XE

MODERN CLI

Cisco IOS XE (Catalyst 9000)

Cisco Catalyst 9300 (Catalyst 9000 family: 9200/9300/9400/9500) running IOS XE 17.18.4 and 26.1.x (26.1.1 and 26.1.2 documented)

IOS XE 26.1.x
Release notes: 26.1.1 on 10 Apr 2026, 26.1.2 on 5 Aug 2026. Extended Support, 48 months. Not in the recommended list (4 Sep 2026: 17.18.4, 17.15.6). 17.18.x supported to 30 Mar 2030.

Lineage

Succeeds the Catalyst 3850 (notice EOL13188); releases had code names: Everest 16.5, Bengaluru 17.6, Cupertino 17.7–17.9, Dublin 17.10–17.12.

CLI style

IOS-style CLI: user EXEC (>), privileged EXEC (#), global and sub-modes; ? gives help; no reverts and default restores the default.

How the config is saved

running-config (active) and startup-config (used at boot); persist with copy running-config startup-config or write memory.

Console

RJ-45 console (RJ-45-to-DB9 cable); default 9600 bps, 8 data bits, no parity; the boot loader is also reachable only at 9600 bps.

Out-of-band management

Gi0/0 (Ethernet management port), a VRF interface; the guide uses vrf forwarding Mgmt-vrf. No routing between it and the network ports.

First login

No factory credentials: enable secret and Telnet password are undefined; create a user and an enable secret at first access.

Lab image

CAT 9000v in CML 2.10 (beta, no formal support): UADP and Q200 nodes, IOS XE 17.18.02; no default user/password; dataplane ~250 Kbps.

Milestones

IOS XE 17.18.x: end-of-sale/life announced 30 Oct 2026 (support until 30 Mar 2030); Cisco advises moving to IOS XE 26.1.1 or later.

VENDOR PROFILE

Aruba AOS-CX

MODERN CLI

HPE Aruba Networking AOS-CX

HPE Aruba Networking CX 6300M (CX 6300/6400 series), AOS-CX 10.16 (LSR; latest build listed: 10.16.2010)

Lineage

Official docs: 'HPE Aruba Networking CX Switch Operating System (AOS-CX)'; HPE pages already say 'HPE Networking CX Switches'.

CLI style

Hierarchical IOS-like CLI: operator (>), manager (#) and config contexts; 'do' runs manager commands from config; '?' for help.

How the config is saved

running-config and startup-config; 'write memory' (alias of copy running-config startup-config); checkpoints with rollback.

Console

Guide default: 115200 bps, 8 data bits, 1 stop bit, no parity, no flow control.

Out-of-band management

OOBM port ('interface mgmt', VRF mgmt): DHCP by default; SSH enabled on the management VRF; static IP with 'ip static'.

First login

User 'admin' with no password; at first login the switch prompts you to set one.

Lab image

AOS-CX Switch Simulator (OVA): learning only, no support; the download needs a support-portal account and an additional licence.

Milestones

2025: HPE acquired Juniper Networks; 'HPE Networking' groups HPE Aruba Networking and HPE Juniper Networking. 10.16 docs: Jan 2026.

VENDOR PROFILE

Juniper Junos

MODERN CLI

HPE Juniper Networking Junos OS (EX Series, ELS)

EX2300/EX3400/EX4100/EX4400 switches (ELS), Junos OS 23.4R2-Sx

Lineage

Juniper Networks, acquired by HPE (closed 2 Jul 2025); now HPE Juniper Networking. The Junos OS name is unchanged.

CLI style

Junos CLI: operational mode (show, ping, request) and configuration mode (configure; set, delete, edit on a statement hierarchy).

How the config is saved

Candidate/commit model: edit the candidate configuration; commit checks, saves and activates; commit confirmed auto-rolls back; up to 49 rollbacks.

Console

RJ-45 CON port: 9600 baud, 8 data bits, no parity, 1 stop bit, no flow control, DCD disregard (EX4400 guide).

Out-of-band management

Out-of-band port me0 (labeled MGMT, rear panel on the EX4400): set interfaces me0 unit 0 family inet address <ip>/<len>.

First login

On the console: log in as root with no password; then cli, configure, set system root-authentication (mandatory in the factory config).

Lab image

vJunos-switch: free KVM VM, lab use only (not production), up to 100 Mbps, no multicast, no in-place upgrade, ge- ports only (up to 96).

Milestones

HPE announced the deal on 9 Jan 2024 (US$40 per share) and closed it on 2 Jul 2025. Junos docs are now served from networkingdocs.hpe.com.

VENDOR PROFILE

MikroTik RouterOS

MODERN CLI

MikroTik RouterOS v7

MikroTik CRS354-48G-4S+2Q+RM (Marvell 98DX3257 switch chip) running RouterOS 7.23.8 (long-term); documentation: manual.mikrotik.com
7.24

Lineage

RouterOS v7 (7.x since 2021); v6 is still maintained (6.49.23, 2026-10-09). On CRS3xx it coexists with SwOS (dual boot).

CLI style

Tree of menus: type the full path (/interface bridge port add ...) or move into a menu; ? and Tab help.

How the config is saved

Safe Mode (Ctrl+X/F4): Ctrl+D exits and undoes changes; a dropped session undoes them too. /system history has undo and redo. /export is text, backup is binary.

Console

RJ45 serial, 115200 bit/s, 8 data bits, 1 stop bit, no parity. The RouterBOOT page suggests disabling hardware flow control.

Out-of-band management

ETH MGMT 10/100 port on the CPU, outside the chip; the MTU table lists ether49 (MTU 2028) apart from ether1 to ether48. Keep it for management only.

First login

User admin; blank password or the one on the device sticker. Switch default: 192.168.88.1/24 on the bridge, admin has a password.

Lab image

CHR (x86-64 VM): free license with no expiry, 1 Mbit/s upload per interface, other features unlocked; no switch chip, no offload.

Milestones

RouterOS 7.1 was the first v7 stable (2021-12-01); current long-term line is 7.23 (7.23.8, 2026-10-09); v6 continues as 6.49.x.

VENDOR PROFILE

Arista EOS

MODERN CLI

Arista EOS

Arista EOS 4.36.x (EOS User Manual 4.36.2F): campus 720XP and 7000-series switches

Lineage

EOS = Extensible Operating System. The former 'vEOS Router' documentation is now titled CloudEOS (it is not the lab image).

CLI style

IOS-like modal CLI: exec (>), enable (#), configure terminal; configuration sessions (commit/abort) and checkpoints.

How the config is saved

'write' (or 'copy running-config startup-config') saves the startup-config. In a configuration session, 'commit' applies and 'abort' discards.

Console

Console port at 9600 baud, 8 data bits, no parity, 1 stop bit, no flow control (9600/N/8/1).

Out-of-band management

'interface management 1' (Management1), a 10/100/1000BASE-T port; its IP is set on the interface itself with 'ip address'.

First login

On the console: user admin with no password. With no startup-config the switch enters ZTP until 'zerotouch cancel' is run.

Lab image

vEOS-lab (VM) and cEOS-lab (container), Arista's official names. Download needs a login; free use and file names are not on any public page.

Milestones

EOS 4.36.x released 08-Apr-2026 (support to 08-Apr-2029). EOS 4.31 reached End of Support on 13-Oct-2026 (notice dated 23-Oct-2026).

VENDOR PROFILE

Huawei VRP

MODERN CLI

Huawei VRP (S5700/S6700 campus switches; CloudEngine S3700/S5700/S6700)

Huawei S5700/S6700 campus switches running VRP (documentation S5700 and S6700 V200R025C00). Newer train: V600R026C00 (CloudEngine S3700/S5700/S6700).

Lineage

Quidway S5700/S6700 (name in Huawei's 2015 security advisory). V200R025C00 docs: S5700 and S6700; V600R026C00: CloudEngine S3700/S5700/S6700.

CLI style

Prompt <HUAWEI> (user view); system-view enters configuration; undo negates, display shows; quit and return leave views.

How the config is saved

Changes apply to the running config at once; 'save' writes the next-startup file (display startup). Unsaved changes are lost on restart.

Console

RJ45 console port (DB9 or USB-serial cable): 9600 bit/s, 8 data bits, no parity, 1 stop bit, no flow control.

Out-of-band management

MEth0/0/1 is the dedicated management interface: it takes an IP, carries no service traffic and is isolated from service ports by default.

First login

First console login makes you set the password (8 to 16 characters); the auth mode becomes password and the user level 15.

Lab image

eNSP Pro (Huawei simulator, VM image) is released only to certified partners, not to common registered users or customers.

Milestones

Quidway S5700/S6700 in Huawei's 2015 advisory; now S5700/S6700. New simulator eNSP Pro released June 2023; the old one not updated since 2019.

NEW

CLI families: what you learn here carries over to other vendors

Like Cisco IOS XE

Dell PowerSwitch
SmartFabric OS10
```
switchport access vlan 10
```

Dell N-series (campus)
Dell Networking OS6
```
switchport access vlan 10
```

RUCKUS ICX
FastIron
```
untagged ethernet 1/1/1
```

NETGEAR M4250/M4300/M4350
CLI no OS brand name
```
switchport access vlan 10
```

Ruijie RG-S
RGOS
```
switchport access vlan 10
```

Hirschmann
HiOS
```
vlan participation include 10
```

Cisco Catalyst IE3400/IE9300
Cisco IOS XE
```
switchport access vlan 10
```

TP-Link Omada
Omada (standalone CLI)
```
switchport pvid 10
```

Like Huawei VRP

H3C
Comware
```
port access vlan 10
```

Like Extreme Switch Engine

ALE OmniSwitch
AOS Release 8
```
vlan 10 members port 1/1/1 untagged
```

Own family

NVIDIA Spectrum
Cumulus Linux (NVUE)
```
nv set interface swp1 bridge domain br_default access 10
```

SONiC
SONiC
```
sudo config vlan member add -u 10 Ethernet0
```

Dashboard / controller

Cisco Meraki MS
Meraki Dashboard
```
(no CLI) Type: Access, VLAN: 10
```

Ubiquiti UniFi
UniFi Network
```
(no CLI) Native VLAN: 10
```

NEW

Cloud-managed switching: the other operating model

CLI (this course)

- Per-device configuration, command by command
- Local save, backup and rollback
- Automation through device scripts and APIs

Dashboard / controller

- Central configuration per network, templates and groups
- The device receives its configuration from the dashboard
- Automation through the dashboard API

Central management platforms (course vendors in grey, cloud-native vendors in blue)

Extreme Networks
ExtremeCloud IQ
Public or private cloud, or ExtremeCloud Edge; REST API (OpenAPI)

Cisco Catalyst
Meraki dashboard · Catalyst Center
Selected Catalyst 9000 in the Meraki dashboard (cloud-native IOS XE)

HPE Aruba Networking
HPE Aruba Central
Cloud, on-prem, as a service or virtual private cloud

HPE Juniper Networking
Juniper Mist
Switches configured through the Mist portal (Wired Assurance)

Arista
CloudVision
SaaS or on-prem appliance; OpenConfig, gRPC and REST APIs

Huawei
iMaster NCE-Campus
Plug-and-play, multi-tenant and 170+ RESTful APIs

MikroTik
CMR · REST API
CMR: fleet server on a RouterOS device; REST API since v7

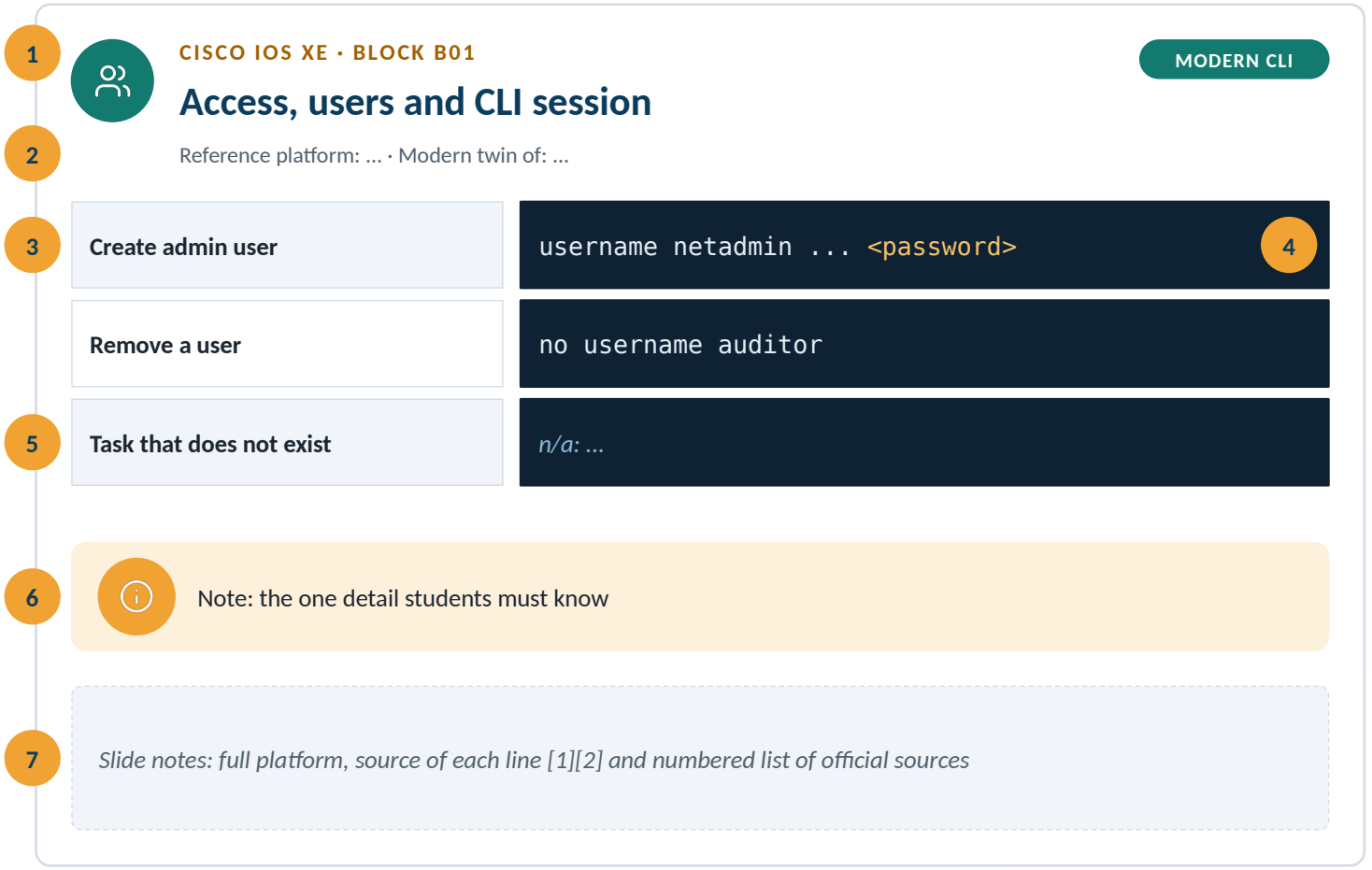
Cisco Meraki MS
Meraki Dashboard
Configured in the Dashboard; the local page covers only the basics

Ubiquiti UniFi
UniFi Network
App on a Cloud Gateway, CloudKey, UniFi OS Server or official hosting

TP-Link Omada
Omada Controller
Cloud, hardware or software; once adopted, local configuration is disabled

MODERN CLI

How to read the MODERN CLI slides



1. Vendor and topic block (the same block appears for all 8 vendors)
2. Reference platform and release, and the 2007 slide this twin modernizes
3. Task on the left, exact command on the right; values from the class lab
4. In amber: values you replace (<password>, <secret>)
5. n/a: the platform lacks the feature; the reason is shown
6. Amber callout: the one detail students must know
7. Sources: always in the notes, with the URL of the official documentation
8. Each task starts in global configuration mode: leave the sub-mode before the next one (exit, quit or top)
9. Cisco: valid for IOS XE 17.18.4 and 26.1.x; a "! IOS XE 26.1.x" line appears only where 26.1 differs

NEW

Pre-course assessment (1/2): Ethernet, switching, STP and addressing

1 **According to RFC 894, what is the maximum size of an IP datagram in an Ethernet frame?**

- A 46 octets
- B 1500 octets
- C 1518 octets
- D 9000 octets

2 **A switch receives a unicast frame whose destination MAC is not in its address table. What does it do?**

- A Drops the frame
- B Floods the frame, as it would a broadcast
- C Sends the frame to the default gateway
- D Sends an ARP request and waits for the reply

3 **In a Spanning Tree domain, which switch becomes the root bridge?**

- A The one that has been up the longest
- B The one with the most active ports
- C The highest numerical priority value
- D Lowest priority value; on a tie, the lowest MAC

4 **Which of these addresses is NOT in the RFC 1918 private ranges?**

- A 10.255.255.254
- B 172.31.0.1
- C 172.32.0.1
- D 192.168.200.1

NEW

Pre-course assessment (2/2): addressing, routing and management

5 How many usable host addresses does the 10.1.10.0/24 network have?

- A 256
- B 255
- C 254
- D 253

6 The table has 10.0.0.0/8 via R1, 10.1.0.0/16 via R2, 10.1.10.0/24 via R3 and 0.0.0.0/0 via R4. Where does a packet to 10.1.10.77 go?

- A R1
- B R2
- C R3
- D R4

7 The TTL of a unicast IP packet reaches zero in a router. What must the router do?

- A Forward it anyway, one last time
- B Discard; send ICMP Time Exceeded (type 11) to the source
- C Always discard silently
- D Return the packet to the previous hop

8 What does SSH (RFC 4253) provide for remote switch management?

- A Encryption, server authentication and integrity protection
- B Only traffic compression
- C Automatic neighbor discovery
- D Automatic configuration backup

1.2

Basic concepts



Switch access, users, CLI session, clock and identity

- Switch access: console and remote management
- Access privileges and user management
- CLI session: history, paging and shortcuts
- Clock, time zone and NTP
- Prompt, name, banner and management IP

Chapter 1.2 agenda

1. Switch access: console and remote management
2. Access privileges and user management
3. CLI session: history, paging and shortcuts
4. Clock, time zone and NTP
5. Prompt, name, banner and management IP

MODERN CLI

- B01 Access, users and CLI session
- B02 Clock, identity and connectivity

Lab

- Lab: basic configuration

LEGACY 2007

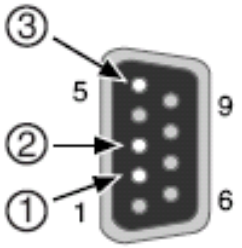
Accessing the switch

- Accessing the switch: RS-232 serial console
- Access through a standard VT100 terminal emulator;
- Default: **9600** baud, **8** bits, **N** (no parity), **1** stop bit

SecureStack

- DB9 console port on the device
- Serial cable, DB9 female to DB9 female (SSCON-CAB)
- Today: Tera Term, PuTTY or OpenSSH; USB or RJ-45 console with a USB-to-serial adapter

Figure 3-8 DB9 Male Console Port Pinout Assignments



1 Pin 2, Received Data (input)
2 Pin 3, Transmitted Data (output)
3 Pin 5, Signal Ground
All other pins not connected.

DB9 console pinout (SecureStack guide, 2007)

LEGACY 2007

Accessing the switch

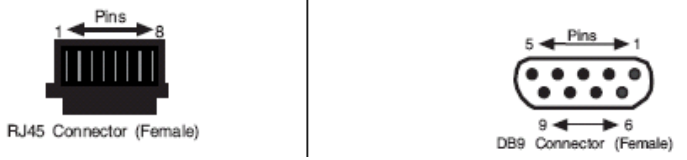
- Accessing the switch: RS-232 serial console

DFE: Distributed Forwarding Engine

- 9600, 8, N, 1 - RJ-45 console port on the module
- Straight-through UTP cable with RJ-45 connectors (Ethernet)
- Standard Enterasys RJ-45 to DB-9 adapter

COM Port Adapter Wiring and Signal Diagram

RJ45		DB9	
Pin	Conductor	Pin	Signal
1	Blue	2	Receive (RX)
4	Red	3	Transmit (TX)
5	Green	5	Ground (GRD)
2	Orange	7	Request to Send (RTS)
6	Yellow	8	Clear to Send (CTS)

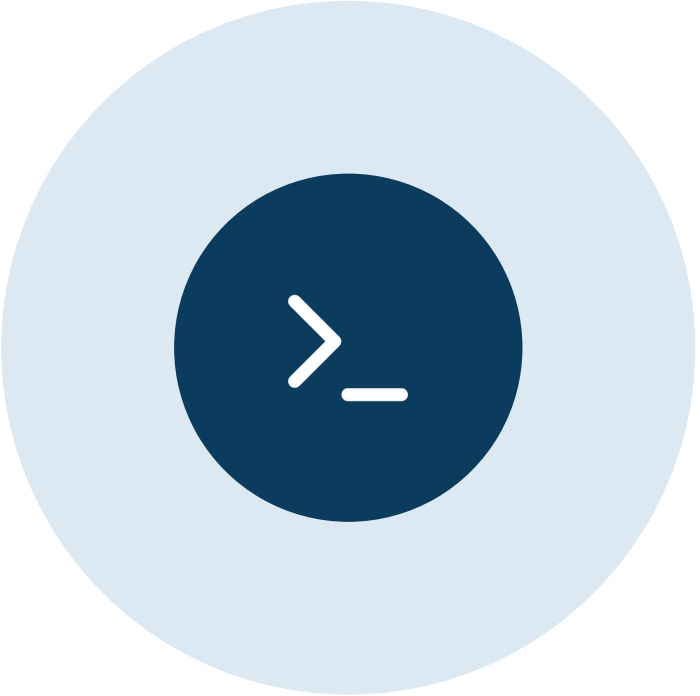


DFE console RJ45 to DB9 adapter (DFE guide, 2007)

LEGACY 2007

Accessing the switch

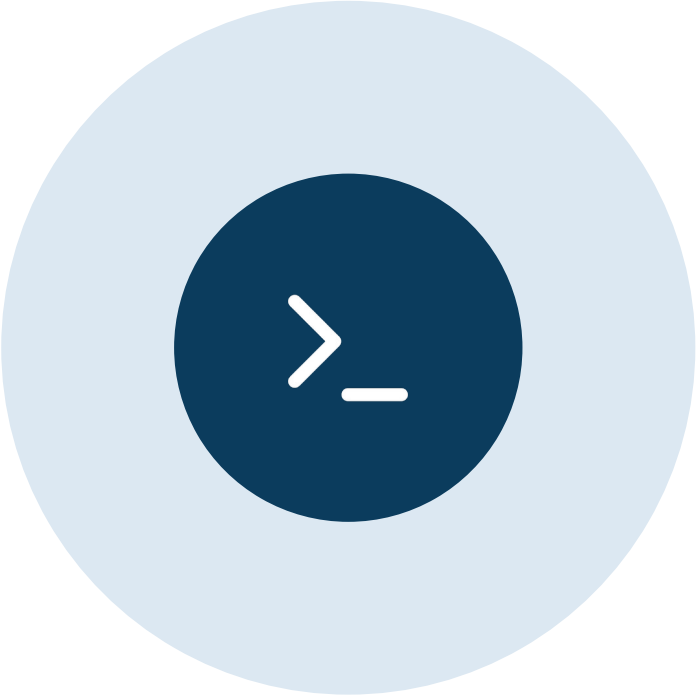
- Accessing the switch: over the network
- **Telnet** (unencrypted; disable in production)
- **SSH** (encrypted)
- **NetSight:** graphical management platform (today: ExtremeCloud IQ Site Engine)
- (unencrypted: via SNMP v1 and v2c)
- (encrypted: via SNMP v3 with privacy, authPriv level)



LEGACY 2007

Accessing the switch

- CLI modes
- Access privileges are configured per user (ro, rw and su)
- The **?** is the operator's best tool
- Remember the syntax of every command on every device?
- Or "build" the commands with the help of the question mark?
- Screen scroll size
- Can be adjusted with `set length xx` → number of lines
- In some situations `set length 0` is very useful



Access privileges

- **Super-User** access privilege
- Default user: admin Default password: None
- Change this user’s password during the initial switch configuration
- **Read-Write** access privilege
- Default user: rw Default password: None
- Disable this user during the initial switch configuration
- **Read-Only** access privilege
- Default user: ro Default password: None
- Disable this user during the initial switch configuration



LEGACY 2007

User management

Sample output (Matrix N, 2007):

```
Matrix(su)->show system login
Password history size: 0
Password aging       : disabled

Username        Access       State
admin           super-user   enabled
ro              read-only    enabled
rw              read-write   enabled
```

Create a new user:

```
set system login username {super-user | read-write |
read-only}                {enable | disable}
```

Remove a user:

```
clear system login username
```

Change a user’s password:

```
set password username
```

LEGACY 2007

Working with the CLI

Timeout for inactive console sessions, in minutes (15 min)

```
set logout timeout
show logout
```

List the users connected to the switch

```
show users
```

Send a message to another user logged in to the switch

```
tell {dest | all} message  ← (DFE only)
```

Disconnect a user from the switch

```
disconnect {ip-addr | console}
```

LEGACY 2007

Working with the CLI

List the commands entered in the CLI session (20-line buffer by default, maximum 100)

```
history  ← (show history shows only the buffer size)
```

Change the size of the CLI command buffer

```
set history size [default]
```

Clear the screen

```
cls
```

Disconnect the CLI session

```
exit
quit
```

MODERN CLI

Access, users and CLI session (1/2)

	Create a local admin user	Enable SSH and disable Telnet
Extreme Switch Engine	`create account admin netadmin <password>`	`enable ssh2` `disable telnet`
Extreme Fabric Engine	`username add netadmin level rwa enable`	`boot config flags sshd` `no boot config flags telnetd`

Cisco IOS XE	`username netadmin privilege 15 secret <password>`	`ip ssh version 2`
		`line vty 0 15`
		` transport input ssh`

Aruba AOS-CX	`user netadmin group administrators password`	`ssh server vrf mgmt` `no telnet server vrf mgmt`

Standards: RFC 4253 · RFC 4251 · RFC 854 · IANA-PORTS

MODERN CLI

Access, users and CLI session (2/2)

	Create a local admin user	Enable SSH and disable Telnet

Juniper Junos	`set system login class ADMIN permissions all`
	`set system login user netadmin class ADMIN`
	`set system login user netadmin authentication plain-text-password`
		`set system services ssh`
		`delete system services telnet`

MikroTik RouterOS	`/user add name=netadmin group=full password=<password>`	`/ip service disable telnet` `/ip ssh set strong-crypto=yes`
Arista EOS	`username netadmin role network-admin secret <password>`	`management telnet` `  shutdown`
Huawei VRP	`aaa` `  local-user netadmin password irreversible-cipher <passwd> privilege level 15`	`stelnet server enable`

Standards: RFC 4253 · RFC 4251 · RFC 854 · IANA-PORTS

MODERN CLI

Access, users and CLI session

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	*n/a: No config mode: configure commands run directly at the # prompt*
Create a local admin user	`create account admin netadmin <password>`
Create a read-only user	`create account user auditor <password>`
Change a user's password	`configure account netadmin <password>`
Remove a user	`delete account auditor`
Idle session timeout (15 min)	`configure cli idle-timeout 15`
List active sessions	`show session`
Terminate another user's session	`clear session 4`
Command history	`history`
Disable paging (same as set length 0)	`disable cli paging`

There is no config mode: an admin (# prompt) types configure/create directly. SSH2 is disabled and Telnet enabled by default; the default idle timeout is 20 min.

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 13 to 17 (EOS 2007)

Enable SSH and disable Telnet	`enable ssh2` `disable telnet`

There is no config mode: an admin (# prompt) types configure/create directly. SSH2 is disabled and Telnet enabled by default; the default idle timeout is 20 min.

MODERN CLI

Access, users and CLI session

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	`enable` `configure terminal`
Create a local admin user	`username add netadmin level rwa enable`
Create a read-only user	`username add auditor level ro enable`
Change a user's password	`cli password netadmin read-write-all`
Remove a user	`no username auditor`
Idle session timeout (15 min)	`cli timeout 900`
List active sessions	`show users`
Terminate another user's session	`clear ssh 0`
Command history	`show history`

Levels: ro, rw, rwa (plus l1 to l3). Factory logins are rwa/rwa, rw/rw and ro/ro: change the passwords at first login.

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 13 to 17 (EOS 2007)

Disable paging (same as set length 0)	`terminal more disable`
Enable SSH and disable Telnet	`boot config flags sshd` `no boot config flags telnetd`

Levels: ro, rw, rwa (plus l1 to l3). Factory logins are rwa/rwa, rw/rw and ro/ro: change the passwords at first login.

MODERN CLI

Access, users and CLI session

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	`configure terminal`
Create a local admin user	`username netadmin privilege 15 secret <password>`
Create a read-only user	`username auditor privilege 2 secret <password>`
Change a user's password	`username netadmin secret <password>`
Remove a user	`no username auditor`
Idle session timeout (15 min)	`line vty 0 15` ` exec-timeout 15 0`
List active sessions	`show users`
Terminate another user's session	`clear line vty 2`
Command history	`show history`

The username reference advises a per-user level other than 1 for restricted users (here 2). SSH needs a hostname, ip domain name and RSA keys before ip ssh version 2.

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 13 to 17 (EOS 2007)

Disable paging (same as set length 0)	`terminal length 0`

Enable SSH and disable Telnet	`ip ssh version 2`
	`line vty 0 15`
	` transport input ssh`

The username reference advises a per-user level other than 1 for restricted users (here 2). SSH needs a hostname, ip domain name and RSA keys before ip ssh version 2.

MODERN CLI

Access, users and CLI session

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	`configure terminal`
Create a local admin user	`user netadmin group administrators password`
Create a read-only user	`user auditor group auditors password`
Change a user's password	`user netadmin password`
Remove a user	`no user auditor`
Idle session timeout (15 min)	`cli-session` ` timeout 15`
List active sessions	`show ssh server sessions all-vrfs`
Terminate another user's session	*n/a: No kill command; 'no user <name>' ends that user's sessions*
Command history	`show history`

Local groups: administrators, operators, auditors. Idle timeout defaults to 30 min (cli-session). SSH is on by default (mgmt or default VRF, by model); Telnet is off by default.

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 13 to 17 (EOS 2007)

Disable paging (same as set length 0)	`no page`
Enable SSH and disable Telnet	`ssh server vrf mgmt` `no telnet server vrf mgmt`

Local groups: administrators, operators, auditors. Idle timeout defaults to 30 min (cli-session). SSH is on by default (mgmt or default VRF, by model); Telnet is off by default.

MODERN CLI

Access, users and CLI session

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	`configure`

Create a local admin user	`set system login class ADMIN permissions all`
	`set system login user netadmin class ADMIN`
	`set system login user netadmin authentication plain-text-password`

Create a read-only user	`set system login user auditor class read-only` `set system login user auditor authentication plain-text-password`
Change a user's password	`set system login user netadmin authentication plain-text-password`
Remove a user	`delete system login user auditor`
Idle session timeout (15 min)	`set system login class ADMIN idle-timeout 15`
List active sessions	`show system users`
Terminate another user's session	`request system logout user auditor`

Idle-timeout works only on user-defined login classes (predefined ones such as super-user reject it). The password is asked at the prompt.

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 13 to 17 (EOS 2007)

Command history	`show cli history`
Disable paging (same as set length 0)	`set cli screen-length 0`
Enable SSH and disable Telnet	`set system services ssh` `delete system services telnet`

Idle-timeout works only on user-defined login classes (predefined ones such as super-user reject it). The password is asked at the prompt.

MODERN CLI

Access, users and CLI session

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	*n/a: No equivalent in docs; type the menu path directly (/user ...)*
Create a local admin user	`/user add name=netadmin group=full password=<password>`
Create a read-only user	`/user add name=auditor group=read password=<password>`
Change a user's password	`/user set netadmin password=<password>`
Remove a user	`/user remove auditor`
Idle session timeout (15 min)	`/user set netadmin inactivity-policy=logout inactivity-timeout=15m`
List active sessions	`/user active print`
Terminate another user's session	`/user active request-logout 0`
Command history	`Up arrow / Ctrl+R (history search)` `/system history print`

Built-in groups are read, write and full. Group read still includes reboot and sensitive. Each user has its own inactivity-timeout (default 10 min, CLI only).

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 13 to 17 (EOS 2007)

Disable paging (same as set length 0)	`/interface print without-paging`
Enable SSH and disable Telnet	`/ip service disable telnet` `/ip ssh set strong-crypto=yes`

Built-in groups are read, write and full. Group read still includes reboot and sensitive. Each user has its own inactivity-timeout (default 10 min, CLI only).

MODERN CLI

Access, users and CLI session

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	`configure terminal`
Create a local admin user	`username netadmin role network-admin secret <password>`
Create a read-only user	`username auditor role network-operator secret <password>`
Change a user's password	`username netadmin secret <new-password>`
Remove a user	`no username auditor`
Idle session timeout (15 min)	`management ssh` ` idle-timeout 15`
List active sessions	`show users detail`
Terminate another user's session	*n/a: No documented EOS 4.36 command to end another user's session*
Command history	`show history`

Built-in roles: network-admin (all commands, all modes) and network-operator (EXEC modes only, no configuration). Telnet is disabled by default; SSH: 'management ssh' then 'no shutdown'. Timeout in minutes (0 = none).

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 13 to 17 (EOS 2007)

Disable paging (same as set length 0)	`terminal length 0`
Enable SSH and disable Telnet	`management telnet` ` shutdown`

Built-in roles: network-admin (all commands, all modes) and network-operator (EXEC modes only, no configuration). Telnet is disabled by default; SSH: 'management ssh' then 'no shutdown'. Timeout in minutes (0 = none).

MODERN CLI

Access, users and CLI session

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 13 to 17 (EOS 2007)

Enter configuration mode	`system-view`
Create a local admin user	`aaa` ` local-user netadmin password irreversible-cipher <passwd> privilege level 15`
Create a read-only user	`aaa` ` local-user auditor password irreversible-cipher <passwd> privilege level 1`
Change a user's password	`local-user change-password`
Remove a user	`aaa` ` undo local-user auditor`
Idle session timeout (15 min)	`user-interface vty 0 4` ` idle-timeout 15`
List active sessions	`display users`
Terminate another user's session	`kill user-interface vty 0`

Telnet and STelnet are both off by default; STelnet needs 'ssh server-source' first. A new local user is level 0 with no service type: without 'service-type ssh' (AAA view) it cannot log in.

MODERN CLI

Access, users and CLI session (cont.)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 13 to 17 (EOS 2007)

Command history	`display history-command`
Disable paging (same as set length 0)	`screen-length 0 temporary`
Enable SSH and disable Telnet	`stelnet server enable`

Telnet and STelnet are both off by default; STelnet needs 'ssh server-source' first. A new local user is level 0 with no service type: without 'service-type ssh' (AAA view) it cannot log in.

Clock

LEGACY 2007

Show the current date and time

```
show time
```

Set the date

```
set time [mm/dd/yyyy]
```

Set the time

```
set time [hh:mm:ss]
```

Daylight saving time

```
show summertime
set summertime {enable | disable} [zone]
clear summertime
```

LEGACY 2007

Prompt, name, banner and IP connectivity

Set the switch CLI prompt

```
set prompt “prompt_string”
```

Set the switch name

```
set system name [string]  ← In quotes; Blank = clears
```

Set the switch login message

```
set banner motd message   ← \n = new line; \t = 8 spaces
```

Clear the switch login message

```
clear banner motd
```

Set the IP address for the switch (host IP)

```
set ip address
```

Check the switch’s IP connectivity

```
ping [-s] host [count]  ← (DFE; no SecureStack: ping host)
```

MODERN CLI

Clock, identity and connectivity (1/2)

	NTP server	Device name (hostname)

Extreme Switch Engine	`configure ntp server add 192.0.2.123 vr VR-Mgmt`
	`enable ntp vr VR-Mgmt`
	`enable ntp`
		`configure snmp sysname SW-CORE-01`

Extreme Fabric Engine	`ntp server 192.0.2.123 enable`
	`ntp`
	`show ntp statistics`
		`sys name SW-CORE-01`

Cisco IOS XE	`ntp server vrf Mgmt-vrf 192.0.2.123` `show ntp status`	`hostname SW-CORE-01`

Aruba AOS-CX	`ntp server 192.0.2.123 iburst`
	`ntp vrf mgmt`
	`show ntp status`
		`hostname SW-CORE-01`

Standards: RFC 5905 · RFC 8915 · RFC 792 · IANA-TZDB · RFC 5737 · RFC 4252

MODERN CLI

Clock, identity and connectivity (2/2)

	NTP server	Device name (hostname)
Juniper Junos	`set system ntp server 192.0.2.123` `run show ntp associations`	`set system host-name SW-CORE-01`

MikroTik RouterOS	`/system ntp client set enabled=yes`
	`/system ntp client servers add address=192.0.2.123`
	`/system ntp monitor-peers`
		`/system identity set name=SW-CORE-01`

Arista EOS	`ntp server 192.0.2.123` `show ntp status`	`hostname SW-CORE-01`
Huawei VRP	`ntp-service unicast-server 192.0.2.123` `display ntp-service status`	`sysname SW-CORE-01`

Standards: RFC 5905 · RFC 8915 · RFC 792 · IANA-TZDB · RFC 5737 · RFC 4252

MODERN CLI

Clock, identity and connectivity

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`show time`
Time zone UTC-3, no daylight saving	`configure timezone name BRT -180 noautodst`

NTP server	`configure ntp server add 192.0.2.123 vr VR-Mgmt`
	`enable ntp vr VR-Mgmt`
	`enable ntp`

Device name (hostname)	`configure snmp sysname SW-CORE-01`

Login banner	`configure banner before-login save-to-configuration`
	`Authorized access only`
	`<Enter on an empty line>`

Out-of-band management IP	`configure vlan Mgmt ipaddress 192.0.2.11/24`
Management default gateway	`configure iproute add default 192.0.2.1 vr VR-Mgmt`
Ping from the management context	`ping vr VR-Mgmt 192.0.2.1`

Time zone offset is in minutes (-180) with noautodst; the default is North American autodst. The mgmt port lives in VR-Mgmt: NTP needs enable ntp and enable ntp vr; verify with show ntp association.

MODERN CLI

Clock, identity and connectivity (cont.)

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 18 and 19 (EOS 2007)

Traceroute	`traceroute vr VR-Mgmt 192.0.2.20`

Time zone offset is in minutes (-180) with noautodst; the default is North American autodst. The mgmt port lives in VR-Mgmt: NTP needs enable ntp and enable ntp vr; verify with show ntp association.

MODERN CLI

Clock, identity and connectivity

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`show clock`
Time zone UTC-3, no daylight saving	`clock time-zone America Sao_Paulo`

NTP server	`ntp server 192.0.2.123 enable`
	`ntp`
	`show ntp statistics`

Device name (hostname)	`sys name SW-CORE-01`
Login banner	`banner custom` `banner "Authorized access only"`

Out-of-band management IP	`mgmt oob`
	` ip address 192.0.2.11/24`
	` enable`

Management default gateway	`mgmt oob` ` ip route 0.0.0.0/0 next-hop 192.0.2.1`

The management port is the 'mgmt oob' instance: it has its own IP and route, and ping/traceroute use 'mgmt oob'. The time zone comes from the zone database (no DST to configure).

MODERN CLI

Clock, identity and connectivity (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 18 and 19 (EOS 2007)

Ping from the management context	`ping 192.0.2.1 mgmt oob`
Traceroute	`traceroute 192.0.2.20 -m 30 mgmt oob`

The management port is the 'mgmt oob' instance: it has its own IP and route, and ping/traceroute use 'mgmt oob'. The time zone comes from the zone database (no DST to configure).

MODERN CLI

Clock, identity and connectivity

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`show clock detail`
Time zone UTC-3, no daylight saving	`clock timezone BRT -3`

NTP server	`ntp server vrf Mgmt-vrf 192.0.2.123`
	`show ntp status`
	`! IOS XE 26.1.x: The 26.x guides show ntp server [vrf]; show ntp status appears in no 26.x guide. The line is the 17.18 one`

Device name (hostname)	`hostname SW-CORE-01`
Login banner	`banner login #Authorized access only#`

Out-of-band management IP	`interface GigabitEthernet0/0`
	` vrf forwarding Mgmt-vrf`
	` ip address 192.0.2.11 255.255.255.0`

Management default gateway	`ip route vrf Mgmt-vrf 0.0.0.0 0.0.0.0 192.0.2.1`

Gi0/0 sits in Mgmt-vrf, so NTP, ping, traceroute and the management default route must name vrf Mgmt-vrf. With no clock summer-time configured there is no DST. IOS XE 26.1.x: In the 26.x guide the NTP version of ntp server and ntp peer ranges 1 to 4 with default 4 (17.18: range 1 to 3, default 3).

MODERN CLI

Clock, identity and connectivity (cont.)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 18 and 19 (EOS 2007)

Ping from the management context	`ping vrf Mgmt-vrf 192.0.2.1`
Traceroute	`traceroute vrf Mgmt-vrf 192.0.2.20`

Gi0/0 sits in Mgmt-vrf, so NTP, ping, traceroute and the management default route must name vrf Mgmt-vrf. With no clock summer-time configured there is no DST. IOS XE 26.1.x: In the 26.x guide the NTP version of ntp server and ntp peer ranges 1 to 4 with default 4 (17.18: range 1 to 3, default 3).

MODERN CLI

Clock, identity and connectivity

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`show clock`
Time zone UTC-3, no daylight saving	`clock timezone America/Sao_Paulo`

NTP server	`ntp server 192.0.2.123 iburst`
	`ntp vrf mgmt`
	`show ntp status`

Device name (hostname)	`hostname SW-CORE-01`

Login banner	`banner motd ^`
	`Authorized access only`
	`^`

Out-of-band management IP	`interface mgmt` ` ip static 192.0.2.11/24`
Management default gateway	`interface mgmt` ` default-gateway 192.0.2.1`

NTP and ping use the default VRF unless you give 'ntp vrf mgmt' or 'vrf mgmt'. The management port is 'interface mgmt'; 'banner motd' shows before login, 'banner exec' after.

MODERN CLI

Clock, identity and connectivity (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 18 and 19 (EOS 2007)

Ping from the management context	`ping 192.0.2.1 vrf mgmt`
Traceroute	`traceroute 192.0.2.20 vrf mgmt`

NTP and ping use the default VRF unless you give 'ntp vrf mgmt' or 'vrf mgmt'. The management port is 'interface mgmt'; 'banner motd' shows before login, 'banner exec' after.

MODERN CLI

Clock, identity and connectivity

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`show system uptime`
Time zone UTC-3, no daylight saving	`set system time-zone America/Sao_Paulo`
NTP server	`set system ntp server 192.0.2.123` `run show ntp associations`
Device name (hostname)	`set system host-name SW-CORE-01`
Login banner	`set system login message "Authorized access only"`
Out-of-band management IP	`set interfaces me0 unit 0 family inet address 192.0.2.11/24`
Management default gateway	`set routing-options static route 0/0 next-hop 192.0.2.1`
Ping from the management context	`ping 192.0.2.20 count 5`
Traceroute	`traceroute 192.0.2.20`

On the EX4400 the management port is me0 (labeled MGMT). The time zone applies to all processes only after the switch is rebooted.

MODERN CLI

Clock, identity and connectivity

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`/system clock print`
Time zone UTC-3, no daylight saving	`/system clock set time-zone-autodetect=no time-zone-name=America/Sao_Paulo`

NTP server	`/system ntp client set enabled=yes`
	`/system ntp client servers add address=192.0.2.123`
	`/system ntp monitor-peers`

Device name (hostname)	`/system identity set name=SW-CORE-01`
Login banner	`/system note set note="Authorized access only" show-at-login=yes`
Out-of-band management IP	`/ip address add address=192.0.2.11/24 interface=ether49`
Management default gateway	`/ip route add gateway=192.0.2.1`
Ping from the management context	`/ping 192.0.2.1`
Traceroute	`/tool traceroute 192.0.2.123`

OOB management uses the 100 Mb/s MGMT port (ether49), wired to the CPU and outside the 48 switch-chip ports. Time-zone autodetect is on by default: turn it off first. The banner is /system note, after login.

MODERN CLI

Clock, identity and connectivity

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`show clock`
Time zone UTC-3, no daylight saving	`clock timezone America/?` `clock timezone <zone_name>`
NTP server	`ntp server 192.0.2.123` `show ntp status`
Device name (hostname)	`hostname SW-CORE-01`

Login banner	`banner login`
	`Authorized access only`
	`EOF`

Out-of-band management IP	`interface management 1` ` ip address 192.0.2.11/24`
Management default gateway	`ip route 0.0.0.0/0 192.0.2.1`

The time zone is a predefined label (e.g. US/Central); 'clock timezone America/?' lists the American ones: pick Brazil's (UTC-3). The banner ends with EOF alone on a line. Management: 'interface management 1'.

MODERN CLI

Clock, identity and connectivity (cont.)

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 18 and 19 (EOS 2007)

Ping from the management context	`ping 192.0.2.1`
Traceroute	`traceroute 192.0.2.20`

The time zone is a predefined label (e.g. US/Central); 'clock timezone America/?' lists the American ones: pick Brazil's (UTC-3). The banner ends with EOF alone on a line. Management: 'interface management 1'.

MODERN CLI

Clock, identity and connectivity

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 18 and 19 (EOS 2007)

Show date and time	`display clock`
Time zone UTC-3, no daylight saving	`clock timezone BRT minus 03:00:00`
NTP server	`ntp-service unicast-server 192.0.2.123` `display ntp-service status`
Device name (hostname)	`sysname SW-CORE-01`
Login banner	`header login information &Authorized access only&`
Out-of-band management IP	`interface MEth 0/0/1` ` ip address 192.0.2.11 24`
Management default gateway	`ip route-static 0.0.0.0 0 192.0.2.1`
Ping from the management context	`ping -a 192.0.2.11 192.0.2.1`
Traceroute	`tracert -a 192.0.2.11 192.0.2.20`

UTC-3 is 'clock timezone BRT minus 03:00:00'; configure no daylight-saving-time (Brazil has none). On a switch with a management interface (MEth0/0/1) the management IP goes there; otherwise on a VLANIF.

LAB

Lab: Basic Configuration

- **a** Access the switch via console
- **b** Change the password of the admin user
- **c** Disable the rw and ro users
- **d** Create a “guest” user with ro privilege
- **e** Set the current date and time
- **f** Change the switch name and prompt
- **g** Set a login message on the switch
- **h** Set the switch management IP address (host IP)
- **i** Set the PC’s IP address
- **j** Check the switch’s IP connectivity

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.



Break

Break between the first and second part of day 01

1.3

Basic maintenance



Save, clear and reboot; recover passwords; files, backups and firmware

- Clearing the configuration and rebooting the switch
- Recovering access after lost passwords
- Configuration files and backups (SCP/SFTP)
- Reconfiguring the switch from files
- Firmware upgrade

Chapter 1.3 agenda

1. Clearing the configuration and rebooting the switch
2. Recovering access after lost passwords
3. Configuration files and backups (SCP/SFTP)
4. Reconfiguring the switch from files
5. Firmware upgrade

MODERN CLI

- B03 Save, erase and reboot
- B04 Password recovery
- B05 Files and configuration backups
- B06 Firmware upgrade

Lab

- Lab: basic maintenance

LEGACY 2007

Clearing the configuration

- The command `clear` usually removes what was configured with the command `set`
- The command `clear config` clears the device configuration (on the DFE, the module is also reset)
- The command `configure <filename>` clears all configuration, reboots the device and uses the specified file to reconfigure the switch (covered later)



Clearing the configuration

LEGACY 2007

SecureStack

Clears the configuration, keeping the Stack ID, IP address and licenses

```
clear config
```

Clears the configuration, keeping the IP address and optional licenses

```
clear config all
```

Removes the stack’s IP address

```
clear ip address
```

Erases optional licenses applied to the switch

```
clear license
```

Clearing the configuration

LEGACY 2007

DFE

Clears the configuration of all modules, keeping the IP address

```
clear config all
```

Clears the system IP address (clear config does not change it)

```
clear ip address
```

Clears the configuration of the module in the specified slot (the IP is kept)

```
clear config mod-num
```

Removes the optional licenses (advanced, user-capacity, port-capacity)

```
clear license advanced | user-capacity |
port-capacity [slot slot]
```

LEGACY 2007

Rebooting the switch

SecureStack

To reboot the entire stack:

```
reset
```

To reboot a single member of the stack:

```
reset stack-id
```

LEGACY 2007

Rebooting the switch

DFE

To reboot the whole system:

```
reset
```

To reboot after a specified delay:

```
reset in hh:mm [reason]
```

To reboot at a specified time:

```
reset at hh:mm [mm/dd] [reason]
```

To check scheduled resets:

```
show reset
```

To cancel scheduled resets:

```
reset cancel
```

Save, erase and reboot

MODERN CLI

	Save the running configuration	Reset to factory defaults
Extreme Switch Engine	`save configuration`	`unconfigure switch all`

Extreme Fabric Engine	`save config`	`unconfigure switch`
		`save config`
		`reset -y`

Cisco IOS XE	`copy running-config startup-config`	`write erase`
		`delete flash:vlan.dat`
		`reload`

Aruba AOS-CX	`write memory` `copy running-config startup-config`	`erase all zeroize`
Juniper Junos	`commit confirmed 5` `commit`	`request system zeroize`
MikroTik RouterOS	*n/a*	`/system reset-configuration` `/system reset-configuration no-defaults=yes`
Arista EOS	`write`	`delete startup-config now` `reload now`
Huawei VRP	`save`	`reset factory-configuration`

MODERN CLI

Save, erase and reboot

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`save configuration`
Show the running configuration	`show configuration`
Compare running vs saved (diff)	`show configuration difference`
Reset to factory defaults	`unconfigure switch all`
Reboot now	`reboot`
Scheduled reboot (in 30 min)	`reboot time 11 2 2026 02 30 00`
Cancel a scheduled reboot	`reboot cancel`

A * in the prompt flags unsaved changes. A scheduled reboot takes an absolute date and time (mm dd yyyy hh mm ss), not a delay. unconfigure switch all wipes everything (users, mgmt IP) and reboots.

MODERN CLI

Save, erase and reboot

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`save config`
Show the running configuration	`show running-config`
Compare running vs saved (diff)	*n/a: No documented diff: compare show running-config with more <file>*

Reset to factory defaults	`unconfigure switch`
	`save config`
	`reset -y`

Reboot now	`reset -y`
Scheduled reboot (in 30 min)	`reset -in 30`
Cancel a scheduled reboot	`reset -cancel`

'reset' restarts from the last saved configuration: run 'save config' first. 'unconfigure switch' returns the switch to factory state (no config files, no user accounts).

MODERN CLI

Save, erase and reboot

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`copy running-config startup-config`
Show the running configuration	`show running-config`
Compare running vs saved (diff)	`show archive config differences running-config startup-config`

Reset to factory defaults	`write erase`
	`delete flash:vlan.dat`
	`reload`

Reboot now	`reload`
Scheduled reboot (in 30 min)	`reload in 30`
Cancel a scheduled reboot	`reload cancel`

write memory also saves. To wipe the config use write erase and delete vlan.dat. factory-reset all also erases the IOS images and leaves the switch in ROMmon; factory-reset config erases only the startup config.

MODERN CLI

Save, erase and reboot

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`write memory` `copy running-config startup-config`
Show the running configuration	`show running-config`
Compare running vs saved (diff)	`checkpoint diff running-config startup-config`
Reset to factory defaults	`erase all zeroize`
Reboot now	`boot system`
Scheduled reboot (in 30 min)	`reload after 00:00:30`
Cancel a scheduled reboot	`reload cancel`

'write memory' is an alias of 'copy running-config startup-config'. 'erase all zeroize' securely wipes everything (minutes to 1 h); 'erase startup-config' only removes the saved config. 'reload' needs 10.16.1000+.

MODERN CLI

Save, erase and reboot

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`commit confirmed 5` `commit`
Show the running configuration	`show configuration | display set`
Compare running vs saved (diff)	`show | compare` `show | compare rollback 1`
Reset to factory defaults	`request system zeroize`
Reboot now	`request system reboot`
Scheduled reboot (in 30 min)	`request system reboot in 30`
Cancel a scheduled reboot	`clear system reboot`

In Junos 'commit' is the save: it activates and stores the config. 'commit confirmed 5' auto-reverts in 5 min without a 2nd commit. 'zeroize' wipes everything and reboots.

MODERN CLI

Save, erase and reboot

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	*n/a: No save command in the docs; Safe Mode (Ctrl+X) undoes on Ctrl+D*
Show the running configuration	`/export`
Compare running vs saved (diff)	*n/a: No startup-config: review changes with /system history print*
Reset to factory defaults	`/system reset-configuration` `/system reset-configuration no-defaults=yes`
Reboot now	`/system reboot`
Scheduled reboot (in 30 min)	`/system script add name=rbt source="/system reboot"` `/system scheduler add name=rbt30 on-event=rbt start-time=<now+30min>`
Cancel a scheduled reboot	`/system scheduler remove rbt30`

Safe Mode (Ctrl+X): Ctrl+D or a lost session undoes the changes; it is ignored for commands that need a reboot. reset-configuration stores a backup first, unless skip-backup=yes.

MODERN CLI

Save, erase and reboot

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`write`
Show the running configuration	`show running-config`
Compare running vs saved (diff)	*n/a: Not documented in EOS 4.36; a file-to-file 'diff' command exists*
Reset to factory defaults	`delete startup-config now` `reload now`
Reboot now	`reload now`
Scheduled reboot (in 30 min)	`reload in 00:30`
Cancel a scheduled reboot	`reload cancel`

'write' equals 'copy running-config startup-config'. With no startup-config the switch enters ZTP: run 'zerotouch cancel' as admin. Plain 'reload' asks first (yes/no/cancel/diff); 'reload in hh:mm' schedules.

MODERN CLI

Save, erase and reboot

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 22 to 26 (EOS 2007)

Save the running configuration	`save`
Show the running configuration	`display current-configuration`
Compare running vs saved (diff)	`compare configuration`
Reset to factory defaults	`reset factory-configuration`
Reboot now	`reboot`
Scheduled reboot (in 30 min)	`schedule reboot delay 00:30`
Cancel a scheduled reboot	`undo schedule reboot`

All of these run in user view (<HUAWEI>). 'reset factory-configuration' deletes all configuration and files except startup, patch, module and license files; 'reboot' warns about unsaved changes.

LEGACY 2007

Recovering access when passwords are lost

SecureStack

Historical procedure *(2007 Enterasys knowledge base; link no longer active)*

⚠ If user authentication via RADIUS is configured:
Disable network access to the RADIUS server first.

a Locate the “Reset Button” on the back of the device (just to the right of the “stack down” connector)

b Hold down the reset button for 5 seconds

c The switch will generate a message on the console and in the log: "Password Reset button has been pressed"

d The passwords of *admin*, *rw* and *ro* will be reset to the factory defaults

LEGACY 2007

Recovering access when passwords are lost

DFE

Historical procedure *(2007 Enterasys knowledge base; link no longer active)*

a. Disconnect all modules from the chassis backplane.

b. Use anti-static protection to avoid damaging the module

c. Select one of the modules and locate the DIP switch bank

d. Change the position of DIP switch 8

e. Re-insert this module into the chassis and wait for it to finish booting

f. Access the module console and verify that the *admin* password has been restored to the factory defaults

g. Reconnect the remaining modules to the chassis

MODERN CLI

Password recovery

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 27 and 28 (EOS 2007)

1	Attach a terminal to the console port and restart the switch (no login needed; recovery is local only). Hold the space bar until the Boot Menu shows (30 s window).	`(hold the space bar)`
2	In the Boot Menu, use the arrow keys and Enter to pick 'EXOS: Primary … with default configuration'. Do not pick 'EXOS: Rescue'. Files and licenses are not deleted.	`EXOS: Primary <version> with default configuration`
3	Log in as admin with an empty password. The old primary.cfg is still on flash: check with ls and do NOT save now, because saving overwrites it with the blank configuration.	`ls`
4	To keep the configuration, create the autoexec.xsf script with vi; it creates a temporary admin account.	`vi autoexec.xsf` `(inside the file) create account admin <user> <password>`
5	Select the old configuration file and reboot WITHOUT saving (answer n to the prompt); the script runs at boot.	`use configuration primary` `reboot`
6	Log in with the temporary account. An account password cannot be read: delete and re-create the account whose password was lost.	`delete account admin` `create account admin admin <password>`

7	Log in with the re-created account, delete the temporary account and the script, then save. To avoid losing access again, configure a failsafe account.	`delete account <user>`
		`rm autoexec.xsf`
		`save configuration`

Console required. 'With default configuration' boots factory defaults and deletes no files or licenses: primary.cfg is lost only if you save before reloading it.

MODERN CLI

Password recovery

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 27 and 28 (EOS 2007)

1. Connect to the serial console (115200, 8 data bits, 1 stop bit, no parity, no flow control). On Universal platforms you may need to press the spacebar to get a console response.
2. Power the switch off for a full 60 seconds, then boot it, to avoid the 'Invalid Password! Try Again.' message.
3. During boot, right after the line 'VSP Linux Uncompressing /boot/a/appfs.tgz ...' (the line for the Universal 5520), type 123! and press Enter.

```
123!
```

4. In the 'Lifecycle recovery menu' choose option 7, Reset all CLI local login passwords (options 6 and 8 go to factory defaults).

```
7
```

5. The menu shows Release#, Serial# and the time. Send them to GTAC and request the privilege password.
6. Enter the privilege password, then leave the menu with q.

```
<privilege-password>
q
```

7. Log in with rwa/rwa (admin/admin when enhanced-security mode is on) and change the password right away.

```
enable
configure terminal
cli password rwa read-write-all
```

Needs console access and a GTAC privilege password. Option 7 resets only the CLI passwords (6 and 8 restore factory defaults). With an rwa session 'cli password' is enough. After reset: rwa/rwa.

MODERN CLI

Password recovery

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 27 and 28 (EOS 2007)

1	Console at 9600 baud. Power the switch off, power on and press Ctrl-C during boot to stop autoboot and reach ROMmon (switch: prompt). *IOS XE 26.1.x: The 26.x guide enters ROMMON with the Mode button (2-3 presses once the System LED blinks), not Ctrl-C.*	`Ctrl-C`
2	In ROMmon, enable manual boot and ignore the startup configuration.	`MANUAL_BOOT=yes` `SWITCH_IGNORE_STARTUP_CFG=1`
3	Boot with packages.conf from flash and answer No to the initial configuration dialog.	`boot flash:packages.conf`
4	Enter privileged EXEC and load the startup configuration into the running configuration (press Enter at the prompts).	`enable` `copy startup-config running-config`

5	Set the new password and undo the ignore-startup setting.	`configure terminal`
		`enable secret <password>`
		`no system ignore startupconfig switch all`

6	Save the configuration and reload.	`end`
		`copy running-config startup-config`
		`reload`

7	Boot again with packages.conf and disable manual boot.	`boot flash:packages.conf`
		`configure terminal`
		`no boot manual`

The config is kept. With 'system disable password recovery' set, ROMmon opens only if you accept a reset to defaults: config.text and vlan.dat are deleted. Console required.

MODERN CLI

Password recovery

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 27 and 28 (EOS 2007)

1	Connect to the console (Service OS is reachable only through the console) and reboot the switch.	
2	At every boot the console shows the boot menu ('Select profile'): enter 0 within 5 seconds to enter Service OS.	`0`
3	At the 'ServiceOS login:' prompt, log in as admin (no password by default). The prompt becomes SVOS>.	`admin`
4	Set the new admin password (enter and confirm; input is masked).	`password`
5	At SVOS>, enter boot and choose profile 1 (primary) or 2 (secondary) to start AOS-CX.	`boot`
6	Log in as admin with the new password and save the configuration: the password takes effect only after boot and save.	`write memory`

Console required. 'password' only sets the password; 'config-clear' renames the startup-config and resets to factory defaults. 'password' is unavailable in enhanced secure mode. No input in 5 s: default profile boots.

MODERN CLI

Password recovery

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 27 and 28 (EOS 2007)

1. Power off the switch, attach the console (9600 bps, 8 data bits, no parity, 1 stop bit, no flow control) and power it on.

2. At the boot prompt, press the space bar to reach the loader prompt.

```
Hit [Enter] to boot immediately, or space bar for
command prompt.
```

3. At the loader prompt, boot in single-user mode.

```
loader> boot -s
```

4. When asked for the shell, type recovery to start root password recovery.

```
recovery
```

5. When the CLI prompt appears, enter configuration mode and set the new root password (typed twice).

```
configure
set system root-authentication plain-text-password
```

6. Commit, exit configuration mode and the CLI, and answer y to reboot the switch.

```
commit
exit
exit
```

Needs console (physical) access. The official steps only set a new root password and commit. It does not work with an 'insecure' console or unattended (U-Boot) mode. Applies to EX2300/3400/4100/4400.

MODERN CLI

Password recovery

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 27 and 28 (EOS 2007)

1. Prerequisite: physical access, and Protected RouterBOOT not enabled (if enabled, the reset button is off and the device cannot be recovered).
2. Turn the switch off by unplugging its power.
3. Hold the reset button and, while holding it, plug the power back in.
4. Keep holding until the LED (usually USR) starts flashing, then release: RouterOS returns to factory defaults. Holding on enters CAPs mode and then Netinstall mode.
5. Log in as admin (empty password, or the sticker password on some models) and set a new password.

```
/user set admin password=<password>
```

6. Restore the configuration from an export or backup kept off the switch.

```
/import file-name=<config>.rsc
/system backup load name=<file>
```

7. Alternative if the button cannot be used: reinstall with Netinstall (it re-formats the drive; configuration and user files are lost).

Destructive: the reset clears the whole configuration and password, so keep backups off the switch. With Protected RouterBOOT on, the reset button is disabled and recovery needs the password.

MODERN CLI

Password recovery

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 27 and 28 (EOS 2007)

1	On the console (9600 8N1) run reload and press Ctrl-C early in boot; enter the Aboot password if one is set.	`reload  (Ctrl-C)`
2	At the Aboot prompt, change to the flash directory.	`Aboot# cd /mnt/flash`
3	Rename the saved configuration so the boot bypasses it.	`Aboot# mv startup-config startup-config.old`
4	Exit Aboot to boot with the default configuration.	`Aboot# exit`
5	Log in as admin (no password) and cancel ZTP.	`zerotouch cancel`

6	Set the enable and admin passwords and save the new configuration.	`configure terminal`
		`username admin secret <password>`
		`write`

The saved config is bypassed at boot (becomes startup-config.old). Console required. Without the Aboot password, 'fullrecover' erases all flash, EOS image included. Only the 'enable password' line edits losslessly.

MODERN CLI

Password recovery

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 27 and 28 (EOS 2007)

1. Connect the terminal to the console port (in a stack, power off the members and work on the master only) and restart the switch.
2. When 'Press Ctrl+B to enter BootLoad menu' appears, press Ctrl+B at once and enter the BootLoad password.

```
Ctrl+B
```

3. In the BootLoad menu, select 'Clear password for console user'.
4. Select 'Boot with default mode' to start the switch. Do not select 'Reboot', or the password is not cleared.
5. Log in on the console (no authentication required) and enter the system view.

```
system-view
```

6. Set the console authentication mode and the new password.

```
user-interface console 0
 authentication-mode password
 set authentication password cipher <password>
```

7. Return to user view and save the configuration.

```
return
save
```

Clearing the console password through the BootLoad menu does not change the configuration, but it requires the BootLoad password. Selecting 'Reboot' instead of 'Boot with default mode' prevents the clearing.

File management on the switch

- Purposes:
- Configuration backup
- Storing configuration versions
- Firmware file management



LEGACY 2007

File management on the switch

List the files stored in the internal *file system*

```
dir
dir [filename]  ← lists the contents of the specified directory
```

View the contents of a file

```
show file filename
```

Delete a file

```
delete filename
```

LEGACY 2007

File management

Copy a file

```
copy source destination
```

Where the *source* and the *destination* can be represented by:

a.) Path of a local file

Valid directories on the **SecureStack**: /images and /configs

Valid directories on the **DFE**: /images and /slot*N*

b.) Path of a remote file (URL)

E.g.: tftp://a.b.c.d/file.ext

E.g.: ftp://user:passwd@a.b.c.d/file.ext (DFE only; SecureStack uses TFTP)

File management

LEGACY 2007

Examples (DFE):

Copy the active configuration to a file:

```
show config outfile filename
```

Transfer firmware from an external server to the filesystem:

```
copy tftp://172.16.89.34/ets-mtxe7-msi newimage
copy ftp://172.16.89.34/ets-mtxe7-msi newimage
copy ftp://user:passwd@172.16.89.34/ets-mtxe7-msi
newimage
```

Copy an external configuration file to the filesystem:

```
copy tftp://172.16.89.34/myconfig slot3/myconfig
```

Copy a configuration file to an external server:

```
copy slot3/myconfig ftp://172.16.89.34/myconfig
```

Copy a configuration file to another slot:

```
copy slot3/myconfig slot5/myconfig
```

File management

LEGACY 2007

Examples (SecureStack):

Copy the active configuration to a file:

```
show config outfile filename
```

Transfer firmware from an external server to the filesystem:

```
copy tftp://10.1.192.34/version01000
system:image
```

Copy a configuration file from an external server to the switch:

```
copy tftp://10.1.192.1/Jan1_2004.cfg
configs/Jan1_2004.cfg
```

List the image and configuration files of the switch:

```
dir
```

LEGACY 2007

Reconfiguring the switch from files

Stored configuration files can be used to reconfigure the switch

```
configure  filename [append]
```

If “append” is **not** used, the switch’s current configuration will be replaced entirely by the configuration in the file.

The switch will reboot automatically.

If “append” is used, the behavior is equivalent to pasting the text of the configuration file into the CLI.

MODERN CLI

Files and configuration backups (1/2)

	Export config to a server (SCP/SFTP)	Checkpoint / rollback

Extreme Switch Engine	`scp2 SW-CORE-01.xsf backup@192.0.2.20:SW-CORE-01.xsf`	`save configuration chk1`
		`use configuration chk1`
		`reboot`

Extreme Fabric Engine	`boot config host user backup`
	`boot config host password <password>`
	`copy /intflash/config.cfg 192.0.2.20:/SW-CORE-01.cfg`
		n/a

Cisco IOS XE	`copy running-config scp://backup@192.0.2.20/SW-CORE-01.cfg vrf Mgmt-vrf`	`archive config`
		`configure replace flash:ckpt-1 time 5`
		`configure confirm`

Aruba AOS-CX	`copy running-config sftp://backup@192.0.2.20/SW-CORE-01.cfg json vrf mgmt`	`checkpoint auto 10` `checkpoint auto confirm`

Standards: DRAFT-SECSH-FILEXFER-13 · OPENSSH-9.0 · RFC 1350 · RFC 6241 · RFC 6242 · IANA-PORTS

MODERN CLI

Files and configuration backups (2/2)

	Export config to a server (SCP/SFTP)	Checkpoint / rollback

Juniper Junos	`file copy /var/tmp/core01.conf scp://backup@192.0.2.20/home/backup/core01.conf`	`run request system configuration rescue save`
		`rollback 1`
		`commit`

MikroTik RouterOS	`/tool fetch url="sftp://192.0.2.20" upload=yes user=backup \`
	`password=<password> src-path=SW-CORE-01.rsc dst-path=SW-CORE-01.rsc`
		`Ctrl+X (Safe Mode)`
		`/system history print`
		`/undo`

Arista EOS	`copy running-config scp://backup@192.0.2.20/SW-CORE-01.cfg`	`configure checkpoint save PRE-CHANGE` `configure checkpoint restore PRE-CHANGE`
Huawei VRP	`system-view` `scp -a 192.0.2.11 flash:/backup.cfg backup@192.0.2.20:backup.cfg`	*n/a*

Standards: DRAFT-SECSH-FILEXFER-13 · OPENSSH-9.0 · RFC 1350 · RFC 6241 · RFC 6242 · IANA-PORTS

MODERN CLI

Files and configuration backups

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`ls`
View a file's content	`cat SW-CORE-01.xsf`
Delete a file	`rm SW-CORE-01-old.xsf`
Save running config to a local file	`save configuration as-script SW-CORE-01.xsf`
Export config to a server (SCP/SFTP)	`scp2 SW-CORE-01.xsf backup@192.0.2.20:SW-CORE-01.xsf`
Import config from a server	`scp2 backup@192.0.2.20:SW-CORE-01.xsf SW-CORE-01.xsf`
Apply a saved config (replace)	`load script SW-CORE-01.xsf`

Checkpoint / rollback	`save configuration chk1`
	`use configuration chk1`
	`reboot`

The .xsf is readable text; load script replays its commands on the running config. There is no automatic rollback: save a named .cfg and go back with use configuration + reboot. scp2 uses VR-Mgmt.

MODERN CLI

Files and configuration backups

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`dir`
View a file's content	`more /intflash/config.cfg`
Delete a file	`delete /intflash/old-config.cfg`
Save running config to a local file	`save config file /intflash/SW-CORE-01.cfg`

Export config to a server (SCP/SFTP)	`boot config host user backup`
	`boot config host password <password>`
	`copy /intflash/config.cfg 192.0.2.20:/SW-CORE-01.cfg`

Import config from a server	`copy 192.0.2.20:/SW-CORE-01.cfg /intflash/SW-CORE-01.cfg`
Apply a saved config (replace)	`boot config /intflash/SW-CORE-01.cfg -y`
Checkpoint / rollback	*n/a: No checkpoint: use backup configure / restore configure (ZIP)*

copy uses FTP (password set in boot config host) or TFTP; SCP/SFTP work only with the switch as server. 'boot config <file> -y' restarts with the saved file. No checkpoint: use backup/restore.

MODERN CLI

Files and configuration backups

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`dir flash:`
View a file's content	`more flash:SW-CORE-01.cfg`
Delete a file	`delete flash:SW-CORE-01.cfg`
Save running config to a local file	`copy running-config flash:SW-CORE-01.cfg`
Export config to a server (SCP/SFTP)	`copy running-config scp://backup@192.0.2.20/SW-CORE-01.cfg vrf Mgmt-vrf`
Import config from a server	`copy scp://backup@192.0.2.20/SW-CORE-01.cfg flash: vrf Mgmt-vrf`
Apply a saved config (replace)	`configure replace flash:SW-CORE-01.cfg force`

Checkpoint / rollback	`archive config`
	`configure replace flash:ckpt-1 time 5`
	`configure confirm`

Rollback first needs 'archive' + 'path flash:ckpt'. With 'time 5', without 'configure confirm' within 5 min the switch reverts. SCP allows only privilege-15 users. IOS XE 26.1.x: 26.1.1 blocks copy over tftp:, ftp: and rcp: by default as insecure; SCP with vrf, as in these rows, is the secure alternative.

MODERN CLI

Files and configuration backups

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`show checkpoint`
View a file's content	`show checkpoint BASE-2026`
Delete a file	`erase checkpoint BASE-2026`
Save running config to a local file	`copy running-config checkpoint BASE-2026`
Export config to a server (SCP/SFTP)	`copy running-config sftp://backup@192.0.2.20/SW-CORE-01.cfg json vrf mgmt`
Import config from a server	`copy sftp://backup@192.0.2.20/SW-CORE-01.cfg running-config vrf mgmt`
Apply a saved config (replace)	`checkpoint rollback BASE-2026`
Checkpoint / rollback	`checkpoint auto 10` `checkpoint auto confirm`

Config 'files' are checkpoints (up to 32 user ones). 'checkpoint auto 10' reverts all unless 'checkpoint auto confirm' follows (1 to 60 min). From a server: running-config takes JSON or CLI; startup-config, JSON only.

MODERN CLI

Files and configuration backups

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`file list /var/tmp detail`
View a file's content	`file show /var/tmp/core01.conf`
Delete a file	`file delete /var/tmp/core01.conf`
Save running config to a local file	`show configuration | save /var/tmp/core01.conf`
Export config to a server (SCP/SFTP)	`file copy /var/tmp/core01.conf scp://backup@192.0.2.20/home/backup/core01.conf`
Import config from a server	`file copy scp://backup@192.0.2.20/home/backup/core01.conf /var/tmp/`
Apply a saved config (replace)	`load override /var/tmp/core01.conf` `commit`

Checkpoint / rollback	`run request system configuration rescue save`
	`rollback 1`
	`commit`

'load override' replaces the whole candidate; it takes effect only after commit. Rollback 0 to 49 returns to earlier commits; the rescue config is a fixed checkpoint you save on demand.

MODERN CLI

Files and configuration backups

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`/file print`
View a file's content	`/file head SW-CORE-01.rsc n=50`
Delete a file	`/file remove SW-CORE-01.rsc`
Save running config to a local file	`/export file=SW-CORE-01` `/system backup save name=SW-CORE-01 password=<password>`
Export config to a server (SCP/SFTP)	`/tool fetch url="sftp://192.0.2.20" upload=yes user=backup \` ` password=<password> src-path=SW-CORE-01.rsc dst-path=SW-CORE-01.rsc`
Import config from a server	`/tool fetch url="sftp://192.0.2.20/SW-CORE-01.rsc" user=backup \` ` password=<password> dst-path=SW-CORE-01.rsc`
Apply a saved config (replace)	`/system backup load name=SW-CORE-01` `/system reset-configuration no-defaults=yes run-after-reset=SW-CORE-01.rsc`

Export (.rsc) is text but leaves out user passwords; backup is binary and device-specific. Where a flash directory exists, files outside it sit in RAM and are lost at reboot.

MODERN CLI

Files and configuration backups (cont.)

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 29 to 34 (EOS 2007)

Checkpoint / rollback	`Ctrl+X (Safe Mode)`
	`/system history print`
	`/undo`

Export (.rsc) is text but leaves out user passwords; backup is binary and device-specific. Where a flash directory exists, files outside it sit in RAM and are lost at reboot.

MODERN CLI

Files and configuration backups

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`dir flash:`
View a file's content	`more flash:SW-CORE-01.cfg`
Delete a file	`delete flash:SW-CORE-01.cfg`
Save running config to a local file	`copy running-config flash:SW-CORE-01.cfg`
Export config to a server (SCP/SFTP)	`copy running-config scp://backup@192.0.2.20/SW-CORE-01.cfg`
Import config from a server	`copy scp://backup@192.0.2.20/SW-CORE-01.cfg flash:`
Apply a saved config (replace)	`configure replace flash:SW-CORE-01.cfg`
Checkpoint / rollback	`configure checkpoint save PRE-CHANGE` `configure checkpoint restore PRE-CHANGE`

Checkpoints live in 'checkpoint:' (list with 'dir checkpoint:'). 'configure replace' swaps the whole config; 'ignore-errors' forces it. 'copy URL running-config' only merges, it does not replace.

MODERN CLI

Files and configuration backups

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 29 to 34 (EOS 2007)

List files	`dir`
View a file's content	`more backup.cfg`
Delete a file	`delete backup.cfg`
Save running config to a local file	`save backup.cfg`
Export config to a server (SCP/SFTP)	`system-view` `scp -a 192.0.2.11 flash:/backup.cfg backup@192.0.2.20:backup.cfg`
Import config from a server	`system-view` `scp -a 192.0.2.11 backup@192.0.2.20:backup.cfg flash:/backup.cfg`
Apply a saved config (replace)	`startup saved-configuration backup.cfg` `reboot`
Checkpoint / rollback	*n/a: Not in V200R025C00 docs; V600R026C00 has config rollback*

File commands run in user view; scp needs system-view. delete moves a file to the recycle bin (reset recycle-bin empties it). A .cfg is plain text; startup saved-configuration applies after a reboot.

Firmware upgrade

LEGACY 2007

SecureStack

a.) Check that there is enough space in flash (maximum 2 files)

```
dir
```

b.) If there are 2 files, delete one of them

```
delete filename
```

c.) Check connectivity to the TFTP server

```
ping a.b.c.d  ← IP address of the TFTP server
```

d.) Check on the server that the firmware is in the correct directory

Check what the root directory is on the TFTP server

e.) Copy the file from the TFTP server to the switch

```
copy tftp://a.b.c.d/filename system:image  ← not a file name
```

f.) Select the file for the next boot, and reboot the switch

```
set boot system filename   ← the original name of the firmware file
```

Firmware upgrade

LEGACY 2007

DFE

a.) Check that there is enough space in flash (see free space in show system hardware)

```
dir
```

b.) If space is short, delete an old image

```
delete filename
```

c.) Check connectivity to the TFTP server

```
ping a.b.c.d  ← IP address of the TFTP server
```

d.) Check on the server that the firmware is in the correct directory

Check what the root directory is on the TFTP server

e.) Copy the file from the TFTP server to the switch

```
copy tftp://a.b.c.d/filename filename   ← give the firmware a name
```

f.) Select the file for the next boot, and reboot the switch

```
set boot system filename   ← the name that was given to the firmware
```

Firmware upgrade (1/2)

MODERN CLI

	Transfer the image	Install / set the boot image
Extreme Switch Engine	`download image 192.0.2.20 summit_arm-<version>.xos vr VR-Mgmt secondary`	`install image summit_arm-<version>.xos secondary` `use image secondary`
Extreme Fabric Engine	`copy 192.0.2.20:/5520.9.4.1.0.voss /intflash/5520.9.4.1.0.voss`	`software add 5520.9.4.1.0.voss` `software activate 9.4.1.0.GA`
Cisco IOS XE	`copy scp://backup@192.0.2.20/cat9k_iosxe.17.18.04.SPA.bin flash: vrf Mgmt-vrf`	`install add file flash:cat9k_iosxe.17.18.04.SPA.bin activate commit`
Aruba AOS-CX	`copy sftp://backup@192.0.2.20/<IMAGE>.swi secondary vrf mgmt`	`boot set-default secondary`

Standards: FIPS180-4 · NIST-HASH · RFC 6151

Firmware upgrade (2/2)

MODERN CLI

	Transfer the image	Install / set the boot image
Juniper Junos	`file copy scp://backup@192.0.2.20/home/backup/<image>.tgz /var/tmp/`	`request system software add /var/tmp/<image>.tgz`
MikroTik RouterOS	`/tool fetch url="sftp://192.0.2.20/routeros-7.23.8-mipsbe.npk" user=backup \ password=<password>`	*n/a*

Arista EOS	`copy scp://backup@192.0.2.20/EOS-<release>.swi flash:`	`boot system flash:EOS-<release>.swi`
		`show boot-config`
		`write`

Huawei VRP	`system-view`
	`scp -a 192.0.2.11 backup@192.0.2.20:<image>.cc flash:/<image>.cc`
	`scp -a 192.0.2.11 backup@192.0.2.20:<image>.cc.asc flash:/<image>.cc.asc`
		`startup system-software <image>.cc`

Standards: FIPS180-4 · NIST-HASH · RFC 6151

MODERN CLI

Firmware upgrade

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`show version`

Check free space / partitions	`show switch`
	`show version images`
	`ls`

Transfer the image	`download image 192.0.2.20 summit_arm-<version>.xos vr VR-Mgmt secondary`
Verify integrity (hash)	*n/a: No hash on the CLI: compare SHA256 offline; signature is verified*
Install / set the boot image	`install image summit_arm-<version>.xos secondary` `use image secondary`
Reboot into the new image	`reboot`
Roll back to the previous image	`use image primary` `reboot`

Install to the inactive partition (see show switch). download image uses TFTP (insecure) and the switch SCP client refuses .xos: for a secure path push the .xos by SFTP from a PC. The 5720 uses the onie prefix.

MODERN CLI

Firmware upgrade

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`show software`
Check free space / partitions	`dir`
Transfer the image	`copy 192.0.2.20:/5520.9.4.1.0.voss /intflash/5520.9.4.1.0.voss`
Verify integrity (hash)	`file-checksum sha512 5520.9.4.1.0.voss`
Install / set the boot image	`software add 5520.9.4.1.0.voss` `software activate 9.4.1.0.GA`
Reboot into the new image	`reset -y` `software commit`
Roll back to the previous image	`software activate 9.3.4.0.GA` `reset -y`

After reset a commit timer runs (default 10 min, auto-commit on); without commit the switch returns to the previous version. The 4220/5320/5420 hold only 2 releases. Check the SHA512.

MODERN CLI

Firmware upgrade

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`show version`
Check free space / partitions	`dir flash:` `show file systems`

Transfer the image	`copy scp://backup@192.0.2.20/cat9k_iosxe.17.18.04.SPA.bin flash: vrf Mgmt-vrf`
	`! IOS XE 26.1.x:`
	`copy scp://backup@192.0.2.20/cat9k_iosxe.26.01.01.SPA.bin flash: vrf Mgmt-vrf`

Verify integrity (hash)	`verify /md5 flash:cat9k_iosxe.17.18.04.SPA.bin`
	`! IOS XE 26.1.x:`
	`verify /md5 flash:cat9k_iosxe.26.01.01.SPA.bin`

Install / set the boot image	`install add file flash:cat9k_iosxe.17.18.04.SPA.bin activate commit`
	`! IOS XE 26.1.x:`
	`install add file flash:cat9k_iosxe.26.01.01.SPA.bin activate commit`

Reboot into the new image	*n/a: Automatic: 'activate' reloads the switch into the new image*
Roll back to the previous image	`install rollback to id <install-ID>`

Install mode only (boot flash:packages.conf). Activate reloads by itself; an uncommitted image does not persist. Free space with install remove inactive. 17.18.x: end-of-sale 30 Apr 2027; Cisco advises IOS XE 26.1.1+.

MODERN CLI

Firmware upgrade

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`show version`
Check free space / partitions	`show system resource-utilization`
Transfer the image	`copy sftp://backup@192.0.2.20/<IMAGE>.swi secondary vrf mgmt`
Verify integrity (hash)	`show images`
Install / set the boot image	`boot set-default secondary`
Reboot into the new image	`boot system secondary`
Roll back to the previous image	`boot system primary`

There are two image slots (primary and secondary). Copy the new one to the idle slot, check its SHA-256 in 'show images' and boot it; to go back, 'boot system primary'. If config changed, boot prompts to save.

MODERN CLI

Firmware upgrade

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`show version`
Check free space / partitions	`show system storage`
Transfer the image	`file copy scp://backup@192.0.2.20/home/backup/<image>.tgz /var/tmp/`
Verify integrity (hash)	`file checksum sha-256 /var/tmp/<image>.tgz`
Install / set the boot image	`request system software add /var/tmp/<image>.tgz`
Reboot into the new image	`request system reboot`
Roll back to the previous image	`request system software rollback` `request system reboot`

On EX the image must be stored in /var/tmp. After rebooting and validating, run 'request system snapshot slice alternate' to copy the new image to the alternate partition.

MODERN CLI

Firmware upgrade

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`/system package print` `/system routerboard print`
Check free space / partitions	`/system resource print`
Transfer the image	`/tool fetch url="sftp://192.0.2.20/routeros-7.23.8-mipsbe.npk" user=backup \` ` password=<password>`
Verify integrity (hash)	*n/a: No file-hash command in docs; /system check-installation tests files*
Install / set the boot image	*n/a: No boot-image setting: the uploaded .npk installs on next reboot*

Reboot into the new image	`/system reboot`
	`/system routerboard upgrade`
	`/system reboot`

Roll back to the previous image	`/system package downgrade`

Upload the .npk (mipsbe architecture on the CRS354) and reboot to install. To go back, upload an older .npk and run package downgrade. After the upgrade, upgrade the bootloader too.

MODERN CLI

Firmware upgrade

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`show version`
Check free space / partitions	`dir flash:`
Transfer the image	`copy scp://backup@192.0.2.20/EOS-<release>.swi flash:`
Verify integrity (hash)	`verify /md5 flash:EOS-<release>.swi`

Install / set the boot image	`boot system flash:EOS-<release>.swi`
	`show boot-config`
	`write`

Reboot into the new image	`reload`

Roll back to the previous image	`boot system flash:EOS-<previous>.swi`
	`write`
	`reload`

Leave room for two copies of the .swi plus 240 MB: EOS copies the image into internal flash at boot. Compare the MD5 with Arista's download page. A downgrade uses the same steps.

MODERN CLI

Firmware upgrade

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 35 and 36 (EOS 2007)

Check the current version	`display version`
Check free space / partitions	`dir`

Transfer the image	`system-view`
	`scp -a 192.0.2.11 backup@192.0.2.20:<image>.cc flash:/<image>.cc`
	`scp -a 192.0.2.11 backup@192.0.2.20:<image>.cc.asc flash:/<image>.cc.asc`

Verify integrity (hash)	`system-view` `check file-integrity <image>.cc <image>.cc.asc`
Install / set the boot image	`startup system-software <image>.cc`
Reboot into the new image	`reboot`
Roll back to the previous image	`rollback`

The .cc image sits in the flash root; check file-integrity compares it with its .asc signature file. rollback works only within 48 h after a reboot-based upgrade and returns to the earlier config.

LAB

Lab: Basic Maintenance

- **a** Copy the current configuration to a file on the switch
- **b** Clear the switch configuration to factory defaults
- **c** Reconfigure the switch with a saved configuration file
- **d** Copy the configuration file to the TFTP server
- **e** Rename the configuration file on the TFTP server
- **f** View the configuration file on the TFTP server
- **g** Copy the configuration file from the TFTP server to the switch
- **h** Perform the procedure to reset the switch password
- **i** Perform a firmware upgrade on the switch

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.



Questions are welcome.

Questions are welcome

LAN Switch Operations and Maintenance

From the Enterasys legacy to the current CLI of eight vendors

MODULE 02 · DAY 2

Layer 2 switching



Rodolfo Nützmann

2026 Edition

Prepared for: [Client] · In partnership with: [Partner]

Course agenda

Day 1

DONE

Access, operation and basic maintenance

- Platforms: 2007 legacy and current
- CLI, users and session
- Clock, identity and connectivity
- Save, erase and reboot
- Password recovery
- Files, backups and firmware

Day 2

TODAY

Layer 2 switching

- Interfaces and naming
- Speed, duplex and jumbo frames
- Storm control, isolation, mirroring
- LACP link aggregation
- 802.1Q VLANs
- Spanning Tree and BPDU protection

Day 3

NEXT

IP routing

- Addressing, masks and subnets
- Route determination and ARP
- Proxy-ARP
- L3 interfaces and management
- Static routes, RIP and OSPF
- Routing table, ECMP and ARP

Day 4

NEXT

Redundancy, security and management

- OSPF and VRRP
- Access control lists
- Syslog, SNMP and AAA
- Multicast: IGMP and PIM
- Troubleshooting and support
- Final lab and assessment

2.1

Switch ports



Port naming, state and parameters; mirroring

- Interface naming and operational state
- Auto-negotiation, speed and duplex
- Flow control and broadcast suppression
- Auto-MDI/MDIX, protected ports and jumbo frames
- Port mirroring

Chapter 2.1 agenda

1. Interface naming and operational state
2. Auto-negotiation, speed and duplex
3. Flow control and broadcast suppression
4. Auto-MDI/MDIX, protected ports and jumbo frames
5. Port mirroring

MODERN CLI

- B07 Interfaces: naming and status
- B08 Speed, duplex, auto-negotiation and jumbo
- B09 Storm control and port isolation
- B10 Port mirroring

LEGACY 2007

Physical Interface Naming

ge . **1** . **1**

port type — slot / stack member — port number

Example: `ge.2.1` = first Gigabit Ethernet port of slot (or stack member) 2.

Port types (Enterasys EOS)

fe	100-Mbps Ethernet
ge	1-Gbps Ethernet
tg	10-Gbps Ethernet
vlan	vlan interfaces
lag	IEEE802.3 link aggregation ports
host.0.1	the host port
rtr	router interfaces
pc	Matrix Security Module
bp	backplane port
lpbk	loopback interfaces

LEGACY 2007

Physical Interface Naming

- Syntax: ***<port type>. <slot> . <port number>***

Slot

- **Matrix N**: Corresponds to the chassis slot number.
- On the N7, the leftmost is 1 and the rightmost is 7
- **SecureStack:** Corresponds to the “device number” in the stack.
- By default unit IDs are assigned in power-on order; powering on the units from top to bottom, the top of the stack is 1.
- Slot 0 refers to the virtual system ports (lag, vlan, host, loopback)



Physical Interface Naming

- Syntax: ***<port type>. <slot> . <port number>***

Port Number

- Corresponds to the physical port on each module or stack member.
- Example: fe.1.1 is the first Fast Ethernet port in slot 1
- Example: ge.1.1 is the first Gigabit Ethernet port in slot 1 (which may be the 25th logical port physically present in slot 1)
- E.g.: NEM modules with Gigabit uplinks on Fast Ethernet cards



Physical Interface Naming

- Using *wildcards*
- **fe.*.*** All 100 Mbps ports in any slot of the chassis
- **ge.2.*** All 1 Gigabit ports in slot or switch 2
- **ge.*.*** All 1 Gigabit ports in the chassis or stack
- ***.*.*** All ports of any type in all slots or switches of the stack
- Using port lists
- **fe.1.1-13** Denotes the 100Mbps ports 1 to 13 in slot 1
- **ge.1-7.10** Denotes port 10 on all Gigabit modules present in slots 1 to 7



LEGACY 2007

Checking the operational state of ports

Check whether a port is enabled or disabled

```
show port [port-string]
```

Check the operating state of ports (speed, duplex)

```
show port status [port-string]
```

Check traffic and error statistics on ports

```
show port counters [port-string] [switch | mib2]
```

Ports showing errors must have their causes investigated.

LEGACY 2007

Managing port parameters

Disable ports

```
set port disable port-string
```

Enable ports

```
set port enable port-string
```

Assign a name to a port

```
set port alias port-string [“name”]  ← To clear the alias, leave name blank
```

List the names assigned to ports

```
show port alias [port-string]
```

MODERN CLI

Interfaces: naming and status (1/2)

	Port naming format (example)	Port status summary
Extreme Switch Engine	`1 = stand-alone port number; 2:1 = slot:port (stack)` `49:4 = channelized port 49, channel 4`	`show ports no-refresh`
Extreme Fabric Engine	`interface gigabitEthernet 1/1` `1/1 = slot/port; channelized: slot/port/sub-port`	`show interfaces gigabitEthernet interface`
Cisco IOS XE	`interface GigabitEthernet1/0/1` `! 1/0/1 = switch/module/port (downlinks module 0, uplinks 1)`	`show interfaces status`
Aruba AOS-CX	`1/1/1 = member/slot/port (6300: slot is always 1)`	`show interface brief`

Standards: RFC 2863

SIDE-BY-SIDE VIEW · BLOCK B07

MODERN CLI

Interfaces: naming and status (2/2)

	Port naming format (example)	Port status summary
Juniper Junos	ge-0/0/0 = type-fpc/pic/port (ports count from 0: lab port 1 = ge-0/0/0)	show interfaces terse
MikroTik RouterOS	ether1..ether48, ether49, sfp-sfpplus1..sfp-sfpplus4 qsfpplus1-1..qsfpplus1-4, qsfpplus2-1..qsfpplus2-4	/interface print
Arista EOS	interface ethernet 1 (Ethernet1 = port 1; Et1 in output) Ethernet4/1 = module 4, port 1 on modular systems Ethernet5/1/1 = module/port/lane on multi-lane (breakout) ports	show interfaces status
Huawei VRP	GigabitEthernet0/0/1 = slot ID/subcard ID/port number (slot is 0) Stacked: stack ID (0 to 8)/subcard ID/port number	display interface brief

Standards: RFC 2863

MODERN CLI

Interfaces: naming and status

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`1 = stand-alone port number; 2:1 = slot:port (stack)` `49:4 = channelized port 49, channel 4`
Port range syntax	`show ports 1-4,24 no-refresh`
Port status summary	`show ports no-refresh`
Detailed port status	`show ports 1 information detail`
Traffic and error counters	`show ports 1 statistics no-refresh` `show ports 1 rxerrors no-refresh`
Clear counters	`clear counters`
Disable a port	`disable port 1`
Enable a port	`enable port 1`
Port description (alias)	`configure ports 1 description-string USERS-PC`

On a stand-alone switch a port is just its number (1, 24); in a stack it is slot:port. The alias (description-string, up to 255 chars) shows in show ports info detail and SNMP; clear counters clears everything.

MODERN CLI

Interfaces: naming and status

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`interface gigabitEthernet 1/1` `1/1 = slot/port; channelized: slot/port/sub-port`
Port range syntax	`interface gigabitEthernet 1/1-1/4,1/10`
Port status summary	`show interfaces gigabitEthernet interface`
Detailed port status	`show interfaces gigabitEthernet config 1/1` `show interfaces gigabitEthernet state 1/1`
Traffic and error counters	`show interfaces gigabitEthernet statistics 1/1` `show interfaces gigabitEthernet error 1/1`
Clear counters	`clear-stats port 1/1`
Disable a port	`interface gigabitEthernet 1/1` ` shutdown`

Port = slot/port (1/1). Ranges combine hyphen and comma (1/1-1/4,1/10); channelized ports add a sub-port (slot/port/sub-port); 'all' and 'slot/all' select every port.

MODERN CLI

Interfaces: naming and status (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 4 to 9 (EOS 2007)

Enable a port	`interface gigabitEthernet 1/1` ` no shutdown`
Port description (alias)	`interface gigabitEthernet 1/1` ` name USERS-ACCESS`

Port = slot/port (1/1). Ranges combine hyphen and comma (1/1-1/4,1/10); channelized ports add a sub-port (slot/port/sub-port); 'all' and 'slot/all' select every port.

MODERN CLI

Interfaces: naming and status

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`interface GigabitEthernet1/0/1` `! 1/0/1 = switch/module/port (downlinks module 0, uplinks 1)`
Port range syntax	`interface range GigabitEthernet1/0/1 - 4`
Port status summary	`show interfaces status`
Detailed port status	`show interfaces GigabitEthernet1/0/1`
Traffic and error counters	`show interfaces GigabitEthernet1/0/1 counters errors`
Clear counters	`clear counters GigabitEthernet1/0/1`
Disable a port	`interface GigabitEthernet1/0/1` ` shutdown`
Enable a port	`interface GigabitEthernet1/0/1` ` no shutdown`

Name = type + switch/module/port. Downlinks use module 0 and uplinks module 1 (e.g. Gi1/1/1). In interface range, put a space before and after the hyphen.

MODERN CLI

Interfaces: naming and status (cont.)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 4 to 9 (EOS 2007)

Port description (alias)	`interface GigabitEthernet1/0/1` ` description USERS-PC`

Name = type + switch/module/port. Downlinks use module 0 and uplinks module 1 (e.g. Gi1/1/1). In interface range, put a space before and after the hyphen.

MODERN CLI

Interfaces: naming and status

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`1/1/1 = member/slot/port (6300: slot is always 1)`
Port range syntax	`interface 1/1/1-1/1/4`
Port status summary	`show interface brief`
Detailed port status	`show interface 1/1/1`
Traffic and error counters	`show interface 1/1/1 statistics` `show interface 1/1/1 error-statistics`
Clear counters	`show interface 1/1/1 monitor` `(key c in monitor view clears statistics)`
Disable a port	`interface 1/1/1` ` shutdown`
Enable a port	`interface 1/1/1` ` no shutdown`

Port = member/slot/port (6300: slot is always 1). The 10.16 interface command list has no 'clear counters': in 'show interface <if> monitor', the c key clears the statistics.

MODERN CLI

Interfaces: naming and status (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port description (alias)	`interface 1/1/1` ` description USERS-PORT-1`

Port = member/slot/port (6300: slot is always 1). The 10.16 interface command list has no 'clear counters': in 'show interface <if> monitor', the c key clears the statistics.

MODERN CLI

Interfaces: naming and status

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`ge-0/0/0 = type-fpc/pic/port (ports count from 0: lab port 1 = ge-0/0/0)`
Port range syntax	`set interfaces interface-range ACCESS member-range ge-0/0/0 to ge-0/0/22`
Port status summary	`show interfaces terse`
Detailed port status	`show interfaces ge-0/0/0 detail`
Traffic and error counters	`show interfaces ge-0/0/0 extensive`
Clear counters	`clear interfaces statistics ge-0/0/0`
Disable a port	`set interfaces ge-0/0/0 disable`
Enable a port	`delete interfaces ge-0/0/0 disable`
Port description (alias)	`set interfaces ge-0/0/0 description "User access port"`

Junos numbers ports from 0: the first front-panel port is ge-0/0/0. To bring a port back up, delete the 'disable' statement (there is no 'enable').

MODERN CLI

Interfaces: naming and status

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`ether1..ether48, ether49, sfp-sfpplus1..sfp-sfpplus4 qsfpplus1-1..qsfpplus1-4, qsfpplus2-1..qsfpplus2-4`
Port range syntax	`/interface ethernet set ether1,ether2,ether3 comment=USERS`
Port status summary	`/interface print`
Detailed port status	`/interface ethernet monitor ether1 once`
Traffic and error counters	`/interface ethernet print stats`
Clear counters	`/interface ethernet reset-counters ether1`
Disable a port	`/interface set ether1 disabled=yes`
Enable a port	`/interface set ether1 disabled=no`
Port description (alias)	`/interface ethernet set ether1 comment=USERS`

Names follow the hardware: ether1-48, sfp-sfpplus1-4 and qsfpplus1-1 to 2-4. There is no 1-48 range: use comma lists. The port alias is the comment field.

MODERN CLI

Interfaces: naming and status

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`interface ethernet 1   (Ethernet1 = port 1; Et1 in output)`
	`Ethernet4/1 = module 4, port 1 on modular systems`
	`Ethernet5/1/1 = module/port/lane on multi-lane (breakout) ports`

Port range syntax	`interface ethernet 1 - 4`
Port status summary	`show interfaces status`
Detailed port status	`show interfaces ethernet 1`
Traffic and error counters	`show interfaces counters rates`
Clear counters	`clear counters`
Disable a port	`interface ethernet 1` ` shutdown`
Enable a port	`interface ethernet 1` ` no shutdown`

Fixed switches name ports EthernetN; modular systems use Ethernet module/port. Output abbreviates to Et. An error-disabled interface is recovered with 'shutdown' then 'no shutdown'.

MODERN CLI

Interfaces: naming and status (cont.)

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 4 to 9 (EOS 2007)

Port description (alias)	`interface ethernet 1` ` description USERS-PORT-01`

Fixed switches name ports EthernetN; modular systems use Ethernet module/port. Output abbreviates to Et. An error-disabled interface is recovered with 'shutdown' then 'no shutdown'.

MODERN CLI

Interfaces: naming and status

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port naming format (example)	`GigabitEthernet0/0/1 = slot ID/subcard ID/port number (slot is 0)` `Stacked: stack ID (0 to 8)/subcard ID/port number`
Port range syntax	`interface range GigabitEthernet0/0/1 to GigabitEthernet0/0/24`
Port status summary	`display interface brief`
Detailed port status	`display interface GigabitEthernet0/0/1`
Traffic and error counters	`display counters interface GigabitEthernet0/0/1` `display counters error interface GigabitEthernet0/0/1`
Clear counters	`reset counters interface GigabitEthernet0/0/1`
Disable a port	`interface GigabitEthernet0/0/1` ` shutdown`
Enable a port	`interface GigabitEthernet0/0/1` ` undo shutdown`

Port = slot/subcard/number (GigabitEthernet0/0/1); in a stack the first number is the stack ID. shutdown and undo shutdown turn a port off and on; cleared counters cannot be restored.

MODERN CLI

Interfaces: naming and status (cont.)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 4 to 9 (EOS 2007)

Port description (alias)	`interface GigabitEthernet0/0/1` ` description USER-PC-01`

Port = slot/subcard/number (GigabitEthernet0/0/1); in a stack the first number is the stack ID. shutdown and undo shutdown turn a port off and on; cleared counters cannot be restored.

Managing port configuration

- Speed and duplex auto-negotiation on Ethernet ports should *never* be enabled on only one side of a link.
- Either both sides have it enabled;
- Or both sides need manual configuration.
- For ports with manual speed and duplex configuration, auto-negotiation must be disabled.
- In situations where it is not possible to manually configure the parameters on the other side of the link, leave auto-negotiation enabled and adjust the advertised capabilities (*advertised ability*)
- Beware of auto-MDI/MDIX!



LEGACY 2007

Auto-negotiation on ports

Check the operational state of ports

```
show port negotiation [port-string]
```

Enable or disable auto-negotiation on ports

```
set port negotiation port-string {enable | disable}
```

List the operating modes negotiated for a port

```
show port advertise [port-string]
```

Add operating modes to auto-negotiation

```
set port advertise {port-string}{10t | 10tfd | 100tx | 100txfd | 1000t | 1000tfd | pause}
```

Remove operating modes from auto-negotiation

```
clear port advertise {port-string}{10t | 10tfd | 100tx | 100txfd | 1000t | 1000tfd | pause}
```

LEGACY 2007

Port speed and duplex configuration

Check the configured speed on ports

```
show port speed [port-string]
```

Manually set the speed on ports (auto-neg must be disabled!)

```
set port speed port-string {10 | 100 | 1000}
```

Check the configured duplex on ports

```
show port duplex [port-string]
```

Manually set the duplex on ports (auto-neg must be disabled!)

```
set port duplex port-string {full | half}
```

LEGACY 2007

Flow control

- IEEE 802.3x flow control prevents network interfaces from being overwhelmed by excessive transmissions from devices during overload conditions.
- This is done by sending messages that ask the sender to pause or reduce its transmission rate.
- To check its operational state: `show flowcontrol` (SecureStack, global) or `show port flowcontrol [port-string]` (Matrix N, per port)
- To enable or disable: `set flowcontrol {enable | disable}` (SecureStack) or `set port flowcontrol port-string {receive | send | both} {enable | disable}` (Matrix N)

LEGACY 2007

Broadcast suppression

- Broadcast suppression limits the number of broadcasts received on a port.
- The default is 14881 pps on each port on SecureStack (on the Matrix N the default is no limit).
- Broadcasts are necessary in networks.
- Limiting them too much, or having too many of them, can be harmful.
- The ideal is to find their sources and eliminate the invalid ones, or fix network design problems.
- To check the value on the ports: `show port broadcast [port-string]`
- To set the value in pps: `set port broadcast port-string threshold-val`
- To return to the default configuration: `clear port broadcast port-string threshold`

LEGACY 2007

Auto-MDI/MDIX

This is the feature that eliminates the need for “crossover” cables to connect a switch to another switch or hub.

Be careful: if a switch port is connected to another port on the same switch, this feature will bring up the link on both ports, and a loop can occur, with the resulting unavailability of the network segment.

DFE (checking the operating state):

```
show port mdix [port-string] {all | auto | mdi | mdix}
```

DFE (configuring port operation):

```
set port mdix [port-string] {auto | mdi | mdix}
```

LEGACY 2007

Protected Ports

- Feature exclusive to SecureStack
- Configuring “protected ports” prevents members of the same VLAN from exchanging data with each other.
- Protected ports belonging to the same group cannot communicate with the other protected ports of the same group.
- They can communicate with all other ports (unprotected by default) and with the protected ports of other groups.
- This feature is limited to the same switch or stack.



Protected Ports

LEGACY 2007

Add protected ports to a group

```
set port protected port-string group-id
```

View information about protected ports

```
show port protected [port-string] | [group-id]
```

Remove protected ports from a group (unprotecting the port)

```
clear port protected [port-string] | [group-id]
```

Manage the name of a protected port group

```
set port protected name group-id name
show port protected name group-id
clear port protected name group-id
```

LEGACY 2007

Jumbo Frames

Gigabit Ethernet ports are able to transmit frames of up to 10KB, larger than the 1500-byte maximum specified for Ethernet and Fast Ethernet.

It is a way to increase traffic efficiency and performance.

To view the port configuration:

```
show port jumbo [port-string]
```

To enable or disable:

```
set port jumbo {enable | disable} [port-string]
```

To return ports to factory defaults (SecureStack: enabled; Matrix N: disabled):

```
clear port jumbo [port-string]
```

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (1/2)

	Force speed to 1 Gb/s	Jumbo frames / MTU 9216
Extreme Switch Engine	`configure ports 1 auto off speed 1000 duplex full`	`configure jumbo-frame-size 9216` `enable jumbo-frame ports all`

Extreme Fabric Engine	`interface gigabitEthernet 1/1`
	` no auto-negotiate`
	` speed 1000`
		`sys mtu 9600`

Cisco IOS XE	`interface GigabitEthernet1/0/1` ` speed 1000`	`system mtu 9198`
Aruba AOS-CX	`interface 1/1/1` ` speed 1000-full`	`interface 1/1/1` ` mtu 9198`

Standards: IEEE 802.3-2022 · IEEE 802.3by-AN-OVERVIEW · IEEE 802.3x-1997 · IEEE 802.3-INTERP-1-07-03 · IEEE 802.3as-2006

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (2/2)

	Force speed to 1 Gb/s	Jumbo frames / MTU 9216
Juniper Junos	`set interfaces ge-0/0/0 ether-options auto-negotiation` `set interfaces ge-0/0/0 speed 1g`	`set interfaces ge-0/0/0 mtu 9216`
MikroTik RouterOS	`/interface ethernet set ether1 advertise=1G-baseT-full` `/interface ethernet set sfp-sfpplus1 auto-negotiation=no speed=1G-baseX`	`/interface ethernet set ether1,ether24 l2mtu=10218 mtu=9216`
Arista EOS	`interface ethernet 1` ` speed 1000full`	`interface vlan 10` ` mtu 9214`

Huawei VRP	`interface GigabitEthernet0/0/1`
	` undo negotiation auto`
	` speed 1000`
		`jumboframe enable 9216`

Standards: IEEE 802.3-2022 · IEEE 802.3by-AN-OVERVIEW · IEEE 802.3x-1997 · IEEE 802.3-INTERP-1-07-03 · IEEE 802.3as-2006

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`show ports 1 configuration no-refresh`
Enable auto-negotiation	`configure ports 1 auto on`
Force speed to 1 Gb/s	`configure ports 1 auto off speed 1000 duplex full`
Force full duplex	`configure ports 1 auto off speed 100 duplex full`
Restrict advertised speeds	`configure ports 1 auto on speed 1000 duplex full`
802.3x flow control	`enable flow-control rx-pause ports 1` `enable flow-control tx-pause ports 1`
Auto-MDI/MDIX	`configure ports 1 auto-polarity on`
Jumbo frames / MTU 9216	`configure jumbo-frame-size 9216` `enable jumbo-frame ports all`

With auto off, speed and duplex go in the same command. Gigabit ports support flow control only with auto-negotiation on. Jumbo frames are disabled by default: enable them per port and set the size.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`show interfaces gigabitEthernet l1-config 1/1`
Enable auto-negotiation	`interface gigabitEthernet 1/1` ` auto-negotiate enable`

Force speed to 1 Gb/s	`interface gigabitEthernet 1/1`
	` no auto-negotiate`
	` speed 1000`

Force full duplex	`interface gigabitEthernet 1/1` ` duplex full`
Restrict advertised speeds	`interface gigabitEthernet 1/1` ` auto-negotiation-advertisements 1000-full`

802.3x flow control	`boot config flags flow-control-mode`
	`interface gigabitEthernet 1/1`
	` tx-flow-control enable`

MTU 9216 does not exist: 'sys mtu' accepts 1522, 1950 or 9600 (9416 on the 7830 only). 802.3x pause needs the flow-control-mode flag, save and reboot before tx-flow-control.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

Auto-MDI/MDIX	*n/a: No command: automatic on fixed copper ports (auto-neg. on)*
Jumbo frames / MTU 9216	`sys mtu 9600`

MTU 9216 does not exist: 'sys mtu' accepts 1522, 1950 or 9600 (9416 on the 7830 only). 802.3x pause needs the flow-control-mode flag, save and reboot before tx-flow-control.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`show interfaces GigabitEthernet1/0/1 status`

Enable auto-negotiation	`interface GigabitEthernet1/0/1`
	` speed auto`
	` duplex auto`

Force speed to 1 Gb/s	`interface GigabitEthernet1/0/1` ` speed 1000`
Force full duplex	`interface GigabitEthernet1/0/1` ` duplex full`
Restrict advertised speeds	`interface GigabitEthernet1/0/1` ` speed auto 100 1000`
802.3x flow control	`interface GigabitEthernet1/0/1` ` flowcontrol receive on`

The maximum MTU on Catalyst 9300 is 9198, not 9216. port-settings (speed/duplex/autoneg) coexists with speed/duplex; the last one applied wins and it shows only in show running-config yang.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (cont.)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

Auto-MDI/MDIX	`interface GigabitEthernet1/0/1` ` mdix auto`
Jumbo frames / MTU 9216	`system mtu 9198`

The maximum MTU on Catalyst 9300 is 9198, not 9216. port-settings (speed/duplex/autoneg) coexists with speed/duplex; the last one applied wins and it shows only in show running-config yang.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`show interface 1/1/1`
Enable auto-negotiation	`interface 1/1/1` ` speed auto`
Force speed to 1 Gb/s	`interface 1/1/1` ` speed 1000-full`
Force full duplex	`interface 1/1/1` ` speed 1000-full`
Restrict advertised speeds	`interface 1/1/1` ` speed auto 100m 1g`
802.3x flow control	`interface 1/1/1` ` flow-control rxtx`
Auto-MDI/MDIX	*n/a: No MDI/MDIX command in the 10.16 interface command list*

Speed and duplex are one command (speed 1000-full). Maximum MTU is 9198 bytes, not 9216. 'speed auto <speed>' limits advertised speeds; the default is auto with all speeds.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

Jumbo frames / MTU 9216

```
interface 1/1/1
 mtu 9198
```

Speed and duplex are one command (speed 1000-full). Maximum MTU is 9198 bytes, not 9216. 'speed auto <speed>' limits advertised speeds; the default is auto with all speeds.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`show interfaces ge-0/0/0 media`
Enable auto-negotiation	`set interfaces ge-0/0/0 ether-options auto-negotiation`
Force speed to 1 Gb/s	`set interfaces ge-0/0/0 ether-options auto-negotiation` `set interfaces ge-0/0/0 speed 1g`
Force full duplex	`set interfaces ge-0/0/0 link-mode full-duplex`
Restrict advertised speeds	*n/a: Advertising one speed only: documented only for QFX5100-48T*
802.3x flow control	`set interfaces ge-0/0/0 ether-options flow-control`
Auto-MDI/MDIX	`show interfaces ge-0/0/0 media | match MDIX`
Jumbo frames / MTU 9216	`set interfaces ge-0/0/0 mtu 9216`

At 1 Gb/s on copper auto-negotiation must stay on; only 10/100 Mb/s are forced with no-auto-negotiation + speed + link-mode. Auto-MDIX is shown in the 'media' output.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`/interface ethernet monitor ether1 once`
Enable auto-negotiation	`/interface ethernet set ether1 auto-negotiation=yes`
Force speed to 1 Gb/s	`/interface ethernet set ether1 advertise=1G-baseT-full` `/interface ethernet set sfp-sfpplus1 auto-negotiation=no speed=1G-baseX`
Force full duplex	`/interface ethernet set ether1 auto-negotiation=no speed=100M-baseT-full`
Restrict advertised speeds	`/interface ethernet set ether1 advertise=100M-baseT-full,1G-baseT-full`
802.3x flow control	`/interface ethernet set ether1 tx-flow-control=on rx-flow-control=on`
Auto-MDI/MDIX	*n/a: Always on: mdix-enable is fixed to yes on this hardware*
Jumbo frames / MTU 9216	`/interface ethernet set ether1,ether24 l2mtu=10218 mtu=9216`

Gigabit and NBASE-T links cannot work with auto-negotiation off: limit speeds with advertise. Forcing speed fits SFP+ (1G-baseX) and up to 100M. The CRS354 jumbo maximum is 10218.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`show interfaces status`
Enable auto-negotiation	`interface ethernet 1` ` speed auto`
Force speed to 1 Gb/s	`interface ethernet 1` ` speed 1000full`
Force full duplex	`interface ethernet 1` ` speed 1000full`
Restrict advertised speeds	`interface ethernet 1` ` speed auto 1000full`
802.3x flow control	`interface ethernet 1` ` flowcontrol receive on`
Auto-MDI/MDIX	*n/a: No documented command in EOS 4.36*

Duplex is part of the speed token (e.g. 1000full). L2 ports already have a fixed 9236 MTU (jumbo always on); on an L3 interface 'mtu' ranges 68 to 9214, not 9216.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (cont.)

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

Jumbo frames / MTU 9216

```
interface vlan 10
 mtu 9214
```

Duplex is part of the speed token (e.g. 1000full). L2 ports already have a fixed 9236 MTU (jumbo always on); on an L3 interface 'mtu' ranges 68 to 9214, not 9216.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

View negotiated speed/duplex	`display interface ethernet brief`
Enable auto-negotiation	`interface GigabitEthernet0/0/1` ` negotiation auto`

Force speed to 1 Gb/s	`interface GigabitEthernet0/0/1`
	` undo negotiation auto`
	` speed 1000`

Force full duplex	`interface GigabitEthernet0/0/1` ` duplex full`
Restrict advertised speeds	`interface GigabitEthernet0/0/1` ` auto speed 100 1000`
802.3x flow control	`interface GigabitEthernet0/0/1` ` flow-control`

Forcing speed or duplex needs undo negotiation auto first (negotiation auto is the default). Flow control must be on at both ends. jumboframe in system view applies to all interfaces.

MODERN CLI

Speed, duplex, auto-negotiation and jumbo (cont.)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 10 to 15 and 18 (EOS 2007)

Auto-MDI/MDIX	`interface GigabitEthernet0/0/1` ` mdi auto`
Jumbo frames / MTU 9216	`jumboframe enable 9216`

Forcing speed or duplex needs undo negotiation auto first (negotiation auto is the default). Flow control must be on at both ends. jumboframe in system view applies to all interfaces.

MODERN CLI

Storm control and port isolation (1/2)

	Limit broadcast (storm control)	Isolate ports from each other (protected ports)
Extreme Switch Engine	`configure ports 1 rate-limit flood broadcast 500`	`configure ports 1,2 isolation on`

Extreme Fabric Engine	`interface gigabitEthernet 1/1`
	` rate-limit broadcast 1000`
		`vlan create 10 name USERS type pvlan-mstprstp 0 secondary 110`
		`interface gigabitEthernet 1/1`
		` private-vlan isolated`

Cisco IOS XE	`interface GigabitEthernet1/0/1` ` storm-control broadcast level 10`	`interface range GigabitEthernet1/0/1 - 2` ` switchport protected`

Aruba AOS-CX	`interface 1/1/1`
	` rate-limit broadcast 10 percent`
		`vlan 110`
		` private-vlan isolated primary-vlan 10`
		`(port: private-vlan port-type secondary)`

Standards: RFC 5517

MODERN CLI

Storm control and port isolation (2/2)

	Limit broadcast (storm control)	Isolate ports from each other (protected ports)

Juniper Junos	`set forwarding-options storm-control-profiles SC all bandwidth-percentage 10`
	`set interfaces ge-0/0/0 unit 0 family ethernet-switching storm-control SC`
		`set vlans ISO private-vlan isolated vlan-id 11`
		`set vlans USERS vlan-id 10 isolated-vlan ISO`
		`set interfaces ge-0/0/0 unit 0 family ethernet-switching vlan members ISO`

MikroTik RouterOS	`/interface ethernet switch port` `set ether1 limit-broadcasts=yes storm-rate=1`	`/interface ethernet switch port-isolation` `set ether1 forwarding-override=ether24`
Arista EOS	`interface ethernet 1` `storm-control broadcast level 5`	`vlan 110` `private-vlan isolated primary vlan 10`

Huawei VRP	`interface GigabitEthernet0/0/1`
	`storm-control broadcast min-rate 3000 max-rate 5000`
	`storm-control action block`
		`interface range GigabitEthernet0/0/1 to GigabitEthernet0/0/22`
		`port-isolate enable group 1`

Standards: RFC 5517

MODERN CLI

Storm control and port isolation

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`configure ports 1 rate-limit flood broadcast 500`
Limit multicast / unknown unicast	`configure ports 1 rate-limit flood multicast 500` `configure ports 1 rate-limit flood unknown-destmac 500`
View storm control	`show ports 1 rate-limit flood no-refresh`
Isolate ports from each other (protected ports)	`configure ports 1,2 isolation on`
View isolation	`show ports 1,2 information detail`

Switch Engine storm control is the per-port rate-limit flood, in pps (0 to 262,144). The multicast keyword also limits known multicast. Ports with isolation on cannot talk to each other.

MODERN CLI

Storm control and port isolation

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`interface gigabitEthernet 1/1` ` rate-limit broadcast 1000`

Limit multicast / unknown unicast	`interface gigabitEthernet 1/1`
	` rate-limit multicast 1000`
	` rate-limit unknown-unicast 1000`

View storm control	`show interfaces gigabitEthernet rate-limit 1/1`

Isolate ports from each other (protected ports)	`vlan create 10 name USERS type pvlan-mstprstp 0 secondary 110`
	`interface gigabitEthernet 1/1`
	` private-vlan isolated`

View isolation	`show interfaces gigabitEthernet private-vlan 1/1` `show vlan private-vlan 10`

rate-limit is in packets per second (1-65535), not percent. In a private VLAN, 'isolated' ports talk only to 'promiscuous' ports (the uplink), never to each other.

MODERN CLI

Storm control and port isolation

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`interface GigabitEthernet1/0/1` ` storm-control broadcast level 10`

Limit multicast / unknown unicast	`interface GigabitEthernet1/0/1`
	` storm-control multicast level 20`
	` storm-control unicast level 20`

View storm control	`show storm-control GigabitEthernet1/0/1`
Isolate ports from each other (protected ports)	`interface range GigabitEthernet1/0/1 - 2` ` switchport protected`
View isolation	`show interfaces GigabitEthernet1/0/1 switchport`

Storm control is off by default. The unicast level covers known and unknown unicast. Protected ports pass no L2 data between each other, only to unprotected ports.

MODERN CLI

Storm control and port isolation

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`interface 1/1/1` ` rate-limit broadcast 10 percent`

Limit multicast / unknown unicast	`interface 1/1/1`
	` rate-limit multicast 10 percent`
	` rate-limit unknown-unicast 10 percent`

View storm control	`show interface 1/1/1 qos`

Isolate ports from each other (protected ports)	`vlan 110`
	` private-vlan isolated primary-vlan 10`
	`(port: private-vlan port-type secondary)`

View isolation	`show private-vlan`

No 'storm-control': use a per-interface 'rate-limit' (broadcast, multicast, unknown-unicast; kbps, pps or percent). Isolation: Private VLAN; VLAN 10 must be 'private-vlan primary' and the ports 'port-type secondary'.

MODERN CLI

Storm control and port isolation

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`set forwarding-options storm-control-profiles SC all bandwidth-percentage 10` `set interfaces ge-0/0/0 unit 0 family ethernet-switching storm-control SC`

Limit multicast / unknown unicast	`set forwarding-options storm-control-profiles SC-MC all bandwidth-level 5000`
	`set forwarding-options storm-control-profiles SC-MC all no-broadcast`
	`set interfaces ge-0/0/0 unit 0 family ethernet-switching storm-control SC-MC`

View storm control	`show configuration forwarding-options storm-control-profiles` `show interfaces ge-0/0/0 extensive`

Isolate ports from each other (protected ports)	`set vlans ISO private-vlan isolated vlan-id 11`
	`set vlans USERS vlan-id 10 isolated-vlan ISO`
	`set interfaces ge-0/0/0 unit 0 family ethernet-switching vlan members ISO`

View isolation	`show vlans private-vlan`

On ELS storm control is on by default (80%) on L2 interfaces. Port isolation uses a private VLAN (PVLAN); the primary VLAN also needs a promiscuous or trunk port (ELS procedure).

MODERN CLI

Storm control and port isolation

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`/interface ethernet switch port set ether1 limit-broadcasts=yes storm-rate=1`
Limit multicast / unknown unicast	`/interface ethernet switch port set ether1 limit-unknown-multicasts=yes limit-unknown-unicasts=yes`
View storm control	`/interface ethernet switch port print`
Isolate ports from each other (protected ports)	`/interface ethernet switch port-isolation set ether1 forwarding-override=ether24`
View isolation	`/interface ethernet switch port-isolation print`

Storm control applies on ingress, in hardware: storm-rate is a % of link speed (default 100). Isolation is per port, not per VLAN, and hardware-offloaded bonding is not in the port-isolation menu.

MODERN CLI

Storm control and port isolation

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`interface ethernet 1` ` storm-control broadcast level 5`

Limit multicast / unknown unicast	`interface ethernet 1`
	` storm-control multicast level 5`
	` storm-control unknown-unicast level 5`

View storm control	`show storm-control`
Isolate ports from each other (protected ports)	`vlan 110` ` private-vlan isolated primary vlan 10`
View isolation	`show vlan private-vlan`

Level is a % of bandwidth or 'level pps N', never both on one interface. No 'switchport protected': isolate with private VLANs (ports with 'switchport access vlan 110'); the manual names the CCS-720XP.

MODERN CLI

Storm control and port isolation

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 14, 16 and 17 (EOS 2007)

Limit broadcast (storm control)	`interface GigabitEthernet0/0/1`
	` storm-control broadcast min-rate 3000 max-rate 5000`
	` storm-control action block`

Limit multicast / unknown unicast	`interface GigabitEthernet0/0/1`
	` storm-control multicast min-rate 3000 max-rate 5000`
	` storm-control unicast min-rate 3000 max-rate 5000`

View storm control	`display storm-control interface GigabitEthernet0/0/1`
Isolate ports from each other (protected ports)	`interface range GigabitEthernet0/0/1 to GigabitEthernet0/0/22` ` port-isolate enable group 1`
View isolation	`display port-isolate group all`

storm-control (pps, min/max) takes action block or error-down and cannot be combined with *-suppression on one port. port-isolate isolates at Layer 2 only; port-isolate mode all adds Layer 3.

Port Mirroring

- Port mirroring is used to replicate traffic to another port that is not part of the original communication
- It is a feature normally used for:
- Troubleshooting problems with a protocol analyzer
- Recording voice conversations or instant messaging
- Security monitoring (IDS)
- Performance monitoring of applications or servers



LEGACY 2007

Port Mirroring

SecureStack

- Port mirror support:
- One to one
- Many to one: up to 8 source ports
- Only one destination port (monitor port) per system
- Bidirectional traffic is mirrored
- Only physical ports can be mirrored
- Mirrored ports cannot belong to LAGs



LEGACY 2007

Port Mirroring

DFE

- Port mirror support:
- One to one
- Many to one: several source ports
- Many to many: up to 16 destination ports per DFE-Platinum module
- Unidirectional or bidirectional traffic mirroring
- Physical ports, LAGs and VLANs can be mirrored



LEGACY 2007

Configuring port mirroring

To check the configured port mirrors:

```
show port mirroring
```

DFE (managing port mirrors):

```
set port mirroring {create | disable | enable} |
igmp-mcast {enable   | disable}source destination
[both | rx | tx]
clear port mirroring {igmp-mcast | source
destination}
```

SecureStack (managing port mirrors):

```
set port mirroring {create | disable | enable}
source destination
clear port mirroring source destination
```

Port mirroring (1/2)

MODERN CLI

	Create a local session (source, both directions)	Show sessions
Extreme Switch Engine	`create mirror SPAN1` `configure mirror SPAN1 add port 1`	`show mirror`
Extreme Fabric Engine	`mirror-by-port 1 in-port 1/1 out-port 1/23` `enable mode both`	`show mirror-by-port` `show mirror-resources`
Cisco IOS XE	`monitor session 1 source interface` `GigabitEthernet1/0/1 both`	`show monitor`
Aruba AOS-CX	`mirror session 1` ` source interface 1/1/1 both`	`show mirror` `show mirror 1`

Standards: DRAFT-FOSCHIANO-ERSPAN-03 RFC 1701

Port mirroring (2/2)

MODERN CLI

	Create a local session (source, both directions)	Show sessions
Juniper Junos	`set forwarding-options analyzer M1 input ingress interface ge-0/0/0.0` `set forwarding-options analyzer M1 input egress interface ge-0/0/0.0`	`show analyzer`
MikroTik RouterOS	`/interface ethernet switch port set ether1 mirror-ingress=yes mirror-egress=yes`	`/interface ethernet switch print detail` `/interface ethernet switch port print detail`
Arista EOS	`monitor session 1 source ethernet 1 both`	`show monitor session`
Huawei VRP	`interface GigabitEthernet0/0/1` ` port-mirroring to observe-port 1 both`	`display port-mirroring` `display observe-port`

Standards: DRAFT-FOSCHIANO-ERSPAN-03 · RFC 1701

MODERN CLI

Port mirroring

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`create mirror SPAN1` `configure mirror SPAN1 add port 1`
Set the destination port	`configure mirror SPAN1 to port 23` `enable mirror SPAN1`
Receive-only (rx) source	`configure mirror SPAN1 delete port 1` `configure mirror SPAN1 add port 1 ingress`
Show sessions	`show mirror`
Remove the session	`disable mirror SPAN1` `delete mirror SPAN1`
Remote mirroring (RSPAN/ERSPAN)	`create mirror RSPAN1 to port 23 remote-tag 1000` `create mirror ERSPAN1 to remote-ip <collector-ip> vr VR-Default`

Here the destination is part of the instance (to port). The monitor port leaves all VLANs. Default is ingress and egress. RSPAN uses a remote-tag; ERSPAN uses GRE to a remote IP that is not in VR-Mgmt.

MODERN CLI

Port mirroring

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`mirror-by-port 1 in-port 1/1 out-port 1/23 enable mode both`

Set the destination port	`no mirror-by-port 1 enable`
	`mirror-by-port monitor-port 1 1/23`
	`mirror-by-port 1 enable`

Receive-only (rx) source	`no mirror-by-port 1 enable`
	`mirror-by-port 1 mode rx`
	`mirror-by-port 1 enable`

Show sessions	`show mirror-by-port` `show mirror-resources`
Remove the session	`no mirror-by-port 1 enable` `no mirror-by-port 1`
Remote mirroring (RSPAN/ERSPAN)	`mirror-by-port 2 in-port 1/1 monitor-isid-offset 2 mode both`

in-port is the source and out-port is the analyzer. Default is rx; 'both' uses 2 of the 4 mirror resources. Remove the destination port from all VLANs. Remote (I-SID RSPAN) needs SPB.

Port mirroring

MODERN CLI

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`monitor session 1 source interface GigabitEthernet1/0/1 both`
Set the destination port	`monitor session 1 destination interface GigabitEthernet1/0/23`
Receive-only (rx) source	`monitor session 1 source interface GigabitEthernet1/0/1 rx`
Show sessions	`show monitor`
Remove the session	`no monitor session 1`
Remote mirroring (RSPAN/ERSPAN)	`monitor session 1 destination remote vlan 100`

Source and destination use the same session number. The destination is a physical port (never an EtherChannel). Without both/rx/tx the source mirrors both directions. ERSPAN has its own chapter.

MODERN CLI

Port mirroring

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`mirror session 1` ` source interface 1/1/1 both`

Set the destination port	`mirror session 1`
	` destination interface 1/1/23`
	` enable`

Receive-only (rx) source	`mirror session 1` ` source interface 1/1/1 rx`
Show sessions	`show mirror` `show mirror 1`
Remove the session	`no mirror session 1`
Remote mirroring (RSPAN/ERSPAN)	`mirror session 1` ` destination tunnel 192.0.2.60 source 10.255.0.1 dscp 0 vrf default`

Sessions 1 to 4, disabled by default ('enable' turns one on). The direction (both, rx or tx) is mandatory. ERSPAN uses 'destination tunnel' and cannot leave through the management port (mgmt VRF).

Port mirroring

MODERN CLI

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`set forwarding-options analyzer M1 input ingress interface ge-0/0/0.0` `set forwarding-options analyzer M1 input egress interface ge-0/0/0.0`
Set the destination port	`set forwarding-options analyzer M1 output interface ge-0/0/22.0`
Receive-only (rx) source	`set forwarding-options analyzer M1 input ingress interface ge-0/0/0.0`
Show sessions	`show analyzer`
Remove the session	`delete forwarding-options analyzer M1`
Remote mirroring (RSPAN/ERSPAN)	`set forwarding-options analyzer M1 output vlan RSPAN`

Mirroring is an 'analyzer' under forwarding-options. EX2300/EX3400 allow at most 4 active sessions, and egress-mirrored copies can carry VLAN tags that differ from the original.

MODERN CLI

Port mirroring

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`/interface ethernet switch port` `  set ether1 mirror-ingress=yes mirror-egress=yes`
Set the destination port	`/interface ethernet switch set switch1 mirror-target=ether23`
Receive-only (rx) source	`/interface ethernet switch port` `  set ether1 mirror-ingress=yes mirror-egress=no`
Show sessions	`/interface ethernet switch print detail` `/interface ethernet switch port print detail`
Remove the session	`/interface ethernet switch port set ether1 mirror-ingress=no mirror-egress=no` `/interface ethernet switch set switch1 mirror-target=none`
Remote mirroring (RSPAN/ERSPAN)	`/interface ethernet switch set switch1 rspan=yes \` `  rspan-egress-vlan-id=999 rspan-ingress-vlan-id=999`

No numbered sessions: the source is set per port (mirror-ingress/egress, since 7.15) and there is one mirror-target per switch, on the same switch chip. RSPAN wraps the copy in a VLAN.

MODERN CLI

Port mirroring

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`monitor session 1 source ethernet 1 both`
Set the destination port	`monitor session 1 destination ethernet 23`
Receive-only (rx) source	`monitor session 1 source ethernet 1 rx`
Show sessions	`show monitor session`
Remove the session	`no monitor session 1`
Remote mirroring (RSPAN/ERSPAN)	`monitor session 1 destination tunnel mode gre`

A session mirrors ingress and egress by default; ACL filtering on tx is unsupported. Remote: GRE tunnel destination (with source/destination); the manual has no RSPAN/ERSPAN. 'destination cpu' captures on the switch.

Port mirroring

MODERN CLI

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 19 to 22 (EOS 2007)

Create a local session (source, both directions)	`interface GigabitEthernet0/0/1` ` port-mirroring to observe-port 1 both`
Set the destination port	`observe-port 1 interface GigabitEthernet0/0/23`
Receive-only (rx) source	`interface GigabitEthernet0/0/1` ` port-mirroring to observe-port 1 inbound`
Show sessions	`display port-mirroring` `display observe-port`

Remove the session	`interface GigabitEthernet0/0/1`
	` undo port-mirroring both`
	`undo observe-port 1`

Remote mirroring (RSPAN/ERSPAN)	`observe-port 1 interface GigabitEthernet0/0/23 vlan <vlan-id>`

Define the observe-port first, then bind the source port to it. Dedicate the observing port to mirroring; the MEth port cannot be one. Layer 2 remote mirroring uses observe-port ... vlan.

2.2

Link Aggregation (LACP)



Link aggregation with LACP

LACP: link aggregation with dynamic control

Group, members, LACP rate and traffic distribution

Chapter 2.2 agenda

1. LACP: link aggregation with dynamic control
2. Group, members, LACP rate and traffic distribution

MODERN CLI

B11 LACP link aggregation

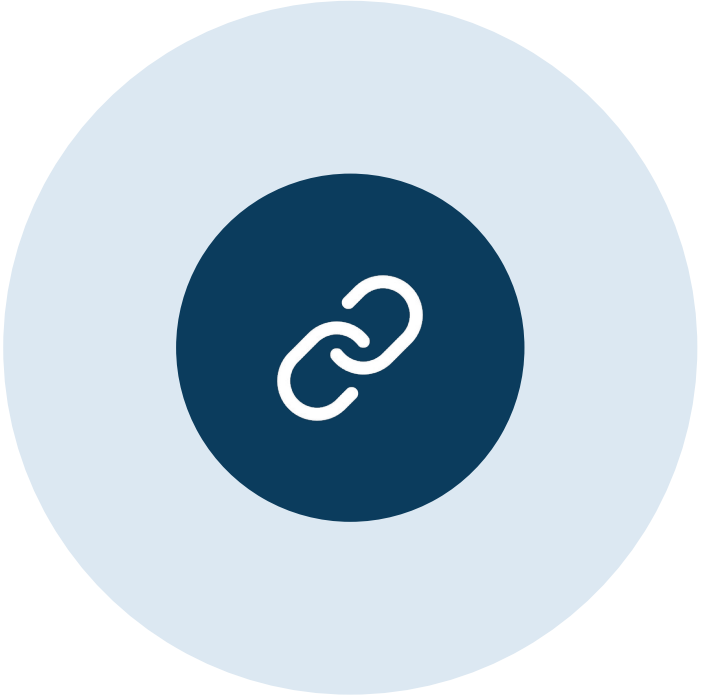
Link Aggregation LACP

- LACP = 802.3ad Link Aggregation Control Protocol
- It consists of grouping multiple links between two devices into a single logical link, adding up the speeds of the physical links.
- The resulting logical interface is called a *Link Aggregation Group (lag)* on the switch
- E.g.: Two Gigabit connections between two switches = a 2Gbps *lag*
- Advantages:
- More bandwidth for congested links
- Fault tolerance through link redundancy



Link Aggregation LACP

- Ports must operate at the same speed and in Full Duplex.
- LACP is enabled by default on all ports.
- If multiple connections are made between two switches with LACP enabled, the LAG will be created automatically.
- Attention:
- Both the physical ports that make up a LAG and the LAG itself must belong to the same VLAN
- Failing to create a LAG for multiple links between two devices can result in a network loop, causing downtime (if Spanning Tree is not enabled)



LEGACY 2007

Link Aggregation LACP

To check the global or port parameters (the state option is Matrix N only):

```
show lacp [state | port-string]
```

To enable or disable LACP globally:

```
set lacp {disable | enable}
```

Check the LACP parameters of ports:

```
show port lacp port port-string {[status
{detail | summary}] | [counters]}
```

Enable or disable LACP per port:

```
set port lacp port port-string {enable |
disable}
```

MODERN CLI

LACP link aggregation (1/2)

	Create a LAG with active LACP	Show LAG status
Extreme Switch Engine	`enable sharing 47 grouping 47 lacp` `configure sharing 47 lacp activity-mode active`	`show sharing`

Extreme Fabric Engine	`lacp enable`
	`mlt 1 enable`
	`interface mlt 1`
	`  lacp enable key 1`
		`show mlt 1`
		`show lacp interface mlt id 1`

Cisco IOS XE	`interface GigabitEthernet1/0/47` `  channel-group 1 mode active`	`show etherchannel summary`

Aruba AOS-CX	`interface lag 1`
	`  no shutdown`
	`  lacp mode active`
		`show lacp aggregates lag1`
		`show interface lag1`

Standards: IEEE 802.1AX-2020 · IEEE 802.3ad-2000 · IANA-IEEE802 · IEEE 802.3-ANNEX57A-RESPONSE · IEEE 8021-AX-MIB

LACP link aggregation (2/2)

MODERN CLI

	Create a LAG with active LACP	Show LAG status

Juniper Junos	`set chassis aggregated-devices ethernet device-count 1`
	`set interfaces ae0 aggregated-ether-options lacp active`
	`set interfaces ae0 unit 0 family ethernet-switching interface-mode trunk`
		`show interfaces ae0 terse`

MikroTik RouterOS	`/interface bonding add name=bond1 mode=802.3ad lacp-mode=active \` ` slaves=ether47,ether48`	`/interface bonding monitor [find]`
Arista EOS	`interface port-channel 1`	`show port-channel summary`
Huawei VRP	`interface Eth-Trunk 1` ` mode lacp`	`display eth-trunk 1`

Standards: IEEE 802.1AX-2020 · IEEE 802.3ad-2000 · IANA-IEEE802 · IEEE 802.3-ANNEX57A-RESPONSE · IEEE 8021-AX-MIB

MODERN CLI

LACP link aggregation

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`enable sharing 47 grouping 47 lacp` `configure sharing 47 lacp activity-mode active`
Add member ports (47 and 48)	`configure sharing 47 add port 48`
LACP fast rate	`configure sharing 47 lacp timeout short`
Hash / load-balancing algorithm	`configure sharing 47 algorithm address-based L3_L4`
Show LAG status	`show sharing`
Show LACP partner	`show lacp lag 47 detail`

Port 47 is the LAG ID: use it in VLANs, and remove member ports from VLANs first. LACP is active by default; short timeout = 3 s (default 90 s). L4 hashing is global: configure forwarding sharing.

MODERN CLI

LACP link aggregation

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`lacp enable`
	`mlt 1 enable`
	`interface mlt 1`
	` lacp enable key 1`

Add member ports (47 and 48)	`interface gigabitEthernet 1/47-1/48`
	` lacp key 1`
	` lacp mode active`
	` lacp aggregation enable`
	` lacp enable`

LACP fast rate	`interface gigabitEthernet 1/47-1/48`
	` no lacp enable`
	` lacp timeout-time short`
	` lacp enable`

Hash / load-balancing algorithm	*n/a: Fixed per-flow hash (IP/L4 or MAC); no command to tune it*

Enable LACP globally first: it is off by default and a global change resets port settings. Same key on MLT and ports; port mode defaults to passive. After a timer change, toggle LACP.

MODERN CLI

LACP link aggregation (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 24 to 26 (EOS 2007)

Show LAG status	`show mlt 1` `show lacp interface mlt id 1`
Show LACP partner	`show lacp partner-oper interface gigabitethernet 1/47-1/48`

Enable LACP globally first: it is off by default and a global change resets port settings. Same key on MLT and ports; port mode defaults to passive. After a timer change, toggle LACP.

MODERN CLI

LACP link aggregation

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`interface GigabitEthernet1/0/47` ` channel-group 1 mode active`
Add member ports (47 and 48)	`interface range GigabitEthernet1/0/47 - 48` ` channel-group 1 mode active`
LACP fast rate	`interface range GigabitEthernet1/0/47 - 48` ` lacp rate fast`
Hash / load-balancing algorithm	`port-channel load-balance src-dst-ip`
Show LAG status	`show etherchannel summary`
Show LACP partner	`show lacp neighbor`

channel-group creates Port-channel1 by itself. LACP takes up to 16 ports (8 active). lacp rate fast uses 1 s instead of 30 s. The default hash is src-mac.

MODERN CLI

LACP link aggregation

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`interface lag 1`
	` no shutdown`
	` lacp mode active`

Add member ports (47 and 48)	`interface 1/1/47-1/1/48`
	` no shutdown`
	` lag 1`

LACP fast rate	`interface lag 1` ` lacp rate fast`
Hash / load-balancing algorithm	`interface lag 1` ` hash l3-src-dst`
Show LAG status	`show lacp aggregates lag1` `show interface lag1`
Show LACP partner	`show lacp interfaces 1/1/47`

The LAG is created with 'interface lag 1' and members join with 'lag 1' on the port. On the 6300 'hash' is set in the LAG context; on 6000/6100/8400 it is the global 'lacp hash'.

MODERN CLI

LACP link aggregation

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`set chassis aggregated-devices ethernet device-count 1`
	`set interfaces ae0 aggregated-ether-options lacp active`
	`set interfaces ae0 unit 0 family ethernet-switching interface-mode trunk`

Add member ports (47 and 48)	`set interfaces ge-0/0/46 ether-options 802.3ad ae0` `set interfaces ge-0/0/47 ether-options 802.3ad ae0`
LACP fast rate	`set interfaces ae0 aggregated-ether-options lacp periodic fast`

Hash / load-balancing algorithm	`set forwarding-options enhanced-hash-key hash-mode layer2-header`
	`set forwarding-options enhanced-hash-key layer2 vlan-id`
	`run show forwarding-options enhanced-hash-key`

Show LAG status	`show interfaces ae0 terse`
Show LACP partner	`show lacp interfaces ae0` `show lacp statistics interfaces ae0`

A LAG needs device-count under chassis and members set with ether-options 802.3ad ae0. At least one end must be LACP active. The default hash mode is Layer 2 payload (IPs and L4 ports).

MODERN CLI

LACP link aggregation

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`/interface bonding add name=bond1 mode=802.3ad lacp-mode=active \ slaves=ether47,ether48`
Add member ports (47 and 48)	`/interface bonding set bond1 slaves=ether47,ether48` `/interface bridge port add bridge=bridge1 interface=bond1 hw=yes`
LACP fast rate	`/interface bonding set bond1 lacp-rate=1sec`
Hash / load-balancing algorithm	*n/a: Chip hash is fixed L2+L3+L4; transmit-hash-policy has no effect*
Show LAG status	`/interface bonding monitor [find]`
Show LACP partner	`/interface bonding monitor-slaves bond1`

Only 802.3ad, balance-xor and active-backup are hardware offloaded on Prestera. Put bond1 in the bridge, never its member ports. Default lacp-rate is 30secs; default lacp-mode is active.

MODERN CLI

LACP link aggregation

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`interface port-channel 1`
Add member ports (47 and 48)	`interface ethernet 47-48` ` channel-group 1 mode active`
LACP fast rate	`interface ethernet 47-48` ` lacp timer fast`
Hash / load-balancing algorithm	`port-channel load-balance trident fields ip src-ip dst-ip`
Show LAG status	`show port-channel summary`
Show LACP partner	`show lacp peer`

There is no 'lacp rate': the command is 'lacp timer fast' (1 s) on the member interface. The LACP mode comes from 'channel-group N mode active'. The hash keyword (trident, arad, fm6000) depends on the switch chip.

MODERN CLI

LACP link aggregation

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 24 to 26 (EOS 2007)

Create a LAG with active LACP	`interface Eth-Trunk 1` ` mode lacp`
Add member ports (47 and 48)	`interface Eth-Trunk 1` ` trunkport GigabitEthernet 0/0/47 to 0/0/48 mode active`
LACP fast rate	`interface Eth-Trunk 1` ` lacp timeout fast`
Hash / load-balancing algorithm	`interface Eth-Trunk 1` ` load-balance src-dst-ip`
Show LAG status	`display eth-trunk 1`
Show LACP partner	`display eth-trunk 1` `display lacp statistics eth-trunk 1`

An Eth-Trunk defaults to manual mode: set mode lacp (members are active by default). Use the same lacp timeout on both ends. load-balance affects outgoing traffic only; the default is src-dst-ip.

2.3

VLAN



802.1Q VLANs, egress and dynamic registration

- 802.1Q VLANs: isolated broadcast domains
- Egress, PVID, tagged and untagged ports
- Dynamic registration: GVRP and MVRP
- VLAN management

Chapter 2.3 agenda

1. 802.1Q VLANs: isolated broadcast domains
2. Egress, PVID, tagged and untagged ports
3. Dynamic registration: GVRP and MVRP
4. VLAN management

MODERN CLI

B12 802.1Q VLANs

Lab

- Labs: ports, LAG, port protect, VLAN and broadcast suppression

802.1Q VLAN

- VLANs are logically isolated L2 broadcast domains.
- Ports in one VLAN do not talk to ports in another VLAN, unless there is an L3 router.
- Every VLAN has a corresponding ID. To extend VLANs across several switches, the ID must be the same on all of them.
- By default, all ports are part of the DEFAULT VLAN, with VLAN ID = 1.
- Multiple VLANs: The switch’s IP management interface, the host port, must be added to the VLAN that corresponds to its IP network.



802.1Q VLAN

- **Egress:** A port will only transmit packets of a given VLAN if that VLAN is configured as egress on that port. A port can have several VLANs as egress, and each egress VLAN can be tagged or untagged.
- **PVID:** This is the “default” VLAN of a port, into which all received (ingress) packets that do not contain an 802.1Q tag are classified.
- **802.1Q tag:** This is the information in the packet indicating which VLAN that packet belongs to. When a switch receives a packet with an 802.1Q tag, it classifies the packet according to the tag, regardless of the PVID of the port on which the packet was received.
- Unless filters are configured.

UPDATED

802.1Q VLAN

- GVRP (GARP VLAN Registration Protocol) is used for the automatic creation of VLANs
- With GVRP, when a port receives a GVRP (GARP) frame declaring a specific VLAN, that VLAN is automatically created on the receiving switch and the port is added, as tagged, to its egress list
- Unless there is specific planning for dynamic VLAN creation in the network topology, GVRP should be disabled on all ports, because indiscriminate VLAN creation can cause accidental disruption



Managing VLANs

LEGACY 2007

Checking information about one or more VLANs (the portinfo option is SecureStack only)

```
show vlan [static] [vlan-list] [portinfo
[vlan vlan-list | vlan-name] [port
portstring]]
Ex: show vlan 1
```

Create a VLAN: `set vlan {create | enable | disable} vlan-list`

Name a VLAN: `set vlan name vlan-list vlan-name`

Remove a VLAN: `clear vlan vlan-list`

Remove a VLAN name: `clear vlan name vlan-list`

Managing VLANs

LEGACY 2007

List the VLAN identifier (PVID) per port.

```
show port vlan [port-string]
```

Change the VLAN identifier (PVID) per port.

```
set port vlan port-string pvid [modify-egress | no-modify-egress]
```

Return the port's PVID to the default (Default VLAN = 1).

```
clear port vlan port-string
```

List the VLAN identifier (PVID) per port.

```
show port vlan [port-string]
```

VLAN Egress

- A port will only transmit traffic of a VLAN if that VLAN is configured as egress on the port.
- If the egress of a VLAN is tagged, its traffic will leave with an 802.1Q tag.
- If the egress of a VLAN is untagged, its traffic will leave without a tag.
- A port can have the egress of several VLANs configured.



VLAN Egress

LEGACY 2007

Check the egress settings configured on switch ports:

```
show port egress [port-string]
```

Configure ports in the VLAN egress list:

In other words, add egress to switch ports.

```
set vlan egress vlan-list port-string
[untagged | forbidden | tagged]
```

Remove ports from the VLAN egress list:

In other words, remove egress from switch ports.

```
clear vlan egress vlan-list port-string
[forbidden]
```

Check all ports that have egress for a VLAN:

```
show vlan vlan-id
```



SIDE-BY-SIDE VIEW · BLOCK B12

802.1Q VLANs (1/2)

MODERN CLI

	Access (untagged) port in VLAN 10	Trunk (tagged) port 10, 20 and 99

Extreme Switch Engine	`configure vlan USERS add ports 1 untagged`	`configure vlan USERS add ports 24 tagged`
		`configure vlan VOICE add ports 24 tagged`
		`configure vlan MGMT99 add ports 24 tagged`

Extreme Fabric Engine	`vlan members add 10 1/1`	`interface gigabitEthernet 1/24`
		`encapsulation dot1q`
		`vlan members add 10 1/24`

Cisco IOS XE	`interface GigabitEthernet1/0/1`
	`switchport access vlan 10`
		`interface GigabitEthernet1/0/24`
		`switchport mode trunk`
		`switchport trunk allowed vlan 10,20,99`

Aruba AOS-CX	`interface 1/1/1` `vlan access 10`	`interface 1/1/24` `vlan trunk allowed 10,20,99`

Standards: IEEE 802.1Q-2022 · IEEE 8021-TC-MIB · IEEE 8021-Q-BRIDGE-MIB · IANA-IEEE802 · IEEE 802.1ak-2007

SIDE-BY-SIDE VIEW · BLOCK B12

802.1Q VLANs (2/2)

MODERN CLI

	Access (untagged) port in VLAN 10	Trunk (tagged) port 10, 20 and 99

Juniper Junos	`set interfaces ge-0/0/0 unit 0 family ethernet-switching interface-mode access`
	`set interfaces ge-0/0/0 unit 0 family ethernet-switching vlan members USERS`
		`edit interfaces ge-0/0/23 unit 0 family ethernet-switching`
		`set interface-mode trunk`
		`set vlan members [ USERS VOICE MGMT ]`
		`top`

MikroTik RouterOS	`/interface bridge port add bridge=bridge1 interface=ether1 pvid=10`
	`/interface bridge vlan set [find vlan-ids=10] untagged=ether1`
		`/interface bridge port add bridge=bridge1 interface=ether24`
		`/interface bridge vlan set [find vlan-ids=10] tagged=ether24`
		`/interface bridge vlan add bridge=bridge1 tagged=ether24 vlan-ids=20,99`

Arista EOS	`interface ethernet 1`
	`switchport mode access`
	`switchport access vlan 10`
		`interface ethernet 24`
		`switchport mode trunk`
		`switchport trunk allowed vlan 10,20,99`

Huawei VRP	`interface GigabitEthernet0/0/1`
	`port link-type access`
	`port default vlan 10`
		`interface GigabitEthernet0/0/24`
		`port link-type trunk`
		`port trunk allow-pass vlan 10 20 99`

Standards: IEEE 802.1Q-2022 · IEEE 8021-TC-MIB · IEEE 8021-Q-BRIDGE-MIB · IANA-IEEE802 · IEEE 802.1ak-2007

MODERN CLI

802.1Q VLANs

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`create vlan USERS tag 10`
	`create vlan VOICE tag 20`
	`create vlan MGMT99 tag 99`

Name a VLAN	*n/a: Name is set by create vlan; extra text: vlan X description*
Access (untagged) port in VLAN 10	`configure vlan USERS add ports 1 untagged`

Trunk (tagged) port 10, 20 and 99	`configure vlan USERS add ports 24 tagged`
	`configure vlan VOICE add ports 24 tagged`
	`configure vlan MGMT99 add ports 24 tagged`

Trunk native/untagged VLAN (999)	`create vlan NATIVE999 tag 999` `configure vlan NATIVE999 add ports 24 untagged`
Remove the port from default VLAN 1	`configure vlan Default delete ports 1,24`
Show VLANs	`show vlan`

VLAN names are not case-sensitive, so MGMT would clash with the built-in Mgmt VLAN: use MGMT99. Untagged ports leave Default with a notice (auto-move inform). MVRP is off by default.

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 28 to 34 (EOS 2007)

Show a port's VLANs	`show ports 24 information detail`
Dynamic registration (GVRP/MVRP)	`show mvrp`

VLAN names are not case-sensitive, so MGMT would clash with the built-in Mgmt VLAN: use MGMT99. Untagged ports leave Default with a notice (auto-move inform). MVRP is off by default.

MODERN CLI

802.1Q VLANs

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`vlan create 10 name USERS type port-mstprstp 0`
	`vlan create 20 name VOICE type port-mstprstp 0`
	`vlan create 99 name MGMT-99 type port-mstprstp 0`

Name a VLAN	`vlan name 10 USERS`
Access (untagged) port in VLAN 10	`vlan members add 10 1/1`

Trunk (tagged) port 10, 20 and 99	`interface gigabitEthernet 1/24`
	` encapsulation dot1q`
	` vlan members add 10 1/24`

Trunk native/untagged VLAN (999)	`interface gigabitEthernet 1/24`
	` default-vlan-id 999`
	` untag-port-default-vlan enable`

Remove the port from default VLAN 1	`vlan members remove 1 1/1`
Show VLANs	`show vlan basic`

VLAN 99 is named MGMT-99: the manual says not to use the name 'Mgmt' (the management VLAN created at boot). An untagged port is in one VLAN only; on the trunk repeat 'vlan members add' and create VLAN 999.

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 28 to 34 (EOS 2007)

Show a port's VLANs	`show interfaces gigabitEthernet vlan 1/24` `show vlan members port 1/24`
Dynamic registration (GVRP/MVRP)	*n/a: No GVRP/MVRP in 9.4; dynamic VLANs come from EAP/RADIUS*

VLAN 99 is named MGMT-99: the manual says not to use the name 'Mgmt' (the management VLAN created at boot). An untagged port is in one VLAN only; on the trunk repeat 'vlan members add' and create VLAN 999.

802.1Q VLANs

MODERN CLI

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`vlan 10`
Name a VLAN	`vlan 10` ` name USERS`
Access (untagged) port in VLAN 10	`interface GigabitEthernet1/0/1` ` switchport access vlan 10`

Trunk (tagged) port 10, 20 and 99	`interface GigabitEthernet1/0/24`
	` switchport mode trunk`
	` switchport trunk allowed vlan 10,20,99`

Trunk native/untagged VLAN (999)	`interface GigabitEthernet1/0/24` ` switchport trunk native vlan 999`
Remove the port from default VLAN 1	`interface GigabitEthernet1/0/24` ` switchport trunk allowed vlan remove 1`
Show VLANs	`show vlan brief`

VLAN 1 cannot be deleted: remove it from trunks with allowed vlan. The default native VLAN is 1. On an access port also use switchport mode access. MVRP is disabled by default (mvrp global enables it).

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 28 to 34 (EOS 2007)

Show a port's VLANs	`show interfaces GigabitEthernet1/0/24 switchport`
Dynamic registration (GVRP/MVRP)	`no mvrp vlan creation`

VLAN 1 cannot be deleted: remove it from trunks with allowed vlan. The default native VLAN is 1. On an access port also use switchport mode access. MVRP is disabled by default (mvrp global enables it).

ARUBA AOS-CX · BLOCK B12

802.1Q VLANs

MODERN CLI

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`vlan 10`
	`vlan 20`
	`vlan 99`

Name a VLAN	`vlan 10` ` name USERS`
Access (untagged) port in VLAN 10	`interface 1/1/1` ` vlan access 10`
Trunk (tagged) port 10, 20 and 99	`interface 1/1/24` ` vlan trunk allowed 10,20,99`

Trunk native/untagged VLAN (999)	`interface 1/1/24`
	` vlan trunk native 999`
	` vlan trunk allowed 10,20,99`

Remove the port from default VLAN 1	*n/a: Implicit: 'vlan access 10' or 'trunk native 999' leaves VLAN 1*

Setting 'vlan trunk native' makes the switch run 'vlan trunk allowed all', so 'allowed 10,20,99' comes after it. Interfaces are L2 by default; only L2 ports take VLANs. MVRP is off by default ('mvrp').

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 28 to 34 (EOS 2007)

Show VLANs	`show vlan`
Show a port's VLANs	`show vlan port 1/1/24`
Dynamic registration (GVRP/MVRP)	`show mvrp config`

Setting 'vlan trunk native' makes the switch run 'vlan trunk allowed all', so 'allowed 10,20,99' comes after it. Interfaces are L2 by default; only L2 ports take VLANs. MVRP is off by default ('mvrp').

802.1Q VLANs

MODERN CLI

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`set vlans USERS vlan-id 10`
	`set vlans VOICE vlan-id 20`
	`set vlans MGMT vlan-id 99`

Name a VLAN	`set vlans USERS description "User hosts"`
Access (untagged) port in VLAN 10	`set interfaces ge-0/0/0 unit 0 family ethernet-switching interface-mode access` `set interfaces ge-0/0/0 unit 0 family ethernet-switching vlan members USERS`

Trunk (tagged) port 10, 20 and 99	`edit interfaces ge-0/0/23 unit 0 family ethernet-switching`
	`set interface-mode trunk`
	`set vlan members [ USERS VOICE MGMT ]`
	`top`

Trunk native/untagged VLAN (999)	`set vlans NATIVE vlan-id 999`
	`set interfaces ge-0/0/23 native-vlan-id 999`
	`set interfaces ge-0/0/23 unit 0 family ethernet-switching vlan members 999`

Remove the port from default VLAN 1	*n/a: No command: configure another VLAN and use it instead of 'default'*

ELS uses interface-mode (access/trunk) and vlan members. By default every port is an access port in VLAN 'default'. A trunk accepts untagged only with native-vlan-id. MVRP is off by default.

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 28 to 34 (EOS 2007)

Show VLANs	`show vlans`
Show a port's VLANs	`show ethernet-switching interface ge-0/0/23.0`
Dynamic registration (GVRP/MVRP)	`set protocols mvrp interface ge-0/0/23`

ELS uses interface-mode (access/trunk) and vlan members. By default every port is an access port in VLAN 'default'. A trunk accepts untagged only with native-vlan-id. MVRP is off by default.

802.1Q VLANs

MODERN CLI

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`/interface bridge add name=bridge1 vlan-filtering=yes` `/interface bridge vlan add bridge=bridge1 vlan-ids=10`
Name a VLAN	`/interface bridge vlan set [find vlan-ids=10] comment=USERS`
Access (untagged) port in VLAN 10	`/interface bridge port add bridge=bridge1 interface=ether1 pvid=10` `/interface bridge vlan set [find vlan-ids=10] untagged=ether1`

Trunk (tagged) port 10, 20 and 99	`/interface bridge port add bridge=bridge1 interface=ether24`
	`/interface bridge vlan set [find vlan-ids=10] tagged=ether24`
	`/interface bridge vlan add bridge=bridge1 tagged=ether24 vlan-ids=20,99`

Trunk native/untagged VLAN (999)	`/interface bridge port set [find interface=ether24] pvid=999` `/interface bridge vlan add bridge=bridge1 untagged=ether24 vlan-ids=999`
Remove the port from default VLAN 1	`/interface bridge set bridge1 frame-types=admit-only-vlan-tagged`
Show VLANs	`/interface bridge vlan print`

VLANs live in the bridge with vlan-filtering=yes, in hardware. Access port: pvid + untagged, one VLAN per entry. Without a pvid the port stays in VLAN 1 (default). MVRP exists and is off by default (mvrp=no).

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 28 to 34 (EOS 2007)

Show a port's VLANs	`/interface bridge port print where interface=ether1`
Dynamic registration (GVRP/MVRP)	`/interface bridge set bridge1 mvrp=no`

VLANs live in the bridge with vlan-filtering=yes, in hardware. Access port: pvid + untagged, one VLAN per entry. Without a pvid the port stays in VLAN 1 (default). MVRP exists and is off by default (mvrp=no).

ARISTA EOS · BLOCK B12

802.1Q VLANs

MODERN CLI

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`vlan 10,20,99`
Name a VLAN	`vlan 10` ` name USERS`

Access (untagged) port in VLAN 10	`interface ethernet 1`
	` switchport mode access`
	` switchport access vlan 10`

Trunk (tagged) port 10, 20 and 99	`interface ethernet 24`
	` switchport mode trunk`
	` switchport trunk allowed vlan 10,20,99`

Trunk native/untagged VLAN (999)	`interface ethernet 24` ` switchport trunk native vlan 999`
Remove the port from default VLAN 1	`interface ethernet 24` ` switchport trunk allowed vlan remove 1`

VLAN 1 exists by default and is every port's default native VLAN. On a trunk, untagged frames go to the native VLAN; with the 'tag' option they are dropped. The chapter mentions neither GVRP nor MVRP.

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 28 to 34 (EOS 2007)

Show VLANs	`show vlan`
Show a port's VLANs	`show interfaces ethernet 1 switchport`
Dynamic registration (GVRP/MVRP)	*n/a: Not in the VLAN chapter; use static VLANs*

VLAN 1 exists by default and is every port's default native VLAN. On a trunk, untagged frames go to the native VLAN; with the 'tag' option they are dropped. The chapter mentions neither GVRP nor MVRP.

802.1Q VLANs

MODERN CLI

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 28 to 34 (EOS 2007)

Create a VLAN	`vlan batch 10 20 99 999`
Name a VLAN	`vlan 10` ` name USERS`

Access (untagged) port in VLAN 10	`interface GigabitEthernet0/0/1`
	` port link-type access`
	` port default vlan 10`

Trunk (tagged) port 10, 20 and 99	`interface GigabitEthernet0/0/24`
	` port link-type trunk`
	` port trunk allow-pass vlan 10 20 99`

Trunk native/untagged VLAN (999)	`interface GigabitEthernet0/0/24` ` port trunk pvid vlan 999`
Remove the port from default VLAN 1	`interface GigabitEthernet0/0/24` ` undo port trunk allow-pass vlan 1`

Create the VLANs first: allow-pass ignores a missing VLAN. A trunk PVID forwards only if it is also in allow-pass. A trunk starts in VLAN 1: undo port trunk allow-pass vlan 1.

MODERN CLI

802.1Q VLANs (cont.)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 28 to 34 (EOS 2007)

Show VLANs	`display vlan`
Show a port's VLANs	`display port vlan GigabitEthernet0/0/24`
Dynamic registration (GVRP/MVRP)	*n/a: GVRP is off by default; check with display gvrp status*

Create the VLANs first: allow-pass ignores a missing VLAN. A trunk PVID forwards only if it is also in allow-pass. A trunk starts in VLAN 1: undo port trunk allow-pass vlan 1.

LAB

Lab: Port configuration

- **a** Name the port connected to the PC and verify it
- **b** Disable the port connected to the PC and check the physical link
- **c** Manually set the port connected to the PC to 100 Full
- **d** Manually set the PC’s network card to 100 Half
- **e** Generate traffic. Check the counters on the port connected to the PC
- **f** Go back to auto-negotiation and check the operational state
- **g** Note the counters on the port and check them again, comparing the change
- **h** Configure the port connected to the PC so that it auto-negotiates 10Mbps half duplex with the PC. Tip: Disconnect and reconnect the network cable so that the new auto-negotiation takes place.
- **i** Check the port’s operational state and counters.
- **j** Return auto-negotiation to normal (all capabilities)

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.

LAB

Lab: Link Aggregation

a. Make multiple connections between two switches

b. Check the operational state of the LAG and the physical ports

c. Connect a PC to each switch, one pinging the other

d. Disconnect one of the LAG cables, check the ports

e. Reconnect the cable. Check the ports and the LAG

f. Disconnect the other cable, check the ports and the LAG

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.

LAB

Lab: Port Protect

a Configure two port protect groups, with names, and add two ports to each.

b Connect two PCs to the ports of one group, and verify that the two cannot communicate with each other (e.g.: ping)

c Move one of the PCs to a port of the other group, and check connectivity between the two PCs.

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.

LAB

Lab: VLAN

a Configure VLANs

b Extend VLANs across switches with 802.1Q

c Observe the differences in behavior with tagged and untagged packets, on ports with different ingress PVIDs

d Configure ports to discard specific tagged packets

e Change the egress and the PVID of ports and observe the changes with the appropriate commands.

f Disable GVRP

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.

LAB

Lab: Broadcast Suppression

a. Start a ping from the PC to the switch, and from the switch to the PC

b. Connect two switch ports to each other with a cable

c. Check the switch CPU utilization (show system ?)

d. If possible: Mirror one of the two ports to the PC port, and capture the traffic. If not, disconnect item “b.” first.

e. Reconnect item “b.” and check the captured traffic.

f. If it is broadcast traffic: Configure broadcast suppression.

g. Disconnect the two switch ports.

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.



Break

second part of module 02

2.4

Spanning Tree



802.1D, 802.1w RSTP and BPDU protection

- 802.1D: root bridge election and port states
- 802.1w RSTP: port roles and topology changes
- Spanning Tree verification commands
- Span Guard: blocking BPDU attacks

Chapter 2.4 agenda

1. 802.1D: root bridge election and port states
2. 802.1w RSTP: port roles and topology changes
3. Spanning Tree verification commands
4. Span Guard: blocking BPDU attacks

MODERN CLI

- B13 Spanning Tree RSTP/MSTP
- B14 BPDU protection (Span Guard)

Lab

- Lab: Spanning Tree and Span Guard

LEGACY 2007

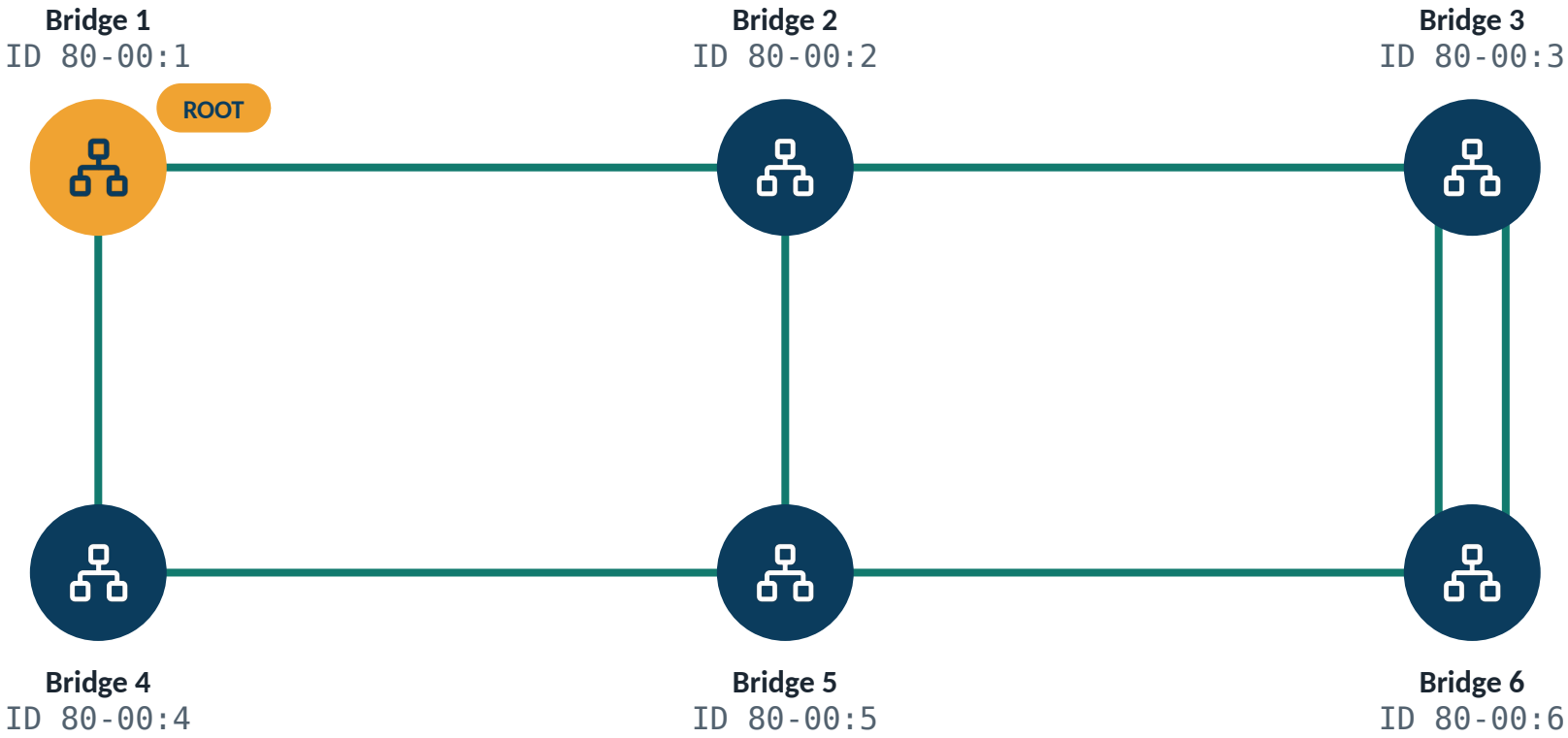
Spanning Tree Protocol

- 802.1d, the original Spanning Tree Protocol, was removed from the IEEE specification in 2004 (802.1D-2004).
- It was replaced by 802.1w RSTP (Rapid Spanning Tree Protocol)
- SecureStack and Matrix N support 802.1w; the default version is MSTP (802.1s), compatible with 802.1w and 802.1d
- 802.1w is compatible with 802.1d, preserving legacy support



IEEE 802.1d Spanning Tree

Typical topology: six bridges, redundant links, equal costs



UPDATED

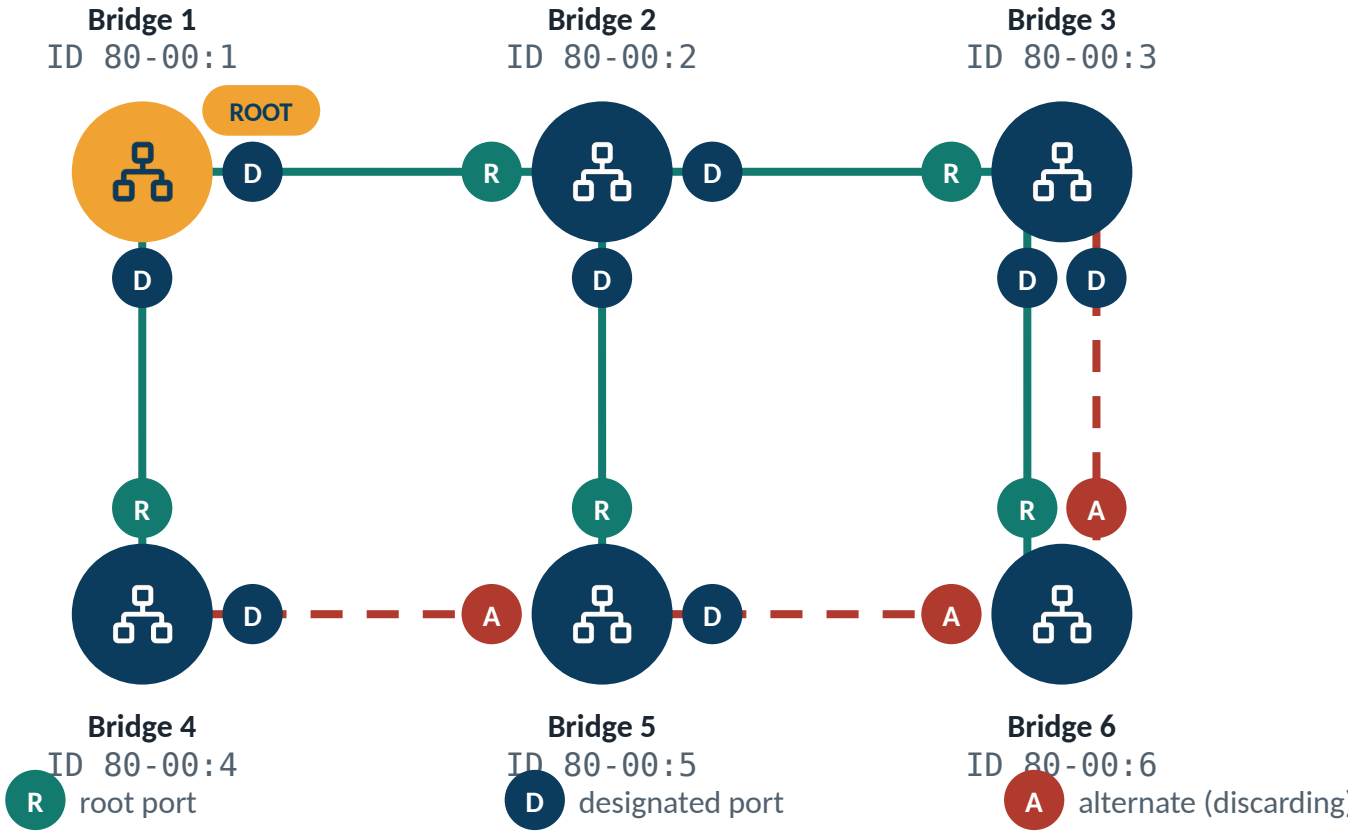
IEEE 802.1d Spanning Tree

802.1d Operation

- Electing a *root bridge* as the reference point for the network
  - Bridge ID = (16-bit Bridge priority, 48-bit Bridge MAC address)
- Assigning *path costs* to the links
  - Port ID = (8-bit Port priority, 8-bit Port number)
- Determining the *designated bridge* for each LAN segment
- Identifying *root ports* and *designated ports* on each bridge
- Resolving loops by placing redundant ports in the *blocking* state
- Maintaining the topology through the constant circulation of BPDU packets, monitoring to make sure no loops occur

IEEE 802.1d Spanning Tree

- Elects the root bridge: *Bridge 1*
- Assigns path costs to the links: *All links have equal path costs*
- Determines the designated bridge:
- *Bridge 1 is the designated bridge on all links*
- *Bridge 2 is the designated bridge on the downstream links*
- *Bridge 4, Bridge 5 and Bridge 3 are the designated bridges on their downstream links*
- Identifies the root and designated ports & blocks redundant links:



802.1w, Rapid Spanning Tree

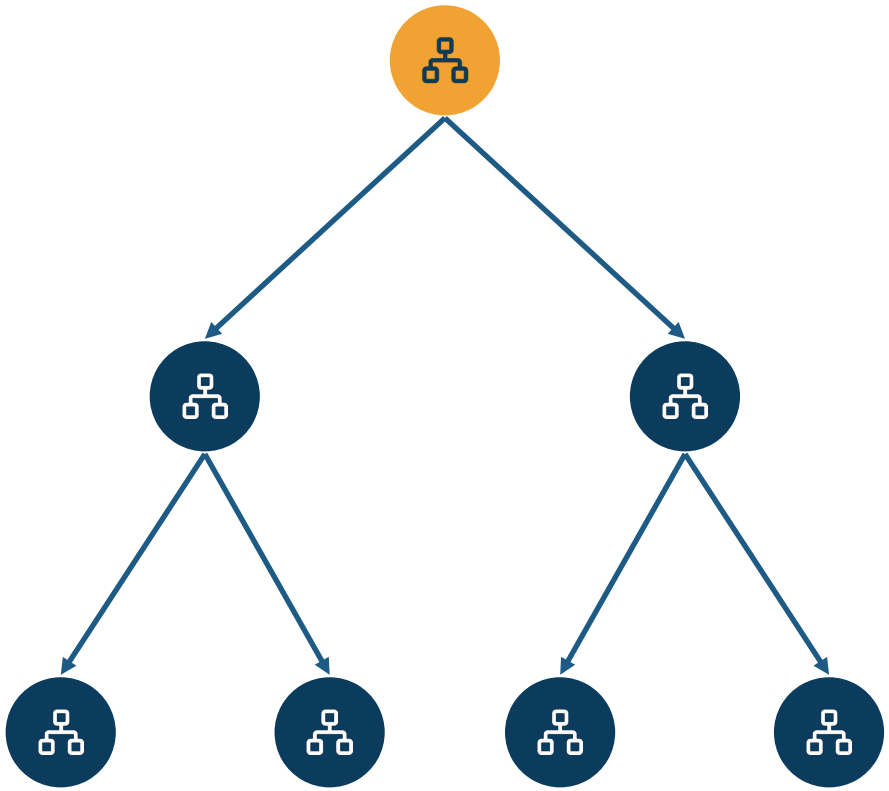
- IEEE 802.1w, Rapid Reconfiguration Spanning Tree (RSTP), is based on the parameters of the original IEEE 802.1d Spanning Tree Protocol.
- The IEEE 802.1w and IEEE 802.1D Spanning Tree algorithms are compatible and interoperate.
- An RSTP switch detects the STP version

when it is connected to
an 802.1d STP switch.

- When the RSTP port is initialized, it transmits

RSTP Bridge Protocol Data Units (BPDUs) for
three seconds, and then starts sending
STP BPDUs if it “hears” STP

- **Spanning Tree**



BPDUs flow downstream from the root

UPDATED

802.1w, Rapid Spanning Tree

RSTP Improvements

- ***Fast reconvergence time, from 30-50s to ~6s (3 × hello) or a few hundred milliseconds***
- Spanning tree per port, instead of per bridge as in 802.1d.
- Topology Change Notifications are propagated in *all directions*, starting from the bridge that detects the change



Spanning Tree

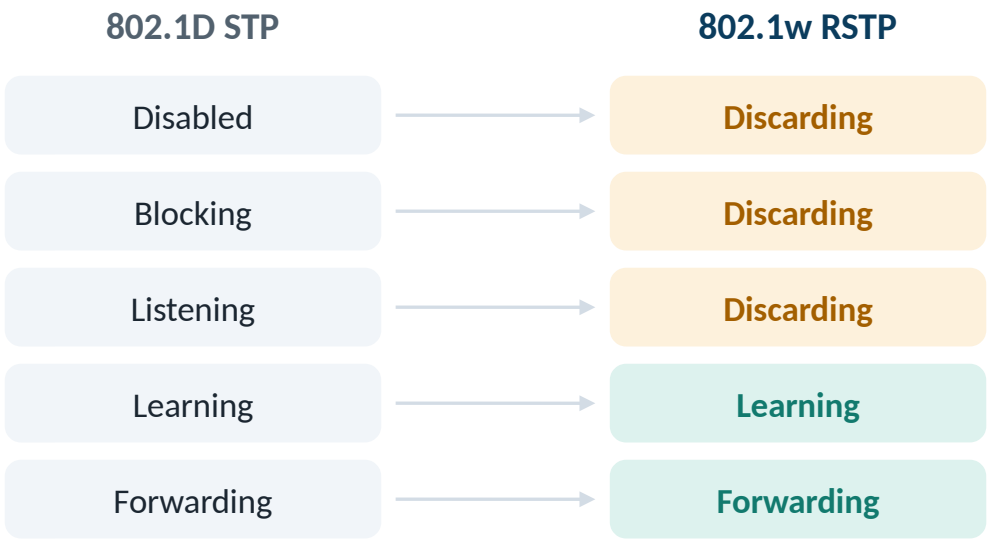
802.1w, Rapid Spanning Tree

- Port Roles:
  - ***Root Port:*** It is **THE** port used to connect to the Root Bridge.
    - The Root Port is elected based on the lowest “path-cost” to the Root Bridge.
  - ***Alternate Port:*** Any other redundant upstream port that provides an alternate path to the Root Bridge
  - ***Designated Port:*** Any downstream port that offers a path to the Root Bridge for a downstream bridge
  - ***Backup Port:*** A redundant Designated Port for a downstream bridge.
  - ***Edge Port:*** A port that has no other bridges connected to it (e.g.: user ports).



802.1w, Rapid Spanning Tree

- Port Roles and Traffic Forwarding
- **Ports with *Root* and *Designated* roles are part of the Spanning Tree Active Topology**
- These ports are in forwarding (passing traffic)
- **Ports with *Alternate* and *Backup* roles are *not* part of the Spanning Tree Active Topology**
- These ports provide redundant fail-over connectivity in case of failure of a Root or Designated port
- Port states
- **RSTP *eliminates* the *Listening* and *Blocking* states of 802.1d STP**
- **Valid port states in RSTP:**
- *Forwarding*, *Learning*, *Discarding*



UPDATED

802.1w, Rapid Spanning Tree

- ***Topology changes* and *address flushing***
  - TCNs in RSTP refer to both **802.1d** & **802.1w** TCNs.
  - If the switch is ***notified*** of a TC (***Rx TCN***), it is propagated on all OTHER active ports (*excluding Edge Ports*).
  - If a port ***detects*** a TC, it is propagated on ALL active ports, including the port that detected the TC (*excluding Edge Ports*).
  - If a port goes from Discarding to Forwarding, all Filter Database Entries are cleared (flushed) on *all other* active ports.
  - All ports that originate the TCNs they transmit (that is, that are not relaying a TCN generated elsewhere in the network) clear all entries in their dynamic filtering database.

LEGACY 2007

802.1w, Rapid Spanning Tree

Matrix N and SecureStack support IEEE 802.1w and 802.1d; the default version is MSTP (802.1s), compatible with both

Enabling/disabling spanning tree (default: enabled)	`set spantree stpmode {none | ieee8021} (Matrix N) / set spantree {disable | enable} (SecureStack)`
Enabling/disabling spanning tree per port	`set spantree portadmin port-string {disable | enable}`
Configuring the spanning tree version	`set spantree version {mstp | stpcompatible | rstp}`
Configuring the spanning tree bridge priority	`set spantree priority priority [sid]`
Configuring the spanning tree port priority	`set spantree portpri port-string priority [sid sid]`
Configuring the spanning tree port cost	`set spantree adminpathcost port-string cost [sid sid]`
Configuring the spanning tree backup root	`set spantree backuproot sid {enable | disable}`

LEGACY 2007

802.1w, Rapid Spanning Tree

Spanning Tree display commands

Per-port details.

Omit the port to get the global parameters.

```
show spantree stats port ge.2.1-4
Spanning tree status         - enabled
Spanning tree instance       - 0
Designated Root MacAddr      - 00-00-5e-00-53-c7
Designated Root Priority     - 0
Designated Root Cost         - 0
Designated Root Port         - 0
Root Max Age                 - 20 sec
Root Hello Time              - 2  sec
Root Forward Delay           - 15 sec
Bridge ID MAC Address        - 00-00-5e-00-53-c7
Bridge ID Priority           - 0
Bridge Max Age               - 20 sec
Bridge Hello Time            - 2  sec
Bridge Forward Delay         - 15 sec
Topology Change Count        - 105
Time Since Top Change        - 00 days 08:04:24
Max Hops                     - 20
SID   Port         State              Role          Cost        Priority
---   ----------   ----------------   -----------   --------    --------
0     ge.2.1       Blocking           Disabled      20000       128
0     ge.2.2       Forwarding         Designated    200000      128
0     ge.2.3       Forwarding         Designated    200000      128
0     ge.2.4       Forwarding         Designated    20000       128
```

LEGACY 2007

802.1w, Rapid Spanning Tree

Spanning Tree display commands

Spanning Tree packet counters (Matrix N)

There should be no TCN increments in a stable network

```
show spantree debug
STP Diagnostic Common Counters
---------------------------------------
Topology Change Count        - 105
Message Expiration Count     - 0
Invalid BPDU Count           - 0
Disputed BPDU Count          - 140
STP BPDU Rx Count            - 0
STP BPDU Tx Count            - 0
STP TCN BPDU Rx Count        - 0
STP TCN BPDU Tx Count        - 0
STP TC BPDU Rx Count         - 0
STP TC BPDU Tx Count         - 0
RST BPDU Rx Count            - 100
RST BPDU Tx Count            - 0
RST TC BPDU Rx Count         - 81
RST TC BPDU Tx Count         - 0
MST BPDU Rx Count            - 1702931
MST BPDU Tx Count            - 22470881
MST CIST TC BPDU Rx Count    - 217
MST CIST TC BPDU Tx Count    - 1651
CIST Regional Root           - 00-00-00-00-5e-00-53-c7
STP Diagnostic Root History Table for SID 0
-------------------------------------------------
Index       Root Bridge ID          Date       Time
-------------------------------------------------
```

Spanning Tree RSTP/MSTP (1/2)

MODERN CLI

	Set the mode (RSTP or MSTP)	Edge port (edge/portfast)
Extreme Switch Engine	`configure stpd s0 mode dot1w`	`configure stpd s0 ports link-type edge 1`
Extreme Fabric Engine	`boot config flags spanning-tree-mode mstp` `save config`	`interface gigabitEthernet 1/1` ` spanning-tree mstp edge-port true`
Cisco IOS XE	`spanning-tree mode rapid-pvst`	`interface GigabitEthernet1/0/1` ` spanning-tree portfast`
Aruba AOS-CX	`spanning-tree mode mstp` `spanning-tree`	`interface 1/1/1` ` spanning-tree port-type admin-edge`

Standards: IEEE 802.1Q-2022 · IEEE 802.1D-1998 · IEEE 802.1w-2001 · IEEE 802.1D-2004 · IEEE 802.1s-2002

Spanning Tree RSTP/MSTP (2/2)

MODERN CLI

	Set the mode (RSTP or MSTP)	Edge port (edge/portfast)

Juniper Junos	`set protocols rstp interface all`
	`set protocols mstp configuration-name region1`
	`set protocols mstp interface all`
		`set protocols rstp interface ge-0/0/0 edge`

MikroTik RouterOS	`/interface bridge set bridge1 protocol-mode=rstp` `/interface bridge set bridge1 protocol-mode=mstp region-name=LAB`	`/interface bridge port set [find interface=ether1] edge=yes`
Arista EOS	`spanning-tree mode mstp`	`interface ethernet 1` `spanning-tree portfast`
Huawei VRP	`stp mode rstp`	`interface GigabitEthernet0/0/1` `stp edged-port enable`

Standards: IEEE 802.1Q-2022 · IEEE 802.1D-1998 · IEEE 802.1w-2001 · IEEE 802.1D-2004 · IEEE 802.1s-2002

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`configure stpd s0 mode dot1w`
Bridge priority (root = 4096)	`configure stpd s0 priority 4096`
Port cost	`configure stpd s0 ports cost 1000 24`
Port priority	`configure stpd s0 ports port-priority 32 24`
Edge port (edge/portfast)	`configure stpd s0 ports link-type edge 1`
Show STP summary	`show stpd s0`
Show STP per port	`show stpd s0 ports 24 detail`
Counters / topology changes	`show stpd s0 ports 24 counters`

STPD s0 starts in MSTP (CIST) mode; dot1w switches to RSTP and mstp cist returns to MSTP. Priority in multiples of 4096. Auto-edge is on by default; link-type edge pins an edge port.

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`boot config flags spanning-tree-mode mstp` `save config`
Bridge priority (root = 4096)	`spanning-tree mstp priority 4096`
Port cost	`interface gigabitEthernet 1/24` ` spanning-tree mstp cost 200000`
Port priority	`interface gigabitEthernet 1/24` ` spanning-tree mstp priority 112`
Edge port (edge/portfast)	`interface gigabitEthernet 1/1` ` spanning-tree mstp edge-port true`
Show STP summary	`show spanning-tree mstp status`
Show STP per port	`show spanning-tree mstp port role 1/24`
Counters / topology changes	`show spanning-tree mstp statistics` `show spanning-tree mstp port statistics 1/24`

The default is MSTP. For RSTP set 'spanning-tree-mode rstp', save and reboot; RSTP keeps the switch out of an SPBM network. Bridge priority goes in steps of 4096.

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`spanning-tree mode rapid-pvst`
Bridge priority (root = 4096)	`spanning-tree vlan 10,20,99 priority 4096`
Port cost	`interface GigabitEthernet1/0/24` ` spanning-tree cost 1000`
Port priority	`interface GigabitEthernet1/0/24` ` spanning-tree port-priority 64`
Edge port (edge/portfast)	`interface GigabitEthernet1/0/1` ` spanning-tree portfast`
Show STP summary	`show spanning-tree summary`
Show STP per port	`show spanning-tree interface GigabitEthernet1/0/24`
Counters / topology changes	`show spanning-tree detail`

Rapid PVST+ is the default mode; use mst for MSTP. Bridge priority: multiples of 4096 (0 to 61440). Port priority: multiples of 16 (0 to 240). The detail output shows topology changes.

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`spanning-tree mode mstp` `spanning-tree`
Bridge priority (root = 4096)	`spanning-tree priority 1`
Port cost	`interface 1/1/24` ` spanning-tree cost 2000`
Port priority	`interface 1/1/24` ` spanning-tree port-priority 4`
Edge port (edge/portfast)	`interface 1/1/1` ` spanning-tree port-type admin-edge`
Show STP summary	`show spanning-tree` `show spanning-tree summary root`
Show STP per port	`show spanning-tree detail`

The default STP mode is MSTP ('no spanning-tree mode' restores it). Priority is a multiplier of 4096 (0 to 15, default 8): 'priority 1' = 4096. Port priority is a multiplier of 16 (0 to 15).

MODERN CLI

Spanning Tree RSTP/MSTP (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 51 to 53 (EOS 2007)

Counters / topology changes

```
show spanning-tree detail
clear spanning-tree statistics
```

The default STP mode is MSTP ('no spanning-tree mode' restores it). Priority is a multiplier of 4096 (0 to 15, default 8): 'priority 1' = 4096. Port priority is a multiplier of 16 (0 to 15).

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`set protocols rstp interface all`
	`set protocols mstp configuration-name region1`
	`set protocols mstp interface all`

Bridge priority (root = 4096)	`set protocols rstp bridge-priority 4k`
Port cost	`set protocols rstp interface ge-0/0/23 cost 1000`
Port priority	`set protocols rstp interface ge-0/0/23 priority 64`
Edge port (edge/portfast)	`set protocols rstp interface ge-0/0/0 edge`
Show STP summary	`show spanning-tree bridge`
Show STP per port	`show spanning-tree interface` `show spanning-tree interface ge-0/0/23`
Counters / topology changes	`show spanning-tree bridge detail`

RSTP is the ELS default; t1 shows RSTP (line 1) or MSTP (lines 2 and 3), never both: with MSTP configured, RSTP cannot be configured (set mstp disable). Bridge: steps of 4096; port: of 16.

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`/interface bridge set bridge1 protocol-mode=rstp` `/interface bridge set bridge1 protocol-mode=mstp region-name=LAB`
Bridge priority (root = 4096)	`/interface bridge set bridge1 priority=4096`
Port cost	`/interface bridge port set [find interface=ether24] path-cost=10`
Port priority	`/interface bridge port set [find interface=ether24] priority=64`
Edge port (edge/portfast)	`/interface bridge port set [find interface=ether1] edge=yes`
Show STP summary	`/interface bridge monitor bridge1`
Show STP per port	`/interface bridge port monitor [find]`
Counters / topology changes	`/interface bridge port monitor [find interface=ether24]`

RSTP is the default (the two t1 lines are alternatives). Priority in steps of 4096. STP, RSTP and MSTP are hardware offloaded on Prestera. Port monitor shows role, cost and topology-changes.

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`spanning-tree mode mstp`
Bridge priority (root = 4096)	`spanning-tree mst 0 priority 4096`
Port cost	`interface ethernet 24` ` spanning-tree cost 20000`
Port priority	`interface ethernet 24` ` spanning-tree port-priority 128`
Edge port (edge/portfast)	`interface ethernet 1` ` spanning-tree portfast`
Show STP summary	`show spanning-tree`
Show STP per port	`show spanning-tree interface ethernet 1`
Counters / topology changes	`show spanning-tree counters`

Modes: mstp (default), rstp, rapid-pvst, backup and none. In rapid-pvst priority is per VLAN: 'spanning-tree vlan-id 10 priority 4096'. In mstp use 'spanning-tree mst 0 priority 4096'.

MODERN CLI

Spanning Tree RSTP/MSTP

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 51 to 53 (EOS 2007)

Set the mode (RSTP or MSTP)	`stp mode rstp`
Bridge priority (root = 4096)	`stp priority 4096`
Port cost	`interface GigabitEthernet0/0/24` ` stp cost 200`
Port priority	`interface GigabitEthernet0/0/24` ` stp port priority 64`
Edge port (edge/portfast)	`interface GigabitEthernet0/0/1` ` stp edged-port enable`
Show STP summary	`display stp brief`
Show STP per port	`display stp interface GigabitEthernet0/0/24`
Counters / topology changes	`display stp topology-change` `display stp tc-bpdu statistics`

The default is MSTP, not RSTP: use stp mode rstp. Bridge priority is a multiple of 4096 (default 32768) and port priority a multiple of 16 (default 128).

Span Guard™

- **Span Guard™ was designed to increase network security and stability**

User devices do not need to run STP

- User ports should never receive or transmit BPDUs
- A malicious device can attack networks using BPDUs

- **Enabling Span Guard ™ on “user” ports**
  - Prevents changes to the spanning tree topology if a BPDU is received on a “user” port
  - Notifies network management, via trap, of the presence of BPDUs on ports where they should not exist.



LEGACY 2007

Blocking BPDU attacks

With Span Guard™ enabled:

If a port with SpanGuard enabled receives a BPDU, the port will be locked and will move to the ***Blocking*** state

The port will resume operation after a time set by the device administrator, or manually

SpanGuard advantages:

Forged BPDUs WILL NOT cause Spanning Tree Topology Changes or “Re-Spans”.

A BPDU attack will be detected and the administrator will be notified:

```
set spantree spanguardtrapenable {disable | enable}
```

Accidentally adding a repeater or a PC with bridging enabled will not make the network unavailable.

LEGACY 2007

Deploying Span Guard ™

It is a global setting, not a per-port one

Default: Disabled

Enable / disable SpanGuard:

```
set spantree spanguard {enable | disable}
```

When enabled, all ports with *adminEdge* = TRUE are treated as “user ports” by Span Guard

```
set spantree adminedge port-string {true | false}
```

Therefore, all links between switches (ISLs: Inter-Switch Links) must have *adminEdge* set to *FALSE before* enabling Span Guard

Ports stay blocked for the time set by a timer

```
set spantree spanguardtimeout timeout
```

Default: 300 seconds. 0 = forever.

SIDE-BY-SIDE VIEW · BLOCK B14

MODERN CLI

BPDU protection (Span Guard) (1/2)

	BPDU guard on an edge port	Automatic recovery (timer)
Extreme Switch Engine	`configure stpd s0 ports edge-safeguard enable 1 bpdu-restrict`	`configure stpd s0 ports bpdu-restrict enable 1-23 recovery-timeout 300`
Extreme Fabric Engine	`interface gigabitEthernet 1/1` ` spanning-tree bpduguard enable`	`interface gigabitEthernet 1/1` ` spanning-tree bpduguard timeout 300`
Cisco IOS XE	`interface GigabitEthernet1/0/1` ` spanning-tree bpduguard enable`	`errdisable recovery cause bpduguard` `errdisable recovery interval 300`

Aruba AOS-CX	`interface 1/1/1`
	` spanning-tree port-type admin-edge`
	` spanning-tree bpdu-guard`
		`spanning-tree bpdu-guard timeout 300`

Standards: IEEE 8021-MSTP-MIB · IEEE 8021-SPANNING-TREE-MIB · IEEE 802.1Q-2022

MODERN CLI

BPDU protection (Span Guard) (2/2)

	BPDU guard on an edge port	Automatic recovery (timer)
Juniper Junos	`set protocols rstp interface ge-0/0/0 edge` `set protocols layer2-control bpdu-block interface ge-0/0/0`	`set protocols layer2-control bpdu-block disable-timeout 300`
MikroTik RouterOS	`/interface bridge port set [find interface=ether1] bpdu-guard=yes`	*n/a*
Arista EOS	`interface ethernet 1` ` spanning-tree bpduguard enable`	`errdisable recovery cause bpduguard` `errdisable recovery interval 300`
Huawei VRP	`interface GigabitEthernet0/0/1` ` stp edged-port enable`	`error-down auto-recovery cause bpdu-protection interval 300`

Standards: IEEE 8021-MSTP-MIB · IEEE 8021-SPANNING-TREE-MIB · IEEE 802.1Q-2022

MODERN CLI

BPDU protection (Span Guard)

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`configure stpd s0 ports edge-safeguard enable 1 bpdu-restrict`
Global BPDU guard (edge ports)	`configure stpd s0 ports edge-safeguard enable 1-23 bpdu-restrict`
Automatic recovery (timer)	`configure stpd s0 ports bpdu-restrict enable 1-23 recovery-timeout 300`
Root guard	`configure stpd s0 ports restricted-role enable 2`
Show blocked ports	`show stpd s0 ports blocked-ports`
Manual recovery	`enable ports 1`

There is no global BPDU guard: use edge-safeguard + bpdu-restrict on a range of edge ports. The port goes down on a BPDU; without recovery-timeout only enable ports restores it. Root guard = restricted-role.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`interface gigabitEthernet 1/1` ` spanning-tree bpduguard enable`
Global BPDU guard (edge ports)	`interface gigabitEthernet 1/1-1/22` ` spanning-tree bpduguard enable`
Automatic recovery (timer)	`interface gigabitEthernet 1/1` ` spanning-tree bpduguard timeout 300`
Root guard	`interface gigabitEthernet 1/24` ` spanning-tree mstp restricted-role enable`
Show blocked ports	`show spanning-tree bpduguard`

Manual recovery	`interface gigabitEthernet 1/1`
	` shutdown`
	` no shutdown`

BPDU Guard is per port (no global switch): on a BPDU the port is shut until the timer ends (default 120 s; 0 = never expires). Root guard is the MSTP 'restricted-role'.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`interface GigabitEthernet1/0/1` ` spanning-tree bpduguard enable`
Global BPDU guard (edge ports)	`spanning-tree portfast edge bpduguard default`
Automatic recovery (timer)	`errdisable recovery cause bpduguard` `errdisable recovery interval 300`
Root guard	`interface GigabitEthernet1/0/24` ` spanning-tree guard root`
Show blocked ports	`show spanning-tree blockedports` `show spanning-tree inconsistentports`

Manual recovery	`interface GigabitEthernet1/0/1`
	` shutdown`
	` no shutdown`

BPDU guard puts the port in err-disabled when it gets a BPDU. Without errdisable recovery the port returns only after shutdown and no shutdown. Global edge bpduguard applies only to PortFast ports.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`interface 1/1/1`
	` spanning-tree port-type admin-edge`
	` spanning-tree bpdu-guard`

Global BPDU guard (edge ports)	*n/a: No global form: 'spanning-tree bpdu-guard' is per interface*
Automatic recovery (timer)	`spanning-tree bpdu-guard timeout 300`
Root guard	`interface 1/1/24` ` spanning-tree root-guard`
Show blocked ports	`show spanning-tree inconsistent-ports` `show interface 1/1/1`

Manual recovery	`interface 1/1/1`
	` shutdown`
	` no shutdown`

BPDU guard disables a port that receives a BPDU ('Bpdu-Error'). The timeout (1 to 65535 s) is global; without it the port stays disabled until re-enabled. Root guard blocks the port on a superior BPDU.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`set protocols rstp interface ge-0/0/0 edge` `set protocols layer2-control bpdu-block interface ge-0/0/0`
Global BPDU guard (edge ports)	`set protocols rstp bpdu-block-on-edge`
Automatic recovery (timer)	`set protocols layer2-control bpdu-block disable-timeout 300`
Root guard	`set protocols rstp interface ge-0/0/23 no-root-port`
Show blocked ports	`show spanning-tree interface` `show layer2-control bpdu-block`
Manual recovery	`clear error bpdu interface ge-0/0/0`

A protected port that gets a BPDU is shut down. Without disable-timeout (10 to 3600 s) it only returns after 'clear error bpdu interface'. 'clear ethernet-switching bpdu-error' is non-ELS only.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`/interface bridge port set [find interface=ether1] bpdu-guard=yes`
Global BPDU guard (edge ports)	*n/a: No global option: bpdu-guard is a per-port property*
Automatic recovery (timer)	*n/a: No timer: bpdu-guard needs the port to be re-enabled manually*
Root guard	`/interface bridge port set [find interface=ether24] restricted-role=yes`
Show blocked ports	`/interface bridge port print` `/interface bridge port monitor [find]`
Manual recovery	`/interface bridge port disable [find interface=ether1]` `/interface bridge port enable [find interface=ether1]`

With bpdu-guard=yes the port goes to the disabled role on a BPDU and returns only after a manual disable and enable. Root guard is restricted-role=yes: the port becomes alternate (discarding) on a superior BPDU.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`interface ethernet 1` ` spanning-tree bpduguard enable`
Global BPDU guard (edge ports)	`spanning-tree edge-port bpduguard default`
Automatic recovery (timer)	`errdisable recovery cause bpduguard` `errdisable recovery interval 300`
Root guard	`interface ethernet 24` ` spanning-tree guard root`
Show blocked ports	`show spanning-tree blockedports`

Manual recovery	`interface ethernet 1`
	` shutdown`
	` no shutdown`

BPDU Guard disables the port on receiving a BPDU (errdisable, cause bpduguard); without 'errdisable recovery' it returns only by manual action. The 'default' option applies to portfast ports.

MODERN CLI

BPDU protection (Span Guard)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 54 to 56 (EOS 2007)

BPDU guard on an edge port	`interface GigabitEthernet0/0/1` ` stp edged-port enable`
Global BPDU guard (edge ports)	`stp bpdu-protection`
Automatic recovery (timer)	`error-down auto-recovery cause bpdu-protection interval 300`
Root guard	`interface GigabitEthernet0/0/24` ` stp root-protection`
Show blocked ports	`display stp abnormal-interface` `display error-down recovery`

Manual recovery	`interface GigabitEthernet0/0/1`
	` shutdown`
	` undo shutdown`

BPDU protection is global (stp bpdu-protection) and acts on edge ports: a BPDU puts the port in error-down. Recover with shutdown then undo shutdown, or configure error-down auto-recovery.

LAB

Lab: Spanning Tree and SpanGuard ™

- **a** Configure RSTP
- **b** Make redundant connections between the switches and predict where blocking will take place, then confirm with the CLI.
- **c** Identify the roles of the ports participating in the spanning tree
- **d** Check the behavior when ports with different roles are disconnected / connected (Wireshark to see TCNs)
- **e** Check reconvergence times and behavior when the root bridge is switched off
- **f** Configure SpanGuard
- **g** Connect a switch with STP configured to a port with SpanGuard active

At the end of the lab, save the configuration named after the lab and the bench number.

Repeat the steps on the bench's current platform, using this topic's MODERN CLI slides.



Questions are welcome

Questions are welcome

LAN Switch Operations and Maintenance

From the Enterasys legacy to the current CLI of eight vendors

MODULE 03 · DAY 3

IP routing



Rodolfo Nützmann

2026 Edition

Prepared for: [Client] · In partnership with: [Partner]

Course agenda

Day 1

DONE

Access, operation and basic maintenance

- Platforms: 2007 legacy and current
- CLI, users and session
- Clock, identity and connectivity
- Save, erase and reboot
- Password recovery
- Files, backups and firmware

Day 2

DONE

Layer 2 switching

- Interfaces and naming
- Speed, duplex and jumbo frames
- Storm control, isolation, mirroring
- LACP link aggregation
- 802.1Q VLANs
- Spanning Tree and BPDU protection

Day 3

TODAY

IP routing

- Addressing, masks and subnets
- Route determination and ARP
- Proxy-ARP
- L3 interfaces and management
- Static routes, RIP and OSPF
- Routing table, ECMP and ARP

Day 4

NEXT

Redundancy, security and management

- OSPF and VRRP
- Access control lists
- Syslog, SNMP and AAA
- Multicast: IGMP and PIM
- Troubleshooting and support
- Final lab and assessment

3.1

IP network communication



Addressing, mask, route decision and ARP

- IP addressing, binary and subnet mask
- Address classes and the bitwise AND
- Route determination and the routing table
- ARP: packet anatomy and operation
- Proxy-ARP

Chapter 3.1 agenda

1. IP addressing, binary and subnet mask
2. Address classes and the bitwise AND
3. Route determination and the routing table
4. ARP: packet anatomy and operation
5. Proxy-ARP

UPDATED

IP Communications: Internet Protocol

- The initial steps for any and all IP communication are:
- **Determining the Route** (Route Determination)
- **Resolving the Address** (Address Resolution)
- Every **IP Host** will always perform these steps, regardless of the network configuration or topology
- In IP, every host makes routing decisions like a router, and every router is also a host
- Explanation: **route determination** and **address resolution** are the two steps always performed by any IP device: PC, router, etc.



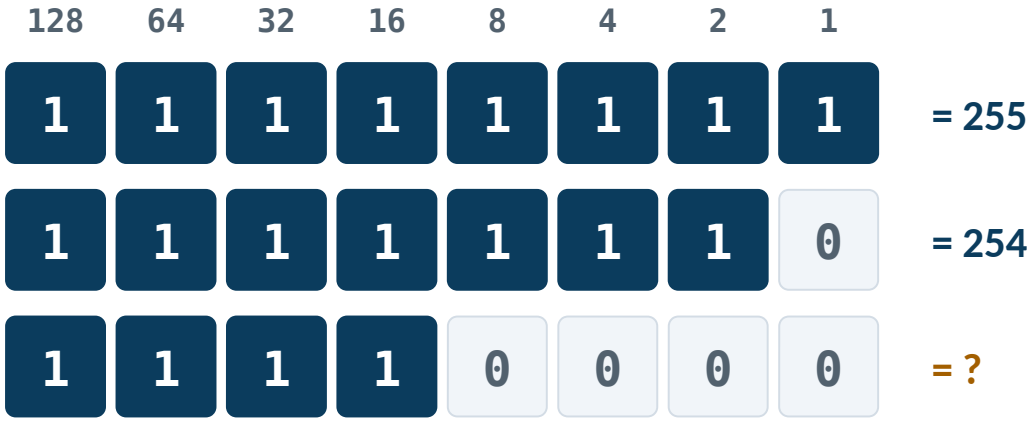
UPDATED

IP Addressing

- A 32-bit address expressed in dotted-decimal notation
- Four numbers separated by dots
- Each number represents an octet
- Octet = 8 bits. Four octets = 32 bits
- Every IP address is made up of two segments:
- The **Network** (Network Number)
- And the **Host** (Node Number)
- Where is the boundary between the two segments?
- It is determined by the mask assigned to each IP address!
- IP is a binary world!

Binary

- Binary = bit
- 1 = YES 0 = NO
- An “Octet” = A string of 8 bits, each one either 0 or 1



UPDATED

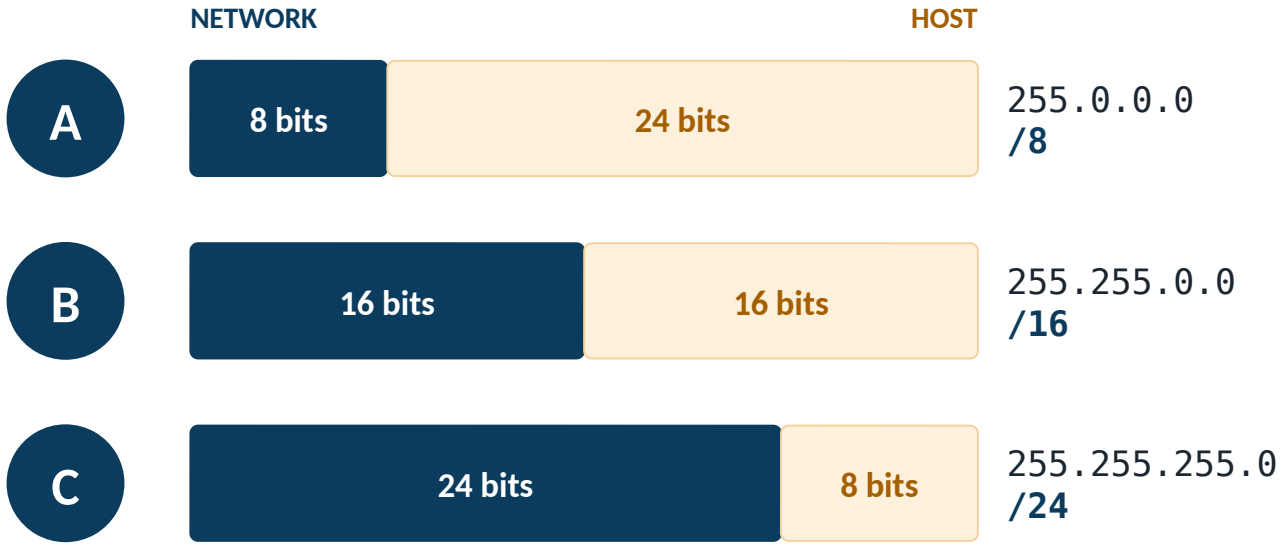
Subnet mask

- The distinction between the NETWORK number and the HOST number
- is determined by the **mask**
- It does not necessarily correspond to the octet boundaries
- To match the octet boundaries, the mask value has to be 255 or 0
- The mask can be expressed in two formats:
- Decimal notation - 255.255.255.0 (255 = Full octet, 8 bits)
  - CIDR notation - /24 (8 is counted for each full octet)
  - The NETWORK portion of an IP address is obtained by a binary AND operation between the IP address and its mask; the remaining bits form the HOST portion.

LEGACY 2007

IP address classes

- **Class A**: 255.0.0.0
- **NETWORK** = First octet. **HOST** = Last three octets.
- **Class B**: 255.255.0.0
- **NETWORK** = First two octets. **HOST** = Last two octets.
- **Class C**: 255.255.255.0
- **NETWORK** = First three octets. **HOST** = Last octet.



The binary AND operation (boolean AND)

- Definition:
- 1 AND X = X
- 0 AND X = 0
- Thus:
- 1 AND 1 = 1
- 1 AND 0 = 0
- 0 AND 1 = 0
- 0 AND 0 = 0

AND	0	1
0	0	0
1	0	1

Determining the network with a “bitwise AND”

- To perform a “bitwise AND”, AND each bit of the address with the bit of equivalent significance in the mask.
- IP is a binary world. Always think in binary!
- (Or download a subnet calculator, but understand what it is calculating).

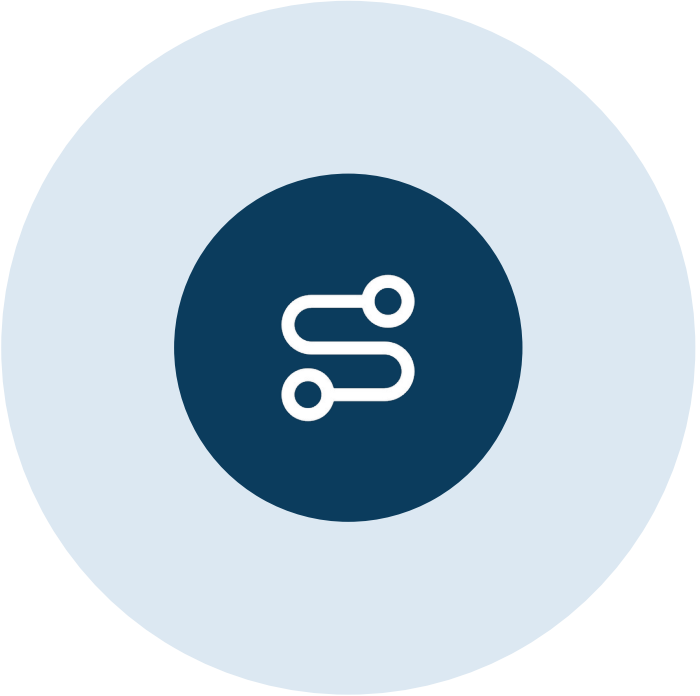
Address	10101100	00010000	10000000	11000011	172.16.128.195
Mask	11111111	11111111	11111111	00000000	/24
					AND
Network	10101100	00010000	10000000	00000000	172.16.128.0

Route Determination

- It is the process used to determine the best “next hop” for a specific packet.
- In conventional IP operation, the route to the packet’s destination IP address is looked up

Same logical network

- If the destination address is on a directly connected IP network (that is, in the same subnet as the host), the destination address itself will always be the “next hop”.



UPDATED

Route Determination

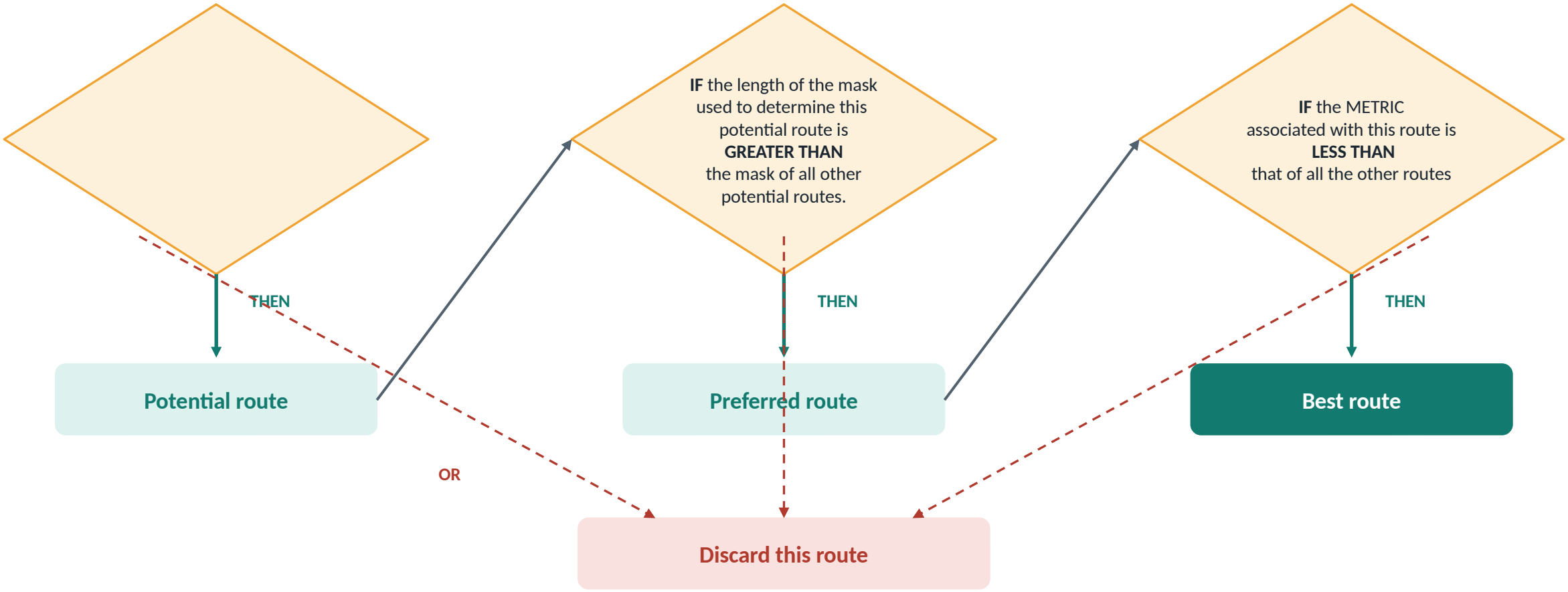
Different logical network

- If the destination address is not on a directly connected IP network (that is, in a subnet other than the host's), the host checks its **routing table** to see whether there is another host (a router, in this case) to which it can forward the packet. The most specific route (longest mask) prevails and, among equivalent routes, the "closest" router will be used, as determined by the **metrics** of each route.

"Default route" or "default gateway"

- It is the router that receives all packets for which no route is known
- IP route determination is a three-step process:
  - Determine potential route(s).
  - Determine preferred route(s).
  - Determine the best route(s).

The Route Selection Process



The "Best Route" used will depend on the specific implementation.

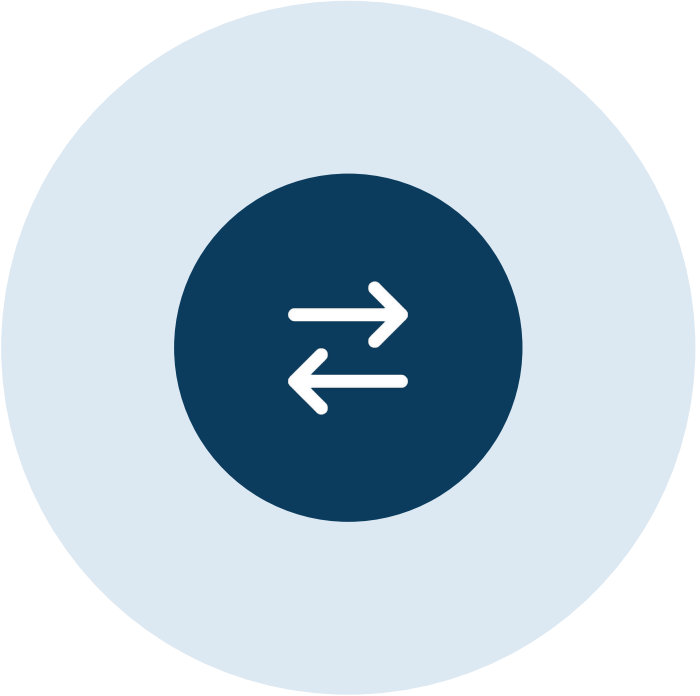
The Routing Table

- Provides the information needed to determine the “next hop”.
- Every IP device uses a routing table (route table).
- The routing table is derived from several sources:
- Manual configuration
- Automatic configuration
- Distributed dynamic routing protocols

Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	172.16.75.1	172.16.75.145	1
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1	1
172.16.0.0	255.255.0.0	172.16.75.145	172.16.75.145	1
172.24.128.0	255.255.255.0	172.16.75.2	172.16.75.145	2
172.24.64.0	255.255.255.0	172.16.75.2	172.16.75.145	2
172.24.192.0	255.255.224.0	172.16.75.3	172.16.75.145	1
172.16.75.145	255.255.255.255	127.0.0.1	127.0.0.1	1
172.16.255.255	255.255.255.255	172.16.75.145	172.16.75.145	1
224.0.0.0	240.0.0.0	172.16.75.145	172.16.75.145	1
255.255.255.255	255.255.255.255	172.16.75.145	172.16.75.145	1

ARP: Address Resolution Protocol

- ARP is defined by IETF RFC 826
- ARP is not part of the IP protocol family
- ARP does not perform route determination
- It does not use the same packet format as IP
- ARP: Ethertype 0x0806



Anatomy of an ARP packet

ARP packet

encapsulated in an

Ethernet II (DIX) Frame

Hardware Type	16 bits
Protocol Type	16 bits
Hardware Address Length	8 bits
Protocol Address Length	8 bits
Opcode	16 bits
Sender Hardware Address	variable
Sender Protocol Address	variable
Target Hardware Address	variable
Target Protocol Address	variable



Ethernet II (DIX) frame, EtherType 0x0806

Field: Hardware Type

Indicates the

physical and data link

technology in use.

Ethernet = 0x0001

Hardware Type 16 bits
Protocol Type 16 bits
Hardware Address Length 8 bits
Protocol Address Length 8 bits
Opcode 16 bits
Sender Hardware Address variable
Sender Protocol Address variable
Target Hardware Address variable
Target Protocol Address variable



Ethernet II (DIX) frame, EtherType 0x0806

Field: Protocol Type

Indicates the network technology in use.

IP = 0x0800

- **Hardware Type** 16 bits
- **Protocol Type** 16 bits
- **Hardware Address Length** 8 bits
- **Protocol Address Length** 8 bits
- **Opcode** 16 bits
- **Sender Hardware Address** variable
- **Sender Protocol Address** variable
- **Target Hardware Address** variable
- **Target Protocol Address** variable



Ethernet II (DIX) frame, EtherType 0x0806

Field: Hardware address length

Indicates the size,

in octets, of the Hardware address.

Ethernet - 6 octets.

Hardware Type	16 bits
Protocol Type	16 bits
Hardware Address Length	8 bits
Protocol Address Length	8 bits
Opcode	16 bits
Sender Hardware Address	variable
Sender Protocol Address	variable
Target Hardware Address	variable
Target Protocol Address	variable



Ethernet II (DIX) frame, EtherType 0x0806

Field: Protocol address length

Indicates the size,

in octets, of the protocol address.

IP - 4 octets.

Hardware Type	16 bits
Protocol Type	16 bits
Hardware Address Length	8 bits
Protocol Address Length	8 bits
Opcode	16 bits
Sender Hardware Address	variable
Sender Protocol Address	variable
Target Hardware Address	variable
Target Protocol Address	variable



Ethernet II (DIX) frame, EtherType 0x0806

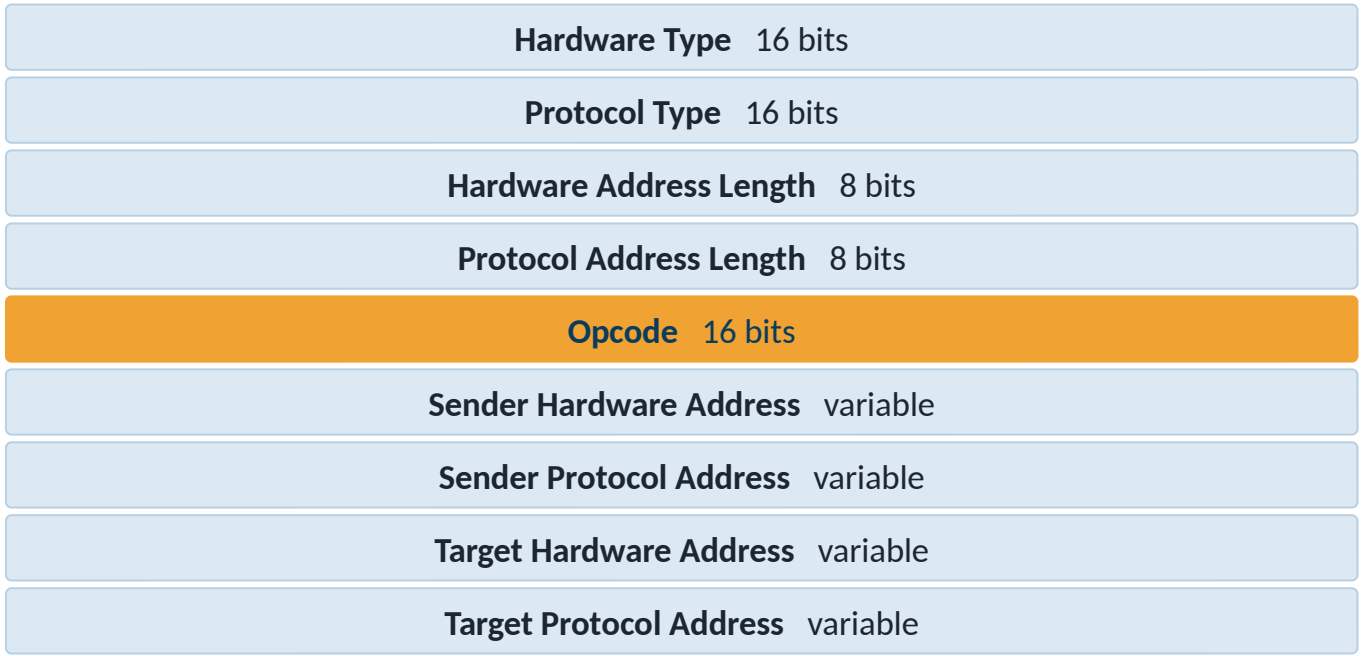
Field: Operational Code (OP_Code)

ARP packet:

Indicates the type of

Request = 0x0001

Response = 0x0002



Ethernet II (DIX) frame, EtherType 0x0806

Fields: Hardware Address

Indicates the source and destination Hardware addresses.

NOTE: The destination address is usually made up of zeros in request packets (because it is unknown).

Hardware Type	16 bits
Protocol Type	16 bits
Hardware Address Length	8 bits
Protocol Address Length	8 bits
Opcode	16 bits
Sender Hardware Address	variable
Sender Protocol Address	variable
Target Hardware Address	variable
Target Protocol Address	variable



Ethernet II (DIX) frame, EtherType 0x0806

Fields: Protocol Address

Indicates the source

and destination protocol addresses.

ARP = The protocol address

is known (e.g.: IP), I want to find the physical address to send the packet to.

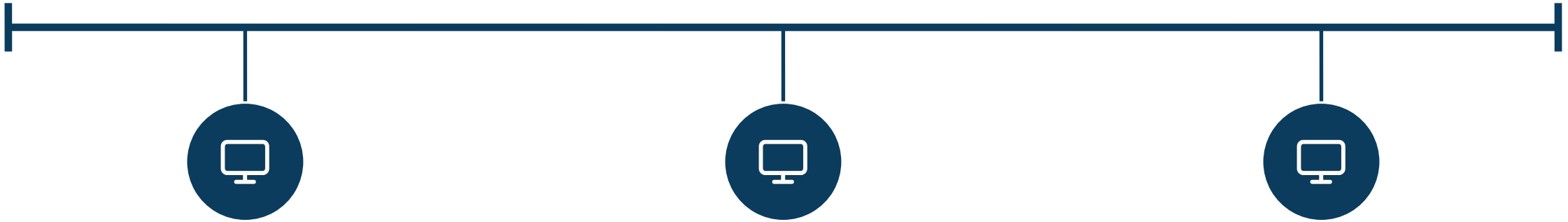
Hardware Type	16 bits
Protocol Type	16 bits
Hardware Address Length	8 bits
Protocol Address Length	8 bits
Opcode	16 bits
Sender Hardware Address	variable
Sender Protocol Address	variable
Target Hardware Address	variable
Target Protocol Address	variable



Ethernet II (DIX) frame, EtherType 0x0806

ARP: Address Resolution Protocol

Station A wants to talk to Station C. For illustration purposes: “A”, “B” and “C” are the MAC addresses of the respective stations



ARP cache A

(empty)

ARP cache B

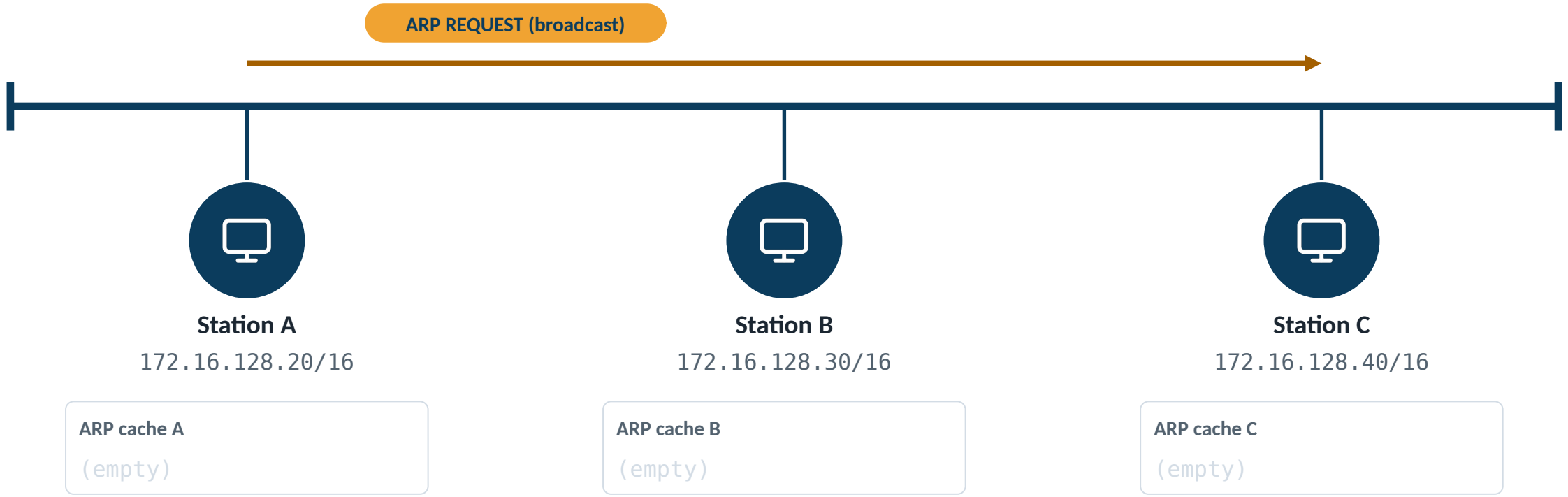
(empty)

ARP cache C

(empty)

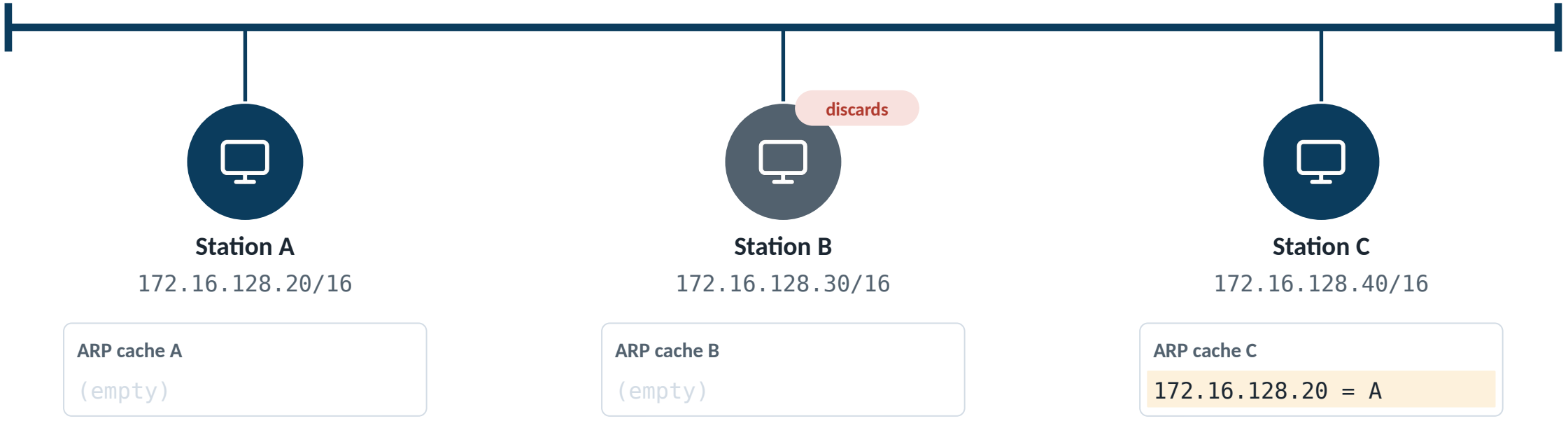
ARP: Address Resolution Protocol

Station A transmits an **ARP-Request** to Station C. Both Station B and Station C receive the ARP-Request.



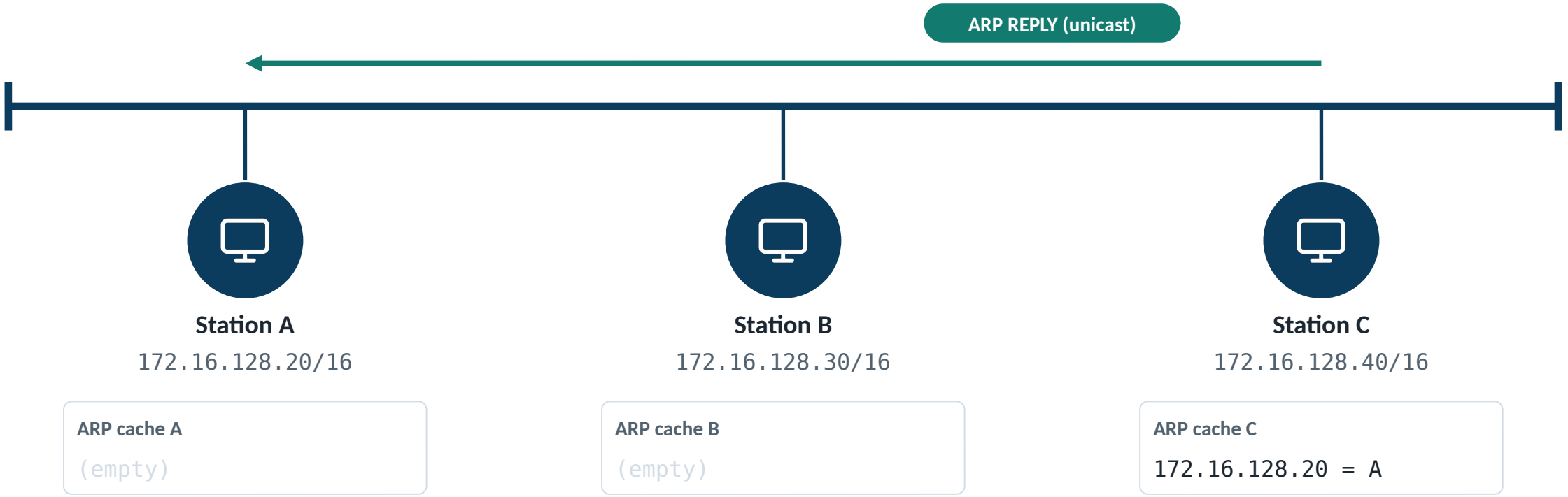
ARP: Address Resolution Protocol

Station C recognizes that the ARP-Request was sent to it, and populates its ARP table (ARP cache) with the information from Station A (the association of the IP address with a MAC address). Station B discards the packet.



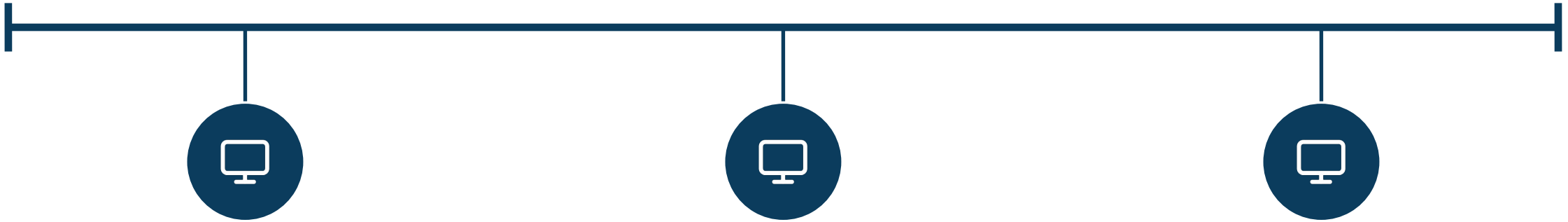
ARP: Address Resolution Protocol

Station C then checks the OP_Code field of the ARP packet received. Since it was a REQUEST, C will send a unicast RESPONSE to A.



ARP: Address Resolution Protocol

Station A recognizes the packet addressed to it, and populates its ARP table (ARP-Cache).



ARP cache A

```
172.16.128.40 = C
```

ARP cache B

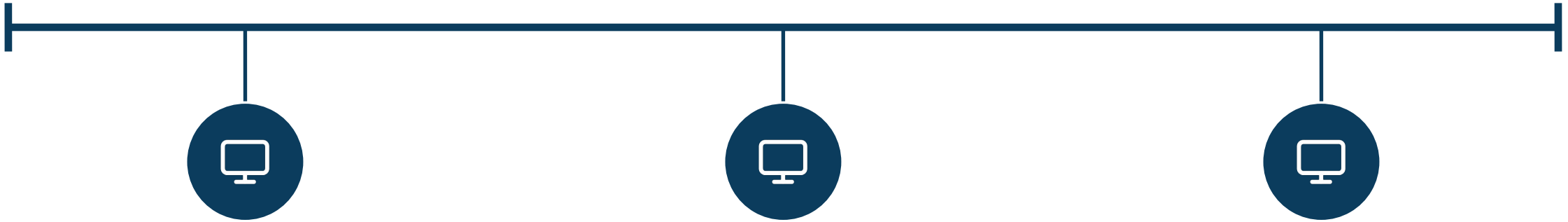
```
(empty)
```

ARP cache C

```
172.16.128.20 = A
```

ARP: Address Resolution Protocol

Station A then checks the OP_Code field of the ARP packet it received. Since it was a RESPONSE, A will not send a response to C.



ARP cache A

```
172.16.128.40 = C
```

ARP cache B

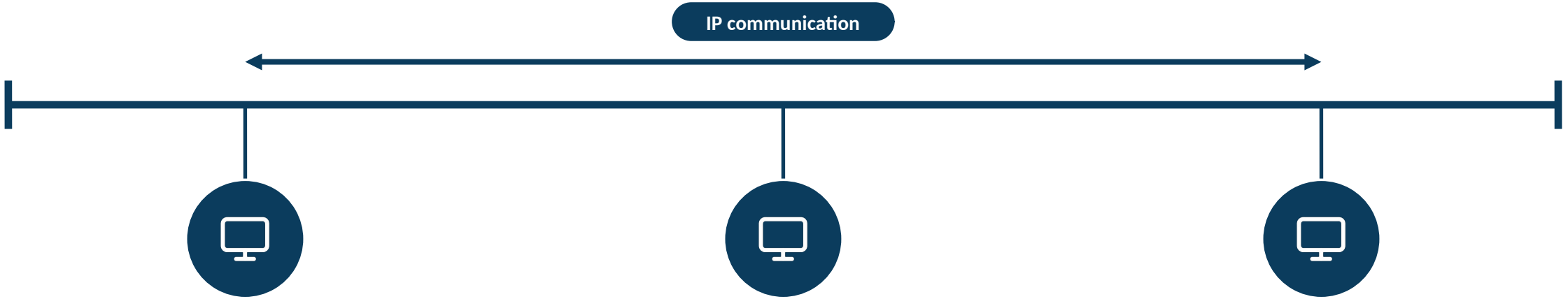
```
(empty)
```

ARP cache C

```
172.16.128.20 = A
```

ARP: Address Resolution Protocol

A and C now have all the information they need to communicate: Protocol address (IP) And hardware address (MAC)



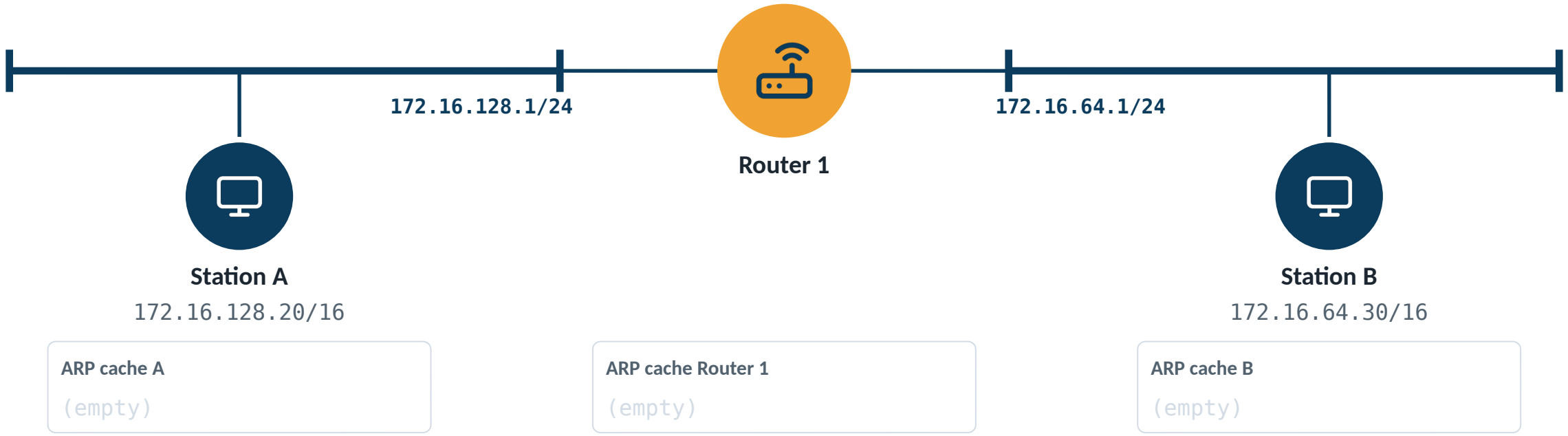
UPDATED

Proxy-ARP

- Described in the mid-1980s (RFC 925, 1984; RFC 1027, 1987) due to the adoption of subnets (*subnetting*).
- Not all IP stacks supported subnets.
- These stacks needed Proxy-ARP in order to work in environments where subnets were used.
- Proxy-ARP is used to allow communication between hosts on different IP subnets that think they are on the same IP network.
- E.g.: **172.16.239.3/16** wants to talk to **172.16.100.9/16**
- They are on different subnets, so they need a router.
- They think they are on the same network, so they will ARP for the destination address.
- Proxy-ARP intercepts and processes these ARPs, allowing the communication.

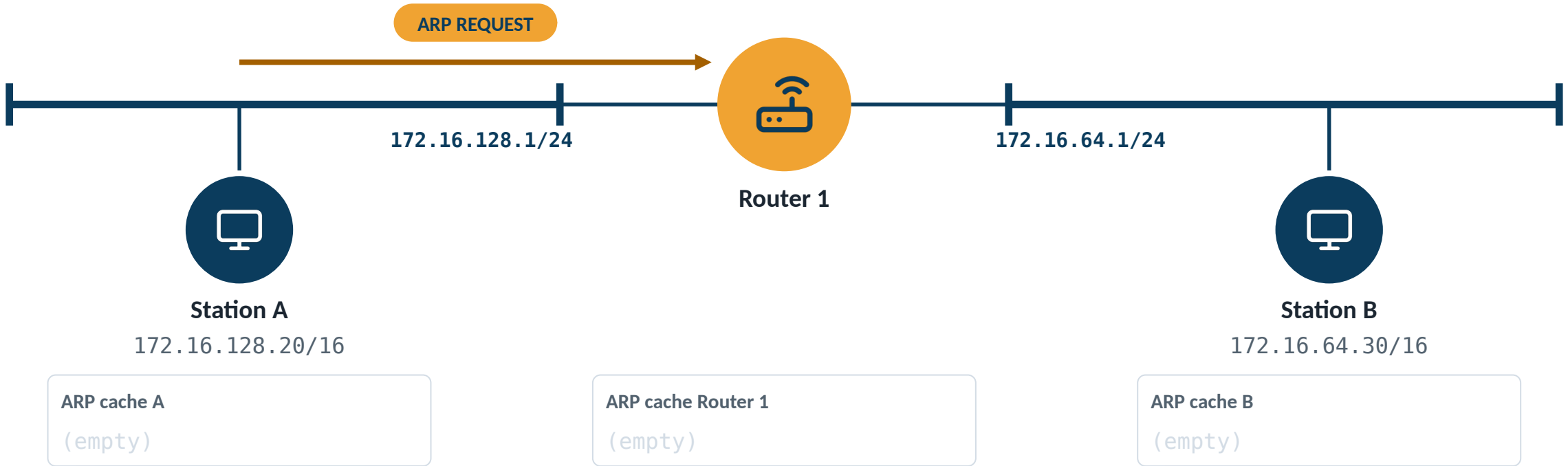
Proxy-ARP

Station A wants to talk to Station B. Based on its routing table, Station A believes that Station B is the ‘Next Hop.’

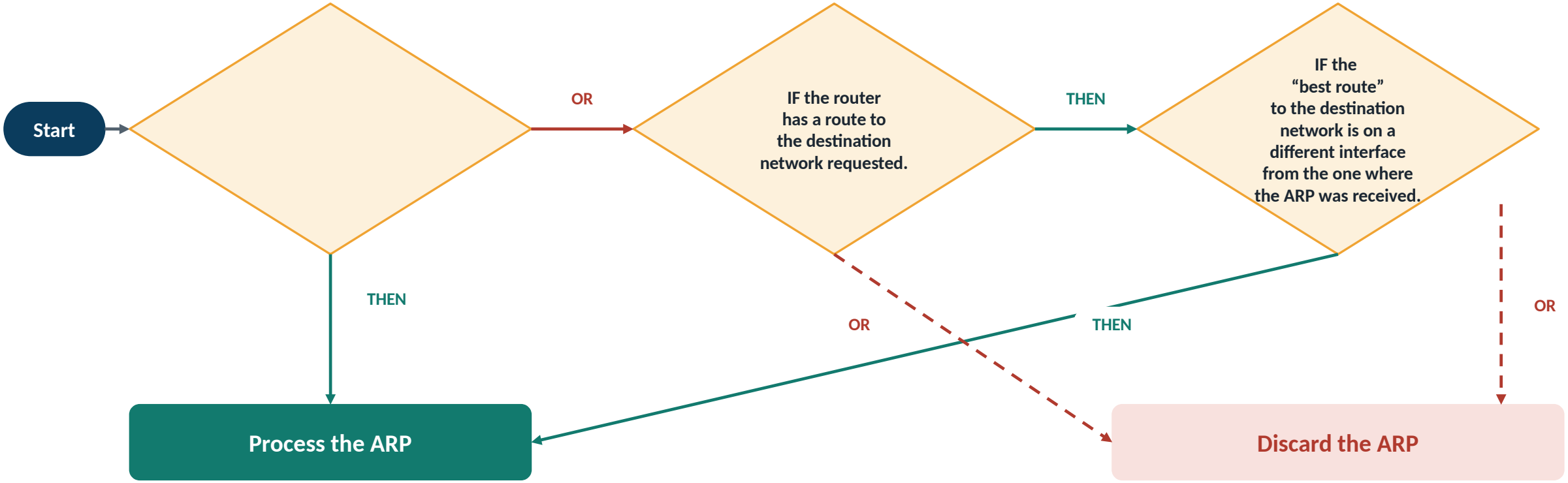


Proxy-ARP

Station A transmits an ARP-Request to Station B. Since it is a Broadcast, the router receives this ARP-Request. The router will go through a three-step process to determine whether it will process this ARP.

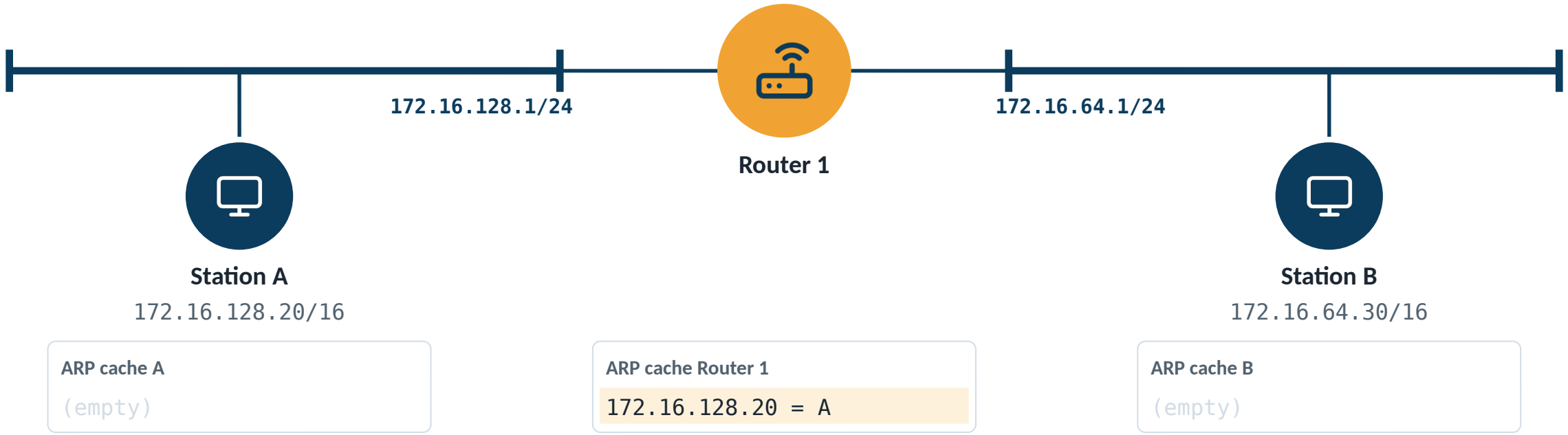


Proxy-ARP


Start
OR
IF the router has a route to the destination network requested.
THEN
IF the "best route" to the destination network is on a different interface from the one where the ARP was received.
THEN
OR
OR
THEN
Process the ARP
Discard the ARP

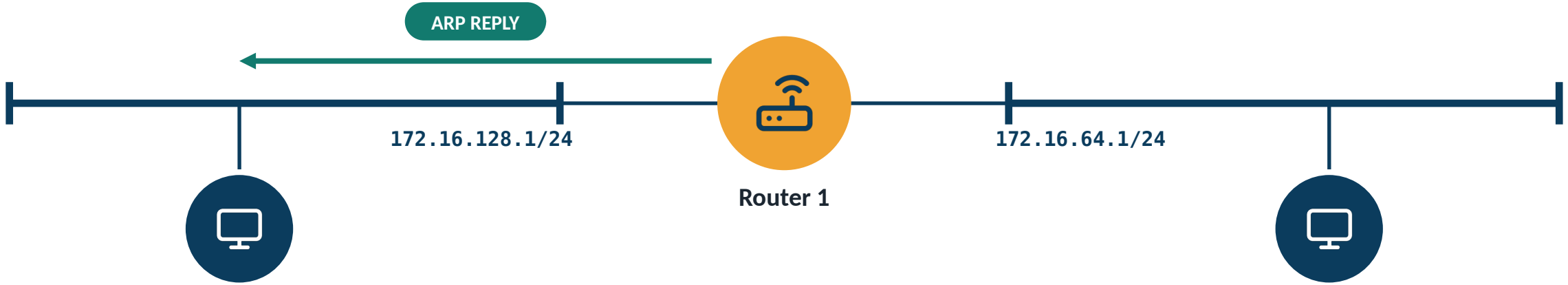
Proxy-ARP

If the router decides to process the ARP, it will populate its ARP table.



Proxy-ARP

The router will inspect the OP_Code in the ARP packet. Since the OP_Code is that of a REQUEST, it will send a RESPONSE to A. The RESPONSE will be sent with B's IP address, but associated with the router's MAC address, instead of B's MAC. The router is replying as a "proxy" for B.



ARP cache A

(empty)

ARP cache Router 1

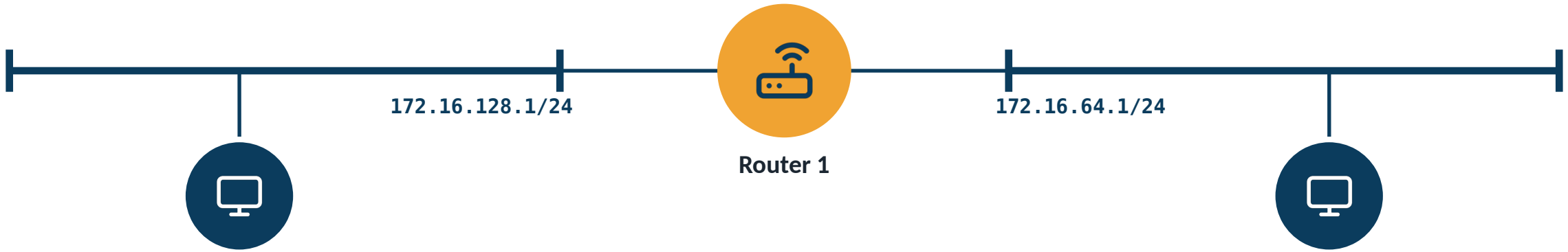
172.16.128.20 = A

ARP cache B

(empty)

Proxy-ARP

A will receive the ARP RESPONSE, and populate its ARP table.



ARP cache A

```
172.16.64.30 = Router 1
```

ARP cache Router 1

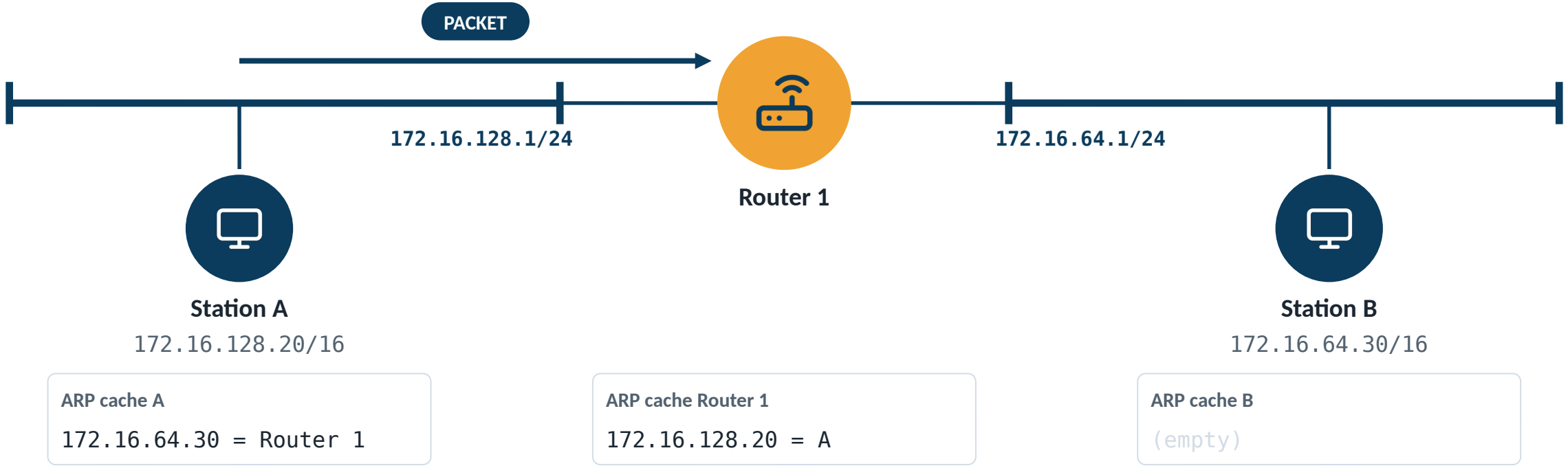
```
172.16.128.20 = A
```

ARP cache B

```
(empty)
```

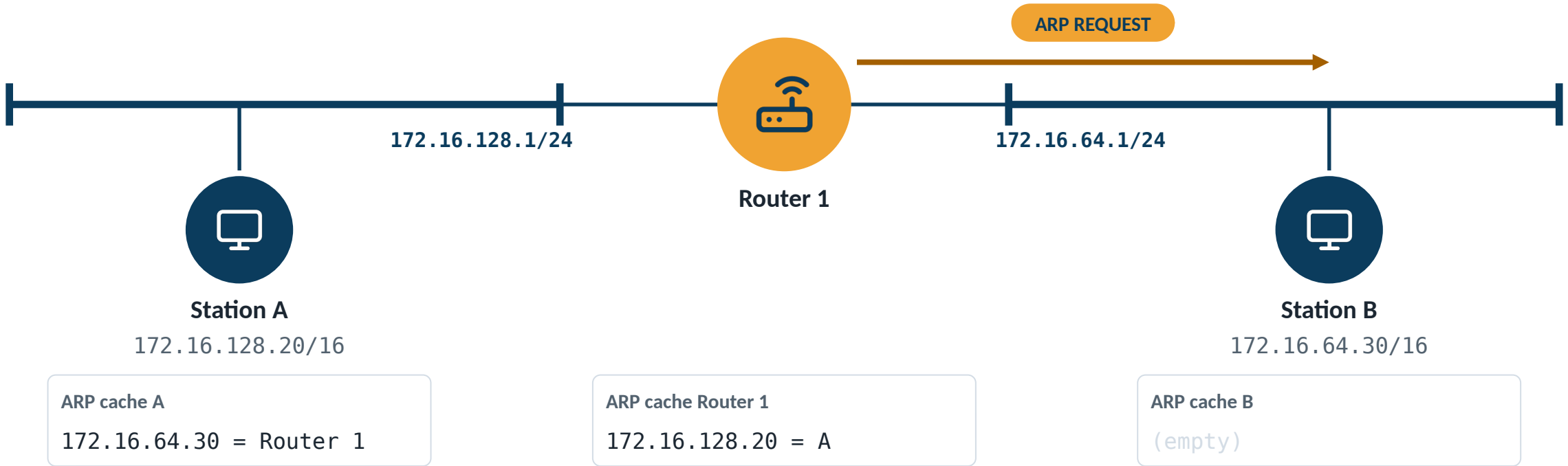
Proxy-ARP

Station A then starts transmitting packets destined for B's IP address, but with the hardware (MAC) address of Router 1. "A" thinks it is communicating directly with "B".



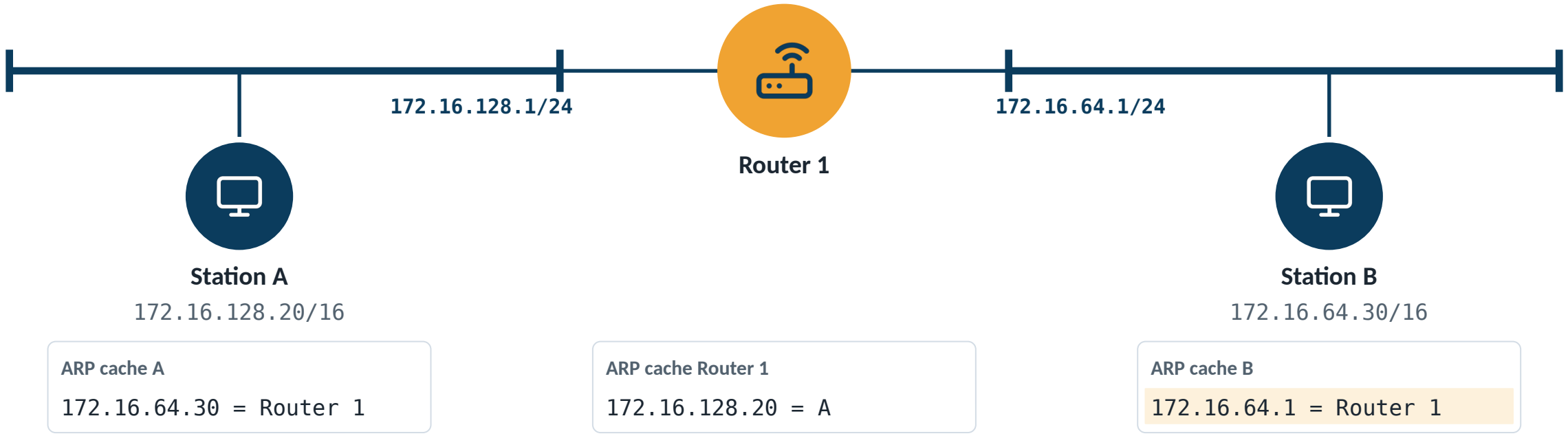
Proxy-ARP

Router 1 will determine, through the Route Determination Process, that B is on its other IP interface. Router 1 will transmit an ARP-Request to B, through its interface 172.16.64.1.



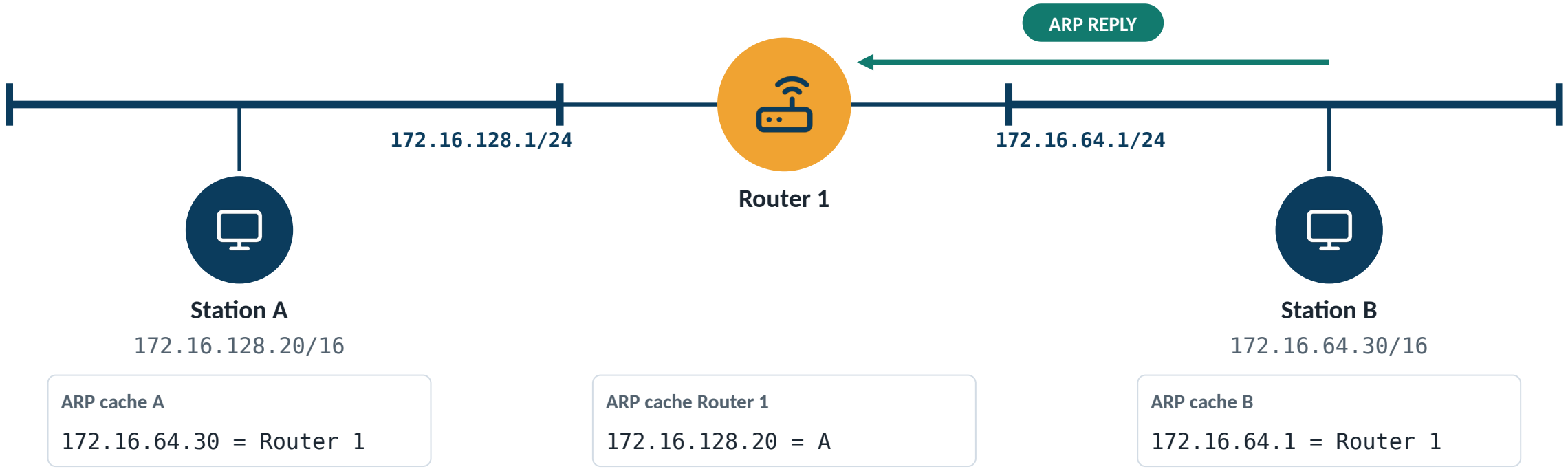
Proxy-ARP

B will receive the ARP and populate its ARP table with the router's IP address associated with the router's MAC address.



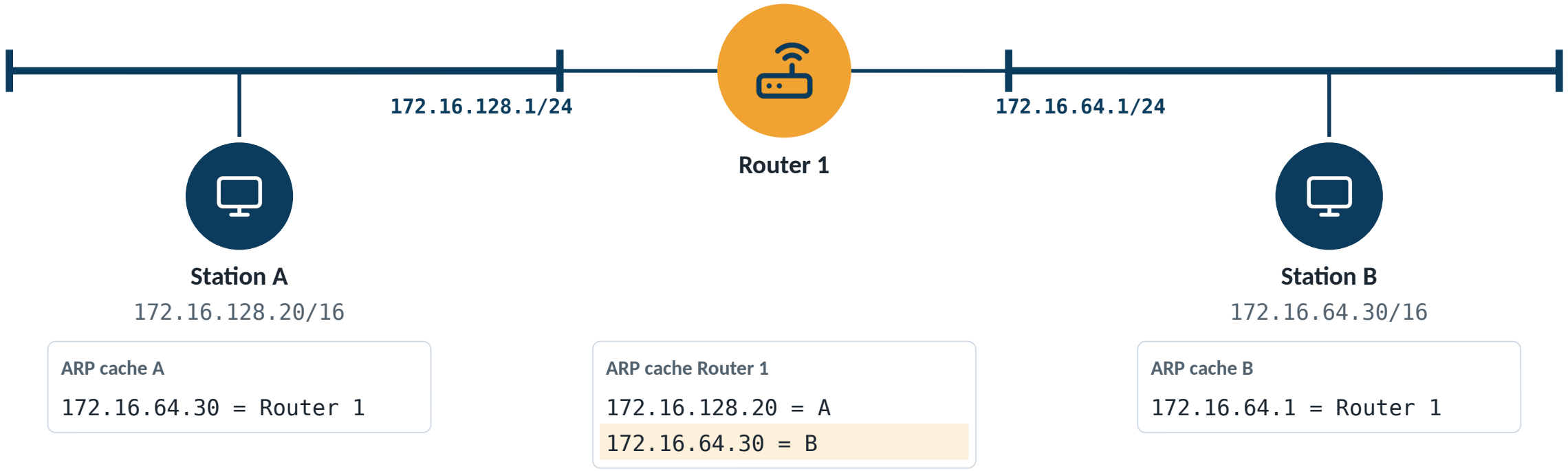
Proxy-ARP

Since it was an ARP REQUEST sent by the router to B, B will send a RESPONSE back to the router.



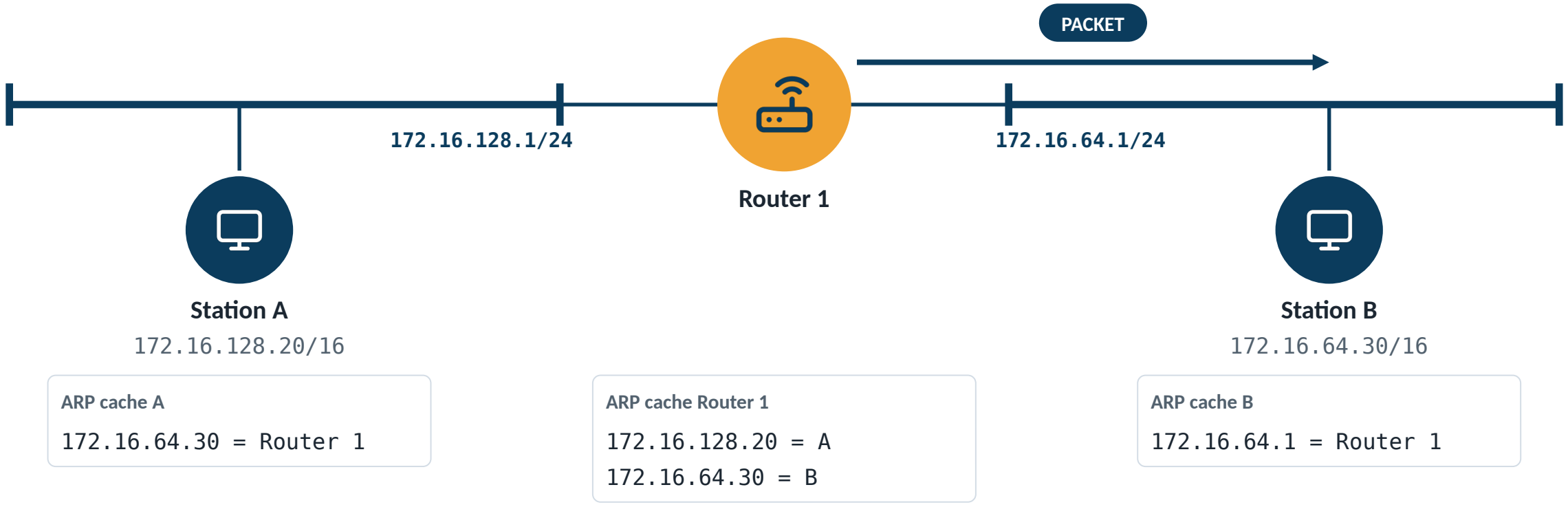
Proxy-ARP

The router receives B’s response and populates its ARP table.



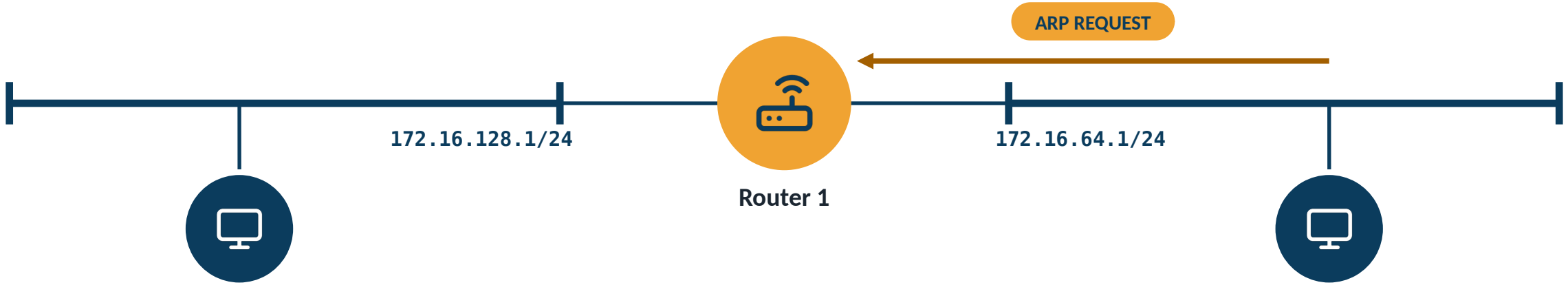
Proxy-ARP

The router then forwards the packet sent by Station A to Station B.



Proxy-ARP

When B decides to reply to A, it will have to perform an ARP for A's MAC address. When B sends the ARP, the router will receive this packet, since it is a Broadcast.



ARP cache A

```
172.16.64.30 = Router 1
```

ARP cache Router 1

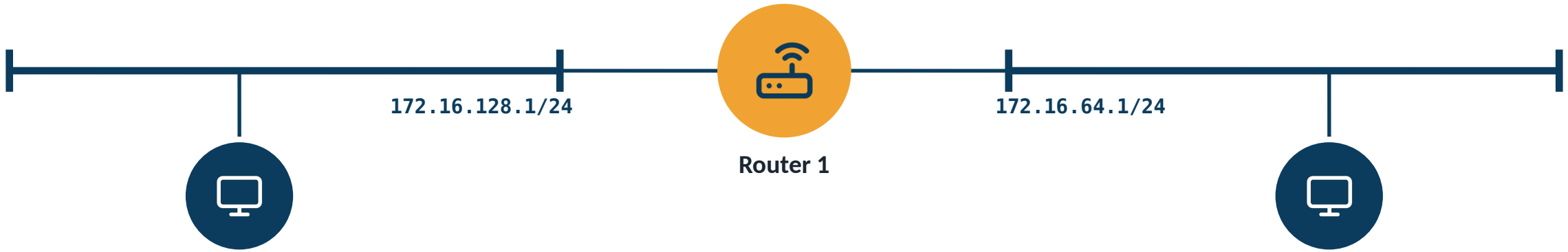
```
172.16.128.20 = A
172.16.64.30 = B
```

ARP cache B

```
172.16.64.1 = Router 1
```

Proxy-ARP

The router will now consider whether it should respond by proxy for Station A, through the same process.



ARP cache A

```
172.16.64.30 = Router 1
```

ARP cache Router 1

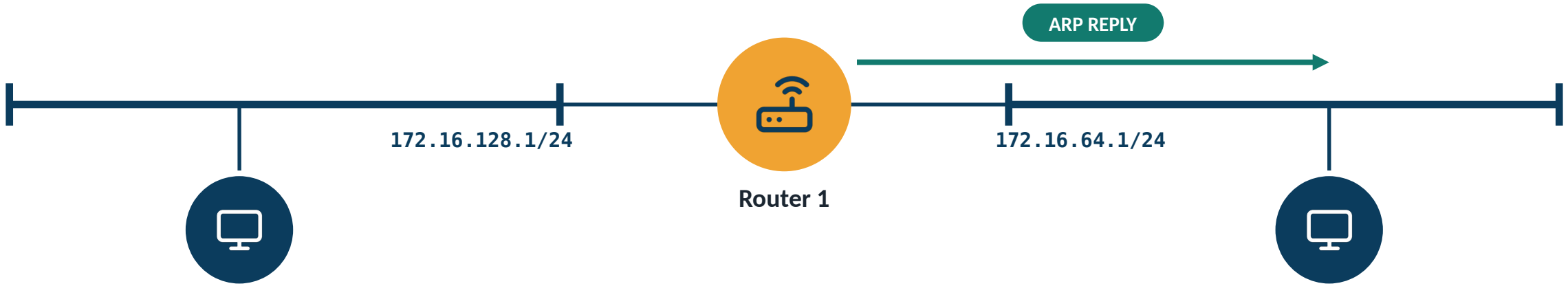
```
172.16.128.20 = A
172.16.64.30 = B
```

ARP cache B

```
172.16.64.1 = Router 1
```

Proxy-ARP

When the router decides to respond by proxy for A, it answers the REQUEST sent by B, with the IP address of A and with the router's MAC address.



ARP cache A

```
172.16.64.30 = Router 1
```

ARP cache Router 1

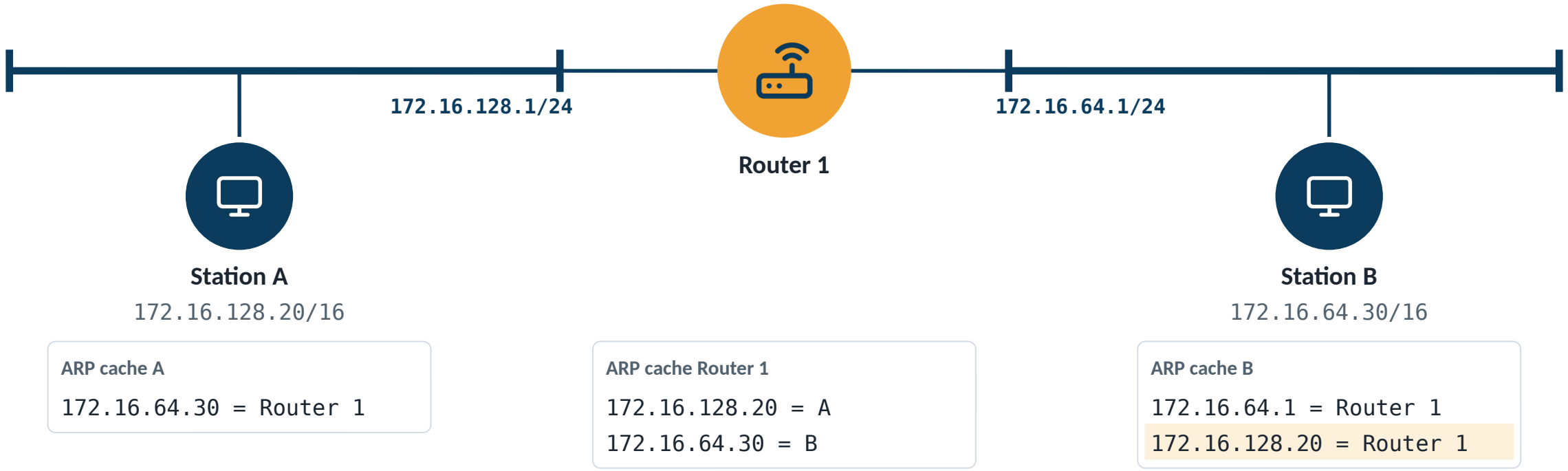
```
172.16.128.20 = A
172.16.64.30 = B
```

ARP cache B

```
172.16.64.1 = Router 1
```

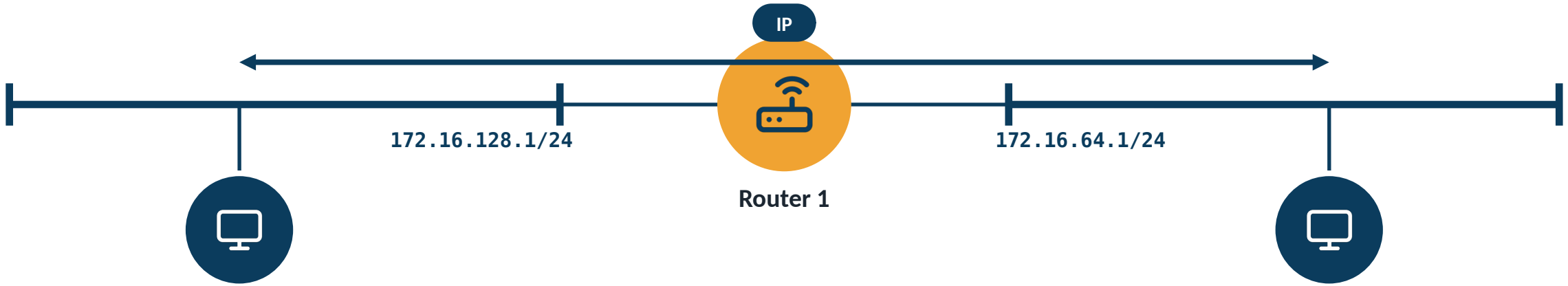
Proxy-ARP

Station B receives the ARP RESPONSE and populates its ARP table.



Proxy-ARP

A and B can now establish communication with each other. Note that each station believes the other is on the same network.



ARP cache A

```
172.16.64.30 = Router 1
```

ARP cache Router 1

```
172.16.128.20 = A
172.16.64.30 = B
```

ARP cache B

```
172.16.64.1 = Router 1
172.16.128.20 = Router 1
```

3.2

IP routing



IP interfaces, static and dynamic routes, precedence and ARP

- Routing devices and features
- IP interfaces, host interface and loopback
- Router mode and basic configuration
- Static routes, RIP and OSPF
- Route precedence and ECMP
- ARP management

Chapter 3.2 agenda

1. Routing devices and features
2. IP interfaces, host interface and loopback
3. Router mode and basic configuration
4. Static routes, RIP and OSPF
5. Route precedence and ECMP
6. ARP management
7. Junos: EX switches vs MX routers

MODERN CLI

- B15 L3 interfaces and management
- B16 Static routes and the routing table
- B17 OSPF: basic enablement
- B18 ARP table

Lab

- Lab: IP interfaces and static routes

LEGACY 2007

Enterasys devices with IP routing

- The following Enterasys devices support features for both **Layer 2 switching** and **Layer 3 IP routing**:
  - Matrix N-Series DFE
  - SecureStack C2/C3
  - Matrix X



LEGACY 2007

Enterasys devices with IP routing

Routing Functionality	Matrix N-series	Matrix E1	SecureStack C2/C3	Matrix X
RIP v1/v2	✓	✓	✓	✓
OSPF	✓*	✓	✓*	✓
BGP-4	n/d	×	×	✓
IS-IS	×	×	×	✓
DVMRP	✓*	✓	✓*	✓
PIM-SM	✓* **	×	✓*	✓
IPv6	n/d	×	✓ (C3)	×
IRDP	✓	✓	✓	✓
VRRP	✓	✓	✓*	✓
LSNAT	✓* **	×	×	n/d
Standard ACLs	✓	✓	✓	✓
Extended ACLs	✓*	✓	✓	✓
PBR	✓	×	×	✓
DoS Prevention	✓	✓	×	✓
DHCP Server	✓	×	✓	✓

* Requires an advanced routing features software license.
** Requires extended memory of 256 MB
n/d = not documented in the available guides

LEGACY 2007

IP interfaces

Each VLAN allows a primary and a secondary IP address

IP interface limit: 256 per module on the Matrix N, 256 on the SecureStack C2 and 1024 on the Matrix X

The Matrix X allows IP addressing on physical ports

```
Attention:  ← VLAN IP interfaces are
administratively      DOWN when created
```

Therefore, the command `no shutdown` is needed to activate them

Configuration changes take effect immediately (right after Enter)

LEGACY 2007

Host IP Interface

The **host** IP interface can be used for in-band management through a VLAN (Matrix N and SecureStack); on the Matrix X, out-of-band management uses the eth0 port

It has its own ARP table, separate from the router's (`show arp` vs `show ip arp`)

On the Matrix N it also has its own routes (`set ip route` and `show ip route` as switch commands)

It is used only for administrative functions (telnet, ping, etc.)

It must be associated with a VLAN (default: VLAN 1)

```
set host vlan vlan-id  ← (SecureStack)
show host vlan  ← (SecureStack)
clear host vlan  ← (SecureStack)
set port vlan host.0.1 vlan-id  ← (Matrix N)
```

On the SecureStack C, the host interface must be in a different subnet from the other IP interfaces that already exist

The Matrix X has a dedicated out-of-band management Ethernet port (eth0), one on each control module

LEGACY 2007

“Router” mode

To access the routing functions, you must enter “router” mode

```
router
```

On the Matrix N, you must first specify in which slot the *router* will be configured

```
set router <slot>
```

You must also specify the slot to access *router* mode

```
router <slot>
```

The Matrix N supports up to two *router* instances in one chassis. Each one is configured independently and creates and maintains its own route tables.

LEGACY 2007

Basic configuration of *router* mode

SecureStack: Enter *router* mode through the CLI.
IP interfaces used for routing must be associated with a **VLAN already configured** on the device.

Prompt Display	CLI Mode	Explanation
`SecureStackC2(su)->`	**Switch command mode**	**Configures layer 2 switching features.** **Shows the switch ARP table (**show arp**).**
`SecureStackC2(su)-> router` `SecureStackC2(su)->router>`	**Enters the CLI *router* mode**	**Limited functionality.** **Some "*show*" commands for router configuration parameters.**
`SecureStackC2(su)->router> enable` `SecureStackC2(su)->router#`	**Privileged mode of the *router***	**Allows clearing the ARP table** clear arp-cache **Or testing connectivity** ping / traceroute
`SecureStackC2(su)->router# configure [terminal]` `SecureStackC2(su)->router(Config)#`	**Global configuration mode of the *router***	**Configures general layer 3 routing features**
`SecureStackC2(su)->router(Config)# interface vlan vid` `SecureStackC2(su)->router(Config-if(Vlan 1))#`	**Interface configuration mode of the *router***	**Configures features specific to layer 3 routing interfaces**
`SecureStackC2(su)->router(Config)# router <protocol>` `SecureStackC2(su)->router(Config-router)#`	**Routing protocol configuration mode**	**Configures features specific to layer 3 routing protocols**

LEGACY 2007

Basic configuration of *router* mode

Matrix N: First, create a *router* instance on the desired module, through the CLI.
IP interfaces used for routing must be associated with a **VLAN already configured** on the device.

Prompt Display	CLI Mode	Explanation
`matrix(su)->`	Switch command mode	Configures layer 2 switching features. Shows only the routing table of the *host*.
`matrix(su)-> set router <slot>`	Create a *router* instance	Done only once. Enables routing on a slot.
`matrix(su)-> router <slot>` `Matrix>Router1>`	Enters the CLI *router* mode	Limited functionality. Some "*show*" commands for router configuration parameters.
`Matrix>Router1> enable` `Matrix>Router1#`	Privileged mode of the *router*	**Allows clearing the ARP table** clear arp-cache **Or viewing IP statistics** show ip traffic **Or testing connectivity** ping / traceroute
`Matrix>Router1# configure [terminal]` `Matrix>Router1(config)#`	Global configuration mode of the *router*	Configures general layer 3 routing features
`Matrix>Router1(config)# interface vlan vid` `Matrix>Router1(config-if(Vlan 1))#`	Interface configuration mode of the *router*	Configures features specific to layer 3 routing interfaces
`Matrix>Router1(config)# router rip` `Matrix>Router1(config-router)#`	Routing protocol configuration mode	Configures features specific to layer 3 routing protocols

LEGACY 2007

Step-by-step creation of an IP interface

The following steps are recommended:

Create the VLAN to be used for IP routing	`set vlan create vid`
Associate physical ports with the VLAN	`set port vlan port-string vlan-id`
Add the ports to the VLAN egress list	`set vlan egress vlan-id port-string untagged`
Remove the ports from the egress list of the default VLAN (*vid* 1)	`clear vlan egress 1 port-string`

In the CLI *router* mode, create an IP interface on the VLAN, and configure an IP address and mask

SecureStackC2(su)-> `router`

SecureStackC2(su)->router> `enable`

SecureStackC2(su)->router# `configure`

SecureStackC2(su)->router(Config)# `interface vlan vid`

SecureStackC2(su)->router(Config-if(Vlan *vid*))# `ip address ip-address ip-mask`

LEGACY 2007

Step-by-step creation of an IP interface (cont.)

SecureStackC2(su)->router(Config-if(Vlan *vid*))# `no shutdown`

LEGACY 2007

Loopback interface configuration

- The *loopback* interface is internal and is not associated with any VLAN
- The following steps are recommended when creating the *loopback*
- Matrix>Router1(config)# `interface loopback <num>`
  - Matrix>Router1(config-if(Lpbk 1))# `ip address ip-address ip-mask`
  - Matrix>Router1(config-if(Lpbk 1))# `no shutdown`
- Remember that IP interfaces are in the *shutdown* state when created, and must be enabled with *no shutdown*.
- *Loopback* interfaces are not associated with any VLAN.
- A *loopback* can be used for *in-band* management of the device, independently of the *host* interface.
- A routing protocol or a static route must be used.
- Best practice: use *loopback* interfaces for the OSPF router-id or the BGP router identifier
- A *loopback* is at no risk of becoming unavailable, as physical interfaces can be when ports are disconnected.

MODERN CLI

L3 interfaces and management (1/2)

	VLAN 10 L3 interface (SVI) with IP	Loopback interface

Extreme Switch Engine	`configure vlan USERS ipaddress 10.1.10.2/24`
	`enable ipforwarding vlan USERS`
		`create vlan LOOP0`
		`enable loopback-mode vlan LOOP0`
		`configure vlan LOOP0 ipaddress 10.255.0.1/32`

Extreme Fabric Engine	`interface vlan 10` ` ip address 10.1.10.2/24`	`interface loopback 1` ` ip address 10.255.0.1/32`

Cisco IOS XE	`interface vlan 10`
	` ip address 10.1.10.2 255.255.255.0`
	` no shutdown`
		`interface loopback 0`
		` ip address 10.255.0.1 255.255.255.255`

Aruba AOS-CX	`interface vlan 10` ` ip address 10.1.10.2/24`	`interface loopback 1` ` ip address 10.255.0.1/32`

Standards: RFC 1918 · RFC 4632 · RFC 3021 · RFC 5737 · RFC 1812

L3 interfaces and management (2/2)

MODERN CLI

	VLAN 10 L3 interface (SVI) with IP	Loopback interface
Juniper Junos	`set interfaces irb unit 10 family inet address 10.1.10.2/24` `set vlans USERS l3-interface irb.10`	`set interfaces lo0 unit 0 family inet address 10.255.0.1/32`
MikroTik RouterOS	`/interface vlan add name=vlan10 interface=bridge1 vlan-id=10` `/ip address add address=10.1.10.2/24 interface=vlan10`	`/interface bridge add name=loopback0` `/ip address add address=10.255.0.1/32 interface=loopback0`
Arista EOS	`interface vlan 10` `ip address 10.1.10.2/24`	`interface loopback 0` `ip address 10.255.0.1/32`
Huawei VRP	`interface Vlanif10` `ip address 10.1.10.2 24`	`interface loopback 0` `ip address 10.255.0.1 32`

Standards: RFC 1918 · RFC 4632 · RFC 3021 · RFC 5737 · RFC 1812

MODERN CLI

L3 interfaces and management

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	`enable ipforwarding`
VLAN 10 L3 interface (SVI) with IP	`configure vlan USERS ipaddress 10.1.10.2/24` `enable ipforwarding vlan USERS`
Secondary address	`configure vlan USERS add secondary-ipaddress 10.1.11.2/24`

Loopback interface	`create vlan LOOP0`
	`enable loopback-mode vlan LOOP0`
	`configure vlan LOOP0 ipaddress 10.255.0.1/32`

Isolated management (mgmt port/VRF)	`configure vlan Mgmt ipaddress 192.0.2.11/24` `configure iproute add default 192.0.2.1 vr VR-Mgmt`
L3 interface summary	`show ipconfig`

The IP lives on the VLAN, and ipforwarding is off on new interfaces. Loopback = a VLAN with no port (loopback-mode). The Mgmt port sits in VR-Mgmt, apart from VR-Default.

MODERN CLI

L3 interfaces and management

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	`ip routing`
VLAN 10 L3 interface (SVI) with IP	`interface vlan 10` ` ip address 10.1.10.2/24`
Secondary address	`interface vlan 10` ` ip address 10.1.11.2/24 secondary`
Loopback interface	`interface loopback 1` ` ip address 10.255.0.1/32`

Isolated management (mgmt port/VRF)	`mgmt oob`
	` ip address 192.0.2.11/24`
	` enable`

L3 interface summary	`show ip interface` `show interfaces vlan ip`

Create the VLAN and enable 'ip routing' before the SVI address; a secondary needs the primary first. 'mgmt oob' is a segmented management instance that does not use the normal OS IP stack.

MODERN CLI

L3 interfaces and management

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	`ip routing`

VLAN 10 L3 interface (SVI) with IP	`interface vlan 10`
	` ip address 10.1.10.2 255.255.255.0`
	` no shutdown`

Secondary address	`interface vlan 10` ` ip address 10.1.11.2 255.255.255.0 secondary`
Loopback interface	`interface loopback 0` ` ip address 10.255.0.1 255.255.255.255`

Isolated management (mgmt port/VRF)	`interface GigabitEthernet0/0`
	` vrf forwarding Mgmt-vrf`
	` ip address 192.0.2.11 255.255.255.0`

L3 interface summary	`show ip interface brief`

By default the switch is L2 only: ip routing turns on L3. The Gi0/0 management port is placed in Mgmt-vrf with vrf forwarding. An extra address uses the secondary keyword.

MODERN CLI

L3 interfaces and management

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	`interface 1/1/22`
	` routing`
	` ip address 10.0.12.1/30`

VLAN 10 L3 interface (SVI) with IP	`interface vlan 10` ` ip address 10.1.10.2/24`
Secondary address	`interface vlan 10` ` ip address 10.1.11.2/24 secondary`
Loopback interface	`interface loopback 1` ` ip address 10.255.0.1/32`

Isolated management (mgmt port/VRF)	`interface mgmt`
	` ip static 192.0.2.11/24`
	` default-gateway 192.0.2.1`

L3 interface summary	`show ip interface vlan 10` `show interface vlan brief`

L3 port: 'routing' on the interface; the SVI comes from 'interface vlan 10' (VLAN first). Loopback IDs run 1 to 256 (there is no loopback 0). 'secondary' works on the 6300 except some SKUs.

MODERN CLI

L3 interfaces and management

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	*n/a: IP routing comes from the IRB: family inet on irb.10 + l3-interface*
VLAN 10 L3 interface (SVI) with IP	`set interfaces irb unit 10 family inet address 10.1.10.2/24` `set vlans USERS l3-interface irb.10`
Secondary address	`set interfaces irb unit 10 family inet address 10.1.11.2/24`
Loopback interface	`set interfaces lo0 unit 0 family inet address 10.255.0.1/32`
Isolated management (mgmt port/VRF)	*n/a: mgmt_junos: docs cite only em0, fxp0, vme; EX uses me0 or vme*
L3 interface summary	`show interfaces irb terse` `show interfaces terse`

The IRB ties the VLAN to L3: an address on irb.10 plus l3-interface on the VLAN. The IRB comes up only with at least one active VLAN port. The mgmt_junos VRF is documented only for em0, fxp0 and vme.

MODERN CLI

L3 interfaces and management

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	`/ip settings set ip-forward=yes` `/interface ethernet switch set switch1 l3-hw-offloading=yes`
VLAN 10 L3 interface (SVI) with IP	`/interface vlan add name=vlan10 interface=bridge1 vlan-id=10` `/ip address add address=10.1.10.2/24 interface=vlan10`
Secondary address	`/ip address add address=10.1.11.2/24 interface=vlan10`
Loopback interface	`/interface bridge add name=loopback0` `/ip address add address=10.255.0.1/32 interface=loopback0`

Isolated management (mgmt port/VRF)	`/ip vrf add name=MGMT interfaces=ether49`
	`/ip route add dst-address=0.0.0.0/0 gateway=192.0.2.1 routing-table=MGMT`
	`/ip service set ssh vrf=MGMT`

L3 interface summary	`/ip address print`

SVI = vlan interface on the bridge (on a physical port it loses hardware L3). Loopback = bridge with no ports, but only one bridge is offloaded and several with L3HW give undefined behavior. VRRP needs L3HW off.

MODERN CLI

L3 interfaces and management

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	`ip routing`
VLAN 10 L3 interface (SVI) with IP	`interface vlan 10` ` ip address 10.1.10.2/24`
Secondary address	`interface vlan 10` ` ip address 10.1.11.2/24 secondary`
Loopback interface	`interface loopback 0` ` ip address 10.255.0.1/32`

Isolated management (mgmt port/VRF)	`vrf instance MGMT`
	`interface management 1`
	` vrf MGMT`

L3 interface summary	`show ip interface brief`

Routed port: 'no switchport' (Ethernet or port-channel). The MGMT VRF is a 'vrf instance' applied with 'vrf MGMT' on the interface (Management1 too); 'vrf' erases the interface IPs, so apply it before 'ip address'.

MODERN CLI

L3 interfaces and management

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 52 to 58 (EOS 2007)

Enable IP routing	*n/a: Enabled by default: IPv4 Layer 3 unicast forwarding*
VLAN 10 L3 interface (SVI) with IP	`interface Vlanif10` ` ip address 10.1.10.2 24`
Secondary address	`interface Vlanif10` ` ip address 10.1.11.2 24 sub`
Loopback interface	`interface loopback 0` ` ip address 10.255.0.1 32`

Isolated management (mgmt port/VRF)	`management-port isolate enable`
	`interface MEth0/0/1`
	` ip address 192.0.2.11 24`

L3 interface summary	`display ip interface brief`

Layer 3 starts with a VLANIF that has an IP address; it comes Up only if the VLAN exists and has an Up port. MEth0/0/1 is isolated from service ports by default (management-port isolate enable).

LEGACY 2007

Static and dynamic routing support

Conventionally, routers use *routing tables* to determine which way, and to which device, to send packets, based on the packet's destination address.
Routing tables can be maintained statically and dynamically.
Static Routes
They are manually configured and programmed into the routing table.
By default, static routes take precedence over routes learned through dynamic routing protocols on the Matrix N and the SecureStack C2 (distance 1); on the Matrix X the default static distance is 60, which loses to internal OSPF (10) and internal IS-IS (11).
Dynamic Routes
They are learned when routers exchange routing information with one another (route tables in RIP, link-state advertisements in OSPF).
The two most widely used forms of dynamic routing are:
Distance Vector (e.g.: RIP) and *Link State (e.g.: OSPF)*.

Switch Router Family	Dynamic Routes	Static Routes
Matrix N-Series Platinum	25,000	n/d
SecureStack C2	n/d	n/d
Matrix X	n/d	2,048

n/d = not documented in the available guides

UPDATED

Managing the routing table (the routes)

- **Lookups** in the routing table: By **destination address**
- Answer provided: The **next hop** address and the **outgoing interface**

Static Routes

- Manually configured and programmed into the routing table

Dynamic Routes

- Learned from information sent by other routers
  - "*Distance Vector*" protocols
  - RIPv1 and RIPv2
  - DVMRP (multicast)
  - "*Link State* " protocols
  - OSPFv2
  - IS-IS
  - "*Path Vector*" protocols
  - BGP4

LEGACY 2007

Route support on the devices

Matrix N-Series DFE Platinum

DFE-Platinum modules support up to 256 routing interfaces, and each interface can be configured for the RIP and/or OSPF routing protocols.

Two router instances in the same chassis work independently, each with its own tables.

OSPF requires activating an optional advanced routing license on the DFE Platinum (*N-EOS-L3*) and on the SecureStack C2 (*C2L3-LIC*)

Only one license is required per chassis or stack.

To activate the license:

```
license advanced license-key  ← (SecureStack C2, in
router global configuration mode) or set license advanced
license-key (Matrix N, in switch mode)
```

LEGACY 2007

Internal route management

Routes are maintained by the **RTM** (**Route Table Manager**), and are held in the **RIB** (**Route Information Base**).

This database (RIB) contains all active static routes, all RIP routes, and, for OSPF, up to the 3 best routes for each destination network when there are multiple routes.

The RTM selects up to three best routes for each destination network, and installs them in the **FIB** (**Forwarding Information Base**).

```
show ip route  ← Shows the router's IP
route information (the routing table).
```

The routes held in the FIB are distributed to each module of the chassis or stack, so that they can be used by the router's distributed routing system (**D**istributed **F**orwarding **E**ngine) right at the *ingress* port, at the moment the frames are received.

LEGACY 2007

Configuring static routes

- Matrix-X (router-config)# `ip route prefix {mask | masklen} {ipv4-address | interface-name | next-hop} [distance] [tag tag] [metric value] [unicast] [multicast] [noinstall] [reject] [retain] [blackhole]`
- SecureStackC2(su)->router(Config)# `ip route network mask nexthop [distance]`
- Static routes take precedence over dynamic routes on the Matrix N and the SecureStack C2, unless the device's default preferences are changed; on the Matrix X the default static distance (60) loses to internal OSPF (10) and internal IS-IS (11).
- **ip route** (Matrix N DFE parameters; on the SecureStack C2 only *prefix*, *mask*, address and *distance*; the Matrix X uses the syntax in the first line)
- ***prefix***: Specifies a prefix for the destination IP address.
- ***mask***: Specifies a mask for the destination.
- ***forward-addr*** | **vlan** *vlan-id:* Specifies the IP address, or the routing interface ID, that will be the *next-hop* (gateway) to which packets for the specified destination will be forwarded.
- ***distance*** (Optional): Specifies an administrative metric (*distance metric*) for this route. Valid values: **1** to **255** (default: **1** on the Matrix N and the SecureStack C2; **60** on the Matrix X). Routes with lower values will be given higher preference in route selection.
- **permanent** (Optional): Specifies a permanent route (present even if *interface down*).
- **tag** *value* (Optional): Specifies a *tag* for this route. Valid values: **1** to **4294967295**.

LEGACY 2007

Configuring the RIP routing protocol

Routing Information Protocol

- A *Distance Vector* protocol, rapidly falling out of use in favor of OSPF.
- Both the **Matrix-N DFE** and the **SecureStack** support the two versions of the protocol, RIPv1 and RIPv2.
- RIPv2 allows "*RIP authentication*" on interfaces, and VLSM.
- To enable RIP: `router rip`
- To specify the interfaces that will participate in RIP (Matrix N DFE; on the SecureStack C2/C3 use `ip rip enable` on the interface):

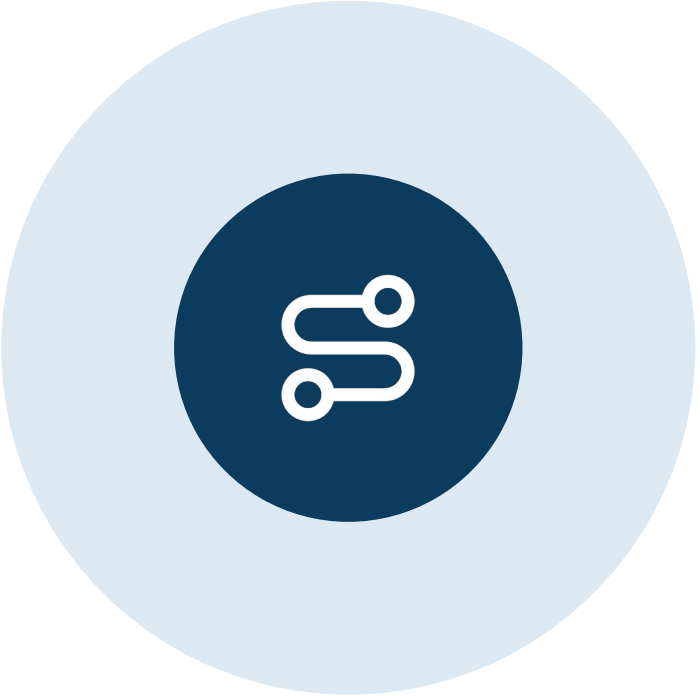
`network ip_address` →

- *ip_address* = Network address of a directly connected interface, which will be included in the *advertisements* sent to the other RIP routers on the network (*neighbors*)

LEGACY 2007

Enabling the OSPF routing protocol

- Open Shortest Path First
- *Link State* routing protocol
- Based on Dijkstra's SPF algorithm
- Extremely efficient:
- Fast reconvergence in case of failure
- Allows load balancing across multiple links
- Only one OSPF process is allowed per DFE or SecureStack
- To enable OSPF: `router ospf process-id`



OSPF: basic enablement (1/2)

MODERN CLI

	Enable OSPF with a router-id	Show neighbors
Extreme Switch Engine	`configure ospf routerid 10.255.0.1` `enable ospf`	`show ospf neighbor`

Extreme Fabric Engine	`router ospf`
	` router-id 10.255.0.1`
	`router ospf enable`
		`show ip ospf neighbor`

Cisco IOS XE	`router ospf 1` ` router-id 10.255.0.1`	`show ip ospf neighbor`
Aruba AOS-CX	`router ospf 1` ` router-id 10.255.0.1`	`show ip ospf neighbors`

Standards: RFC 2328 · IANA-PROTO · IANA-MCAST · RFC 9454

OSPF: basic enablement (2/2)

MODERN CLI

	Enable OSPF with a router-id	Show neighbors
Juniper Junos	`set routing-options router-id 10.255.0.1` `set protocols ospf area 0.0.0.0`	`show ospf neighbor`
MikroTik RouterOS	`/routing ospf instance add name=ospf1 router-id=10.255.0.1`	`/routing ospf neighbor print`
Arista EOS	`router ospf 1` `  router-id 10.255.0.1`	`show ip ospf neighbor`
Huawei VRP	`ospf 1 router-id 10.255.0.1`	`display ospf peer brief`

Standards: RFC 2328 · IANA-PROTO · IANA-MCAST · RFC 9454

MODERN CLI

OSPF: basic enablement

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`configure ospf routerid 10.255.0.1` `enable ospf`
Put uplink and loopback in area 0	`configure ospf add vlan UPLINK area 0.0.0.0` `configure ospf add vlan LOOP0 area 0.0.0.0`
Passive interface (VLAN 10)	`configure ospf add vlan USERS area 0.0.0.0 passive`
Show neighbors	`show ospf neighbor`
Show OSPF routes	`show iproute origin ospf`

In EXOS the OSPF interface is the VLAN (add vlan … area). The router ID is automatic (highest IP): set it by hand. passive stops sending and receiving hellos on the VLAN.

MODERN CLI

OSPF: basic enablement

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`router ospf`
	` router-id 10.255.0.1`
	`router ospf enable`

Put uplink and loopback in area 0	`interface vlan 900`
	` ip ospf area 0.0.0.0`
	` ip ospf enable`

Passive interface (VLAN 10)	`interface vlan 10`
	` ip ospf network passive`
	` ip ospf enable`

Show neighbors	`show ip ospf neighbor`
Show OSPF routes	`show ip route`

Loopback: 'interface loopback 1' + 'ip ospf' (no area parameter in the CLI). OSPF on a VLAN already creates area 0.0.0.0. A passive interface advertises the network as internal, with no hellos or neighbors.

MODERN CLI

OSPF: basic enablement

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`router ospf 1` ` router-id 10.255.0.1`

Put uplink and loopback in area 0	`router ospf 1`
	` network 10.0.12.0 0.0.0.3 area 0`
	` network 10.255.0.1 0.0.0.0 area 0`

Passive interface (VLAN 10)	`router ospf 1`
	` network 10.1.10.0 0.0.0.255 area 0`
	` passive-interface vlan 10`

Show neighbors	`show ip ospf neighbor`
Show OSPF routes	`show ip route ospf`

Style used here: network + wildcard + area. A passive interface shows up as a stub network and neither sends nor receives OSPF. A new router-id applies only after the process restarts (clear ip ospf).

MODERN CLI

OSPF: basic enablement

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`router ospf 1` ` router-id 10.255.0.1`
Put uplink and loopback in area 0	`interface 1/1/22` ` ip ospf 1 area 0.0.0.0`

Passive interface (VLAN 10)	`interface vlan 10`
	` ip ospf 1 area 0.0.0.0`
	` ip ospf passive`

Show neighbors	`show ip ospf neighbors`
Show OSPF routes	`show ip ospf routes`

An interface joins OSPF with 'ip ospf 1 area 0.0.0.0'; repeat the line on loopback 1. Process ID 1 to 65535; OSPFv2 on the 6300 excludes three SKUs (S3L75A, S3L76A, S3L77A).

MODERN CLI

OSPF: basic enablement

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`set routing-options router-id 10.255.0.1` `set protocols ospf area 0.0.0.0`

Put uplink and loopback in area 0	`set interfaces ge-0/0/21 unit 0 family inet address 10.0.12.1/30`
	`set protocols ospf area 0.0.0.0 interface ge-0/0/21.0`
	`set protocols ospf area 0.0.0.0 interface lo0.0`

Passive interface (VLAN 10)	`set protocols ospf area 0.0.0.0 interface irb.10 passive`
Show neighbors	`show ospf neighbor`
Show OSPF routes	`show ospf route`

OSPF is disabled by default and starts with an area and an interface. The router-id is configured under routing-options. A passive interface advertises its network without running OSPF on it.

MODERN CLI

OSPF: basic enablement

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`/routing ospf instance add name=ospf1 router-id=10.255.0.1`

Put uplink and loopback in area 0	`/routing ospf area add name=backbone area-id=0.0.0.0 instance=ospf1`
	`/routing ospf interface-template add area=backbone interfaces=vlan900`
	`/routing ospf interface-template add area=backbone interfaces=loopback0`

Passive interface (VLAN 10)	`/routing ospf interface-template add area=backbone \` `  interfaces=vlan10 passive`
Show neighbors	`/routing ospf neighbor print`
Show OSPF routes	`/ip route print`

In v7 OSPF uses interface-template: it runs on the interfaces that match the template. Uplink = vlan900 (10.0.12.1/30). OSPF routes show the o flag (DAo) in /ip route.

MODERN CLI

OSPF: basic enablement

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`router ospf 1` ` router-id 10.255.0.1`

Put uplink and loopback in area 0	`router ospf 1`
	` network 10.0.12.0/30 area 0`
	` network 10.255.0.1/32 area 0`

Passive interface (VLAN 10)	`router ospf 1` ` passive-interface vlan 10`
Show neighbors	`show ip ospf neighbor`
Show OSPF routes	`show ip route ospf`

Needs 'ip routing' first (without it 'router ospf' warns 'IP routing not enabled'). The process can be disabled with 'shutdown' in router-ospf. The area can also be set per interface with 'ip ospf area'.

MODERN CLI

OSPF: basic enablement

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 64 and 65 (EOS 2007)

Enable OSPF with a router-id	`ospf 1 router-id 10.255.0.1`

Put uplink and loopback in area 0	`ospf 1`
	` area 0`
	`  network 10.0.0.0 0.255.255.255`

Passive interface (VLAN 10)	`ospf 1` ` silent-interface Vlanif10`
Show neighbors	`display ospf peer brief`
Show OSPF routes	`display ospf routing`

The 10.0.0.0/8 wildcard is a lab shortcut: it matches every 10.x interface, and the manual warns about short masks with many addresses. silent-interface blocks Hellos but still advertises the subnet.

UPDATED

Internal route precedence

Internal Route Precedence

- Each route type has an internal *default* **preference**.
- The **precedence** among the protocols determines the preference.
- The lower the precedence, the more preferred the route.
- The **distance** is configurable for each protocol (and for each OSPF route type) and for each individual static route.



LEGACY 2007

Internal route precedence

The LOWER the **precedence** the HIGHER the **preference**

Route Type	X	N	C2
Directly connected	0	0	0
OSPF Intra Area	10	110	8
OSPF Inter Area	10	110	10
ISIS internal	11	n/a	n/a
Static	60	1	1
RIP	100	120	15
OSPF external Type 1	150	110	13
OSPF external Type 2	150	110	150
ISIS external	151	n/a	n/a
BGP	170	n/a	n/a

LEGACY 2007

Routing Table

The *host* interface maintains a routing table that is separate from, and independent of, the VLAN IP interfaces (Matrix N)
Each can be viewed and maintained separately
The two tables can have distinct *default routes*
Viewing the **Host Routes** (from the switch CLI mode, NOT the router!)

```
Matrix(rw)->show ip route
ROUTE TABLE
Destination     Gateway         Mask     TOS Flags Refcnt Use      Interface
-------------------------------------------------------------------------------
default         192.168.0.1      00000000 0     UC    0     0       host0
10.0.0.0        192.168.0.1      ff000000 0    UC           0       host0
127.0.0.1       127.0.0.1        00000000 0     UH    0      104     lo0
-------------------------------------------------------------------------------
```

LEGACY 2007

Routing Table

To view the routing table of the VLAN IP interfaces, you must be in the CLI router mode.

```
SecureStackC2(su)->router> show ip route
Codes: C - connected, S - static, R - RIP, O - OSPF, IA - OSPF interarea
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
* - candidate default
U - per user static route
C    10.1.204.0/26 directly connected, Vlan 1
C    10.1.204.64/27 directly connected, Vlan 1
C    10.1.204.96/27 directly connected, Vlan 1
C    172.16.0.0/24 directly connected, Vlan 123
S    192.168.1.0/24 via 172.16.0.51, Vlan 123
S    192.168.100.0/24 via 172.16.0.37, Vlan 123
```

LEGACY 2007

ECMP: Multiple equal-cost paths

Equal Cost Multi-Path
Load balancing (load sharing)
The DFE-Platinum modules support up to 3 equal-cost paths.
The SecureStack C2 supports up to 3 equal-cost paths.
"*Round Robin*" algorithm: Distributes the load across the links flow by flow (DFE and Matrix X)
"*Hashing*" algorithm: Distributes the load across the links per source/destination pair

Switch-Router Family	Maximum paths	Round Robin	Hashing	RIP	OSPF	Static Routes
Matrix N-Series Platinum	3	✓	✓	×	✓	✓
SecureStack C2	3	n/d	n/d	×	✓	n/d
Matrix X	4 or 8	✓	n/d	✓	✓	✓

n/d = not documented in the available guides

Static routes and the routing table

MODERN CLI

	Static route	Show the routing table
Extreme Switch Engine	`configure iproute add 172.16.0.0/16 10.0.12.2`	`show iproute`
Extreme Fabric Engine	`ip route 172.16.0.0/16 10.0.12.2 weight 1` `ip route 172.16.0.0/16 10.0.12.2 enable`	`show ip route` `show ip route static`
Cisco IOS XE	`ip route 172.16.0.0 255.255.0.0 10.0.12.2`	`show ip route`
Aruba AOS-CX	`ip route 172.16.0.0/16 10.0.12.2`	`show ip route`
Juniper Junos	`set routing-options static route 172.16.0.0/16 next-hop 10.0.12.2`	`show route` `show route protocol static`
MikroTik RouterOS	`/ip route add dst-address=172.16.0.0/16 gateway=10.0.12.2`	`/ip route print`
Arista EOS	`ip route 172.16.0.0/16 10.0.12.2`	`show ip route`
Huawei VRP	`ip route-static 172.16.0.0 16 10.0.12.2`	`display ip routing-table`

Standards: RFC 1812 · RFC 2991 · RFC 2992

MODERN CLI

Static routes and the routing table

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`configure iproute add 172.16.0.0/16 10.0.12.2`
Default route	`configure iproute add default 10.0.12.2`
Floating route (distance 250)	`configure iproute add default 10.1.99.254 250`
Null (blackhole) route	*n/a: The 33.7.1 reference documents blackhole only for IPv6 and default*
Show the routing table	`show iproute`
Route used for a destination	`rtlookup 172.16.5.5`
ECMP (maximum paths)	`configure iproute sharing max-gateways 4` `enable iproute sharing`

No per-route distance: a floating route is just a higher metric (250); priority is per protocol (static 1100). The gateway must be on an attached subnet: build the VLAN 900 uplink first.

MODERN CLI

Static routes and the routing table

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`ip route 172.16.0.0/16 10.0.12.2 weight 1` `ip route 172.16.0.0/16 10.0.12.2 enable`
Default route	`ip route 0.0.0.0 0.0.0.0 10.0.12.2 weight 1` `ip route 0.0.0.0 0.0.0.0 10.0.12.2 enable`
Floating route (distance 250)	`ip route 0.0.0.0 0.0.0.0 10.1.99.254 weight 1 preference 250` `ip route 0.0.0.0 0.0.0.0 10.1.99.254 enable`
Null (blackhole) route	`ip route 10.99.0.0/16 discard weight 1` `ip route 10.99.0.0/16 discard enable`
Show the routing table	`show ip route` `show ip route static`
Route used for a destination	`show ip route 172.16.0.0`
ECMP (maximum paths)	`ip ecmp` `ip ecmp max-path 4`

In 9.4 a static route is created with 'weight' (cost) and takes effect only after 'enable'. 'preference' is the administrative distance (static = 5). A 'discard' route is the blackhole.

MODERN CLI

Static routes and the routing table

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`ip route 172.16.0.0 255.255.0.0 10.0.12.2`
Default route	`ip route 0.0.0.0 0.0.0.0 10.0.12.2`
Floating route (distance 250)	`ip route 0.0.0.0 0.0.0.0 10.1.99.254 250`
Null (blackhole) route	`ip route 10.99.0.0 255.255.0.0 Null0`
Show the routing table	`show ip route`
Route used for a destination	`show ip route 172.16.1.1`
ECMP (maximum paths)	`router ospf 1` ` maximum-paths 8`

Floating route: a static route with a distance higher than the dynamic protocol (OSPF = 110). A route to Null0 drops traffic without an ACL. The hardware never uses more than 16 paths per route.

MODERN CLI

Static routes and the routing table

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`ip route 172.16.0.0/16 10.0.12.2`
Default route	`ip route 0.0.0.0/0 10.0.12.2`
Floating route (distance 250)	`ip route 0.0.0.0/0 10.1.99.254 distance 250`
Null (blackhole) route	`ip route 10.99.0.0/16 nullroute`
Show the routing table	`show ip route`
Route used for a destination	`show ip route 172.16.5.10`
ECMP (maximum paths)	`ip route 172.16.0.0/16 10.0.12.2` `ip route 172.16.0.0/16 10.0.12.6`

Discard with 'nullroute' (silent) or 'reject' (sends an ICMP error). Default static distance is 1 (1 to 255); up to 32 next hops per route, but some 6300 SKUs accept only one.

MODERN CLI

Static routes and the routing table

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`set routing-options static route 172.16.0.0/16 next-hop 10.0.12.2`
Default route	`set routing-options static route 0.0.0.0/0 next-hop 10.0.12.2`

Floating route (distance 250)	`edit routing-options static route 0.0.0.0/0`
	` set qualified-next-hop 10.1.99.254 preference 250`
	`top`

Null (blackhole) route	`set routing-options static route 10.99.0.0/16 discard`
Show the routing table	`show route` `show route protocol static`
Route used for a destination	`show route best 172.16.5.5`

ECMP (maximum paths)	`set routing-options static route 172.16.0.0/16 next-hop 10.1.99.254`
	`set policy-options policy-statement LB then load-balance per-packet`
	`set routing-options forwarding-table export LB`

Static route: preference 5 (global). Floating route: qualified-next-hop with preference 250. With several next hops Junos installs one at random; an exported load-balance per-packet policy installs all.

MODERN CLI

Static routes and the routing table

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`/ip route add dst-address=172.16.0.0/16 gateway=10.0.12.2`
Default route	`/ip route add dst-address=0.0.0.0/0 gateway=10.0.12.2`
Floating route (distance 250)	`/ip route add dst-address=0.0.0.0/0 gateway=10.1.99.254 distance=250`
Null (blackhole) route	`/ip route add dst-address=10.99.0.0/16 blackhole`
Show the routing table	`/ip route print`
Route used for a destination	`/ip route check dst-ip=172.16.5.5` `/tool traceroute 172.16.5.5`
ECMP (maximum paths)	`/ip route add dst-address=172.16.0.0/16 gateway=10.1.99.254`

In RouterOS the route with the lowest distance becomes active. Routes to one destination with equal distance merge into one ECMP route (+ flag). Static, blackhole and ECMP can go to hardware (H = candidate).

MODERN CLI

Static routes and the routing table

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`ip route 172.16.0.0/16 10.0.12.2`
Default route	`ip route 0.0.0.0/0 10.0.12.2`
Floating route (distance 250)	`ip route 0.0.0.0/0 10.1.99.254 250`
Null (blackhole) route	`ip route 10.99.0.0/16 Null0`
Show the routing table	`show ip route`
Route used for a destination	`show ip route 172.16.1.1`
ECMP (maximum paths)	`router ospf 1` ` maximum-paths 4`

Static-route administrative distance: default 1, range 1-255. A null route points to null0 and drops traffic. In the MGMT VRF: 'ip route vrf MGMT 0.0.0.0/0 192.0.2.1'. 'ip load-sharing' only sets the hash.

MODERN CLI

Static routes and the routing table

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 59 to 63 and 68 to 70 (EOS 2007)

Static route	`ip route-static 172.16.0.0 16 10.0.12.2`
Default route	`ip route-static 0.0.0.0 0 10.0.12.2`
Floating route (distance 250)	`ip route-static 0.0.0.0 0 10.1.99.254 preference 250`
Null (blackhole) route	`ip route-static 10.99.0.0 16 NULL 0`
Show the routing table	`display ip routing-table`
Route used for a destination	`display ip routing-table 172.16.5.10`
ECMP (maximum paths)	`ip route-static 172.16.0.0 16 10.0.12.2` `ip route-static 172.16.0.0 16 <nexthop-2>`

The default static preference is 60; a higher value (250) makes a floating backup. Equal preference to one destination shares load; NULL 0 is a blackhole. An IP in display ip routing-table shows the longest match.

LEGACY 2007

Managing ARP

ARP Configuration

View the ARP table for the host:

```
SecureStackC2(su)-> show arp
```

View the ARP table for the VLAN IP interfaces:

```
SecureStackC2(su)->router> show ip arp
```

Adding a static ARP entry to the router table: `SecureStackC2(su)->router(Config)# arp <ip-address> <h-h-h>`

Changing the “ARP cache timeout” (how long entries stay in the ARP table)

```
SecureStackC2(su)->router(Config)# arp timeout seconds
```

Clearing the ARP table usually solves problems after IP addresses are changed on the network:

```
SecureStackC2(su)->router# clear arp-cache
```

Limits

ARP Timeout (default): 4 hours on the Matrix N and the SecureStack C2

6 hours on the Matrix X.

LEGACY 2007

Managing ARP

The ARP table of host.0.1

```
SecureStackC2(su)-> show arp
LINK LEVEL ARP TABLE
IP Address       Phys Address       Flags     Interface
-------------------------------------------------------
10.1.204.17      00-01-f4-5f-49-c5  S         host.0.1
10.1.204.65      00-01-f4-5f-49-c5  S         host.0.1
10.1.204.97      00-01-f4-5f-49-c5  S         host.0.1
10.1.204.98      00-00-00-00-00-00            host.0.1
10.1.204.2       00-00-00-00-00-00            host.0.1
10.1.204.15      00-00-00-00-00-00            host.0.1
10.1.204.17      00-01-f4-5f-49-c5  S         host.0.1
10.1.204.20      00-00-00-00-00-00            host.0.1
10.1.204.21      00-00-00-00-00-00            host.0.1
10.1.204.22      00-00-00-00-00-00            host.0.1
10.1.204.34      00-00-00-00-00-00            host.0.1
10.1.204.65      00-01-f4-5f-49-c5  S         host.0.1
10.1.204.66      00-00-00-00-00-00            host.0.1
10.1.204.67      00-00-00-00-00-00            host.0.1
10.1.204.97      00-01-f4-5f-49-c5  S         host.0.1
10.1.204.98      00-00-00-00-00-00            host.0.1
-------------------------------------------------------
```

LEGACY 2007

Managing ARP

The ARP table of the VLAN IP interfaces

```
SecureStackC2(su)->router> show ip arp
Protocol      Address           Age(min)  Hardware Addr     Interface
-------------------------------------------------------------------------
Dynamic      10.1.204.2           0m    000D:883C:5A4B      Vlan1
Dynamic      10.1.204.15          0m    000D:883E:10E3      Vlan1
Internet     10.1.204.17           -    0001:F45F:49C5      Vlan1
Dynamic      10.1.204.20          0m    0011:1136:6B0B      Vlan1
Dynamic      10.1.204.21          0m    0009:6B99:814D      Vlan1
Dynamic      10.1.204.31         141m   00D0:B7B6:6597      Vlan1
Dynamic      10.1.204.32          0m    00D0:B7A7:7159      Vlan1
Dynamic      10.1.204.34          0m    0006:1BDA:A1A6      Vlan1
Internet     10.1.204.65           -    0001:F45F:49C5      Vlan1
Dynamic      10.1.204.66          0m    0010:A4E6:513B      Vlan1
Dynamic      10.1.204.67          0m    000D:883C:97CC      Vlan1
Internet     10.1.204.97           -    0001:F45F:49C5      Vlan1
Dynamic      10.1.204.98          3m    0002:B32F:B563      Vlan1
Arp entry count = 13.
* - Static
```

ARP table

MODERN CLI

	Show the ARP table	Clear ARP (all or one)
Extreme Switch Engine	`show iparp`	`clear iparp` `clear iparp 10.1.10.50`
Extreme Fabric Engine	`show ip arp`	`clear ip arp interface vlan 10` `clear ip arp interface gigabitethernet all`
Cisco IOS XE	`show ip arp`	`clear arp-cache` `clear arp-cache 10.1.10.50`
Aruba AOS-CX	`show arp`	`clear arp` `clear arp 1/1/1`
Juniper Junos	`show arp`	`clear arp` `clear arp hostname 10.1.10.50`
MikroTik RouterOS	`/ip arp print`	`/ip arp remove [find dynamic]` `/ip arp remove [find address=10.1.10.50]`
Arista EOS	`show ip arp`	`clear arp-cache`
Huawei VRP	`display arp all`	`reset arp dynamic` `reset arp dynamic ip 10.1.10.50`

Standards: RFC 826 · IANA-IEEE802 · RFC 4861 · RFC 1122 · RFC 5227 · RFC 1027

MODERN CLI

ARP table

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`show iparp`
ARP entry for one IP	`show iparp 10.1.10.50`
Static ARP entry	`configure iparp add 10.1.10.50 00:11:22:33:44:55`
ARP timeout	`configure iparp timeout 20`
Clear ARP (all or one)	`clear iparp` `clear iparp 10.1.10.50`

The ARP table is per VR: without the vr keyword the current VR applies. Default timeout: 20 min, 0 disables. clear iparp removes dynamic entries only; static (permanent) ones stay.

MODERN CLI

ARP table

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`show ip arp`
ARP entry for one IP	`show ip arp 10.1.10.50`
Static ARP entry	`ip arp 10.1.10.50 00-11-22-33-44-55 1/1`
ARP timeout	`ip arp timeout 1200`
Clear ARP (all or one)	`clear ip arp interface vlan 10` `clear ip arp interface gigabitethernet all`

ARP aging is tied to MAC aging: an ARP entry goes only when its MAC ages out and the host stops answering. 'clear ip arp' works per port or VLAN, not per IP; the timeout is in seconds.

MODERN CLI

ARP table

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`show ip arp`
ARP entry for one IP	`show ip arp 10.1.10.50`
Static ARP entry	`arp 10.1.10.50 0011.2233.4455 arpa`
ARP timeout	`interface vlan 10` ` arp timeout 7200`

Clear ARP (all or one)	`clear arp-cache`
	`clear arp-cache 10.1.10.50`
	`! IOS XE 26.1.x: The 26.x guide shows clear arp-cache with no argument; the IP-address form is not shown. Line 2 is the 17.18 one`

The default ARP timeout is 4 hours; avoid values below 3600 s. With no argument, clear arp-cache clears the whole dynamic cache; with an IP it acts on that entry only.

MODERN CLI

ARP table

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`show arp`
ARP entry for one IP	`show arp | include "10.1.10.50"`
Static ARP entry	`interface vlan 10` ` arp ip 10.1.10.50 mac 00:11:22:33:44:55`
ARP timeout	*n/a: No ARP timeout setter documented; only 'show arp timeout' (read)*
Clear ARP (all or one)	`clear arp` `clear arp 1/1/1`

'show arp' lists IP, MAC, port and state (reachable, stale). With no argument 'clear arp' clears the default VRF; with a port it clears that port only. 'show arp timeout' shows the age-out.

ARP table

MODERN CLI

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`show arp`
ARP entry for one IP	`show arp hostname 10.1.10.50`

Static ARP entry	`edit interfaces irb unit 10 family inet address 10.1.10.2/24`
	` set arp 10.1.10.50 mac 00:00:5e:00:53:55`
	`top`

ARP timeout	`set system arp aging-timer 10`
Clear ARP (all or one)	`clear arp` `clear arp hostname 10.1.10.50`

ARP ages out after 20 min by default (range 1 to 240). A new value applies only to entries refreshed after the commit. On an IRB, show arp lists irb.10 with the L2 port in square brackets.

MODERN CLI

ARP table

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`/ip arp print`
ARP entry for one IP	`/ip arp print where address=10.1.10.50`
Static ARP entry	`/ip arp add address=10.1.10.60 mac-address=00:11:22:33:44:55 \` ` interface=vlan10`
ARP timeout	`/ip settings set arp-timeout=1m`
Clear ARP (all or one)	`/ip arp remove [find dynamic]` `/ip arp remove [find address=10.1.10.50]`

The timeout is global (/ip settings arp-timeout); initial validity is drawn from timeout/2 to 3*timeout/2, default 15s to 45s. Clearing is a remove in /ip arp; dynamic entries carry the D flag.

MODERN CLI

ARP table

Reference platform: Arista EOS 4.36.x · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`show ip arp`
ARP entry for one IP	`show ip arp 10.1.10.50`
Static ARP entry	`arp 10.1.10.50 0011.2233.4455 arpa`
ARP timeout	`interface vlan 10` ` arp aging timeout 7200`
Clear ARP (all or one)	`clear arp-cache`

ARP timeout is set per interface, in seconds (default 14400; manual: 'arp aging timeout 7200' on a VLAN). 'clear arp-cache' refreshes the active VRF, per interface; the static entry takes MAC xxxx.xxxx.xxxx.

ARP table

MODERN CLI

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Slides 71 to 73 (EOS 2007)

Show the ARP table	`display arp all`
ARP entry for one IP	`display arp all | include 10.1.10.50`
Static ARP entry	`arp static 10.1.10.50 00e0-fc12-3456`
ARP timeout	`interface Vlanif10` ` arp expire-time 600`
Clear ARP (all or one)	`reset arp dynamic` `reset arp dynamic ip 10.1.10.50`

Dynamic ARP entries age out after 1200 s by default. reset arp is a user-view command: dynamic clears all dynamic entries, ip x clears one; static entries go only with reset arp static.

MODERN CLI

Junos: EX switches vs MX routers

Block B02
Clock, identity and connectivity

On MX the out-of-band management port is fxp0 (or em0), not me0. With two REs also set a backup-router (set system backup-router 192.0.2.1 destination 0.0.0.0/0), because the backup RE does not run rpd.

```
set interfaces fxp0 unit 0 family inet address
192.0.2.11/24
set routing-options static route 0/0 next-hop
192.0.2.1 retain no-readvertise
```

Block B09
Storm control and port isolation

On MX the storm control profile is bound under 'family bridge' (not ethernet-switching) and is off by default, unlike EX with ELS (80%). PVLAN on MX needs enhanced LAN mode (chassis network-services lan, with reboot) and a virtual-switch with isolated-vlan.

```
set forwarding-options storm-control-profiles
SC all bandwidth-level 100
set interfaces ge-0/0/0 unit 0 family bridge
storm-control SC
```

Block B10
Port mirroring

The analyzer is the same (forwarding-options analyzer), but ports use 'family bridge' with interface-mode and vlan-id, and the check is 'show forwarding-options analyzer'. A standalone trunk port cannot be the MX destination; use a bridge domain.

```
set interfaces ge-0/0/0 unit 0 family bridge
interface-mode access vlan-id 10
set forwarding-options analyzer M1 input ingress
interface ge-0/0/0.0
show forwarding-options analyzer
```

Block B12
802.1Q VLANs

On MX the VLAN is a bridge-domain and the port uses 'family bridge': interface-mode access with vlan-id, or trunk with vlan-id-list [10 20 99]. Untagged on a trunk needs flexible-vlan-tagging and native-vlan-id on the physical interface.

```
set bridge-domains USERS domain-type bridge
vlan-id 10
set interfaces ge-0/0/0 unit 0 family bridge
interface-mode access vlan-id 10
```

Block B15
L3 interfaces and management

On MX the link is 'bridge-domains USERS routing-interface irb.10' (EX: vlans USERS l3-interface irb.10); only one routing interface fits per bridge domain. Isolated management: management-instance puts fxp0/em0 in the mgmt_junos VRF (routes in the same commit).

```
set interfaces irb unit 10 family inet address
10.1.10.2/24
set bridge-domains USERS routing-interface
irb.10
set system management-instance
```

LAB

Lab: IP interfaces and static routes

1. Confirm VLANs 10, 99 and 900; the uplink is a routed port or, where routing is VLAN-based (C2/C3), VLAN 900.
2. VLAN 10 IP interface: 10.1.10.2/24, plus the secondary address 10.1.11.2/24.
3. VLAN 99 IP interface: 10.1.99.2/24.
4. Uplink interface: 10.0.12.1/30 (neighbor 10.0.12.2).
5. Loopback 0: 10.255.0.1/32 (it becomes the OSPF router-id).
6. Static route 172.16.0.0/16 via 10.0.12.2.
7. Default route 0.0.0.0/0 via 10.0.12.2.
8. Floating default route via 10.1.99.254, with administrative distance 250.
9. Discard (blackhole) route for 10.99.0.0/16.

Goal

Put SW-CORE-01 on the L3 network: IP interfaces, static route, default route, floating route and discard route.

Topology

SW-CORE-01 with VLAN 10 and 99 SVIs, a loopback and a 10.0.12.1/30 uplink to the router 10.0.12.2.

Verification

Routing table: connected (C) and static (S); the floating route is absent.
Ping 10.0.12.2; traceroute to 172.16.1.1 leaves via 10.0.12.2.
Bring the uplink down: the floating route takes over.
Restore it: the primary returns.

End of day 03

Questions are welcome



LAN Switch Operations and Maintenance

From the Enterasys legacy to the current CLI of eight vendors

MODULE 04 · DAY 4

Redundancy, security and management



Rodolfo Nützmann
2026 Edition
Prepared for: [Client] · In partnership with: [Partner]

Course agenda

Day 1

DONE

Access, operation and basic maintenance

- Platforms: 2007 legacy and current
- CLI, users and session
- Clock, identity and connectivity
- Save, erase and reboot
- Password recovery
- Files, backups and firmware

Day 2

DONE

Layer 2 switching

- Interfaces and naming
- Speed, duplex and jumbo frames
- Storm control, isolation, mirroring
- LACP link aggregation
- 802.1Q VLANs
- Spanning Tree and BPDU protection

Day 3

DONE

IP routing

- Addressing, masks and subnets
- Route determination and ARP
- Proxy-ARP
- L3 interfaces and management
- Static routes, RIP and OSPF
- Routing table, ECMP and ARP

Day 4

TODAY

Redundancy, security and management

- OSPF and VRRP
- Access control lists
- Syslog, SNMP and AAA
- Multicast: IGMP and PIM
- Troubleshooting and support
- Final lab and assessment



Redundancy, security and management

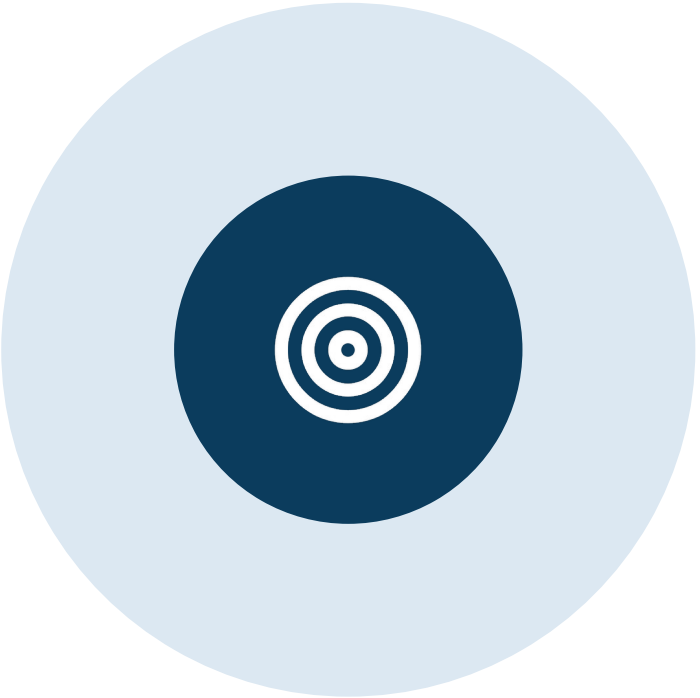
Day 4: OSPF, VRRP, ACL, syslog, SNMP, AAA, multicast and troubleshooting

OSPF | VRRP | ACL | Syslog | SNMP | AAA | Multicast | Troubleshooting

NEW

Objectives of the day

- Configure and verify OSPFv2 on an L3 switch: cost, authentication and redistribution
- Build gateway redundancy with VRRP: election, priority and preemption
- Protect a VLAN with ACLs and interpret the hit counters
- Send logs to a syslog server and monitor the switch with SNMPv3
- Centralize administrative access with RADIUS and TACACS+
- Explain IGMP snooping and PIM-SM and check multicast groups
- Troubleshoot methodically and gather evidence for support



4.1

OSPF: dynamic routing



Link state, areas, neighbors, DR and BDR, authentication

- OSPF: link state and areas
- OSPF: neighbor states, DR and BDR
- OSPF: network types, cost and LSAs
- OSPF authentication: from MD5 to HMAC-SHA
- OSPF on the Enterasys CLI (C2/C3)

Chapter 4.1 agenda

1. OSPF: link state and areas
2. OSPF: neighbor states, DR and BDR
3. OSPF: network types, cost and LSAs
4. OSPF authentication: from MD5 to HMAC-SHA
5. OSPF on the Enterasys CLI (C2/C3)

MODERN CLI

B19 OSPF: configuration and verification

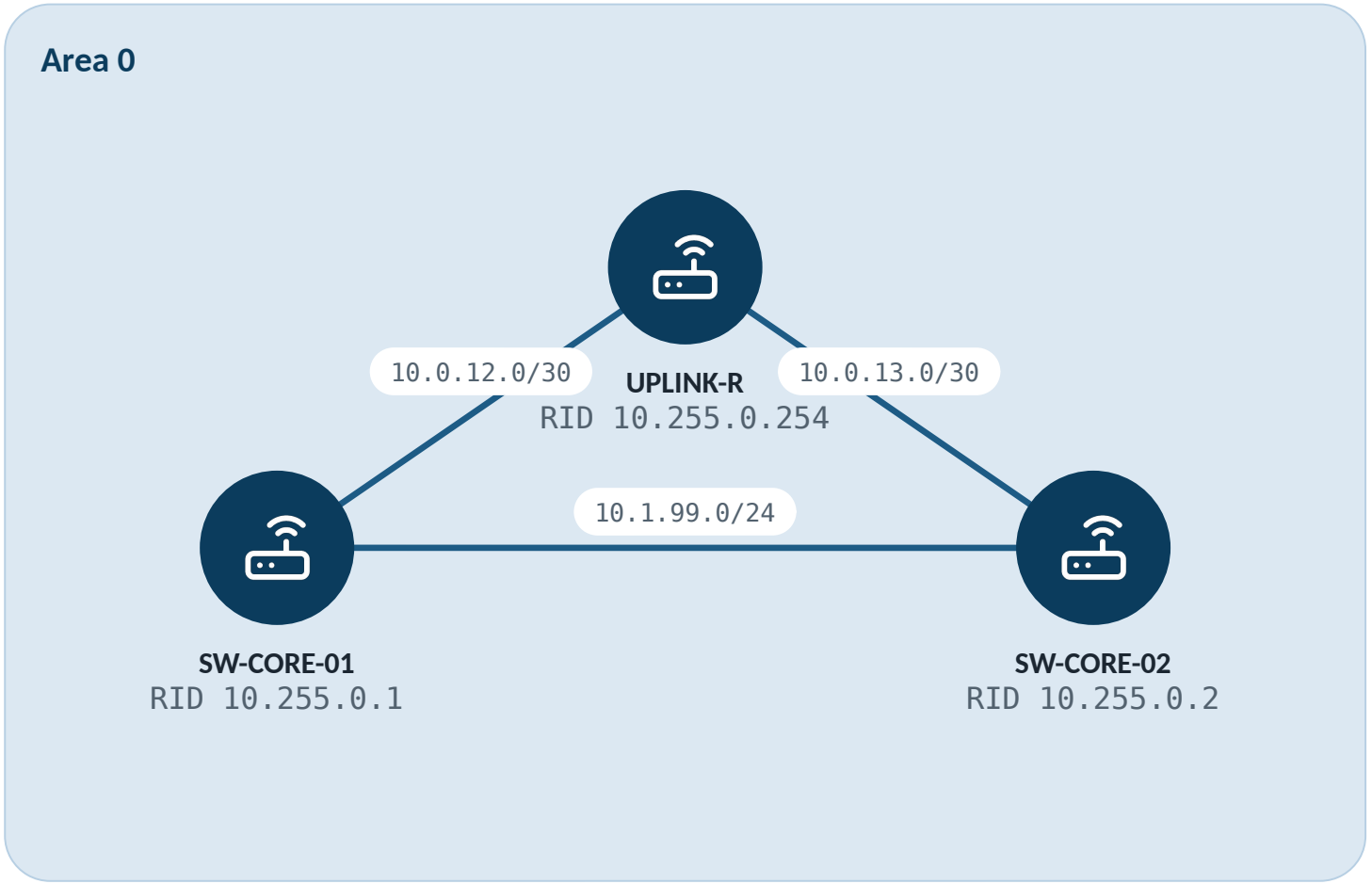
Lab

- Lab: OSPF between the core and the uplink

UPDATED

OSPF: link state and areas

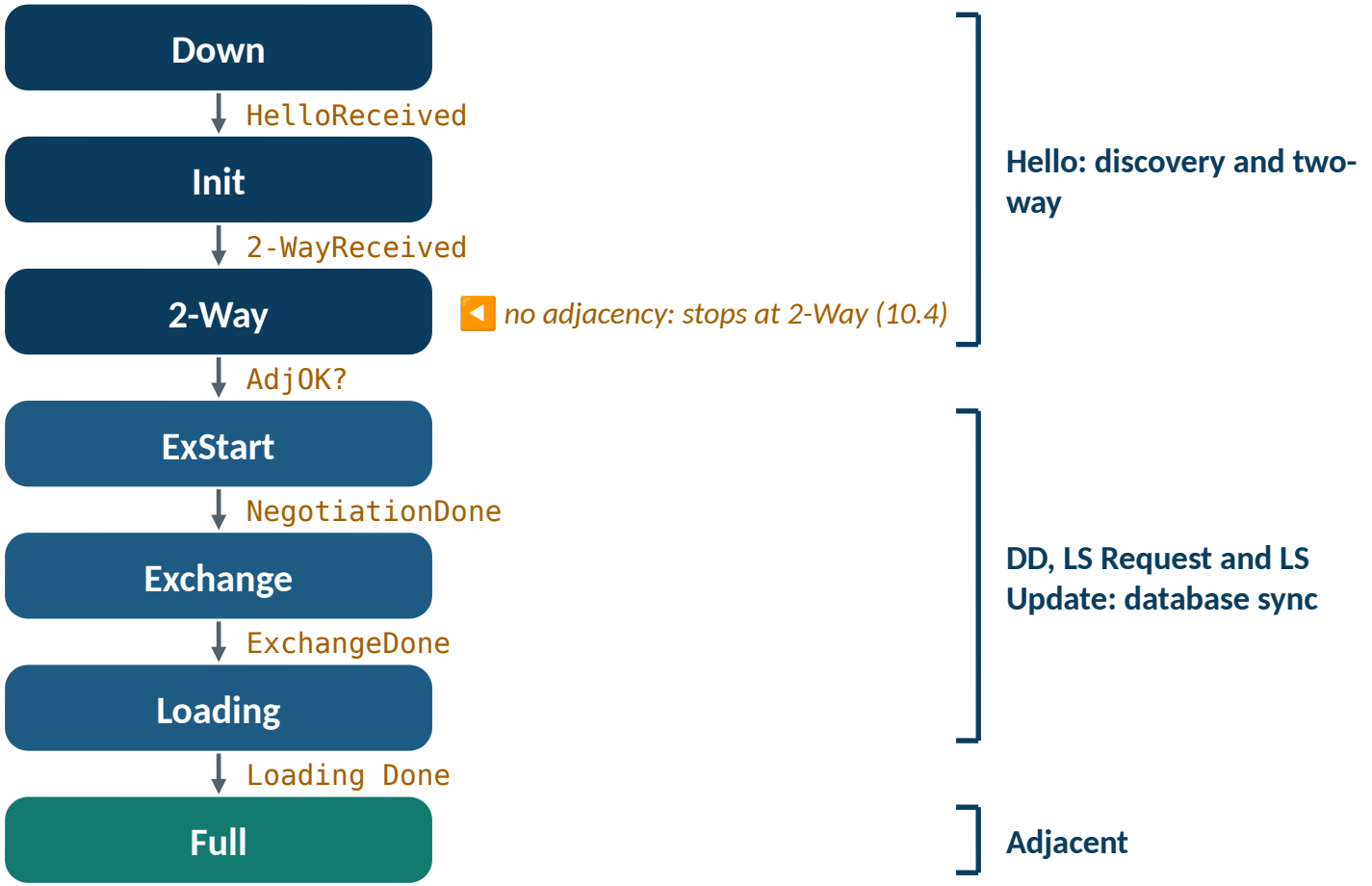
Link-state IGP running directly over IP (protocol 89); Hellos go to 224.0.0.5



- Each router keeps a link-state database; inside an area all copies are identical
- All run SPF and build a shortest-path tree with themselves as the root
- An area hides its internal topology and cuts traffic; area 0 is the backbone
- A router with interfaces in several areas is an ABR and keeps one database per area

NEW

OSPF: neighbor states, DR and BDR



- 2-Way: the router sees itself in the neighbor's Hello (two-way communication)
- ExStart, Exchange and Loading: pick the master, describe the database, request the rest
- DR and BDR (broadcast and NBMA): highest priority; tie, highest Router ID
- Priority 0 is never DR; on broadcast only the DR and BDR go Full with the others

NEW

OSPF: network types, cost and LSAs

Item	What it is	Note
Point-to-point	Joins two routers only	Always adjacent; no DR
Broadcast (Ethernet)	Neighbors found by multicast Hello	Elects DR and BDR
NBMA	Many routers, no broadcast	Configured neighbors; elects DR
Point-to-multipoint	Non-broadcast network seen as point-to-point links	No DR or BDR
Cost	Dimensionless metric; greater than 0	C3: 10 by default; equal costs share load
LSA 1 and 2	Router-LSA (interfaces) and Network-LSA (from the DR)	Stay inside the area
LSA 3 and 4	Summary: networks (3) and ASBRs (4), from the ABR	Condense topology at the border
LSA 5 and 7	Routes external to the AS (5) or inside an NSSA (7)	Type 5 stays out of stub areas

UPDATED

OSPF authentication: from MD5 to HMAC-SHA

2007: MD5 and simple password

- Simple password (type 1) travels in the clear: vulnerable to passive attacks
- Cryptographic (type 2): per-packet digest; the key never travels
- C3: simple password up to 8 characters; MD5 with key-id 1 to 255
- MD5 is no longer acceptable where collision resistance is required
- A packet that fails authentication is discarded

Today: HMAC-SHA (RFC 5709)

- Adds HMAC-SHA-1, -256, -384 and -512 to cryptographic authentication
- HMAC-SHA-256 mandatory; SHA-1 and MD5 recommended for compatibility
- RFC 7474 closes replay with manual keys and covers the IP header
- New designs should not employ HMAC-MD5 (RFC 6151)
- Lab: SHA where supported; otherwise MD5

LEGACY 2007

OSPF on the Enterasys CLI (C2/C3)

Advanced routing license; one OSPF process per stack

Historical platform (2007): SecureStack C2/C3 (2007)

Enable the process and set the router-id

```
router ospf 1
router id 10.255.0.1
```

Enable OSPF on the uplink interface

```
interface vlan 900
ip ospf enable
```

Interface area (default 0.0.0.0)

```
ip ospf areaid 0.0.0.0
```

Interface cost (default 10)

```
ip ospf cost 10
```

MD5 authentication (key-id 1 to 255)

```
ip ospf message-digest-key 1 md5 <key>
```

Redistribute static routes (router mode)

```
router ospf 1
redistribute static subnets
```

Check neighbors, interface and database

```
show ip ospf neighbor detail
show ip ospf interface vlan 900
show ip ospf database
```

MODERN CLI

OSPF: configuration and verification (1/2)

	Authentication (SHA or MD5)	Show neighbors in detail
Extreme Switch Engine	`configure ospf vlan UPLINK authentication md5 1 <key>`	`show ospf neighbor detail`

Extreme Fabric Engine	`interface vlan 900`
	` ip ospf digest-key 1 key <key>`
	` ip ospf authentication-type sha-2 primary-digest-key 1`
		`show ip ospf neighbor`

Cisco IOS XE	`interface GigabitEthernet1/0/2`
	` ip ospf authentication message-digest`
	` ip ospf message-digest-key 1 md5 <key>`
		`show ip ospf neighbor detail`

Aruba AOS-CX	`interface 1/1/22`
	` ip ospf authentication hmac-sha-256`
	` ip ospf sha-key 1 sha plaintext <key>`
		`show ip ospf neighbors detail`

Standards: RFC 2328 · RFC 5709 · RFC 7474

MODERN CLI

OSPF: configuration and verification (2/2)

	Authentication (SHA or MD5)	Show neighbors in detail

Juniper Junos	`edit protocols ospf area 0.0.0.0 interface ge-0/0/21.0`
	` set authentication md5 1 key <key>`
	`top`
		`show ospf neighbor detail`

MikroTik RouterOS	`/routing ospf interface-template set [find interfaces=vlan900] \` ` auth=sha256 auth-id=1 auth-key=<key>`	`/routing ospf neighbor print detail`

Arista EOS	`interface port-channel 1`
	` ip ospf authentication message-digest`
	` ip ospf message-digest-key 1 md5 0 <key>`
		`show ip ospf neighbor detail`

Huawei VRP	`interface Vlanif900` ` ospf authentication-mode hmac-sha256 1 cipher <key>`	`display ospf peer`

Standards: RFC 2328 · RFC 5709 · RFC 7474

MODERN CLI

OSPF: configuration and verification

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`configure ospf add vlan UPLINK area 0.0.0.0 link-type point-to-point`
Interface cost (10)	`configure ospf vlan UPLINK cost 10`
Authentication (SHA or MD5)	`configure ospf vlan UPLINK authentication md5 1 <key>`
Redistribute static routes	`enable ospf export static cost 10 ase-type-2`
Originate a default route	`enable ospf originate-default cost 10 ase-type-2`
Show neighbors in detail	`show ospf neighbor detail`
Show the LSDB	`show ospf lsdb`
Show OSPF interfaces	`show ospf interfaces`

OSPFv2 authenticates with a simple password or MD5 only; SHA-256 exists in OSPFv3 (IPsec). export static makes the router an ASBR. Without always, the default is advertised only if present in the table.

MODERN CLI

OSPF: configuration and verification

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`interface vlan 900` ` ip ospf network p2p`
Interface cost (10)	`interface vlan 900` ` ip ospf cost 10`

Authentication (SHA or MD5)	`interface vlan 900`
	` ip ospf digest-key 1 key <key>`
	` ip ospf authentication-type sha-2 primary-digest-key 1`

Redistribute static routes	`router ospf`
	` as-boundary-router enable`
	` redistribute static enable`

Originate a default route	*n/a: No OSPF default origination; nearest: redistribute static*
Show neighbors in detail	`show ip ospf neighbor`

Redistribution takes effect only after 'ip ospf apply redistribute static' (exec or config mode). 'sha-2' means SHA-256 in this release; use the same key on both sides. 'ip ospf' commands go on the interface.

MODERN CLI

OSPF: configuration and verification (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy OSPF slide

Show the LSDB	`show ip ospf lsdb` `show ip ospf lsdb detail`
Show OSPF interfaces	`show ip ospf interface`

Redistribution takes effect only after 'ip ospf apply redistribute static' (exec or config mode). 'sha-2' means SHA-256 in this release; use the same key on both sides. 'ip ospf' commands go on the interface.

MODERN CLI

OSPF: configuration and verification

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`interface GigabitEthernet1/0/2` ` ip ospf network point-to-point`
Interface cost (10)	`interface GigabitEthernet1/0/2` ` ip ospf cost 10`

Authentication (SHA or MD5)	`interface GigabitEthernet1/0/2`
	` ip ospf authentication message-digest`
	` ip ospf message-digest-key 1 md5 <key>`
	`! IOS XE 26.1.x: 26.x shows ip ospf message-digest-key; ip ospf authentication message-digest does not appear. The line is the 17.18 one`

Redistribute static routes	`router ospf 1` ` redistribute static subnets`
Originate a default route	`router ospf 1` ` default-information originate`
Show neighbors in detail	`show ip ospf neighbor detail`

Authentication here is MD5 (fits in 3 lines). For SHA use ip ospf authentication key-chain, defining the key chain with its cryptographic-algorithm first. Without a local default route, use originate always.

MODERN CLI

OSPF: configuration and verification (cont.)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy OSPF slide

Show the LSDB	`show ip ospf database`
Show OSPF interfaces	`show ip ospf interface`

Authentication here is MD5 (fits in 3 lines). For SHA use ip ospf authentication key-chain, defining the key chain with its cryptographic-algorithm first. Without a local default route, use originate always.

MODERN CLI

OSPF: configuration and verification

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`interface 1/1/22` ` ip ospf network point-to-point`
Interface cost (10)	`interface 1/1/22` ` ip ospf cost 10`

Authentication (SHA or MD5)	`interface 1/1/22`
	` ip ospf authentication hmac-sha-256`
	` ip ospf sha-key 1 sha plaintext <key>`

Redistribute static routes	`router ospf 1` ` redistribute static`
Originate a default route	`router ospf 1` ` default-information originate`
Show neighbors in detail	`show ip ospf neighbors detail`
Show the LSDB	`show ip ospf lsdb`

Authentication: hmac-sha-256 + sha-key (preferred); MD5: 'message-digest' + 'message-digest-key 1 md5 plaintext <key>'. 'default-information originate' advertises the default route only if it exists.

MODERN CLI

OSPF: configuration and verification (cont.)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy OSPF slide

Show OSPF interfaces	`show ip ospf interface brief`

Authentication: hmac-sha-256 + sha-key (preferred); MD5: 'message-digest' + 'message-digest-key 1 md5 plaintext <key>'. 'default-information originate' advertises the default route only if it exists.

MODERN CLI

OSPF: configuration and verification

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`set protocols ospf area 0.0.0.0 interface ge-0/0/21.0 interface-type p2p`
Interface cost (10)	`set protocols ospf area 0.0.0.0 interface ge-0/0/21.0 metric 10`

Authentication (SHA or MD5)	`edit protocols ospf area 0.0.0.0 interface ge-0/0/21.0`
	` set authentication md5 1 key <key>`
	`top`

Redistribute static routes	`set policy-options policy-statement EXP-STATIC term 1 from protocol static`
	`set policy-options policy-statement EXP-STATIC term 1 then accept`
	`set protocols ospf export EXP-STATIC`

Originate a default route	`set policy-options policy-statement EXP-DEF from route-filter 0.0.0.0/0 exact`
	`set policy-options policy-statement EXP-DEF then accept`
	`set protocols ospf export EXP-DEF`

Show neighbors in detail	`show ospf neighbor detail`
Show the LSDB	`show ospf database`

Static routes and 0/0 enter OSPF by export policy (0/0 is exported only while active). On EX, 'authentication keychain' (HMAC-SHA1) exists since 23.4R1; t3 shows MD5.

MODERN CLI

OSPF: configuration and verification (cont.)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy OSPF slide

Show OSPF interfaces	`show ospf interface`

Static routes and 0/0 enter OSPF by export policy (0/0 is exported only while active). On EX, 'authentication keychain' (HMAC-SHA1) exists since 23.4R1; t3 shows MD5.

MODERN CLI

OSPF: configuration and verification

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`/routing ospf interface-template set [find interfaces=vlan900] type=ptp`
Interface cost (10)	`/routing ospf interface-template set [find interfaces=vlan900] cost=10`
Authentication (SHA or MD5)	`/routing ospf interface-template set [find interfaces=vlan900] \ auth=sha256 auth-id=1 auth-key=<key>`
Redistribute static routes	`/routing ospf instance set ospf1 redistribute=static`
Originate a default route	`/routing ospf instance set ospf1 originate-default=if-installed`
Show neighbors in detail	`/routing ospf neighbor print detail`
Show the LSDB	`/routing ospf lsa print`
Show OSPF interfaces	`/routing ospf interface print`

Redistribute and originate-default live in the instance; type, cost and authentication live in the interface-template. Authentication supports md5 and SHA (sha1 to sha512); key and auth-id must match on neighbors.

MODERN CLI

OSPF: configuration and verification

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`interface port-channel 1` ` ip ospf network point-to-point`
Interface cost (10)	`interface port-channel 1` ` ip ospf cost 10`

Authentication (SHA or MD5)	`interface port-channel 1`
	` ip ospf authentication message-digest`
	` ip ospf message-digest-key 1 md5 0 <key>`

Redistribute static routes	`router ospf 1` ` redistribute static`
Originate a default route	`router ospf 1` ` default-information originate`
Show neighbors in detail	`show ip ospf neighbor detail`
Show the LSDB	`show ip ospf database`

The documented authentication is MD5 (message-digest-key N md5); SHA does not appear in the OSPFv2 chapter. In 'show ip ospf neighbor … detail' and 'show ip ospf database' the interface and area are optional.

MODERN CLI

OSPF: configuration and verification (cont.)

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy OSPF slide

Show OSPF interfaces	`show ip ospf interface brief`

The documented authentication is MD5 (message-digest-key N md5); SHA does not appear in the OSPFv2 chapter. In 'show ip ospf neighbor … detail' and 'show ip ospf database' the interface and area are optional.

MODERN CLI

OSPF: configuration and verification

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy OSPF slide

Point-to-point network on the uplink	`interface Vlanif900` ` ospf network-type p2p`
Interface cost (10)	`interface Vlanif900` ` ospf cost 10`
Authentication (SHA or MD5)	`interface Vlanif900` ` ospf authentication-mode hmac-sha256 1 cipher <key>`
Redistribute static routes	`ospf 1` ` import-route static`
Originate a default route	`ospf 1` ` default-route-advertise always`
Show neighbors in detail	`display ospf peer`
Show the LSDB	`display ospf lsdb`
Show OSPF interfaces	`display ospf interface`

On Ethernet the default network type is broadcast; p2p changes it. HMAC-SHA256 is the recommended authentication (md5 and simple carry risk) and the key-id must match the neighbor's.

LAB

Lab: OSPF between the core and the uplink

1. Start from the Module 03 route lab and enable OSPF with router-id 10.255.0.1.
2. Put the uplink and the loopback in area 0.
3. Uplink as a point-to-point network with cost 10 (where the platform allows).
4. Make VLAN 10 passive: advertised but forming no neighbors (where supported).
5. Uplink authentication: HMAC-SHA where available, otherwise MD5, same key on both sides.
6. Redistribute the static routes (172.16.0.0/16) and originate the default route.
7. Compare neighbors, interfaces and the database on both sides.
8. Break it on purpose: a different key on one side; then a different Hello. Observe and fix.

Goal

Form the OSPF adjacency between SW-CORE-01 and the uplink router, authenticated, and advertise the static routes.

Topology

SW-CORE-01 (RID 10.255.0.1) and the uplink router (10.0.12.2) on link 10.0.12.0/30, area 0.

Verification

Neighbor in Full; 172.16.0.0/16 reaches the uplink as an external route.
Wrong key: the neighbor does not come up (packets discarded).
Different Hello or dead interval: the adjacency does not form.

4.2

VRRP: gateway redundancy



One virtual IP, two switches, failover without touching the hosts

VRRP: virtual IP, election and preemption

VRRP: timers and versions 2 and 3

VRRP on the Enterasys CLI (C3)

Chapter 4.2 agenda

1. VRRP: virtual IP, election and preemption
2. VRRP: timers and versions 2 and 3
3. VRRP on the Enterasys CLI (C3)

MODERN CLI

B20 VRRP

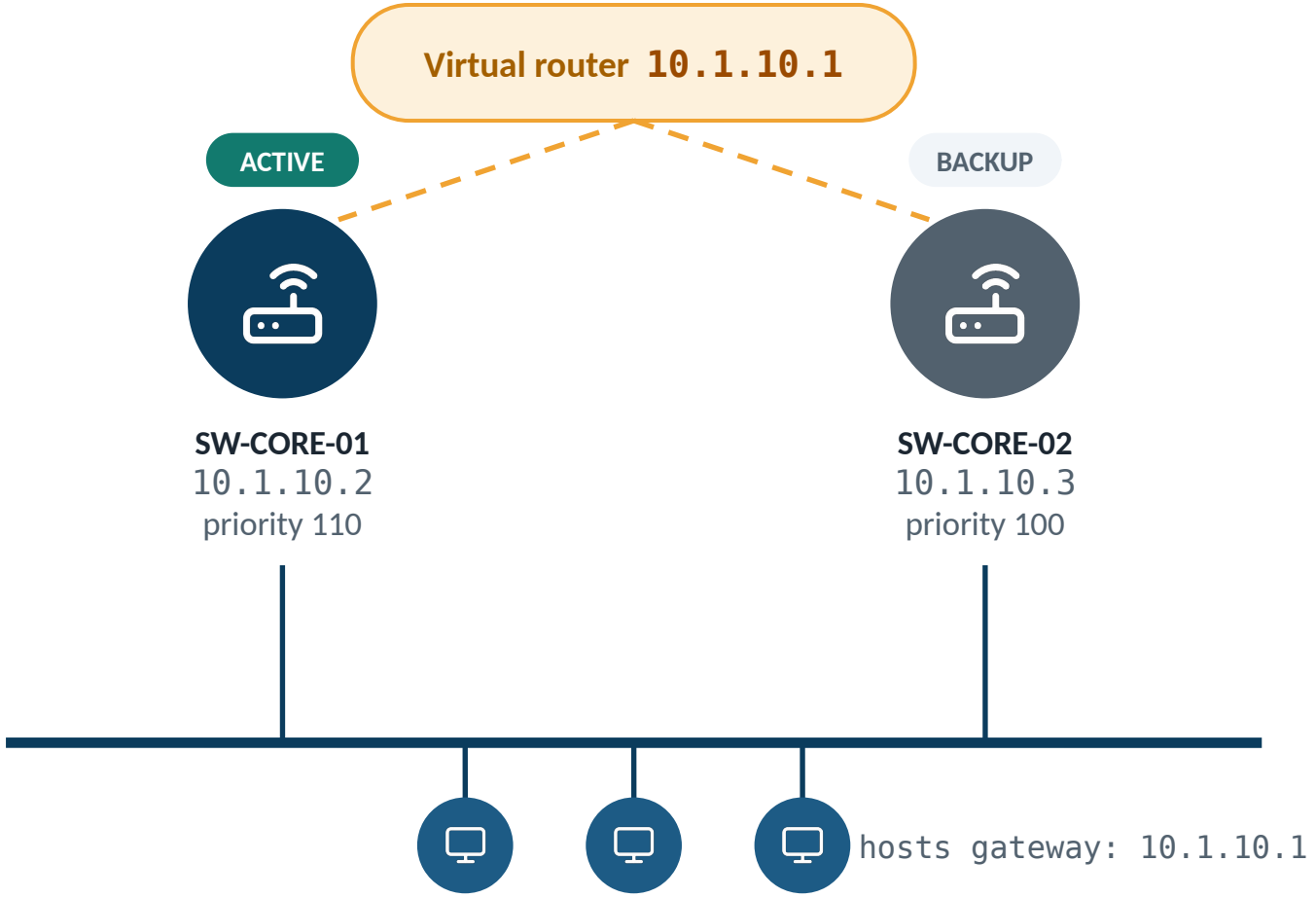
Lab

- Lab: redundant gateway on VLAN 10

UPDATED

VRRP: virtual IP, election and preemption

VRID 10 on VLAN 10: hosts use 10.1.10.1 as their gateway



- The Active answers ARP for the virtual IP and forwards; the Backup waits
- Election: highest priority (1 to 254, default 100); tie, higher IP
- 255 = address owner and always takes over; 0 = the Active stepped down
- Preemption is on by default: the higher priority router retakes the role

UPDATED

VRRP: timers and versions 2 and 3

Item	VRRPv2 (RFC 3768)	VRRPv3 (RFC 9568)
Families	IPv4 only	IPv4 and IPv6, as independent instances
Advertisement interval	Seconds; default 1 s	Centiseconds; default 100 (1 s)
Skew_Time	(256 - prio) / 256 s	((256 - prio) x intervalo) / 256
Declare down	3 x interval + Skew_Time	3 x interval + Skew_Time
Packets	IP 112; 224.0.0.18; TTL 255	Same; IPv6: ff02::12, Hop Limit 255
Virtual MAC	00-00-5E-00-01-VRID	IPv4 same; IPv6: 00-00-5E-00-02-VRID
Forwarding role	Master	Active
Status	Replaced by RFC 5798	From Apr. 2024; replaces RFC 5798

LEGACY 2007

VRRP on the Enterasys CLI (C3)

Advanced routing license; router mode; one VRID per VLAN

Historical platform (2007): SecureStack C3 (2007)

Enter VRRP mode (advanced license)	`router vrrp`
Create the session: VLAN 10, VRID 10	`create vlan 10 10`
Virtual IP (last number 0: not the owner)	`address vlan 10 10 10.1.10.1 0`
Priority (1 to 254)	`priority vlan 10 10 110`
Advertisement interval, in seconds	`advertise-interval vlan 10 10 1`
Preemption (on by default)	`preempt vlan 10 10`
Enable the session	`enable vlan 10 10`
Check state, owner and priority	`show ip vrrp`

VRRP (1/2)

MODERN CLI

	Create VRRP group 10 on VLAN 10	Show VRRP status
Extreme Switch Engine	`create vrrp vlan USERS vrid 10`	`enable vrrp vlan USERS vrid 10` `show vrrp`

Extreme Fabric Engine	`interface vlan 10`
	` ip vrrp address 10 10.1.10.1`
	` ip vrrp 10 enable`
		`show ip vrrp interface vlan 10 verbose`

Cisco IOS XE	`interface vlan 10` ` vrrp 10 address-family ipv4`	`show vrrp detail`
Aruba AOS-CX	`interface vlan 10` ` vrrp 10 address-family ip`	`show vrrp`

Standards: RFC 9568 · RFC 5798 obsoleted · RFC 3768 obsoleted · IANA-PROTO · IANA-MCAST

VRRP (2/2)

MODERN CLI

	Create VRRP group 10 on VLAN 10	Show VRRP status
Juniper Junos	`edit interfaces irb unit 10 family inet address 10.1.10.2/24 set vrrp-group 10 top`	`show vrrp brief show vrrp detail`
MikroTik RouterOS	`/interface ethernet switch set switch1 l3-hw-offloading=no /interface vrrp add name=vrrp10 interface=vlan10 vrid=10`	`/interface vrrp print detail`
Arista EOS	`interface vlan 10 vrrp 10 ipv4 10.1.10.1`	`show vrrp brief`
Huawei VRP	`interface Vlanif10 vrrp vrid 10 virtual-ip 10.1.10.1`	`display vrrp`

Standards: RFC 9568 · RFC 5798 obsoleted · RFC 3768 obsoleted · IANA-PROTO · IANA-MCAST

VRRP

MODERN CLI

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`create vrrp vlan USERS vrid 10`
Virtual IP 10.1.10.1	`configure vrrp vlan USERS vrid 10 add ipaddress 10.1.10.1`
Priority 110	`configure vrrp vlan USERS vrid 10 priority 110`
Preemption	`configure vrrp vlan USERS vrid 10 preempt`
Version 3	`configure vrrp vlan USERS vrid 10 version v3`
Show VRRP status	`enable vrrp vlan USERS vrid 10` `show vrrp`

Create the instance before configuring it, and activate it with enable vrrp. Preempt is on by default; the default version is v3 with v2 compatibility. The virtual IP owner gets priority 255, fixed.

VRRP

MODERN CLI

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`interface vlan 10`
	` ip vrrp address 10 10.1.10.1`
	` ip vrrp 10 enable`

Virtual IP 10.1.10.1	`interface vlan 10` ` ip vrrp address 10 10.1.10.1`
Priority 110	`interface vlan 10` ` ip vrrp 10 priority 110`
Preemption	`interface vlan 10` ` ip vrrp 10 preempt-mode enable`
Version 3	`interface vlan 10` ` ip vrrp version 3`
Show VRRP status	`show ip vrrp interface vlan 10 verbose`

Set 'ip vrrp version 3' first: changing the version erases the interface's IPv4 VRRP. The group is created by 'ip vrrp address'. Do not use the SVI IP as the virtual IP (here .2 and .1).

VRRP

MODERN CLI

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`interface vlan 10` ` vrrp 10 address-family ipv4`

Virtual IP 10.1.10.1	`interface vlan 10`
	` vrrp 10 address-family ipv4`
	`  address 10.1.10.1 primary`

Priority 110	`interface vlan 10`
	` vrrp 10 address-family ipv4`
	`  priority 110`

Preemption	`interface vlan 10`
	` vrrp 10 address-family ipv4`
	`  preempt delay minimum 30`

Version 3	`fhrp version vrrp v3`
Show VRRP status	`show vrrp detail`

Run fhrp version vrrp v3 (global) first. Priority ranges from 1 to 254 (default 100). Preemption is enabled by default; the minimum delay is optional.

VRRP

MODERN CLI

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`interface vlan 10` ` vrrp 10 address-family ip`

Virtual IP 10.1.10.1	`interface vlan 10`
	` vrrp 10 address-family ip`
	`  address 10.1.10.1 primary`

Priority 110	`interface vlan 10`
	` vrrp 10 address-family ip`
	`  priority 110`

Preemption	`interface vlan 10`
	` vrrp 10 address-family ip`
	`  preempt`

Version 3	`interface vlan 10`
	` vrrp 10 address-family ip`
	`  version 3`

Show VRRP status	`show vrrp`

Turn VRRP on with 'router vrrp enable'. The group is created with 'vrrp 10 address-family ip' ('ipv4' is deprecated since 10.14). Preemption is on by default; the default version is 2 (use 'version 3').

VRRP

MODERN CLI

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`edit interfaces irb unit 10 family inet address 10.1.10.2/24`
	` set vrrp-group 10`
	`top`

Virtual IP 10.1.10.1	`edit interfaces irb unit 10 family inet address 10.1.10.2/24`
	` set vrrp-group 10 virtual-address 10.1.10.1`
	`top`

Priority 110	`edit interfaces irb unit 10 family inet address 10.1.10.2/24`
	` set vrrp-group 10 priority 110`
	`top`

Preemption	`edit interfaces irb unit 10 family inet address 10.1.10.2/24`
	` set vrrp-group 10 preempt`
	`top`

Version 3	`set protocols vrrp version-3`
Show VRRP status	`show vrrp brief` `show vrrp detail`

In Junos, preempt is already the default. VRRPv3 is enabled once under protocols vrrp and does not interoperate with VRRPv2: enable it on every router. Priority 255 is not allowed on an RVI.

VRRP

MODERN CLI

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`/interface ethernet switch set switch1 l3-hw-offloading=no` `/interface vrrp add name=vrrp10 interface=vlan10 vrid=10`
Virtual IP 10.1.10.1	`/ip address add address=10.1.10.1/32 interface=vrrp10`
Priority 110	`/interface vrrp set vrrp10 priority=110`
Preemption	`/interface vrrp set vrrp10 preemption-mode=yes`
Version 3	`/interface vrrp set vrrp10 version=3`
Show VRRP status	`/interface vrrp print detail`

VRRP is N/A with hardware L3: it needs l3-hw-offloading=no on the switch (routing moves to the CPU, undoing B15's L3HW). Virtual IP as /32 on the vrrp interface. Defaults: version 3, preemption yes, priority 100.

VRRP

MODERN CLI

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`interface vlan 10` ` vrrp 10 ipv4 10.1.10.1`
Virtual IP 10.1.10.1	`interface vlan 10` ` vrrp 10 ipv4 10.1.10.1`
Priority 110	`interface vlan 10` ` vrrp 10 priority-level 110`
Preemption	`interface vlan 10` ` vrrp 10 preempt`
Version 3	`interface vlan 10` ` vrrp 10 ipv4 version 3`
Show VRRP status	`show vrrp brief`

EOS defaults to VRRPv2; version 3 is chosen per group with 'vrrp N ipv4 version 3'. Priority ranges 1 to 254 (default 100) and preemption is on by default. 'show vrrp' without 'brief' details each group.

VRRP

MODERN CLI

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy VRRP slide

Create VRRP group 10 on VLAN 10	`interface Vlanif10` ` vrrp vrid 10 virtual-ip 10.1.10.1`
Virtual IP 10.1.10.1	`interface Vlanif10` ` vrrp vrid 10 virtual-ip 10.1.10.1`
Priority 110	`interface Vlanif10` ` vrrp vrid 10 priority 110`
Preemption	`interface Vlanif10` ` vrrp vrid 10 preempt-mode timer delay 0`
Version 3	`vrrp version v3`
Show VRRP status	`display vrrp`

The same vrrp vrid virtual-ip line creates the group and sets the virtual IP. Immediate preemption is already the default. vrrp version v3 is system-wide (default v2); default priority is 100.

LAB

Lab: redundant gateway on VLAN 10

1. On both switches, create the VLAN 10 IP interface (10.1.10.2/24 and 10.1.10.3/24).
2. Create VRRP VRID 10 on VLAN 10 with virtual IP 10.1.10.1 on both (version 3 where configurable).
3. SW-CORE-01: priority 110. SW-CORE-02: priority 100.
4. Preemption on and the same advertisement interval (1 s) on both.
5. On the VLAN 10 host, gateway 10.1.10.1 and a continuous ping to a destination beyond the core.
6. Check the state on both and the gateway MAC in the host ARP table.
7. Cut SW-CORE-01's link: SW-CORE-02 takes over. Measure the switchover time.
8. Restore: SW-CORE-01 becomes Active again. Repeat with preemption off and compare.

Goal

Give VLAN 10 a redundant gateway, 10.1.10.1, between SW-CORE-01 and SW-CORE-02, and test failure and preemption.

Topology

SW-CORE-01 (10.1.10.2, priority 110) and SW-CORE-02 (10.1.10.3, priority 100) on VLAN 10; host with gateway 10.1.10.1.

Verification

Active on SW-CORE-01 and Backup on SW-CORE-02. Gateway MAC on the host: 00-00-5E-00-01-0A. Switchover in about 3.6 s (link cut); faster on an orderly shutdown.

4.3

ACLs: traffic filtering



Who may talk to whom: rules evaluated in order

- ACL: first match and implicit deny
- ACL: standard or extended, wildcard or prefix
- ACL on the Enterasys CLI (C3)

Chapter 4.3 agenda

1. ACL: first match and implicit deny
2. ACL: standard or extended, wildcard or prefix
3. ACL on the Enterasys CLI (C3)

MODERN CLI

B21 Access control lists (ACL)

Lab

- Lab: protect the management VLAN

NEW

ACL: first match and implicit deny

MGMT-PROTECT protects VLAN 99 (management) from VLAN 10



Top-down evaluation: the first matching rule decides.

- The first matching rule decides; the following ones are not consulted
- Specific before general: rule order changes the outcome
- There is an implicit deny at the end: what was not permitted is denied
- Applied inbound on the VLAN 10 interface (on the C3, inbound only)

NEW

ACL: standard or extended, wildcard or prefix

- Standard (1 to 99): source only. Extended (100 to 199): protocol, source, destination, port
- Wildcard = bits to ignore (inverse mask); prefix = significant bits
- Stateless: decides on fields of the IP and transport headers

Wildcard	Prefix	Matches
0.0.0.0	/32	a single address (host)
0.0.0.3	/30	4 addresses (uplink link)
0.0.0.255	/24	256 addresses (VLAN 10, VLAN 99)
0.0.255.255	/16	65,536 addresses
0.255.255.255	/8	16,777,216 addresses
255.255.255.255	/0	any address (any)

LEGACY 2007

ACL on the Enterasys CLI (C3)

Router mode; numbered ACL applied per interface, inbound only

Historical platform (2007): SecureStack C3 (2007)

Permit TCP from VLAN 10 to VLAN 99 (no port)

```
access-list 101 permit tcp 10.1.10.0 0.0.0.255 10.1.99.0
0.0.0.255
```

Deny the rest to VLAN 99

```
access-list 101 deny ip any 10.1.99.0 0.0.0.255
```

Permit everything else

```
access-list 101 permit ip any any
```

Apply inbound on VLAN 10

```
interface vlan 10
ip access-group 101 in
```

Show the numbered entries

```
show access-lists 101
```

Remove the ACL from the interface

```
no ip access-group 101 in
```

Standard ACL (1 to 99): source only

```
access-list 1 permit 192.5.34.0 0.0.0.255
```

MODERN CLI

Access control lists (ACL) (1/2)

	Permit SSH from 10.1.10.0/24 to management	Apply inbound on VLAN 10
Extreme Switch Engine	`create access-list MGMT-PROTECT-1 "protocol tcp;source-address 10.1.10.0/24;\ destination-address 10.1.99.0/24;destination-port 22" "permit;count ssh_ok"`	`configure access-list add MGMT-PROTECT-1 first vlan USERS ingress` `configure access-list add MGMT-PROTECT-2 last vlan USERS ingress`

Extreme Fabric Engine	`filter acl 1 type inVlan name MGMT-PROTECT`
	`filter acl ace 1 10`
	`filter acl ace ethernet 1 10 ether-type eq ip`
	`filter acl ace ip 1 10 src-ip mask 10.1.10.0 24`
	`filter acl ace ip 1 10 dst-ip mask 10.1.99.0 24`
	`filter acl ace ip 1 10 ip-protocol-type eq tcp`
	`filter acl ace protocol 1 10 dst-port eq 22`
	`filter acl ace action 1 10 permit count`
	`filter acl ace 1 10 enable`
		`filter acl vlan 1 10`

Cisco IOS XE	`ip access-list extended MGMT-PROTECT` ` permit tcp 10.1.10.0 0.0.0.255 10.1.99.0 0.0.0.255 eq 22`	`interface vlan 10` ` ip access-group MGMT-PROTECT in`
Aruba AOS-CX	`access-list ip MGMT-PROTECT` ` 10 permit tcp 10.1.10.0/24 10.1.99.0/24 eq 22 count`	`interface vlan 10` ` apply access-list ip MGMT-PROTECT routed-in`

Standards: RFC 8519

Access control lists (ACL) (2/2)

MODERN CLI

	Permit SSH from 10.1.10.0/24 to management	Apply inbound on VLAN 10

Juniper Junos	`edit firewall family inet filter MGMT-PROTECT term SSH`
	` set from source-address 10.1.10.0/24`
	` set from destination-address 10.1.99.0/24`
	` set from protocol tcp`
	` set from destination-port 22`
	` set then accept`
	`top`
		`set interfaces irb unit 10 family inet filter input MGMT-PROTECT`

MikroTik RouterOS	`/interface ethernet switch rule add switch=switch1 disabled=yes \`
	` mac-protocol=ip protocol=tcp src-address=10.1.10.0/24 dst-port=22 \`
	` dst-address=10.1.99.0/24 comment=MGMT-PROTECT`
		`/interface ethernet switch rule set [find comment=MGMT-PROTECT] \`
		` ports=ether1 disabled=no`

Arista EOS	`ip access-list MGMT-PROTECT`
	` 10 permit tcp 10.1.10.0/24 10.1.99.0/24 eq 22`
	` counters per-entry`
		`interface vlan 10`
		` ip access-group MGMT-PROTECT in`

Huawei VRP	`acl name MGMT-PROTECT 3000` ` rule permit tcp source 10.1.10.0 0.0.0.255 destination 10.1.99.0 0.0.0.255 destination-port eq 22`	`traffic-filter vlan 10 inbound acl name MGMT-PROTECT` `traffic-statistic vlan 10 inbound acl name MGMT-PROTECT`

Standards: RFC 8519

MODERN CLI

Access control lists (ACL)

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`create access-list MGMT-PROTECT-1 "protocol tcp;source-address 10.1.10.0/24;\ destination-address 10.1.99.0/24;destination-port 22" "permit;count ssh_ok"`
Deny everything else to 10.1.99.0/24	`create access-list MGMT-PROTECT-2 "destination-address \ 10.1.99.0/24" "deny;count blocked"`
Permit everything else	*n/a: Implicit: a packet matching no rule is permitted*
Apply inbound on VLAN 10	`configure access-list add MGMT-PROTECT-1 first vlan USERS ingress` `configure access-list add MGMT-PROTECT-2 last vlan USERS ingress`
Show the ACL and counters	`show access-list dynamic` `show access-list dynamic counter`
Remove the ACL from the interface	`configure access-list delete MGMT-PROTECT-1 vlan USERS` `configure access-list delete MGMT-PROTECT-2 vlan USERS`

I used dynamic ACLs (one rule per create access-list), the CLI default; the alternative is a .pol policy file with configure access-list. An ACL on a VLAN applies to all switch ports.

MODERN CLI

Access control lists (ACL)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`filter acl 1 type inVlan name MGMT-PROTECT`
	`filter acl ace 1 10`
	`filter acl ace ethernet 1 10 ether-type eq ip`
	`filter acl ace ip 1 10 src-ip mask 10.1.10.0 24`
	`filter acl ace ip 1 10 dst-ip mask 10.1.99.0 24`
	`filter acl ace ip 1 10 ip-protocol-type eq tcp`
	`filter acl ace protocol 1 10 dst-port eq 22`
	`filter acl ace action 1 10 permit count`
	`filter acl ace 1 10 enable`

Deny everything else to 10.1.99.0/24	`filter acl ace 1 20`
	`filter acl ace ethernet 1 20 ether-type eq ip`
	`filter acl ace ip 1 20 dst-ip mask 10.1.99.0 24`
	`filter acl ace action 1 20 deny count`
	`filter acl ace 1 20 enable`

Permit everything else	`filter acl set 1 default-action permit`
Apply inbound on VLAN 10	`filter acl vlan 1 10`

Create the ACL before its ACEs; each ACE needs an action and 'enable' (created disabled), lower ID wins; official examples include ether-type ip. Default action: UG says permit, CR says deny; set it before applying.

MODERN CLI

Access control lists (ACL) (cont.)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy ACL slide

Show the ACL and counters	`show filter acl config 1` `show filter acl statistics 1`
Remove the ACL from the interface	`no filter acl vlan 1 10`

Create the ACL before its ACEs; each ACE needs an action and 'enable' (created disabled), lower ID wins; official examples include ether-type ip. Default action: UG says permit, CR says deny; set it before applying.

MODERN CLI

Access control lists (ACL)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`ip access-list extended MGMT-PROTECT` ` permit tcp 10.1.10.0 0.0.0.255 10.1.99.0 0.0.0.255 eq 22`
Deny everything else to 10.1.99.0/24	`ip access-list extended MGMT-PROTECT` ` deny ip any 10.1.99.0 0.0.0.255`
Permit everything else	`ip access-list extended MGMT-PROTECT` ` permit ip any any`
Apply inbound on VLAN 10	`interface vlan 10` ` ip access-group MGMT-PROTECT in`
Show the ACL and counters	`show ip access-lists MGMT-PROTECT` `show platform software fed switch active acl counters hardware`

Remove the ACL from the interface	`interface vlan 10`
	` no ip access-group MGMT-PROTECT in`
	`! IOS XE 26.1.x: 26.x shows ip access-group {number|name} {in|out} and the general no rule; no ip access-group is not shown. 17.18 line`

Every ACL ends with an implicit deny. New entries go at the end of the list. The show ip access-lists counter omits traffic handled in hardware; use the fed counters command.

MODERN CLI

Access control lists (ACL)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`access-list ip MGMT-PROTECT` ` 10 permit tcp 10.1.10.0/24 10.1.99.0/24 eq 22 count`
Deny everything else to 10.1.99.0/24	`access-list ip MGMT-PROTECT` ` 20 deny any any 10.1.99.0/24 count`
Permit everything else	`access-list ip MGMT-PROTECT` ` 30 permit any any any count`
Apply inbound on VLAN 10	`interface vlan 10` ` apply access-list ip MGMT-PROTECT routed-in`
Show the ACL and counters	`show access-list ip MGMT-PROTECT` `show access-list hitcounts ip MGMT-PROTECT vlan 10`
Remove the ACL from the interface	`interface vlan 10` ` no apply access-list ip MGMT-PROTECT routed-in`

The lowest sequence number has the highest priority, and only ACEs with 'count' keep hit counters. On an SVI use 'routed-in'; on a physical port or LAG, 'apply access-list ip <NAME> in' (or 'out').

MODERN CLI

Access control lists (ACL)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`edit firewall family inet filter MGMT-PROTECT term SSH`
	` set from source-address 10.1.10.0/24`
	` set from destination-address 10.1.99.0/24`
	` set from protocol tcp`
	` set from destination-port 22`
	` set then accept`
	`top`

Deny everything else to 10.1.99.0/24	`edit firewall family inet filter MGMT-PROTECT term DENY`
	` set from destination-address 10.1.99.0/24`
	` set then count DENIED`
	` set then discard`
	`top`

Permit everything else	`set firewall family inet filter MGMT-PROTECT term PERMIT then accept`
Apply inbound on VLAN 10	`set interfaces irb unit 10 family inet filter input MGMT-PROTECT`

A filter is evaluated term by term, in order, and ends in an implicit discard, so t3 permits the rest. Every row starts at [edit]: a row that uses edit ends with 'top', which returns to the root.

MODERN CLI

Access control lists (ACL) (cont.)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy ACL slide

Show the ACL and counters	`show firewall filter MGMT-PROTECT` `show configuration firewall family inet filter MGMT-PROTECT`
Remove the ACL from the interface	`delete interfaces irb unit 10 family inet filter input`

A filter is evaluated term by term, in order, and ends in an implicit discard, so t3 permits the rest. Every row starts at [edit]: a row that uses edit ends with 'top', which returns to the root.

MODERN CLI

Access control lists (ACL)

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`/interface ethernet switch rule add switch=switch1 disabled=yes \ mac-protocol=ip protocol=tcp src-address=10.1.10.0/24 dst-port=22 \ dst-address=10.1.99.0/24 comment=MGMT-PROTECT`
Deny everything else to 10.1.99.0/24	`/interface ethernet switch rule add switch=switch1 disabled=yes \ mac-protocol=ip dst-address=10.1.99.0/24 new-dst-ports="" \ comment=MGMT-PROTECT`
Permit everything else	`/interface ethernet switch rule add switch=switch1 disabled=yes \ comment=MGMT-PROTECT`
Apply inbound on VLAN 10	`/interface ethernet switch rule set [find comment=MGMT-PROTECT] \ ports=ether1 disabled=no`
Show the ACL and counters	`/interface ethernet switch rule print`
Remove the ACL from the interface	`/interface ethernet switch rule remove [find comment=MGMT-PROTECT]`

The hardware ACL is /interface ethernet switch rule: the first matching rule wins; no action = permit; new-dst-ports="" = drop. It filters on ingress, per ports=. Packets/bytes counters exist only since 7.25.

MODERN CLI

Access control lists (ACL)

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`ip access-list MGMT-PROTECT`
	` 10 permit tcp 10.1.10.0/24 10.1.99.0/24 eq 22`
	` counters per-entry`

Deny everything else to 10.1.99.0/24	`ip access-list MGMT-PROTECT` ` 20 deny ip any 10.1.99.0/24`
Permit everything else	`ip access-list MGMT-PROTECT` ` 30 permit ip any any`
Apply inbound on VLAN 10	`interface vlan 10` ` ip access-group MGMT-PROTECT in`
Show the ACL and counters	`show ip access-lists MGMT-PROTECT`
Remove the ACL from the interface	`interface vlan 10` ` no ip access-group MGMT-PROTECT in`

Per-rule counters appear only with 'counters per-entry' inside the ACL; 'clear ip access-lists counters NAME' resets them. Rules carry sequence numbers (10, 20, 30). Masks use CIDR (/24).

MODERN CLI

Access control lists (ACL)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy ACL slide

Permit SSH from 10.1.10.0/24 to management	`acl name MGMT-PROTECT 3000` ` rule permit tcp source 10.1.10.0 0.0.0.255 destination 10.1.99.0 0.0.0.255 destination-port eq 22`
Deny everything else to 10.1.99.0/24	`acl name MGMT-PROTECT 3000` ` rule deny ip destination 10.1.99.0 0.0.0.255`
Permit everything else	`acl name MGMT-PROTECT 3000` ` rule permit ip`
Apply inbound on VLAN 10	`traffic-filter vlan 10 inbound acl name MGMT-PROTECT` `traffic-statistic vlan 10 inbound acl name MGMT-PROTECT`
Show the ACL and counters	`display acl name MGMT-PROTECT` `display traffic-statistics vlan 10 inbound`
Remove the ACL from the interface	`undo traffic-filter vlan 10 inbound acl name MGMT-PROTECT`

A packet matching no rule is forwarded. The per-VLAN filter is a system-view command; on a Vlanif it works only on the listed S5731/S5732/S6730 models. Counters need traffic-statistic.

LAB

Lab: protect the management VLAN

1. Create the MGMT-PROTECT ACL (on the C3, numbered extended ACL 101).
2. Rule 10: permit TCP from 10.1.10.0/24 to 10.1.99.0/24, port 22 (SSH).
3. Rule 20: deny IP from any source to 10.1.99.0/24.
4. Rule 30: permit everything else.
5. Apply the ACL inbound on the VLAN 10 interface.
6. From a VLAN 10 host: SSH to a VLAN 99 host, then ping the same host.
7. Check the hit counters (or the log) of each rule.
8. Swap rules 10 and 20 and repeat the SSH. Restore the order.
9. Remove the ACL from the interface and confirm the ping works again.

Goal

Permit SSH from VLAN 10 to management (VLAN 99), deny the rest to VLAN 99 and let other traffic through.

Topology

SW-CORE-01 routes VLAN 10 (10.1.10.0/24) and VLAN 99 (10.1.99.0/24); the MGMT-PROTECT ACL goes inbound on the VLAN 10 interface.

Verification

SSH from VLAN 10 to VLAN 99: permitted (rule 10).
Ping from VLAN 10 to VLAN 99: denied (rule 20).
With the order swapped, SSH is also denied.

4.4

Syslog: the central event record



What the device saw, when it saw it and where it sent it

- Syslog: from device to server
- Syslog: the eight severities
- Syslog: formats and transports
- Syslog on the Enterasys CLI (C3)

Chapter 4.4 agenda

1. Syslog: from device to server
2. Syslog: the eight severities
3. Syslog: formats and transports
4. Syslog on the Enterasys CLI (C3)

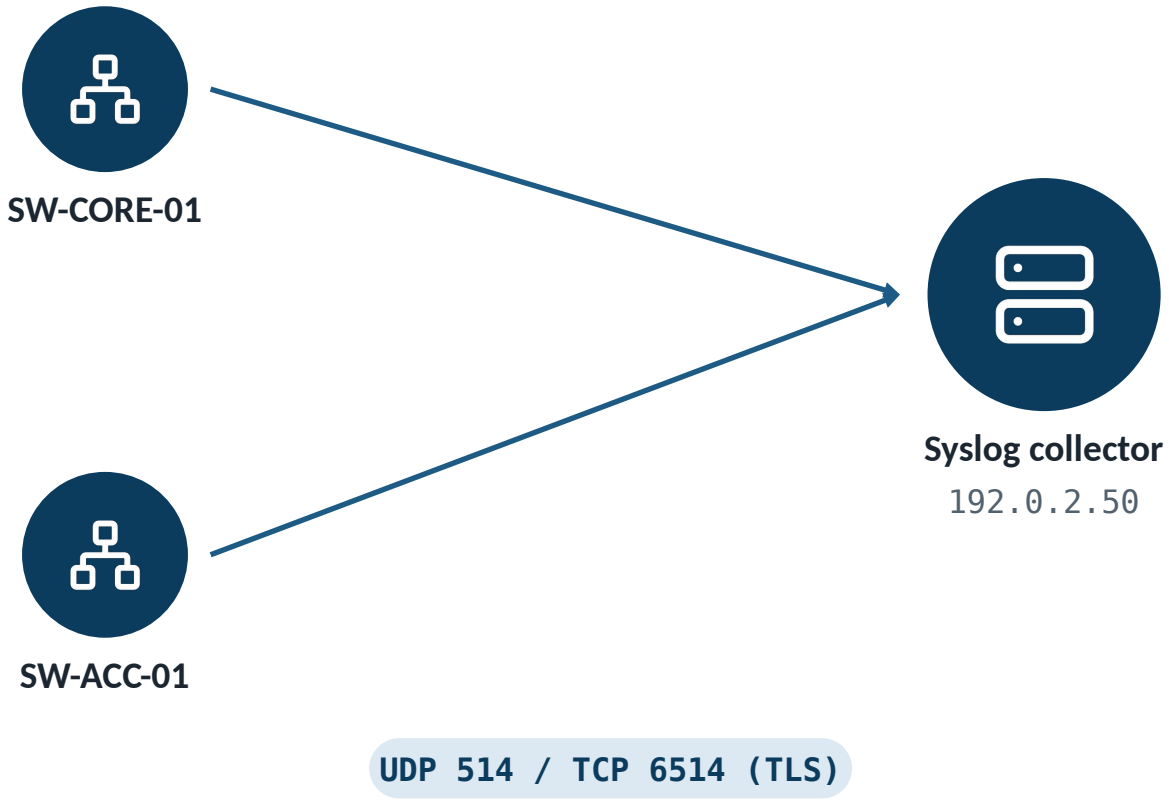
MODERN CLI

B22 Syslog

NEW

Syslog: from device to server

Each message carries priority, time and origin



- PRI = facility × 8 + severity (local4 + notice = 165)
- UDP 514: no acknowledgement, messages can be lost
- TLS (TCP 6514): confidentiality, integrity, authentication
- Time: 5424 accepts a zone; the BSD format has no year or zone

UPDATED

Syslog: the eight severities

Careful: the 2007 guide numbers 1 to 8, the RFC numbers 0 to 7

- PRI = facility × 8 + severity; facilities run 0 to 23 (local0 to local7 = 16 to 23)
- Check the severity name: on the C3 the number is always the RFC number plus 1

RFC	Name	Meaning	C3 (2007)
0	Emergency	system is unusable	1
1	Alert	action must be taken immediately	2
2	Critical	critical conditions	3
3	Error	error conditions	4
4	Warning	warning conditions	5
5	Notice	normal but significant condition	6
6	Informational	informational messages	7
7	Debug	debug-level messages	8

UPDATED

Syslog: formats and transports

UDP is still common, but only TLS protects the message

Transport	Port	Delivery and security	Status
UDP	514	no acknowledgement; may lose messages	RFC 5426
Plain TCP	no standard port	no strong security; not recommended by the IESG	RFC 6587 (Historic)
TLS over TCP	6514	confidentiality, integrity, authentication	RFC 5425, 9662
Message format	n/a	5424 (current) replaces the BSD format of 3164	RFC 5424

LEGACY 2007

Syslog on the Enterasys CLI (C3)

Up to 8 servers; facility local0 to local7; UDP port

Historical platform (2007): SecureStack C3 (2007)

Syslog server, informational (7), UDP 514	`set logging server 1 ip-addr 192.0.2.50 facility local4 severity 7 port 514 state enable`
Show the configured servers	`show logging server`
Show default facility, severity and port	`show logging default`
Level per application (example: SNMP)	`set logging application SNMP level 4`
Show the last 256 messages	`show logging buffer`
Local log: console yes, file no	`set logging local console enable file disable`
Clock: unicast SNTP client	`set sntp client unicast`
Clock: lab NTP server	`set sntp server 192.0.2.123`

Syslog

MODERN CLI

	Remote syslog server	Show the local log
Extreme Switch Engine	`configure syslog add 192.0.2.50 vr VR-Mgmt local0` `enable syslog`	`show log`

Extreme Fabric Engine	`syslog enable`
	`syslog host 1 address 192.0.2.50`
	`syslog host 1 enable`
		`show logging file tail`

Cisco IOS XE	`logging host 192.0.2.50`	`show logging`
Aruba AOS-CX	`logging 192.0.2.50 vrf mgmt`	`show events -r -n 50`
Juniper Junos	`set system syslog host 192.0.2.50 any any`	`show log messages`
MikroTik RouterOS	`/system logging action add name=syslog1 target=remote \` `  remote=192.0.2.50 remote-log-format=syslog`	`/log print`
Arista EOS	`logging host 192.0.2.50`	`show logging`
Huawei VRP	`info-center loghost 192.0.2.50`	`display logbuffer`

Standards: RFC 5424 · RFC 5426 · RFC 5425 · RFC 9662 · RFC 6587 · RFC 3164 obsoleted · IANA-PORTS

MODERN CLI

Syslog

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`configure syslog add 192.0.2.50 vr VR-Mgmt local0` `enable syslog`
Severity sent (informational)	`configure log target syslog 192.0.2.50 vr VR-Mgmt local0 severity info`
Source interface	`configure log target syslog 192.0.2.50 vr VR-Mgmt local0 from 192.0.2.11`
Show the local log	`show log`
Precise timestamps in logs	`configure log target syslog all format timestamp hundredths date yyyy-mm-dd`

Remote syslog is disabled by default: configure syslog add + enable syslog (up to 4 servers). Without vr, VR-Mgmt is used. The syslog default timestamp is seconds; hundredths is more precise.

MODERN CLI

Syslog

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`syslog enable`
	`syslog host 1 address 192.0.2.50`
	`syslog host 1 enable`

Severity sent (informational)	`syslog host 1 severity info warning error fatal`
Source interface	*n/a: No source command; the Management Instance (CLIP, VLAN, OOB) decides*
Show the local log	`show logging file tail`
Precise timestamps in logs	`boot config flags syslog-rfc5424-format`

No source command: the source IP follows the Management Instance. Defaults conflict (Cmd Ref: syslog and RFC 5424 off; guide: on), so the commands are explicit. Run 'save config'.

MODERN CLI

Syslog

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`logging host 192.0.2.50`
Severity sent (informational)	`logging trap informational`
Source interface	`logging source-interface loopback 0` `! IOS XE 26.1.x: 26.x only prints the field Logging Source-Interface in show logging; the logging source-interface command is not shown`
Show the local log	`show logging`
Precise timestamps in logs	`service timestamps log datetime msec localtime show-timezone`

Syslog servers get messages up to informational by default. Without service timestamps, messages carry no date or time. A source interface (a loopback is best) keeps one stable IP in every message.

Syslog

MODERN CLI

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`logging 192.0.2.50 vrf mgmt`
Severity sent (informational)	`logging 192.0.2.50 severity info vrf mgmt`
Source interface	`ip source-interface syslog 192.0.2.11 vrf mgmt`
Show the local log	`show events -r -n 50`
Precise timestamps in logs	*n/a: No timestamp setting in 'logging' or in the guide*

'show events' shows the local log since the last boot ('-a' adds earlier boots). Remote 'logging' takes up to 4 servers; default severity is info. The source address must exist in the VRF used for sending.

Syslog

MODERN CLI

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`set system syslog host 192.0.2.50 any any`
Severity sent (informational)	`set system syslog host 192.0.2.50 any info`
Source interface	`set system syslog source-address 192.0.2.11`
Show the local log	`show log messages`
Precise timestamps in logs	`set system syslog time-format year millisecond`

Severity means the level you set and higher. time-format does not affect a remote host destination, only file, console and user. source-address is an IP configured on an interface, not the interface name.

MODERN CLI

Syslog

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`/system logging action add name=syslog1 target=remote \ remote=192.0.2.50 remote-log-format=syslog`
Severity sent (informational)	`/system logging add topics=info,warning,error,critical action=syslog1`
Source interface	`/system logging action set syslog1 src-address=192.0.2.11`
Show the local log	`/log print`
Precise timestamps in logs	*n/a: No ms setting in docs; only syslog-time-format (bsd-syslog, iso8601)*

An action (target=remote) says where to send and a logging rule says what, by topics. 'info' is a topic, not a level: list info,warning,error,critical. Source is a src-address; in a VRF use vrf=.

MODERN CLI

Syslog

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`logging host 192.0.2.50`
Severity sent (informational)	`logging trap informational`
Source interface	`logging source-interface loopback 0`
Show the local log	`show logging`
Precise timestamps in logs	`logging format timestamp high-resolution`

Inside a VRF use 'logging vrf MGMT host 192.0.2.50' and 'logging vrf MGMT source-interface ...'. The default port is 514; 'protocol tcp|tls|udp' changes the transport. 'logging on' enables syslog.

MODERN CLI

Syslog

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy Syslog slide

Remote syslog server	`info-center loghost 192.0.2.50`
Severity sent (informational)	`info-center source default channel loghost log level informational`
Source interface	`info-center loghost source loopback 0`
Show the local log	`display logbuffer`
Precise timestamps in logs	`info-center timestamp log date precision-time millisecond`

By default no logs go to a remote host. The level is set per channel with info-center source. The source interface only supplies the IP carried in logs; it need not be the sending interface.

4.5

SNMP: layered management



From the clear-text community to SNMPv3 with authentication and privacy

- SNMP: manager, agent and notifications
- SNMP: v1, v2c and v3
- SNMPv3: security levels and algorithms (USM)
- SNMP: MIB, OID and notifications
- RMON: remote monitoring (2007 heritage)
- SNMPv3 on the Enterasys CLI (C3)

Chapter 4.5 agenda

1. SNMP: manager, agent and notifications
2. SNMP: v1, v2c and v3
3. SNMPv3: security levels and algorithms (USM)
4. SNMP: MIB, OID and notifications
5. RMON: remote monitoring (2007 heritage)
6. SNMPv3 on the Enterasys CLI (C3)

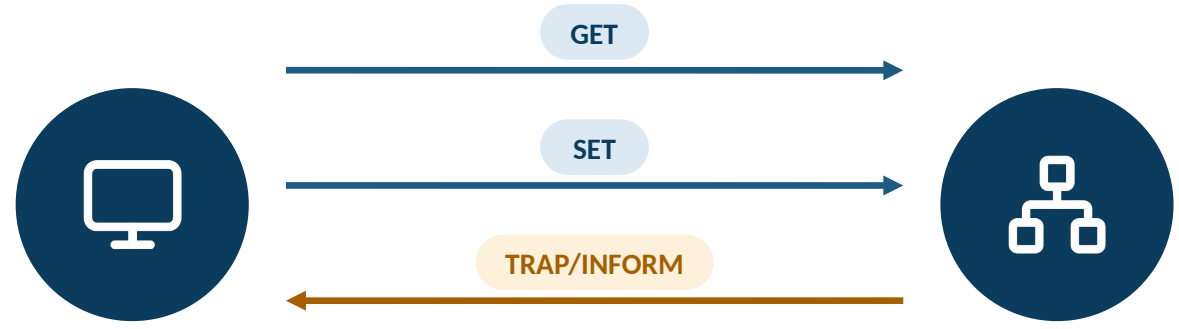
MODERN CLI

B23 SNMP (v3 and legacy v2c)

UPDATED

SNMP: manager, agent and notifications

The manager asks on UDP 161; the agent notifies the receiver on UDP 162



- Manager: command generator and notification receiver; agent: command responder
- GET, GETNEXT, GETBULK and SET requests require a response
- Trap: no confirmation. Inform: acknowledged by a response
- UDP 161 on the agent; UDP 162 on the notification receiver

`UDP 161 · UDP 162`

UPDATED

SNMP: v1, v2c and v3

Only version 3 authenticates the user and protects the content

- v1 and v2c: clear-text community, fundamentally insecure (RFC 3410)
- v3 = SNMPv2 plus security and administration: user, authentication, privacy

Version	Identification	Encryption	IETF status
SNMPv1	clear-text community	none	Historic (RFC 1157)
SNMPv2c	clear-text community	none	Historic (RFC 1901)
SNMPv3	user; MD5 or SHA (C3, 2007)	DES (C3, 2007)	Standard (STD 62)

UPDATED

SNMPv3: security levels and algorithms (USM)

Lab: authPriv with SHA and AES-128, where the device supports it

Level	Authentication	Privacy	Use
noAuthNoPriv	user name only	none	avoid
authNoPriv	user HMAC	none	integrity
authPriv	user HMAC	encrypted	lab
Authentication	MD5, SHA (RFC 3414)	SHA-2 (RFC 7860)	SHA-256 mandatory
Privacy	DES (RFC 3414)	AES-128 (RFC 3826)	C3: DES only

NEW

SNMP: MIB, OID and notifications

MIB, OID and views

- Each managed object has an OID, a numeric path in a tree
- 1.3.6.1.2.1.1.1 = sysDescr (iso, org, dod, internet, mgmt, mib-2, system)
- View = set of included or excluded subtrees
- A group gets read, write and notify views (VACM)

Notifications on the C3

- notify: name + tag + type (trap or inform)
- targetaddr: receiver IP, tag and target parameters
- targetparams: user, security model and level
- A trap is not confirmed; an inform is acknowledged by a response

LEGACY 2007

RMON: remote monitoring (2007 heritage)

- RFC 2819 (STD 59): MIB with 10 groups, all optional
- C3 (2007): statistics, history, alarm, event, filter, capture
- Alarm: compares a variable with thresholds and generates an event
- The management station reads the results over SNMP



LEGACY 2007

SNMPv3 on the Enterasys CLI (C3)

User, group, view, access and notification target

Historical platform (2007): SecureStack C3 (2007)

User nms: SHA and privacy (DES-56 on the C3)

```
set snmp user nms authentication sha <auth-pass>
privacy <priv-pass>
```

Read-only group with the user

```
set snmp group NMS-RO user nms security-model usm
```

View that includes the whole tree

```
set snmp view viewname NMS-VIEW subtree 1 included
```

Access: privacy level, read only

```
set snmp access NMS-RO security-model usm privacy
read NMS-VIEW
```

Target parameters (user and level)

```
set snmp targetparams NMSv3 user nms security-model usm
message-processing v3 privacy
```

Inform-type notification with the tag

```
set snmp notify NMS-NOTIFY tag NMS-TAG inform
```

Notification receiver

```
set snmp targetaddr NMS 192.0.2.60 param NMSv3
taglist NMS-TAG
```

Check users and groups

```
show snmp user
show snmp group
```

SNMP (v3 and legacy v2c) (1/2)

MODERN CLI

	SNMPv3 user (SHA + AES)	Show SNMP configuration

Extreme Switch Engine	`configure snmpv3 add user nms authentication sha <auth-pass> \`
	`privacy aes 128 <priv-pass>`
	`configure snmpv3 add group NMS-RO user nms`
		`show management`
		`show snmpv3 user`

Extreme Fabric Engine	`snmp-server user nms group NMS-RO sha aes`	`show snmp-server user`
		`show snmp-server group`
		`show snmp-server host`

Cisco IOS XE	`snmp-server user nms NMS-RO v3 auth sha <auth-pass> priv aes 128 <priv-pass>`	`show snmp user` `show snmp group`

Aruba AOS-CX	`snmpv3 user nms auth sha256 auth-pass plaintext <auth-pass> priv aes priv-pass`
	`snmpv3 user nms view RO-ALL`
		`show snmpv3 users`
		`show snmp trap`
		`show snmp system`

Standards: RFC 3411 · RFC 3414 · RFC 3415 · RFC 3826 · RFC 7860 · RFC 3410 · RFC 1901 · RFC 1157 · RFC 3417

SNMP (v3 and legacy v2c) (2/2)

MODERN CLI

	SNMPv3 user (SHA + AES)	Show SNMP configuration

Juniper Junos	`set snmp engine-id use-mac-address`
	`commit`
	`edit snmp v3 usm local-engine user nms`
	`  set authentication-sha256 authentication-password <auth-pass>`
	`  set privacy-aes128 privacy-password <priv-pass>`
	`top`
		`show snmp v3`

MikroTik RouterOS	`/snmp community add name=nms security=private \`
	`  authentication-protocol=SHA1 authentication-password=<auth-pass> \`
	`  encryption-protocol=AES encryption-password=<priv-pass>`
		`/snmp print`
		`/snmp community print`

Arista EOS	`snmp-server user nms NMS-RO v3 auth sha <auth-pass> priv aes <priv-pass>`	`show snmp user`
		`show snmp group`
		`show snmp notification host`

Huawei VRP	`snmp-agent usm-user v3 nms group NMS-RO`
	`snmp-agent usm-user v3 nms authentication-mode sha2-256`
	`snmp-agent usm-user v3 nms privacy-mode aes128`
		`display snmp-agent sys-info`
		`display snmp-agent usm-user`
		`display snmp-agent target-host`

Standards: RFC 3411 · RFC 3414 · RFC 3415 · RFC 3826 · RFC 7860 · RFC 3410 · RFC 1901 · RFC 1157 · RFC 3417

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`enable snmp access snmpv3`
	`configure snmpv3 add mib-view allMIB subtree 1.3`
	`configure snmpv3 add access NMS-RO sec-level priv read-view allMIB`

SNMPv3 user (SHA + AES)	`configure snmpv3 add user nms authentication sha <auth-pass> \`
	`privacy aes 128 <priv-pass>`
	`configure snmpv3 add group NMS-RO user nms`

v3 trap receiver	`configure snmpv3 add target-params P1 user nms mp-model snmpv3 \`
	`sec-model usm sec-level priv`
	`configure snmpv3 add target-addr A1 param P1 ipaddress 192.0.2.60 vr VR-Mgmt`

v2c RO community (legacy, avoid)	`configure snmp add community readonly <community>`
Location and contact	`configure snmp syslocation "Datacenter A, rack 12"` `configure snmp syscontact "noc@example.com"`
Show SNMP configuration	`show management` `show snmpv3 user`

SNMP is disabled by default (enable snmp access). v3 authentication is md5 or sha (no SHA-256 in the syntax); privacy aes 128. Default communities public/private exist: delete them.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`snmp-server view NMS-ALL 1.3.6.1` `snmp-server group NMS-RO "" auth-priv notify-view NMS-ALL read-view NMS-ALL`
SNMPv3 user (SHA + AES)	`snmp-server user nms group NMS-RO sha aes`
v3 trap receiver	`snmp-server host 192.0.2.60 v3 authPriv nms`
v2c RO community (legacy, avoid)	`snmp-server community <community> index v2ro secname readview`
Location and contact	`snmp-server location Lab-Rack-01` `snmp-server contact noc@example.com`

Show SNMP configuration	`show snmp-server user`
	`show snmp-server group`
	`show snmp-server host`

<auth-pass> and <priv-pass> are typed at prompts, not in the command. The guide lists only MD5 and SHA-1 (no SHA-256). The group uses one view for notify-view and read-view. Secname 'readview' is the default RO one.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`snmp-server view ALL-VIEW iso included` `snmp-server group NMS-RO v3 priv read ALL-VIEW`
SNMPv3 user (SHA + AES)	`snmp-server user nms NMS-RO v3 auth sha <auth-pass> priv aes 128 <priv-pass>`
v3 trap receiver	`snmp-server host 192.0.2.60 traps version 3 priv nms` `snmp-server enable traps`

v2c RO community (legacy, avoid)	`snmp-server community <community> RO`
	`! IOS XE 26.1.x:`
	`system mode insecure`
	`snmp-server community <community> RO`

Location and contact	`snmp-server location Datacenter-SP` `snmp-server contact noc@example.com`
Show SNMP configuration	`show snmp user` `show snmp group`

Use SNMPv3 priv (SHA + AES). In the 17.18 guide, MD5, DES and 3DES require disabling the compliance shield. v1 and v2c are the least secure models: keep the community string as legacy only.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`snmp-server view RO-ALL 1.3.6.1 included`
SNMPv3 user (SHA + AES)	`snmpv3 user nms auth sha256 auth-pass plaintext <auth-pass> priv aes priv-pass` `snmpv3 user nms view RO-ALL`
v3 trap receiver	`snmp-server host 192.0.2.60 trap version v3 user nms vrf mgmt`
v2c RO community (legacy, avoid)	`snmp-server community <community>` ` access-level ro`
Location and contact	`snmp-server system-location DC1` `snmp-server system-contact noc`

Show SNMP configuration	`show snmpv3 users`
	`show snmp trap`
	`show snmp system`

No SNMPv3 group: a view is bound to the user instead. The SNMP agent listens on no VRF until 'snmp-server vrf mgmt'. Creating a community deletes the default 'public'.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`set snmp view ALL oid .1 include`
	`edit snmp v3 vacm`
	`  set security-to-group security-model usm security-name nms group NMS-RO`
	`  edit access group NMS-RO default-context-prefix security-model usm`
	`   set security-level privacy read-view ALL`
	`   set security-level privacy notify-view ALL`
	`top`

SNMPv3 user (SHA + AES)	`set snmp engine-id use-mac-address`
	`commit`
	`edit snmp v3 usm local-engine user nms`
	`  set authentication-sha256 authentication-password <auth-pass>`
	`  set privacy-aes128 privacy-password <priv-pass>`
	`top`

Set and commit the engine ID before creating USM users: the keys depend on it. The group needs a notify-view for traps. A row that uses edit ends with 'top'. The v2c community is legacy.

MODERN CLI

SNMP (v3 and legacy v2c) (cont.)

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy SNMP slide

v3 trap receiver	`edit snmp v3`
	` set notify N1 type trap tag TRAP`
	` set target-address NMS address 192.0.2.60 tag-list TRAP target-parameters TP`
	` edit target-parameters TP parameters`
	`  set message-processing-model v3 security-model usm security-level privacy`
	`  set security-name nms`
	`top`

v2c RO community (legacy, avoid)	`set snmp community <community> authorization read-only`
Location and contact	`set snmp location "<location>"` `set snmp contact "<contact>"`
Show SNMP configuration	`show snmp v3`

Set and commit the engine ID before creating USM users: the keys depend on it. The group needs a notify-view for traps. A row that uses edit ends with 'top'. The v2c community is legacy.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	*n/a: No group/view: access is read-access/write-access on the community*

SNMPv3 user (SHA + AES)	`/snmp community add name=nms security=private \`
	` authentication-protocol=SHA1 authentication-password=<auth-pass> \`
	` encryption-protocol=AES encryption-password=<priv-pass>`

v3 trap receiver	`/snmp set trap-target=192.0.2.60 trap-version=3 trap-community=nms`

v2c RO community (legacy, avoid)	`/snmp community add name=<community> addresses=192.0.2.60/32 \`
	` security=none`
	`/snmp community remove [find name=public]`

Location and contact	`/snmp set enabled=yes contact=noc@example.com location="DC-SP rack 12"`
Show SNMP configuration	`/snmp print` `/snmp community print`

In RouterOS the SNMPv3 user is a community with security=private (no group/view). Only MD5/SHA1 and DES/AES exist, no SHA-256. The default 'public' community is insecure; the v3 trap uses the trap-community.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`snmp-server view ALL iso include` `snmp-server group NMS-RO v3 priv read ALL`
SNMPv3 user (SHA + AES)	`snmp-server user nms NMS-RO v3 auth sha <auth-pass> priv aes <priv-pass>`
v3 trap receiver	`snmp-server host 192.0.2.60 version 3 priv nms`
v2c RO community (legacy, avoid)	`snmp-server community <community> ro`
Location and contact	`snmp-server location DC-SP-01` `snmp-server contact netadmin`

Show SNMP configuration	`show snmp user`
	`show snmp group`
	`show snmp notification host`

Documented authentication: md5 or sha; encryption: aes (128) or des. SHA-256 and AES-192/256 do not appear in the manual. The v2c community is optional; 'ro' (read-only) is the default.

MODERN CLI

SNMP (v3 and legacy v2c)

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy SNMP slide

Read-only SNMPv3 group/view	`snmp-agent mib-view included NMS-VIEW 1.3.6.1` `snmp-agent group v3 NMS-RO privacy read-view NMS-VIEW notify-view NMS-VIEW`

SNMPv3 user (SHA + AES)	`snmp-agent usm-user v3 nms group NMS-RO`
	`snmp-agent usm-user v3 nms authentication-mode sha2-256`
	`snmp-agent usm-user v3 nms privacy-mode aes128`

v3 trap receiver	`snmp-agent trap enable` `snmp-agent target-host trap address udp-domain 192.0.2.60 params securityname nms v3 privacy`
v2c RO community (legacy, avoid)	`snmp-agent sys-info version v2c v3` `snmp-agent community read <community>`
Location and contact	`snmp-agent sys-info location Lab Rack 01` `snmp-agent sys-info contact noc@example.com`

Show SNMP configuration	`display snmp-agent sys-info`
	`display snmp-agent usm-user`
	`display snmp-agent target-host`

The usm-user passwords are typed at the prompt, not on the command line. The agent defaults to SNMPv3 only; v2c needs sys-info version, and the community must pass a complexity check.

4.6

AAA: RADIUS and TACACS+



Who you are, what you may do and what you did

RADIUS: the path of a management login

RADIUS or TACACS+?

RADIUS on the Enterasys CLI (C3)

Chapter 4.6 agenda

1. RADIUS: the path of a management login
2. RADIUS or TACACS+?
3. RADIUS on the Enterasys CLI (C3)

MODERN CLI

B24 AAA: RADIUS and TACACS+

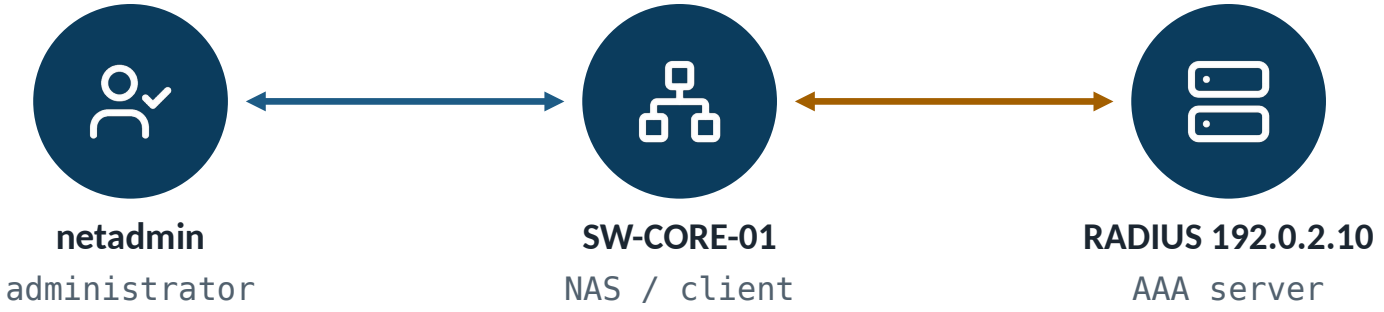
Lab

- Lab: syslog, SNMPv3 and RADIUS

NEW

RADIUS: the path of a management login

The switch is the client (NAS); the server decides and the switch obeys



1. netadmin opens SSH and enters user name and password
2. The switch sends an Access-Request (password hidden)
3. The server answers Access-Accept or Access-Reject
4. Accepted: access is granted at the level indicated

- The switch (NAS) is the client; the server keeps users and rules
- The shared secret is never sent over the network
- Without a reply, the client re-sends or uses the alternate server
- Authentication on UDP 1812; accounting on UDP 1813

NEW

RADIUS or TACACS+?

Both do AAA; the differences are in transport and in what is protected

Feature	RADIUS	TACACS+
Transport and port	UDP 1812 (authentication), 1813 (accounting)	TCP 49
What is protected	user password, hidden with MD5	packet body obfuscated; not encryption
AAA model	authentication, authorization and configuration together	authentication, authorization and accounting separated
Use case in the RFCs	network access server (NAS) authenticating links and users	device administration, with per-command authorization and accounting
Accounting	RFC 2866: Start, Stop, Interim-Update	Start, Stop, Update
TLS channel	RFC 6614 and RFC 9765 (experimental)	RFC 9887, TLS 1.3, TCP 300

LEGACY 2007

RADIUS on the Enterasys CLI (C3)

The management-access realm forces every management login through the server

Historical platform (2007): SecureStack C3 (2007)

Register the server (index, IP, port, secret)	`set radius server 1 192.0.2.10 1812 <secret>`
Timeout of 5 seconds	`set radius timeout 5`
Enable the RADIUS client	`set radius enable`
Check the status and the servers	`show radius`
Management goes through the server (disables local admin)	`set radius realm management-access all`
Accounting server (prompts for the secret)	`set radius accounting server 192.0.2.10 1813`
Enable accounting	`set radius accounting enable`
Check accounting	`show radius accounting`

MODERN CLI

AAA: RADIUS and TACACS+ (1/2)

	RADIUS server	AAA login with local fallback

Extreme Switch Engine	`configure radius mgmt-access primary server 192.0.2.10 client-ip 192.0.2.11`
	`configure radius mgmt-access primary shared-secret "<secret>"`
	`enable radius mgmt-access`
		n/a

Extreme Fabric Engine	`radius server host 192.0.2.10 key <secret> used-by cli enable`	`radius enable`
		`tacacs protocol enable`
		`tacacs authentication cli`

Cisco IOS XE	`radius server RAD-1`
	` address ipv4 192.0.2.10 auth-port 1812 acct-port 1813`
	` key <secret>`
		`aaa new-model`
		`aaa authentication login default group tacacs+ local`

Aruba AOS-CX	`radius-server host 192.0.2.10 key plaintext <secret> vrf mgmt`	`aaa authentication login default group radius local`

Standards: RFC 2865 · RFC 2866 · RFC 6614 · RFC 9765 · RFC 8907 · RFC 9887 · IANA-PORTS

AAA: RADIUS and TACACS+ (2/2)

MODERN CLI

	RADIUS server	AAA login with local fallback
Juniper Junos	`set system radius-server 192.0.2.10 port 1812 secret <secret>`	`set system authentication-order [ radius tacplus password ]` `set system login user remote class read-only`
MikroTik RouterOS	`/radius add address=192.0.2.10 service=login secret=<secret>`	`/user aaa set use-radius=yes`
Arista EOS	`radius-server host 192.0.2.10 key <secret>`	`aaa authentication login default group radius local`

Huawei VRP	`radius-server template RAD-1`
	` radius-server shared-key cipher <secret>`
	` radius-server authentication 192.0.2.10 1812`
		`aaa`
		` authentication-scheme default`
		`  authentication-mode radius local`

Standards: RFC 2865 · RFC 2866 · RFC 6614 · RFC 9765 · RFC 8907 · RFC 9887 · IANA-PORTS

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`configure radius mgmt-access primary server 192.0.2.10 client-ip 192.0.2.11`
	`configure radius mgmt-access primary shared-secret "<secret>"`
	`enable radius mgmt-access`

TACACS+ server	`configure tacacs primary server 192.0.2.12 client-ip 192.0.2.11`
	`configure tacacs primary shared-secret <secret>`
	`enable tacacs`

AAA login with local fallback	*n/a: No command: local fallback is automatic*

Command accounting	`configure tacacs-accounting primary server 192.0.2.12 client-ip 192.0.2.11`
	`configure tacacs-accounting primary shared-secret <secret>`
	`enable tacacs-accounting`

Test / show AAA	`show radius mgmt-access` `show tacacs`

Client uses VR-Mgmt by default. RADIUS and TACACS+ cannot both run for management. The local account works only if servers do not answer; a reject is final. Audit: enable cli config-logging.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`radius server host 192.0.2.10 key <secret> used-by cli enable`
TACACS+ server	`tacacs server host 192.0.2.12 key <secret>`

AAA login with local fallback	`radius enable`
	`tacacs protocol enable`
	`tacacs authentication cli`

Command accounting	`radius accounting enable` `radius accounting include-cli-commands`

Test / show AAA	`show radius-server`
	`show radius reachability`
	`show tacacs`

RADIUS and TACACS+ fall back to the local user if no server answers. Enable one only: precedence between the two is not documented. Command accounting is RADIUS; TACACS+ logs only sessions.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`radius server RAD-1`
	` address ipv4 192.0.2.10 auth-port 1812 acct-port 1813`
	` key <secret>`

TACACS+ server	`tacacs server TAC-1`
	` address ipv4 192.0.2.12`
	` key <secret>`

AAA login with local fallback	`aaa new-model` `aaa authentication login default group tacacs+ local`
Command accounting	`aaa accounting commands 15 default start-stop group tacacs+`

Test / show AAA	`test aaa group radius netadmin <password> new-code`
	`show aaa servers`
	`! IOS XE 26.1.x: 26.x shows test aaa group SG1 user password new-code (named group); the radius group form is not shown. 17.18 line`

The next method (local) is used only if the server fails to respond, not if it rejects the login. The local method needs a user in the local database: create it before enabling aaa new-model.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`radius-server host 192.0.2.10 key plaintext <secret> vrf mgmt`
TACACS+ server	`tacacs-server host 192.0.2.12 key plaintext <secret> vrf mgmt`
AAA login with local fallback	`aaa authentication login default group radius local`
Command accounting	`aaa accounting all-mgmt default start-stop group tacacs`

Test / show AAA	`show aaa authentication`
	`show radius-server`
	`show tacacs-server`

Key and VRF are set per server ('radius-server host', 'tacacs-server host'). 'local' last in the group list is the fallback. With TACACS+ the accounting includes the commands. 'aaa troubleshoot test-server' tests RADIUS.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`set system radius-server 192.0.2.10 port 1812 secret <secret>`
TACACS+ server	`set system tacplus-server 192.0.2.12 secret <secret>`
AAA login with local fallback	`set system authentication-order [ radius tacplus password ]` `set system login user remote class read-only`
Command accounting	`set system accounting events interactive-commands` `set system accounting destination tacplus`
Test / show AAA	`show configuration system authentication-order` `show configuration system radius-server`

There are two radius-server hierarchies: system (admin login) and access (802.1X). A remote user without a local account needs the 'remote' template or login fails. 'password' in the order is the local fallback.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`/radius add address=192.0.2.10 service=login secret=<secret>`
TACACS+ server	*n/a: No TACACS+ in the docs; remote login AAA uses RADIUS only*
AAA login with local fallback	`/user aaa set use-radius=yes`
Command accounting	*n/a: RADIUS accounting covers login/logout only, not commands*

Test / show AAA	`/user aaa print`
	`/radius monitor 0`
	`/user active print`

The local database is checked first and RADIUS only afterwards, if the user is not found locally. Remote login = /radius service=login + /user aaa use-radius=yes. The docs describe RADIUS only, not TACACS+.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`radius-server host 192.0.2.10 key <secret>`
TACACS+ server	`tacacs-server host 192.0.2.12 key <secret>`
AAA login with local fallback	`aaa authentication login default group radius local`
Command accounting	`aaa accounting commands all default start-stop group tacacs+`
Test / show AAA	`show aaa session`

The local fallback follows 'group radius' in the same list. EOS supports TACACS+, RADIUS and LDAP. Built-in roles: network-admin and network-operator. There is also 'show aaa authentication lockout'.

MODERN CLI

AAA: RADIUS and TACACS+

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy RADIUS slide

RADIUS server	`radius-server template RAD-1`
	` radius-server shared-key cipher <secret>`
	` radius-server authentication 192.0.2.10 1812`

TACACS+ server	`hwtacacs-server template TAC-1`
	` hwtacacs-server authentication 192.0.2.12 49`
	` hwtacacs-server shared-key cipher <secret>`

AAA login with local fallback	`aaa`
	` authentication-scheme default`
	`  authentication-mode radius local`

Command accounting	`aaa`
	` recording-scheme REC-1`
	`  recording-mode hwtacacs TAC-1`

Test / show AAA	`test-aaa netadmin <password> radius-template RAD-1`
	`display radius-server configuration template RAD-1`
	`display hwtacacs-server template TAC-1`

HWTACACS is compatible with Cisco's TACACS+ (proprietary attributes may differ). The default_admin domain has no RADIUS template by default: apply radius-server. Command logging: HWTACACS only.

LAB

Lab: syslog, SNMPv3 and RADIUS

1. Synchronize with NTP 192.0.2.123; time zone America/Sao_Paulo (UTC-03:00, no daylight saving).
2. Send syslog to 192.0.2.50 at the informational level; generate a login and watch the message arrive.
3. Create the group NMS-RO with a read-only view of the whole tree.
4. Create the SNMPv3 user nms (SHA and AES-128, or the best the platform has) in group NMS-RO.
5. Point notifications to 192.0.2.60 with user nms; bring a link down and watch the alert.
6. Query sysDescr (1.3.6.1.2.1.1.1) over SNMPv3 authPriv from the manager.
7. Register RADIUS 192.0.2.10 with the secret [secret] and test it with a test user.
8. With a session already open, require RADIUS for management access and open a new session.
9. On the syslog server, check the login, the link event and consistent times.

Goal

Give SW-CORE-01 correct time, central logging, SNMPv3 monitoring and a management login validated by RADIUS.

Topology

SW-CORE-01 (192.0.2.11) with NTP 192.0.2.123, syslog 192.0.2.50, SNMP receiver 192.0.2.60 and RADIUS 192.0.2.10.

Verification

The server log shows the login and the link going down with the right time.
The SNMPv3 query answers with the right credentials and fails with a wrong password.
The new management login only succeeds with a user from the RADIUS server.

4.7

Multicast: IGMP, snooping and PIM



One stream, many receivers, without flooding the network

- IPv4 multicast: the address blocks
- IGMP: the three versions
- IGMP snooping: deliver only to those who asked
- PIM-SM: the delivery tree and the rendezvous point (RP)
- DVMRP: flood-and-prune multicast (legacy)
- Multicast on the Enterasys CLI (C3)

Chapter 4.7 agenda

1. IPv4 multicast: the address blocks
2. IGMP: the three versions
3. IGMP snooping: deliver only to those who asked
4. PIM-SM: the delivery tree and the rendezvous point (RP)
5. DVMRP: flood-and-prune multicast (legacy)
6. Multicast on the Enterasys CLI (C3)

MODERN CLI

B25 Multicast: IGMP snooping and PIM

Lab

- Lab: multicast with IGMP snooping and PIM-SM

NEW

IPv4 multicast: the address blocks

Class D: 224.0.0.0 to 239.255.255.255 (prefix 1110, or 224.0.0.0/4)

- Outside the control blocks: 232/8 (SSM) and 239/8 (administrative scope)
- Ethernet uses 23 of the 28 group bits: different groups can share a MAC

Block	Use	Note
224.0.0.0/4	the whole IPv4 multicast space	class D (prefix 1110)
224.0.0.0/24	local link control; not forwarded off link	224.0.0.1 = all hosts
224.0.1.0/24	internetwork control	224.0.1.1 = NTP
232.0.0.0/8	SSM: only from the sources asked for	RFC 4607
233.252.0.0/24	MCAST-TEST-NET: documentation	never on the public Internet
239.0.0.0/8	administrative scope: local use in the domain	lab: 239.1.1.1

UPDATED

IGMP: the three versions

The receiver tells the local router; the router elects a querier

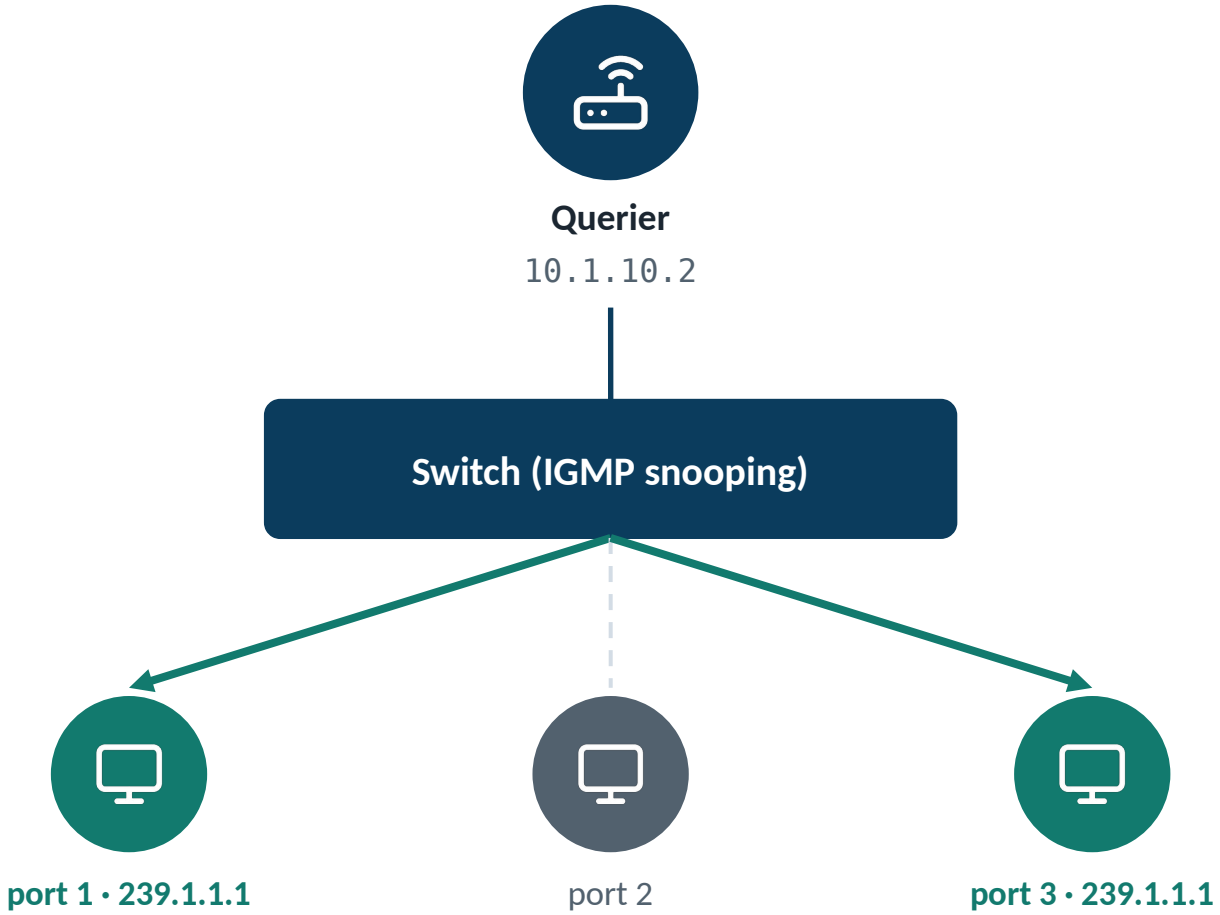
- Querier: the router with the lowest IP asks; the others stay quiet (v2)
- v3 router with a v2-only snooping switch: groups flooded or cut off (RFC 4541)

Version	What it adds	Standard
IGMPv1	Query to 224.0.0.1 and Reports from hosts	RFC 1112, Appendix I
IGMPv2	Leave Group, querier election, group-specific Query	RFC 2236
IGMPv3	source filtering (include and exclude); Reports to 224.0.0.22	RFC 9776 (STD 100)

UPDATED

IGMP snooping: deliver only to those who asked

The switch listens to IGMP and builds the group table per port

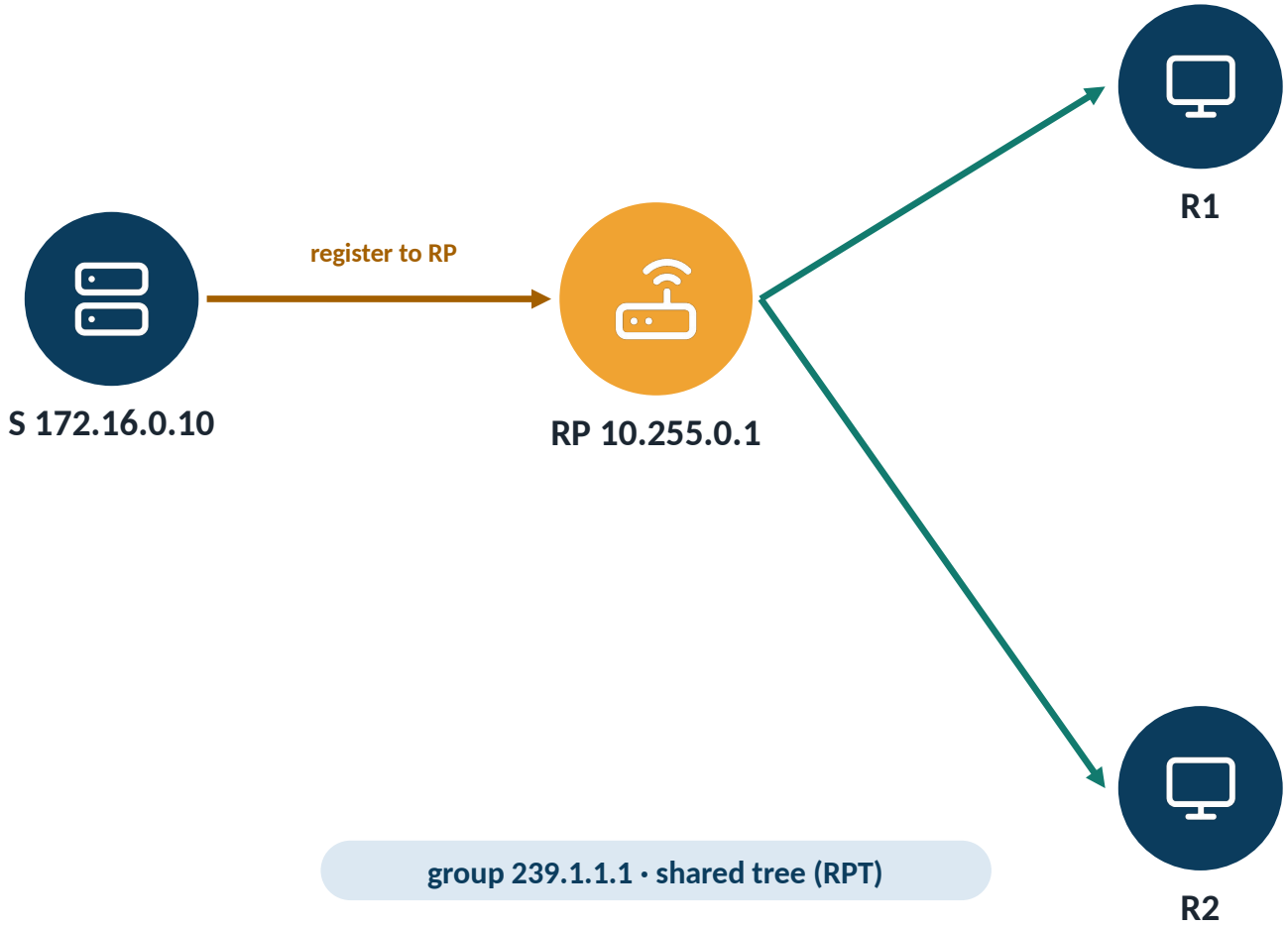


- Without snooping, multicast floods every port of the VLAN
- Reports go only to router ports; data goes to member and router ports
- 224.0.0.x is always sent to all ports
- The group entry expires: it does not depend only on Leave

NEW

PIM-SM: the delivery tree and the rendezvous point (RP)

Receivers join the RP tree; they may later move to the shortest path



- Receiver: IGMP Join; the DR sends a (*,G) Join to the RP
- Source: the DR encapsulates it in Register and sends it to the RP
- Optional: move to the shortest-path tree (S,G)
- RP by static configuration, bootstrap or embedded-RP

LEGACY 2007

DVMRP: flood-and-prune multicast (legacy)

DVMRP on the C3 (2007)

- Floods out of all paths except the one back to the source (RPF)
- No members: the router sends a prune up the tree
- Refloods periodically to reach new receivers
- Needs an advanced license and IGMP on every VLAN running DVMRP

Why PIM-SM replaced it

- RFC 1075: experimental protocol, implementation not recommended
- C3: DVMRP suffices with many members and plentiful bandwidth
- PIM-SM suits sparse members and limited bandwidth
- PIM-SM uses the unicast routing table (or its own)

LEGACY 2007

Multicast on the Enterasys CLI (C3)

Snooping in switch mode; IGMP and PIM-SM in router mode

Historical platform (2007): SecureStack C3 (2007)

Snooping: enable globally (switch mode)

```
set igmpsnooping adminmode enable
```

Snooping: enable on ports

```
set igmpsnooping interfacemode ge.1.1-24 enable
```

Show the multicast forwarding table

```
show igmpsnooping mfdb
```

IGMP on the router and on the VLAN 10 interface

```
ip igmp
interface vlan 10
ip igmp enable
```

Show the groups and client ports

```
show ip igmp groups
```

Global PIM-SM and static RP for all multicast

```
ip pimsm
ip pimsm staticrp 10.255.0.1 224.0.0.0 240.0.0.0
```

PIM-SM on the VLAN 10 interface

```
interface vlan 10
ip pimsm enable
```

Show the PIM neighbors

```
show ip pimsm neighbor
```

MODERN CLI

Multicast: IGMP snooping and PIM (1/2)

	IGMP snooping on VLAN 10	PIM-SM on the L3 interface

Extreme Switch Engine	`enable igmp snooping vlan USERS`	`enable ipmcforwarding`
		`configure pim add vlan USERS sparse`
		`configure pim add vlan UPLINK sparse`

Extreme Fabric Engine	`interface vlan 10`
	` ip igmp snooping`
		`ip pim enable`
		`interface vlan 900`
		` ip pim enable`

Cisco IOS XE	`ip igmp snooping vlan 10`	`ip multicast-routing`
		`interface vlan 10`
		` ip pim sparse-mode`

Aruba AOS-CX	`vlan 10`
	` ip igmp snooping enable`
		`interface vlan 10`
		` ip igmp enable`
		` ip pim-sparse enable`

Standards: RFC 9776 · RFC 3376 obsoleted · RFC 9777 · RFC 4541 · RFC 7761 · IANA-PROTO · IANA-MCAST

MODERN CLI

Multicast: IGMP snooping and PIM (2/2)

	IGMP snooping on VLAN 10	PIM-SM on the L3 interface
Juniper Junos	`set protocols igmp-snooping vlan USERS`	`set protocols pim interface irb.10 mode sparse` `set protocols pim interface ge-0/0/21.0 mode sparse`

MikroTik RouterOS	`/interface bridge set bridge1 igmp-snooping=yes`	`/routing pimsm instance add name=pim1`
		`/routing pimsm interface-template add instance=pim1 interfaces=vlan10,vlan900`
		`/routing pimsm igmp-interface-template add instance=pim1 interfaces=vlan10`

Arista EOS	`ip igmp snooping vlan 10`	`interface vlan 10` `  pim ipv4 sparse-mode`

Huawei VRP	`igmp-snooping enable`
	`igmp-snooping enable vlan 10`
		`multicast routing-enable`
		`interface Vlanif10`
		`  pim sm`

Standards: RFC 9776 · RFC 3376 obsoleted · RFC 9777 · RFC 4541 · RFC 7761 · IANA-PROTO · IANA-MCAST

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`enable igmp snooping vlan USERS`
IGMP querier	`enable igmp vlan USERS`
Show learned groups	`show igmp snooping vlan USERS` `show igmp group vlan USERS`

PIM-SM on the L3 interface	`enable ipmcforwarding`
	`configure pim add vlan USERS sparse`
	`configure pim add vlan UPLINK sparse`

Static RP 10.255.0.1	`enable pim` `configure pim crp static 10.255.0.1 rp-list`
Show PIM neighbors / mroutes	`show pim neighbor` `show mcast cache`

Snooping is on by default but needs a querier: the SVI (primary IP) sends the queries. PIM needs enable ipmcforwarding; the static RP uses a policy file (nlri 224.0.0.0/4).

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`interface vlan 10` ` ip igmp snooping`

IGMP querier	`interface vlan 10`
	` ip igmp snoop-querier`
	` ip igmp snoop-querier-addr 10.1.10.2`

Show learned groups	`show ip igmp group` `show ip igmp snooping`

PIM-SM on the L3 interface	`ip pim enable`
	`interface vlan 900`
	` ip pim enable`

Static RP 10.255.0.1	`ip pim static-rp` `ip pim static-rp 224.0.0.0/240.0.0.0 10.255.0.1`
Show PIM neighbors / mroutes	`show ip pim neighbor` `show ip pim mroute terse`

Guide: do not enable PIM on IGMP snooping interfaces; PIM only on uplink VLAN 900. PIM needs 'no boot config flags spbm-config-mode' (default on), save, reset. Add 'ip pim' on loopback 1 (RP); static-rp asks y.

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`ip igmp snooping vlan 10`
IGMP querier	`ip igmp snooping vlan 10 querier` `ip igmp snooping vlan 10 querier address 10.1.10.2`
Show learned groups	`show ip igmp snooping groups vlan 10`

PIM-SM on the L3 interface	`ip multicast-routing`
	`interface vlan 10`
	` ip pim sparse-mode`

Static RP 10.255.0.1	`ip pim rp-address 10.255.0.1`
Show PIM neighbors / mroutes	`show ip pim neighbor` `show ip mroute`

With no RP configured, the group is treated as dense. Configure the RP on every device and repeat ip pim sparse-mode on the uplink. The querier disables itself if it sees IGMP from a multicast router.

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`vlan 10` ` ip igmp snooping enable`
IGMP querier	`interface vlan 10` ` ip igmp querier`
Show learned groups	`show ip igmp snooping groups vlan 10` `show ip igmp groups`

PIM-SM on the L3 interface	`interface vlan 10`
	` ip igmp enable`
	` ip pim-sparse enable`

Static RP 10.255.0.1	`router pim`
	` enable`
	` rp-address 10.255.0.1`

Show PIM neighbors / mroutes	`show ip pim neighbor` `show ip mroute`

PIM needs a global 'enable' under 'router pim' and 'ip pim-sparse enable' on the interface. The same static RP must be configured on every PIM-SM router in the domain.

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`set protocols igmp-snooping vlan USERS`
IGMP querier	`set protocols igmp interface irb.10`
Show learned groups	`show igmp snooping membership vlan USERS`
PIM-SM on the L3 interface	`set protocols pim interface irb.10 mode sparse` `set protocols pim interface ge-0/0/21.0 mode sparse`
Static RP 10.255.0.1	`set protocols pim rp static address 10.255.0.1`
Show PIM neighbors / mroutes	`show pim neighbors` `show multicast route`

IGMP snooping is on for all VLANs by default (EX docs). IGMP starts on PIM interfaces; the querier uses the IRB address. PIM-SM is listed for EX4100 (22.2R1) and EX4400 (21.1R1). On the RP use 'rp local address'.

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`/interface bridge set bridge1 igmp-snooping=yes`
IGMP querier	`/interface bridge set bridge1 multicast-querier=yes`
Show learned groups	`/interface bridge mdb print`

PIM-SM on the L3 interface	`/routing pimsm instance add name=pim1`
	`/routing pimsm interface-template add instance=pim1 interfaces=vlan10,vlan900`
	`/routing pimsm igmp-interface-template add instance=pim1 interfaces=vlan10`

Static RP 10.255.0.1	`/routing pimsm static-rp add address=10.255.0.1 instance=pim1`
Show PIM neighbors / mroutes	`/routing pimsm neighbor print` `/routing pimsm uib-sg print`

The bridge querier sends untagged queries only: source 0.0.0.0 up to 7.23 and the bridge's own IP since 7.24 (querier-uses-bridge-address). On VLAN 10, PIM-SM igmp-interface-template acts as the IGMP querier.

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`ip igmp snooping vlan 10`
IGMP querier	`ip igmp snooping vlan 10 querier` `ip igmp snooping vlan 10 querier address 10.1.10.2`
Show learned groups	`show ip igmp snooping groups`
PIM-SM on the L3 interface	`interface vlan 10` `  pim ipv4 sparse-mode`

Static RP 10.255.0.1	`router pim sparse-mode`
	`  ipv4`
	`   rp address 10.255.0.1`

Show PIM neighbors / mroutes	`show ip pim neighbor` `show ip mroute 239.1.1.1`

Snooping is on by default. Before PIM, enable multicast routing: 'router multicast', 'ipv4', 'routing'. PIM is enabled per interface with 'pim ipv4 sparse-mode'; the static RP uses 'rp address'.

MODERN CLI

Multicast: IGMP snooping and PIM

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 legacy IGMP slide

IGMP snooping on VLAN 10	`igmp-snooping enable` `igmp-snooping enable vlan 10`

IGMP querier	`igmp-snooping send-query source-address 10.1.10.2`
	`vlan 10`
	`  igmp-snooping querier enable`

Show learned groups	`display igmp-snooping port-info vlan 10`

PIM-SM on the L3 interface	`multicast routing-enable`
	`interface Vlanif10`
	`  pim sm`

Static RP 10.255.0.1	`pim` `  static-rp 10.255.0.1`
Show PIM neighbors / mroutes	`display pim neighbor` `display pim routing-table`

Snooping must be enabled globally before per VLAN. The query source address is global (system view); the querier is per VLAN. PIM needs multicast routing-enable before pim sm and static-rp.

LAB

Lab: multicast with IGMP snooping and PIM-SM

1. Enable IGMP snooping on VLAN 10 and on the receiver ports.
2. Configure the VLAN 10 querier with source 10.1.10.2, where the platform allows.
3. Join group 239.1.1.1 with R1 and see the group and port in the snooping table.
4. Join with R2 and check both ports; then make R1 leave and see its port leave the table.
5. Enable PIM-SM on the VLAN 10 interface and on the uplink (IGMP already active).
6. Configure the static RP 10.255.0.1 for 224.0.0.0/4 on all PIM routers.
7. Send the stream from the source to group 239.1.1.1 and check PIM neighbors and multicast routes.
8. Turn snooping off on purpose and watch the stream flood the ports without receivers. Turn it back on.

Goal

Deliver group 239.1.1.1 only to the receivers on VLAN 10, with IGMP snooping and PIM-SM with a static RP.

Topology

Source behind the uplink (172.16.0.10); SW-CORE-01 is the RP (10.255.0.1) and the VLAN 10 querier (10.1.10.2); receivers R1 and R2 on VLAN 10.

Verification

With snooping, only the receiver ports (and the router port) get the group.
The PIM neighbor is up and RP 10.255.0.1 shows on the routers.
Without snooping, the ports without receivers also receive the stream.

4.8

Troubleshooting and support



From symptom to cause, and from cause to a well-opened case

- Troubleshooting: climb the layers, bottom-up
- Troubleshooting: the six-step method
- Common symptoms and where to start
- Before opening a case: what to collect
- Data collection on the Enterasys CLI
- Final knowledge assessment

Chapter 4.8 agenda

1. Troubleshooting: climb the layers, bottom-up
2. Troubleshooting: the six-step method
3. Common symptoms and where to start
4. Before opening a case: what to collect
5. Data collection on the Enterasys CLI
6. Final knowledge assessment
7. Satisfaction survey

MODERN CLI

B26 Troubleshooting toolkit

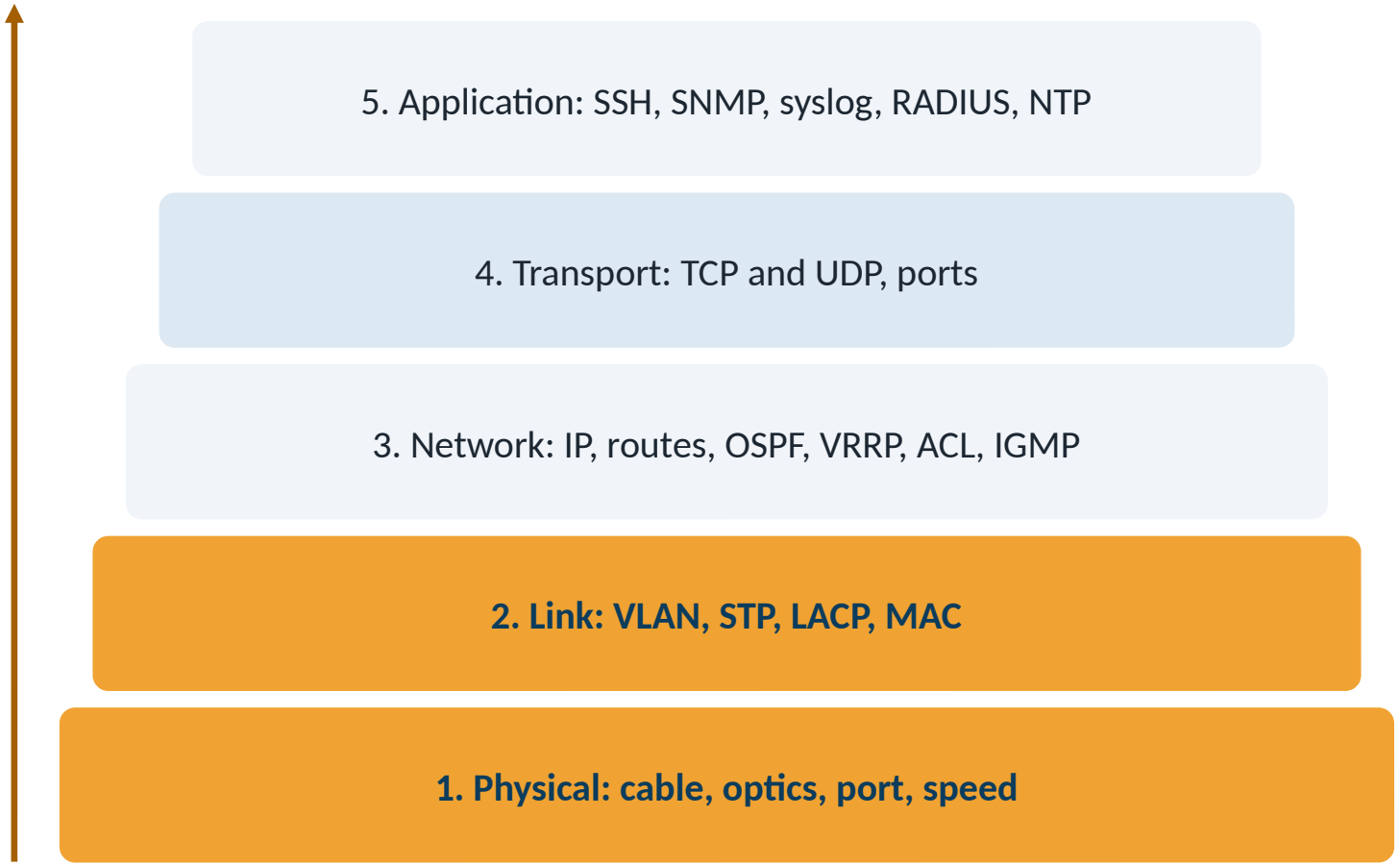
Lab

- Final lab A: redundant core
- Final lab B: protection and management

NEW

Troubleshooting: climb the layers, bottom-up

Each layer depends on the one below: there is no point checking the route if the port has no link

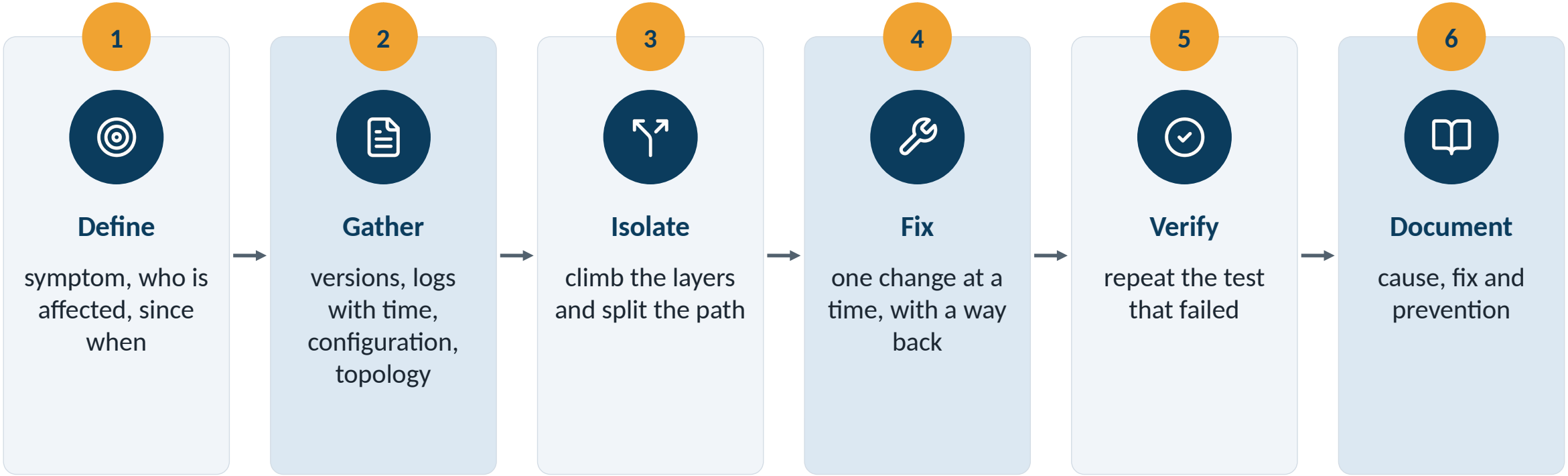


- Start at the base: port with link, correct VLAN and STP
- OSPF (IP protocol 89) and VRRP (112) travel over IP
- Syslog, SNMP and RADIUS use UDP
- IEEE 802.3 defines the physical layer and the MAC

NEW

Troubleshooting: the six-step method

The course's working method, not a standard: repeat it for every incident



NEW

Common symptoms and where to start

Each row links a symptom to a topic of this day

Symptom	Layer	Likely cause	Start with
OSPF neighbor does not reach Full	Network	different key or Hello/dead (on a LAN, 2-Way between non-DR is normal)	neighbors, authentication and timers
Two VRRP routers active	Network	the Backup takes over without hearing advertisements; VRRP has no authentication	advertisements between the two and the VLAN path
SSH denied or allowed too much	Network	rule order: the first match decides	the order and the hits of each rule
Logs do not reach the server	Application	UDP does not acknowledge; silent loss	local log, route and port to the server
Wrong time or no zone in logs	Application	clock without NTP; BSD format without year or zone	NTP, time zone and message format
SNMPv3 not answering	Application	different user, password or level: USM drops and counts	USM counters: digest, decryption, user
RADIUS login times out	Application	client without a secret on the server: request silently discarded	client entry and secret on the server
Multicast receiver without stream	Link and network	no querier; v2 snooping with a v3 router; unknown RP	querier, groups, IGMP version, PIM neighbors



SIDE-BY-SIDE VIEW · BLOCK B26

Troubleshooting toolkit

MODERN CLI

	Collect tech-support	LLDP neighbors
Extreme Switch Engine	`show tech-support logto file` `upload debug 192.0.2.20 vr VR-Mgmt`	`show lldp neighbors`
Extreme Fabric Engine	`terminal more disable` `show tech`	`show lldp neighbor`
Cisco IOS XE	`show tech-support > flash:tech.txt`	`show lldp neighbors detail`

Aruba AOS-CX	`show tech local-file`
	`copy show-tech local-file`
	`sftp://backup@192.0.2.20/tech.txt vrf mgmt`
		`show lldp neighbor-info`

Juniper Junos	`request support information | save` `/var/tmp/rsi.txt`	`show lldp neighbors`
MikroTik RouterOS	`/system sup-output name=supout.rif`	`/ip neighbor print`
Arista EOS	`show tech-support`	`show lldp neighbors`
Huawei VRP	`display diagnostic-information SW-CORE-01.txt`	`display lldp neighbor brief`

Standards: IEEE 802.1AB-2016 · IEEE 802.1AB-2026 · IEEE 802.1AB-2016-MIB · IANA-IEEE802 · SFF-8472 · DRAFT-OPSAWG-PCAP-09

MODERN CLI

Troubleshooting toolkit

Reference platform: Extreme Switch Engine 33.7.1 · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`show tech-support logto file` `upload debug 192.0.2.20 vr VR-Mgmt`
Recent logs	`show log severity warning`
LLDP neighbors	`show lldp neighbors`
CPU and memory	`show cpu-monitoring` `show memory`
Packet capture	*n/a: No local capture: mirror a port to an analyzer*
Optics: power levels (DOM)	`show ports 24 transceiver information detail`
Cable test (TDR)	`run diagnostics cable ports 1`

Use upload debug only with Extreme Support guidance. TDR works on copper ports only and briefly disrupts traffic. DOM power can be off by +/-3 dB: confirm with a power meter.

MODERN CLI

Troubleshooting toolkit

Reference platform: Extreme 5520-48T, Fabric Engine 9.4.1.0 · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`terminal more disable` `show tech`
Recent logs	`show logging file tail`
LLDP neighbors	`show lldp neighbor`
CPU and memory	`show khi performance cpu` `show khi performance memory`
Packet capture	*n/a: No on-box capture on 5520; use port mirroring (TPVM: 5720/7x20)*
Optics: power levels (DOM)	`show pluggable-optical-modules detail`
Cable test (TDR)	`tdr-test port 1/1` `show tdr-test port 1/1`

'show tech' is an alias of 'show fulltech'; run 'terminal more disable' first. TDR affects traffic and runs on copper only; the 5520-48T supports it (not the 5520-24X or 48SE).

MODERN CLI

Troubleshooting toolkit

Reference platform: Cisco Catalyst 9300, IOS XE 17.18.4 and 26.1.x · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`show tech-support > flash:tech.txt`
Recent logs	`show logging`
LLDP neighbors	`show lldp neighbors detail`
CPU and memory	`show processes cpu sorted` `show processes memory sorted`

Packet capture	`monitor capture CAP interface GigabitEthernet1/0/1 both`
	`monitor capture CAP match any`
	`monitor capture CAP start`

Optics: power levels (DOM)	`show interfaces transceiver detail`
Cable test (TDR)	`test cable-diagnostics tdr interface GigabitEthernet1/0/1` `show cable-diagnostics tdr interface GigabitEthernet1/0/1`

Redirect show tech-support with > file to send it to TAC. EPC capture names are 8 characters at most; end with 'monitor capture CAP stop'. The transceiver command works only on DOM-capable SFPs.

MODERN CLI

Troubleshooting toolkit

Reference platform: HPE Aruba Networking CX 6300M, AOS-CX 10.16 · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`show tech local-file` `copy show-tech local-file sftp://backup@192.0.2.20/tech.txt vrf mgmt`
Recent logs	`show events -r -n 20`
LLDP neighbors	`show lldp neighbor-info`
CPU and memory	`show system resource-utilization` `top cpu`

Packet capture	`diag utilities tcpdump command -c 100 -w cap.pcap`
	`diag utilities tcpdump list-files`
	`copy tcpdump-pcap cap.pcap sftp://backup@192.0.2.20/cap.pcap`

Optics: power levels (DOM)	`show interface transceiver detail`
Cable test (TDR)	`diag cable-diagnostic test 1/1/1` `diag cable-diagnostic show 1/1/1`

tcpdump captures only traffic mirrored to the CPU, not transit traffic. The cable test drops the port's link for several seconds.

MODERN CLI

Troubleshooting toolkit

Reference platform: EX2300/EX3400/EX4100/EX4400 switches, Junos OS 23.4R2-Sx · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`request support information | save /var/tmp/rsi.txt`
Recent logs	`show log messages | last 50`
LLDP neighbors	`show lldp neighbors`
CPU and memory	`show chassis routing-engine`
Packet capture	`monitor traffic interface irb.10 no-resolve count 20`
Optics: power levels (DOM)	`show interfaces diagnostics optics ge-0/0/46`
Cable test (TDR)	`request diagnostics tdr start interface ge-0/0/0` `show diagnostics tdr interface ge-0/0/0`

monitor traffic captures only Routing Engine traffic; use mirroring for transit traffic. TDR is listed for EX2300, 3400, 4100 and 4400, RJ-45 ports only. Save RSI output to a file with '| save'.

MODERN CLI

Troubleshooting toolkit

Reference platform: MikroTik CRS354-48G-4S+2Q+RM, RouterOS 7.23.8 · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`/system sup-output name=supout.rif`
Recent logs	`/log print`
LLDP neighbors	`/ip neighbor print`
CPU and memory	`/system resource print` `/system resource cpu print`

Packet capture	`/tool sniffer start interface=ether1`
	`/tool sniffer stop`
	`/tool sniffer save file-name=cap1.pcap`

Optics: power levels (DOM)	`/interface ethernet monitor sfp-sfpplus1`
Cable test (TDR)	*n/a: cable-test is not listed for CRS3xx in the docs' device list*

The sniffer does not see traffic of a hardware-offloaded bridge (flooded frames only). SFP DOM comes from /interface ethernet monitor. cable-test is not in the supported-device list for CRS3xx.

MODERN CLI

Troubleshooting toolkit

Reference platform: Arista EOS 4.36.x · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`show tech-support`
Recent logs	`show logging`
LLDP neighbors	`show lldp neighbors`
CPU and memory	`show processes top once`
Packet capture	`monitor session CAP source ethernet 1 rx` `monitor session CAP destination cpu`
Optics: power levels (DOM)	`show interfaces transceiver dom`
Cable test (TDR)	*n/a: Only in a login-only TOI (4.27.2F); no command in the public manual*

Local capture: the 'cpu' destination creates the mirror0 interface ('Cpu : active (mirror0)'); from the shell, 'bash tcpdump -i mirror0' captures (manual example). Traffic mirrored to the CPU is rate limited.

MODERN CLI

Troubleshooting toolkit

Reference platform: Huawei S5700/S6700 campus switches running VRP, VRP V200R025C00 · Modern twin of: Module 04 troubleshooting concepts

Collect tech-support	`display diagnostic-information SW-CORE-01.txt`
Recent logs	`display logbuffer size 50`
LLDP neighbors	`display lldp neighbor brief`
CPU and memory	`display cpu-usage` `display memory-usage`
Packet capture	`capture-packet interface GigabitEthernet0/0/1 destination terminal` `capture-packet stop`
Optics: power levels (DOM)	`display transceiver diagnosis interface GigabitEthernet0/0/24`
Cable test (TDR)	`interface GigabitEthernet0/0/1` ` virtual-cable-test`

With a file name (.txt, flash:/), diagnostic-information is not shown on screen and loads the CPU. The capture is not saved in the config (default 64 kbit/s limit). The cable test briefly stops service on the port.

MODERN CLI

Junos: EX switches vs MX routers

Block B21
Access control lists (ACL)

A Layer 2 filter on MX uses 'family bridge' (matches ip-source-address, ip-protocol, destination-port) and is applied on the port (family bridge filter input) or on the bridge domain (forwarding-options filter input, as in the virtual-switch example).

```
edit firewall family bridge filter MGMT-PROTECT term
SSH
 set from destination-port 22
set interfaces ge-0/0/0 unit 0 family bridge filter
input MGMT-PROTECT
```

Block B25
Multicast: IGMP snooping and PIM

On MX IGMP snooping lives in the bridge-domain (protocols igmp-snooping) and the MX syntax has no l2-querier. With interface-mode trunk plus an irb in one bridge domain, snooping does not work (commit passes with a message); Juniper uses encapsulation vlan-bridge.

```
set bridge-domains USERS protocols igmp-
snooping
show igmp snooping membership bridge-domain
USERS
```

NEW

Before opening a case: what to collect

The course's practical list; the vendor or the [Partner] may ask for more

- Identification: model, serial number, software version and licenses
- Support output (tech-support) and current configuration, without passwords or secrets
- Logs with correct date, time and zone, covering before and after the event
- Topology: diagram, links, VLANs, routing and what changed before the problem
- Exact time of the event, frequency and impact on the [Client] users
- Steps to reproduce and what was already tested (a screen recording helps)
- Valid support (contract) and technical contact for the [Partner] or the TAC

LEGACY 2007

Data collection on the Enterasys CLI

The C3 commands and the Matrix DFE show support

Historical platform (2007): SecureStack C3 (2007) e Matrix DFE

Hardware and firmware version	`show version`
System hardware configuration	`show system hardware`
CPU, memory and processes	`show system utilization`
System clock time	`show time`
SNMP counters	`show snmp counters`
Last 256 log messages	`show logging buffer`
Current configuration	`show config`
Matrix DFE: everything at once, to a file	`show support slot1/support3.txt`

LAB

Final lab A: redundant core

VLANs, LACP, RSTP, VRRP and OSPF in a single topology

1. Create VLANs 10 (USERS), 20 (VOICE) and 99 (MGMT) on the three switches.
2. Trunk on port 24 of each core: VLANs 10, 20 and 99 tagged; native VLAN 999 (unused).
3. Form the LAG with LACP on ports 47 and 48 between the cores, with the same settings on both sides.
4. Enable RSTP on the three switches; SW-CORE-01 with the lowest priority (root) and SW-CORE-02 just above.
5. Create the VLAN 10 interfaces: 10.1.10.2/24 on SW-CORE-01 and 10.1.10.3/24 on SW-CORE-02.
6. VRRP VRID 10, virtual IP 10.1.10.1, priorities 110 and 100, preemption on, 1 s advertisement.
7. OSPF in area 0 with the same key on the uplink links; VLAN 10 passive; check the neighbors in Full.
8. From the VLAN 10 host (gateway 10.1.10.1), run a continuous ping beyond the uplink; cut SW-CORE-01's link and measure.
9. Save each switch configuration named after the lab and the bench number.

Goal

Build the redundant core: VRRP gateway on VLAN 10, OSPF to the uplink, a LAG between the cores and RSTP with a defined root.

Topology

SW-CORE-01 (10.1.10.2, RID 10.255.0.1, link 10.0.12.0/30) and SW-CORE-02 (10.1.10.3, RID 10.255.0.2, link 10.0.13.0/30), LAG on ports 47 and 48, SW-ACC-01 trunked to each core.

Verification

The LAG is up, SW-CORE-01 is the root and SW-CORE-02 is the VRRP Backup.
OSPF neighbors in Full on both uplink links; the host reaches 172.16.0.0/16.
With the link cut, the switchover takes about 3.6 s; with one LAG cable removed, the ping does not drop.

LAB

Final lab B: protection and management

ACL on VLAN 99, time, syslog, SNMPv3 and RADIUS on top of lab A's core

1. Synchronize the three switches with NTP 192.0.2.123; time zone America/Sao_Paulo (UTC-03:00, no daylight saving).
2. ACL MGMT-PROTECT: permit tcp 10.1.10.0/24 to 10.1.99.0/24 port 22; deny the rest to VLAN 99; permit everything else.
3. Apply the ACL inbound on the VLAN 10 interface and check that the rule order is the right one.
4. Send syslog from the three switches to 192.0.2.50 at the informational level.
5. SNMPv3: read-only group NMS-RO, user nms with SHA and AES-128 (authPriv), notifications to 192.0.2.60.
6. Register RADIUS 192.0.2.10 with the secret [secret]; validate a test user before requiring RADIUS.
7. Failure 1: from the VLAN 10 host, test SSH and ping to SW-CORE-01 on VLAN 99 (10.1.99.2) and read the ACL counters, where available.
8. Failures 2 and 3: SNMPv3 query with a wrong password; then take RADIUS access down and observe the login.
9. Save each switch configuration named after the lab and the bench number.

Goal

Protect VLAN 99 with an ACL and give the core correct time, central logging, SNMPv3 monitoring and a management login validated by RADIUS.

Topology

Lab A's topology, plus NTP 192.0.2.123, syslog 192.0.2.50, SNMP receiver 192.0.2.60 and RADIUS 192.0.2.10 on the management network (192.0.2.0/24).

Verification

SSH from VLAN 10 to VLAN 99 works, the ping is denied by rule 20 and the log shows the event with the right time.
The SNMPv3 query answers with the right credentials and fails with a wrong password.
The management login only succeeds with a RADIUS user; without the server, you know what the device does.

NEW

Final assessment (1/3): Ethernet, switching, STP and addressing

The same questions as at the start of the course: compare with the initial result

1 **According to RFC 894, what is the maximum size of an IP datagram in an Ethernet frame?**

- **A** 46 octets
- **B** 1500 octets
- **C** 1518 octets
- **D** 9000 octets

2 **A switch receives a unicast frame whose destination MAC is not in its address table. What does it do?**

- **A** Drops the frame
- **B** Floods the frame, as it would a broadcast
- **C** Sends the frame to the default gateway
- **D** Sends an ARP request and waits for the reply

3 **In a Spanning Tree domain, which switch becomes the root bridge?**

- **A** The one that has been up the longest
- **B** The one with the most active ports
- **C** The highest numerical priority value
- **D** Lowest priority value; on a tie, the lowest MAC

4 **Which of these addresses is NOT in the RFC 1918 private ranges?**

- **A** 10.255.255.254
- **B** 172.31.0.1
- **C** 172.32.0.1
- **D** 192.168.200.1

NEW

Final assessment (2/3): addressing, routing and management

Second half of the questions from the start of the course

5 **How many usable host addresses does the 10.1.10.0/24 network have?**

- **A** 256
- **B** 255
- **C** 254
- **D** 253

6 **The table has 10.0.0.0/8 via R1, 10.1.0.0/16 via R2, 10.1.10.0/24 via R3 and 0.0.0.0/0 via R4. Where does a packet to 10.1.10.77 go?**

- **A** R1
- **B** R2
- **C** R3
- **D** R4

7 **The TTL of a unicast IP packet reaches zero in a router. What must the router do?**

- **A** Forward it anyway, one last time
- **B** Discard; send ICMP Time Exceeded (type 11) to the source
- **C** Always discard silently
- **D** Return the packet to the previous hop

8 **What does SSH (RFC 4253) provide for remote switch management?**

- **A** Encryption, server authentication and integrity protection
- **B** Only traffic compression
- **C** Automatic neighbor discovery
- **D** Automatic configuration backup

NEW

Final assessment (3/3): OSPF, VRRP, ACL and SNMPv3

Four new questions about this module's topics

9 **On an Ethernet LAN with a DR and a BDR elected, two OSPF routers that are neither DR nor BDR reach which state with each other?**

- **A** Full
- **B** 2-Way
- **C** ExStart
- **D** Down

10 **SW-CORE-01 (priority 110) fails and comes back; SW-CORE-02 (100) is Active and preemption is on. What happens?**

- **A** SW-CORE-02 stays Active until it is shut down
- **B** SW-CORE-01 becomes Active again
- **C** Both become Active at the same time
- **D** The higher IP wins: SW-CORE-02 (10.1.10.3)

11 **The ACL has: 10 permits tcp 10.1.10.0/24 to 10.1.99.0/24 port 22; 20 denies ip to 10.1.99.0/24; 30 permits ip any any. What about a ping from 10.1.10.50 to 10.1.99.2?**

- **A** Permitted by rule 10
- **B** Denied by rule 20
- **C** Permitted by rule 30
- **D** Denied by the implicit deny at the end

12 **The SNMPv3 user nms authenticates with SHA and encrypts with AES-128. What is its security level?**

- **A** noAuthNoPriv
- **B** authNoPriv
- **C** authPriv
- **D** v2c with a community

Satisfaction survey

Your opinion guides the next edition of the course

- Mark one option per row
- Free comments and suggestions are welcome at the end

Criterion	Poor	Fair	Good	Excellent
Instructor	[ ]	[ ]	[ ]	[ ]
Course material	[ ]	[ ]	[ ]	[ ]
Overall rating of the course	[ ]	[ ]	[ ]	[ ]



Closing

Thank you! What we take from the day and the next steps

- Redundancy: OSPF and VRRP
- Security: ACLs and AAA with RADIUS
- Management: syslog and SNMPv3
- Multicast: IGMP, snooping and PIM-SM
- Troubleshooting and how to open a case